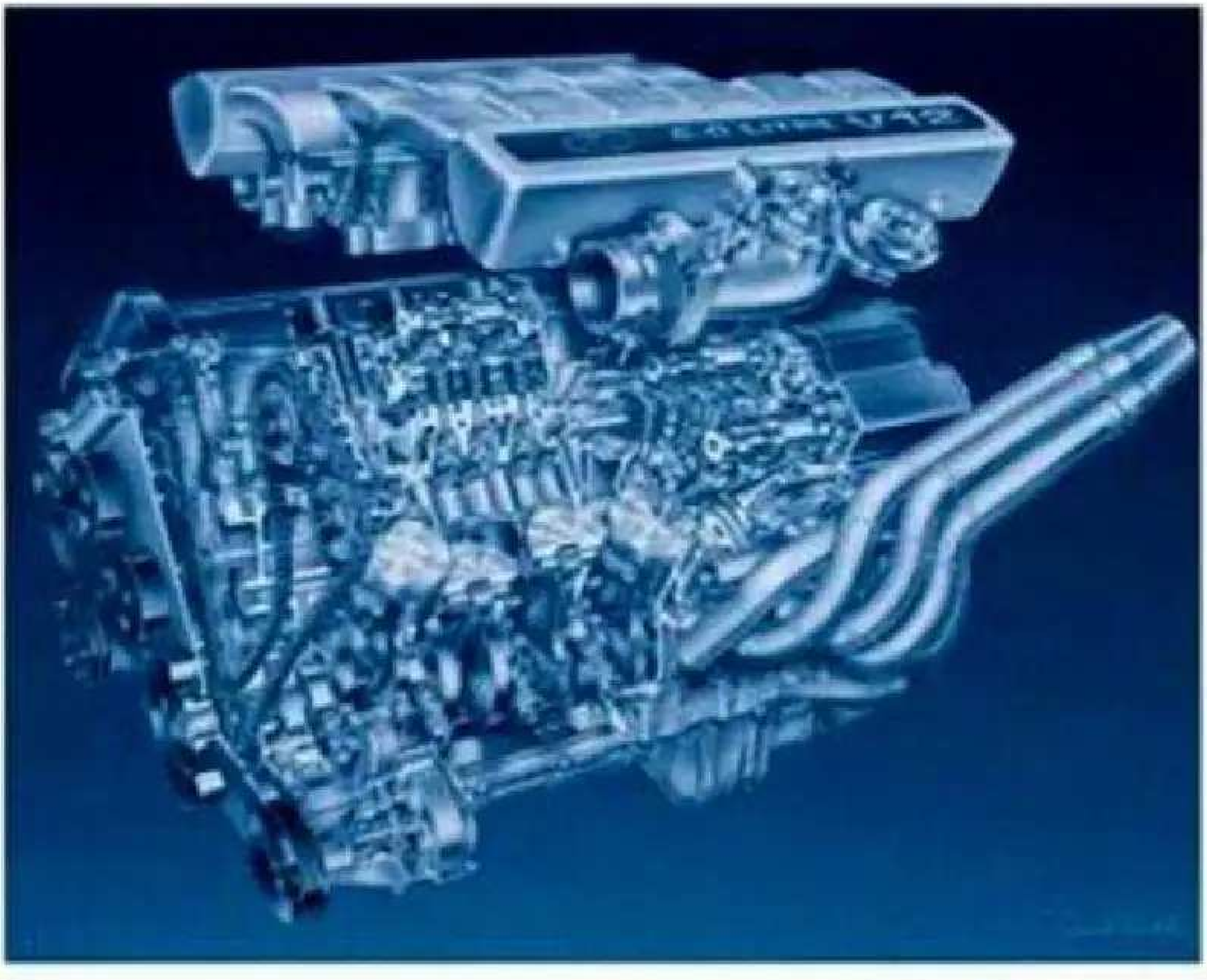
Engineering Fundamentals of the Internal Combustion Engine

Second Edition



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Contents

Preface

This book was written to be used as an applied thermoscience textbook in a one-semester, college-level, undergraduate engineering course on internal combustion engines. It provides the material needed for a basic understanding of the operation of internal combustion engines. Students are assumed to have knowledge of fundamental thermodynamics, heat transfer, and fluid mechanics as a prerequisite to get maximum benefit from the text. This book can also be used for self-study and/or as a reference book in the field of engines.

Contents include the fundamentals of most types of internal combustion engines, with a major emphasis on reciprocating engines. Both spark ignition and compression ignition engines are covered, as are those operating on four-stroke and two-stroke cycles, and ranging in size from small model airplane engines to the largest stationary engines. Rocket engines and jet engines are not included. Because of the large number of engines that are used in automobiles and other vehicles, a major emphasis is placed on these.

The book is divided into eleven chapters. Chapters 1 and 2 give an introduction, terminology, definitions, and basic operating characteristics. This is followed in Chapter 3 with a detailed analysis of basic engine cycles. Chapter 4 reviews fundamental thermochemistry as applied to engine operation and engine fuels. Chapters 5 through 9 follow the air-fuel charge as it passes sequentially through an engine, including intake, motion within a cylinder, combustion, exhaust, and emis-

sions. Engine heat transfer, friction, and lubrication are covered in Chapters 10 and 11. Each chapter includes solved example problems and historical notes followed by a set of unsolved review problems. Also included at the end of each chapter are open-ended problems that require limited design application. This is in keeping with the modern engineering education trend of emphasizing design. These design problems can be used as a minor weekly exercise or as a major group project. Included in the Appendix is a table of solutions to selected review problems.

Fueled by intensive commercial competition and stricter government regulations on emissions and safety, the field of engine technology is forever changing. It is difficult to stay knowledgeable of all advancements in engine design, materials, controls, and fuel development that are experienced at an ever-increasing rate. As the outline for this text evolved over the past few years, continuous changes were required as new developments occurred. Those advancements, which are covered in this book, include Miller cycle, lean burn engines, two-stroke cycle automobile engines, variable valve timing, and thermal storage. Advancements and technological changes will continue to occur, and periodic updating of this text will be required.

Information in this book represents an accumulation of general material collected by the author over a period of years while teaching courses and working in research and development in the field of internal combustion engines at the Mechanical Engineering Department of the University of Wisconsin-Platteville. During this time, information has been collected from many sources: conferences, newspapers, personal communication, books, technical periodicals, research, product literature, television, etc. This information became the basis for the outline and notes used in the teaching of a class about internal combustion engines. These class notes, in turn, have evolved into the general outline for this textbook. A list of references from the technical literature from which specific information for this book was taken is included in the Appendix in the back of the book. This list will be referred to at various points throughout the text. A reference number in brackets will refer to that numbered reference in the Appendix list.

Several references were of special importance in the development of these notes and are suggested for additional reading and more in-depth study. For keeping up with information about the latest research and development in automobile and internal combustion engine technology at about the right technical level, publications by SAE (Society of Automotive Engineers) are highly recommended; Reference [11] is particularly appropriate for this. For general information about most engine subjects, [40,58,100,116] are recommended. On certain subjects, some of these go into much greater depth than what is manageable in a one-semester course. Some of the information is slightly out of date but, overall, these are very informative references. For historical information about engines and automobiles in general, [29, 45, 97, 102] are suggested. General data, formulas, and principles of engineering thermodynamics and heat transfer are used at various places throughout this text. Most undergraduate textbooks on these subjects would supply the needed information. References [63] and [90] were used by the author.

Keeping with the trend of the world, SI units are used throughout the book, often supplemented with English units. Most research and development of engines is done using SI units, and this is found in the technical literature. However, in the non-technical consumer market, English units are still common, especially with automobiles. Horsepower, miles per gallon, and cubic inch displacement are some of the English terminology still used. Some example problems and some review problems are done with English units. A conversion table of SI and English units of common parameters used in engine work is induded in the Appendix at the back of the book.

I would like to express my gratitude to the many people who have influenced me and helped in the writing of this book. First I thank Dorothy with love for always being there, along with John, Tim, and Becky. I thank my Mechanical Engineering Department colleagues Ross Fiedler and Jerry Lolwing for their assistance on many occasions. I thank engineering students Pat Horihan and Jason Marcott for many of the computer drawings that appear in the book. I thank the people who reviewed the original book manuscript and offered helpful suggestions for additions and improvements. Although I have never met them, I am indebted to authors J. B. Heywood, C. R. Ferguson, E. F. Obert, and R. Stone. The books these men have written about internal combustion engines have certainly influenced the content of this textbook. I thank my father, who many years ago introduced me to the field of automobiles and generated a lifelong interest. I thank Earl of Capital City Auto

Electric for carrying on the tradition.

ACKNOWLEDGMENTS

The author wishes to thank and acknowledge the following organizations for permission to reproduce photographs, drawings, and tables from their publications in this text: Carnot Press, Fairbanks Morse Engine Division of Coltec Industries, Ford Motor Company, General Motors, Harley Davidson, Prentice-Hall Inc., SAE International, Th~ Combustion Institute, and Tuescher Photography.

Notation

Notation, symbols and abbreviations

...ation, symbols, and abbreviations used in this text. Common units are given in brackets, both SI and English.

A	Cross-section area of flow [cm^2] [in.^{2}]
A_c	Flow area of fuel capillary tube [mm^2] [in.^{2}]
A_{cc}	Surface area of combustion chamber [cm^2] [in.^{2}]
A_{ch}	Cylinder head surface area [cm^2] [in.^{2}]
A_{ex}	Area of exhaust valve [cm^2] [in.^{2}]
A_i	Area of intake valve [cm^2] [in.^{2}]
A_p	Piston face area of flat-faced piston, and/or cross-section area of cylinder [cm^2] [in.^{2}]
A_t	Throat area of carburetor [cm^2] [in.^{2}]
AF	Air–fuel ratio [kg$_a$/kg$_f$] [lbm$_a$/lbm$_f$]
AKI	Anti-knock index
AON	Aviation octane number
B	Cylinder bore [cm] [in.]
C	Constant
C_D	Discharge coefficient
C_{Dc}	Discharge coefficient of capillary tube

C_{Dt}	Discharge coefficient of carburetor throat
CI	Cetane index
CN	Cetane number
EGR	Exhaust gas recycle [%]
F	Force [N] [lbf]
F_f	Friction force [N] [lbf]
F_r	Force of connecting rod [N] [lbf]
F_x	Forces in the X direction [N] [lbf]
F_y	Forces in the Y direction [N] [lbf]
F_{1-2}	View factor
FA	Fuel-air ratio [kgf/kga] [lbmf/lbma]
FS	Fuel sensitivity
I	Moment of inertia [kg-m^2] [lbm-ft^2]
ID	Ignition delay [sec]
K_e	Chemical equilibrium constant
M	Molecular weight (molar mass) [kg/kgmole] [lbm/lbmmole]
MON	Motor octane number
N	Engine speed [RPM]
N	Number of moles
N_c	Number of cylinders
N_v	Moles of vapor
Nu	Nusselt number
ON	Octane number
P	Pressure [kPa] [atm] [psi]
P_a	Air pressure [kPa] [atm] [psi]
P_{ex}	Exhaust pressure [kPa] [atm] [psi]
P_{EVO}	Pressure when the exhaust valve opens [kPa] [psi]
P_f	Fuel pressure [kPa] [atm] [psi]
P_i	Intake pressure [kPa] [atm] [psi]

P_{inj} Injection pressure [kPa] [atm] [psi]
P_o Standard pressure [kPa] [atm] [psi]
P_I Pressure in carburetor throat [kPa] [atm] [psi]
P_v Vapor pressure [kPa] [atm] [psi]
Q Heat transfer [kJ] [BTU]
$\dot{Q}$ Heat transfer rate [kW] [hp] [BTU/sec]
Q_{HHV} Higher heating value [kJ/kg] [BTU/lbm]
Q_{HV} Heating value of fuel [kJ/kg] [BTU/lbm]
Q_{LHV} Lower heating value [kJ/kg] [BTU/lbm]
R Ratio of connecting rod length to crank offset
R Gas constant [kJ/kg-K] [ft-Ibf/lbm-OR] [BTU/lbm-OR]
Re Reynolds number
RON Research octane number
S Stroke length [cm] [in.]
S_g Specific gravity

SOF Soluble organic fraction
SR Swirl ratio
T Temperature [°C] [K] [°F] [°R]
T_a Temperature of air [°C] [K] [°F] [°R]
T_c Temperature of coolant [°C] [K] [°F] [°R]
T_{EVO} Temperature when the exhaust valve opens [°C] [K] [°F] [°R]
T_{ex} Temperature of exhaust [°C] [K] [°F] [°R]
T_g Temperature of gas [°C] [K] [°F] [°R]
T_i Intake temperature [°C] [K] [°F] [°R]
T_m Temperature of mixture [°C] [K] [°F] [°R]
T_{mp} Midpoint boiling temperature in °F [°F]
T_o Standard temperature [°C] [°F]
T_w Wall temperature [°C] [K] [°F] [°R]
U_p Piston speed [m/sec] [ft/sec]
$\overline{U}_p$ Average piston speed [m/sec] [ft/sec]
V Cylinder volume [L] [cm^3] [$in.^3$]
V_{BDC} Cylinder volume at bottom-dead-center [L] [cm^3] [$in.^3$]
V_c Clearance volume [L] [cm^3] [$in.^3$]
V_d Displacement volume [L] [cm^3] [$in.^3$]
V_{TDC} Cylinder volume at top-dead-center [L] [cm^3] [$in.^3$]
W Work [kJ] [ft-lbf] [BTU]
W_b Brake work [kJ] [ft-lbf] [BTU]
W_f Friction work [kJ] [ft-lbf] [BTU]
W_i Indicated work [kJ] [ft-lbf] [BTU]
$\dot{W}$ Power [kW] [hp]
$\dot{W}_b$ Brake power [kW] [hp]
$\dot{W}_c$ Power to drive compressor [kW] [hp]
$\dot{W}_f$ Friction power [kW] [hp]
$\dot{W}_i$ Indicated power [kW] [hp]
$\dot{W}_m$ Power to motor engine [kW] [hp]
$\dot{W}_{sc}$ Power to drive supercharger [kW] [hp]
$\dot{W}_t$ Turbine power [kW] [hp]

Turbine power [kW] [hp]

a Crank offset (crank radius) [cm] [in.]
c Speed of sound [m/sec] [ft/sec]
c_{ex} Speed of sound at exhaust conditions [m/sec] [ft/sec]
c_i Speed of sound at inlet conditions [m/sec] [ft/sec]
c_o Speed of sound at ambient conditions [m/sec] [ft/sec]
c_p Specific heat at constant pressure [kJ/kg-K] [BTU/lbm-°R]
c_v Specific heat at constant volume [kJ/kg-K] [BTU/lbm-°R]
d_v Valve diameter [cm] [in.]
g Acceleration of gravity [m/sec^2] [ft/sec^2]
h Height differential in fuel capillary tube [cm] [in.]
h Convection heat transfer coefficient [kW/m^2-K] [BTU/hr-ft^2-°R]
h Specific enthalpy [kJ/kg] [BTU/lbm]

h_a Specific enthalpy of air [kJ/kg] [BTU/lbm]
h_c Convection heat transfer coefficient on the coolant side [kW/m^2-K] [BTU/hr-ft^2-°R]
h_{ex} Specific enthalpy of exhaust [kJ/kg] [BTU/lbm]
h_g Convection heat transfer coefficient on the gas side [kW/m^2-K] [BTU/hr-ft^2-°R]
h_m Specific enthalpy of mixture [kJ/kg] [BTU/lbm]
h_f° Enthalpy of formation [kJ/kgmole] [BTU/lbmmole]
Δh Change in enthalpy from standard conditions [kJ/kgmole]
k Ratio of specific heats
k Thermal conductivity [kW/m-K] [BTU/hr-ft-°R]
k_g Thermal conductivity of gas [kW/m-K] [BTU/hr-ft-°R]
l Valve lift [cm] [in.]
m Mass [kg] [lbm]
m_a Mass of air [kg] [lbm]
m_{ex} Mass of exhaust [kg] [lbm]
m_f Mass of fuel [kg] [lbm]
m_m Mass of gas mixture [kg] [lbm]
m_{mi} Mass of mixture ingested [kg] [lbm]
m_{mt} Mass of mixture trapped [kg] [lbm]
m_{tc} Mass of total charge trapped [kg] [lbm]
$\dot{m}$ Mass flow rate [kg/sec] [lbm/sec]
$\dot{m}_a$ Mass flow rate of air [kg/sec] [lbm/sec]
$\dot{m}_{EGR}$ Mass flow rate of exhaust gas recycle [kg/sec] [lbm/sec]
$\dot{m}_f$ Mass flow rate of fuel [kg/sec] [lbm/sec]
$\dot{m}_i$ Mass flow into the cylinder [kg/sec] [lbm/sec]
mep Mean effective pressure [kPa] [atm] [psi]
n Number of revolutions per cycle
q Heat transfer per unit mass [kJ/kg] [BTU/lbm]
q Heat transfer per unit area [kJ/m^2] [BTU/ft^2]
$\dot{q}$ Heat transfer rate per unit mass [kW/kg] [BTU/hr-lbm]
$\dot{q}$ Heat transfer rate per unit area [kW/m^2] [BTU/hr-ft^2]
r Connecting rod length [cm] [in.]
r_c Compression ratio
r_e Expansion ratio
rh Relative humidity [%]

s	Distance between wrist pin and crankshaft axis	[cm] [in.]
t	Time	[sec]
u	Specific internal energy	[kJ/kg] [BTU/lbm]
u_t	Swirl tangential speed	[m/sec] [ft/sec]
v	Specific volume	[m^3/kg] [ft^3/lbm]
v_{BDC}	Specific volume at bottom-dead-center	[m^3/kg] [ft^3/lbm]
v_{ex}	Specific volume of exhaust	[m^3/kg] [ft^3/lbm]
v_{TDC}	Specific volume at top-dead-center	[m^3/kg] [ft^3/lbm]

W	Specific work	[kJ/kg] [ft-Ibf/lbm] [BTU/lbm]
Wb	Brake-specific work	[kJ/kg] [ft-Ibf/lbm] [BTU/lbm]
wf	Friction-specific work	[kJ/kg] [ft-Ibf/lbm] [BTU/lbm]
Wi	Indicated-specific work	[kJ/kg] [ft-Ibf/lbm] [BTU/lbm]
x	Distance	[em] [m] [in.] [ft]
Xex	Fraction of exhaust	
Xr	Exhaust residual	
Xv	Mole fraction of water vapor	
a	Pressure ratio	
a	Ratio of valve areas	
13	Cutoff ratio	
r	Angular momentum	[kg-m^2/sec] [lbm-ft^2/sec]
eg	Emissivity of gas	
ew	Emissivity of wall	
T]c	Combustion efficiency	[%]
T]f	Fuel conversion efficiency	[%]
T]m	Mechanical efficiency	[%]
T]s	Isentropic efficiency	[%]
T]t	Thermal efficiency	[%]
T]v	Volumetric efficiency of the engine	[%]
9	Crank angle measured from TDC	[0]
Ace	Charging efficiency	
Adr	Delivery ratio	
Arc	Relative charge	
Ase	Scavenging efficiency	
Ate	Trapping efficiency	
/L	Dynamic viscosity	[kg/m-sec] [lbm/ft-sec]
/.lg	Dynamic viscosity of gas	[kg/m-sec] [lbm/ft-see]
v	Stoichiometric coefficients	
P	Density	[kg/m^3] [lbm/ft^3]
Pa	Density of air	[kg/m^3] [lbm/ft^3]
Po	Density of air at standard conditions	[kg/m^3] [lbm/ft^3]
Pf	Density of fuel	[kg/m^3] [lbm/ft^3]
CJ'	Stefan-Boltzmann constant	[W/m2-K4] [BTU/hr-ft2-OR4]
T	Torque	[N-m] [lbf-ft]
Ts	Shear force per unit area	[N/m^2] [lbf/ft^2]
‹I›	Equivalence ratio	
‹I›	Angle between connecting rod and centerline of the cylinder	
W	Angular velocity of swirl	[rev/see]
W	Specific humidity	[kgv/kga] [grainsv/lbm$_a$]

1

Introduction

This chapter introduces and defines the internal combustion engine. It lists ways of classifying engines and terminology used in engine technology. Descriptions are given of many common engine components and of basic four-stroke and two-stroke cycles for both spark ignition and compression ignition engines.

1-1 INTRODUCTION

The **internal combustion engine** (Ie) is a heat engine that converts chemical energy in a fuel into mechanical energy, usually made available on a rotating output shaft. Chemical energy of the fuel is first converted to thermal energy by means of combustion or oxidation with air inside the engine. This thermal energy raises the temperature and pressure of the gases within the engine, and the high-pressure gas then expands against the mechanical mechanisms of the engine. This expansion is converted by the mechanical linkages of the engine to a rotating crankshaft, which is the output of the engine. The crankshaft, in turn, is connected to a transmission and/or power train to transmit the rotating mechanical energy to the desired final use. For engines this will often be the propulsion of a vehicle (i.e., automobile, truck, locomotive, marine vessel, or airplane). Other applications include stationary

engines to drive generators or pumps, and portable engines for things like chain saws and lawn mowers.

Most internal combustion engines are **reciprocating engines** having pistons that reciprocate back and forth in cylinders internally within the engine. This book concentrates on the thermodynamic study of this type of engine. Other types of IC engines also exist in much fewer numbers, one important one being the rotary engine [104]. These engines will be given brief coverage. Engine types not covered by this book include steam engines and gas turbine engines, which are better classified as **external combustion engines** (i.e., combustion takes place outside the mechanical engine system). Also not included in this book, but which could be classified as internal combustion engines, are rocket engines, jet engines, and firearms.

Reciprocating engines can have one cylinder or many, up to 20 or more. The cylinders can be arranged in many different geometric configurations. Sizes range from small model airplane engines with power output on the order of 100 watts to large multicylinder stationary engines that produce thousands of kilowatts per cylinder.

There are so many different engine manufacturers, past, present, and future, that produce and have produced engines which differ in size, geometry, style, and operating characteristics that no absolute limit can be stated for any range of engine characteristics (i.e., size, number of cylinders, strokes in a cycle, etc.). This book will work within normal characteristic ranges of engine geometries and operating parameters, but there can always be exceptions to these.

Early development of modern internal combustion engines occurred in the latter half of the 1800s and coincided with the development of the automobile. History records earlier examples of crude internal combustion engines and self-propelled road vehicles dating back as far as the 1600s [29]. Most of these early vehicles were steam-driven prototypes which never became practical operating vehicles. Technology, roads, materials, and fuels were not yet developed enough. Very early examples of heat engines, including both internal combustion and external combustion, used gun powder and other solid, liquid, and gaseous fuels. Major development of the modern steam engine and, consequently, the railroad locomotive occurred in the latter half of the 1700s and early 1800s. By the 1820s and 1830s, railroads were present in several countries around the world.

HISTORIC-ATMOSPHERIC ENGINES

Most of the very earliest internal combustion engines of the 17th and 18th centuries can be classified as atmospheric engines. These were large engines with a single piston and cylinder, the cylinder being open on the end. Combustion was initiated in the open cylinder using any of the various fuels which were available. Gunpowder was often used as the fuel. Immediately after combustion, the cylinder would be full of hot exhaust gas at atmospheric pressure. At this time, the cylinder end was closed and the trapped gas was allowed to cool. As the gas cooled, it cre-

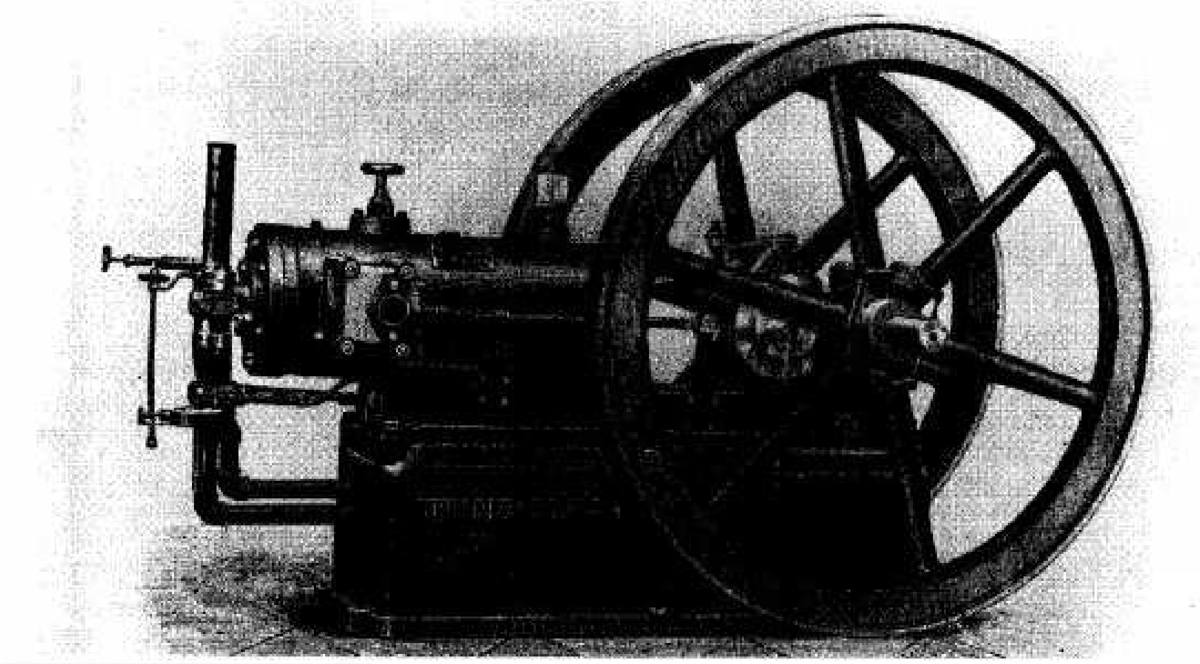


Figure 1-1 The Charter Engine made in 1893 at the Beloit works of Fairbanks, Morse & Company was one of the first successful gasoline engine offered for sale in the United States. Printed with permission, Fairbanks Morse Engine Division, Coltec Industries.

ated a vacuum within the cylinder. This caused a pressure differential acrossthe piston, atmospheric pressure on one side and a vacuum on the other. As the piston moved becauseof this pressure differential, it would do work by being connected to an external system, such as raising a weight [29].

Some early steam engines also were atmospheric engines. Instead of combustion, the open cylinder was filled with hot steam. The end was then closed and the steam was allowed to cool and condense. This created the necessary vacuum.

In addition to a great amount of experimentation and development in Europe and the United States during the middle and latter half of the 1800s, two other technological occurrences during this time stimulated the emergence of the internal combustion engine. In 1859, the discovery of crude oil in Pennsylvania finally made available the development of reliable fuels which could be used in these newly developed engines. Up to this time, the lack of good, consistent fuels was a major drawback in engine development. Fuels like whale oil, coal gas, mineral oils, coal, and gun powder which were available before this time were less than ideal for engine use and development. It still took many years before products of the petroleum industry evolved from the first crude oil to gasoline, the automobile fuel of the 20th century. However, improved hydrocarbon products began to appear as early

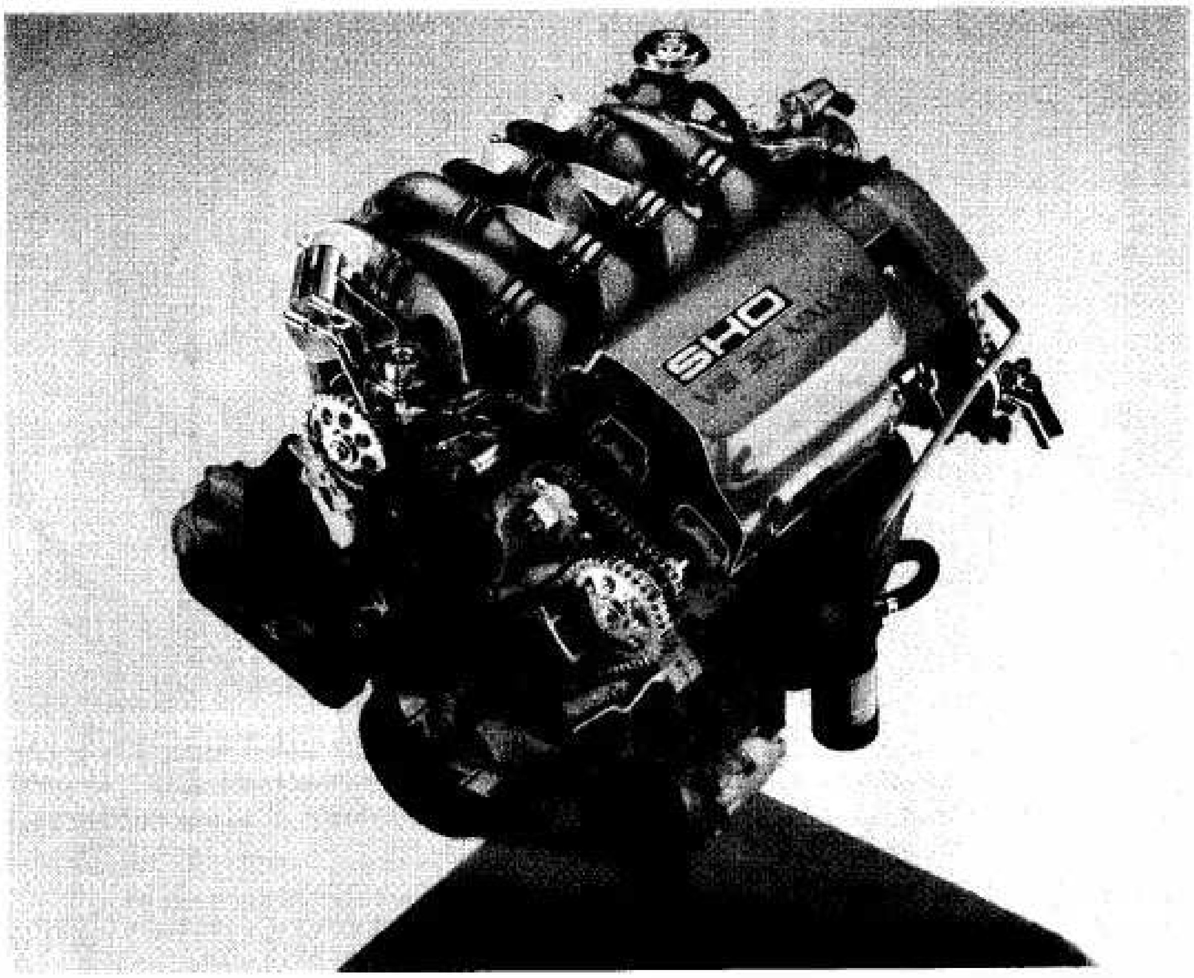


Figure 1·2 Ford Taurus SHO 3.4 liter (208 in.^{3}), spark ignition, four-stroke cycle engine. The engine is rated at 179 kW at 6500 RPM (240 hp) and develops 305 N-m of torque at 4800 RPM (225 lbf-ft). It is a 60°V8 with 8.20 cm bore (3.23 in.), 7.95 cm stroke (3.13 in.), and a compression ratio of 10:1. The engine has four chain driven camshafts mounted in aluminum heads with four valves per cylinder and coil-on-plug ignition. Each spark plug has a separate high-voltage coil and is fired by Ford's Electronic Distributorless Ignition System (EDIS). Courtesy of Ford Motor Company.

as the 1860s and gasoline, lubricating oils, and the internal combustion engine evolved together.

The second technological invention that stimulated the development of the internal combustion engine was the pneumatic rubber tire, which was first marketed by John B. Dunlop in 1888 [141]. This invention made the automobile much more practical and desirable and thus generated a large market for propulsion systems, including the internal combustion engine.

During the early years of the automobile, the internal combustion engine competed with electricity and steam engines as the basic means of propulsion. Early in the 20th century, electricity and steam faded from the automobile picture-electricity because of the limited range it provided, and steam because of the long start-up time needed. Thus, the 20th century is the period of the internal combustion engine and the automobile powered by the internal combustion engine. Now, at the end of the century, the internal combustion engine is again being challenged by electricity and other forms of propulsion systems for automobiles and other applications. What goes around comes around.

1-2 EARLY HISTORY

During the second half of the 19th century, many different styles of internal combustion engines were built and tested. Reference [29] is suggested as a good history of this period. These engines operated with variable success and dependability using many different mechanical systems and engine cycles.

The first fairly practical engine was invented by J.J.E. Lenoir (1822-1900) and appeared on the scene about 1860 (Fig. 3-19). During the next decade, several hundred of these engines were built with power up to about 4.5 kW (6 hp) and mechanical efficiency up to 5%. The Lenoir engine cycle is described in Section 3-13. In 1867 the Otto-Langen engine, with efficiency improved to about 11%, was first introduced, and several thousand of these were produced during the next decade. This was a type of atmospheric engine with the power stroke propelled by atmospheric pressure acting against a vacuum. Nicolaus A. Otto (1832-1891) and Eugen Langen (1833-1895) were two of many engine inventors of this period.

During this time, engines operating on the same basic four-stroke cycle as the modern automobile engine began to evolve as the best design. Although many people were working on four-stroke cycle design, Otto was given credit when his prototype engine was built in 1876.

In the 1880s the internal combustion engine first appeared in automobiles [45]. Also in this decade the two-stroke cycle engine became practical and was manufactured in large numbers.

By 1892, Rudolf Diesel (1858-1913) had perfected his compression ignition engine into basically the same diesel engine known today. This was after years of development work which included the use of solid fuel in his early experimental engines. Early compression ignition engines were noisy, large, slow, single-cylinder engines. They were, however, generally more efficient than spark ignition engines. It wasn't until the 1920s that multicylinder compression ignition engines were made small enough to be used with automobiles and trucks.

1-3 ENGINE CLASSIFICATIONS

Internal combustion engines can be classified in a number of different ways:

1. Types of Ignition

(a) Spark Ignition (SI). An SI engine starts the combustion process in each cycle by use of a spark plug. The spark plug gives a high-voltage electrical

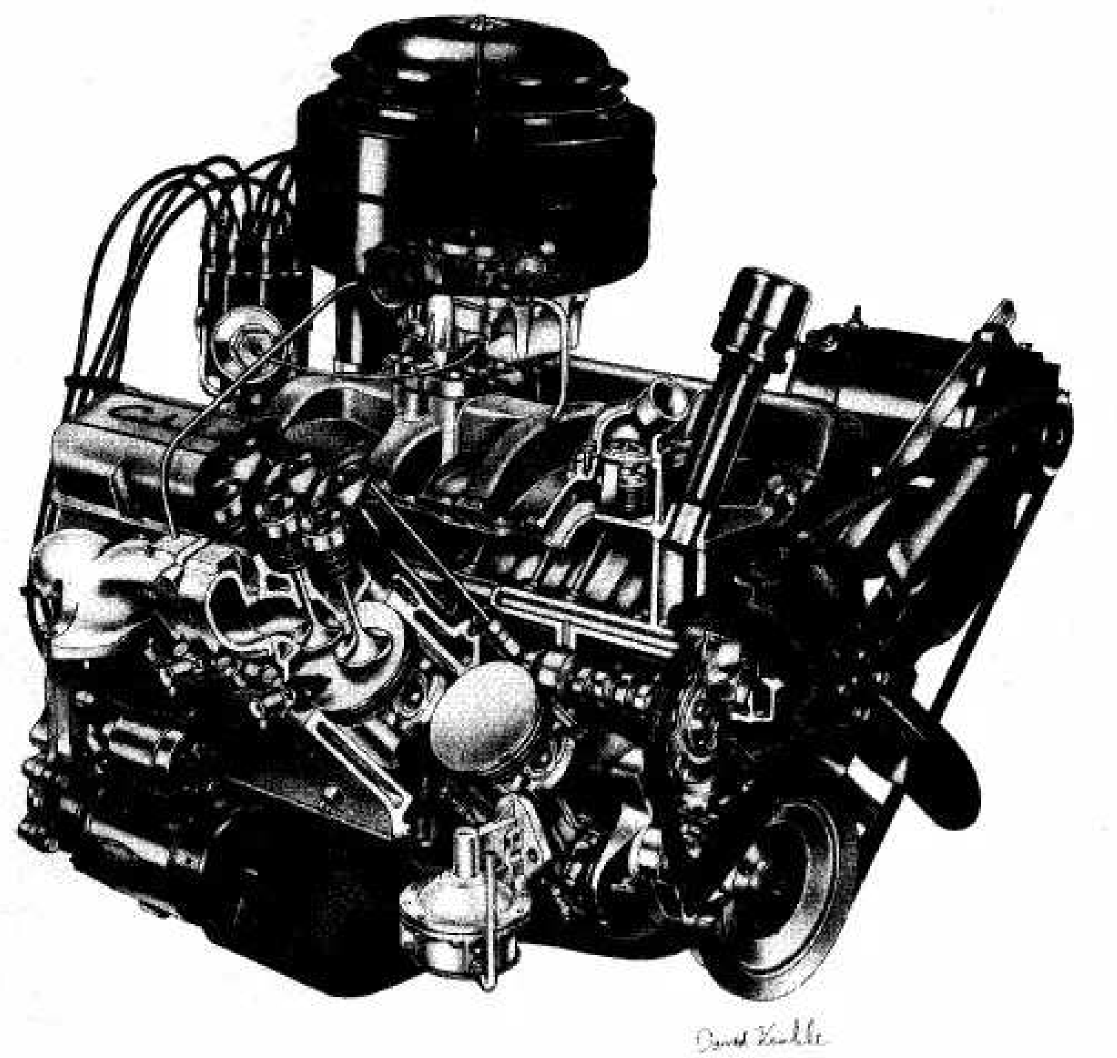


Figure 1-3 1955 Chevrolet "small block" V8 engine with 265 $in.^3$ (4.34 L) displacement. The four-stroke cycle, spark ignition engine was equipped with a carburetor and overhead valves. Copyright General Motors Corp., used with permission.

discharge between two electrodes which ignites the air-fuel mixture in the combustion chamber surrounding the plug. In early engine development, before the invention of the electric spark plug, many forms of torch holes were used to initiate combustion from an external flame.

(b) Compression Ignition (CI). The combustion process in a CI engine starts when the air-fuel mixture self-ignites due to high temperature in the combustion chamber caused by high compression.

2. Engine Cycle

(a) Four-Stroke Cycle. A four-stroke cycle experiences four piston movements over two engine revolutions for each cycle.

(b) Two-Stroke Cycle. A two-stroke cycle has two piston movements over one revolution for each cycle.

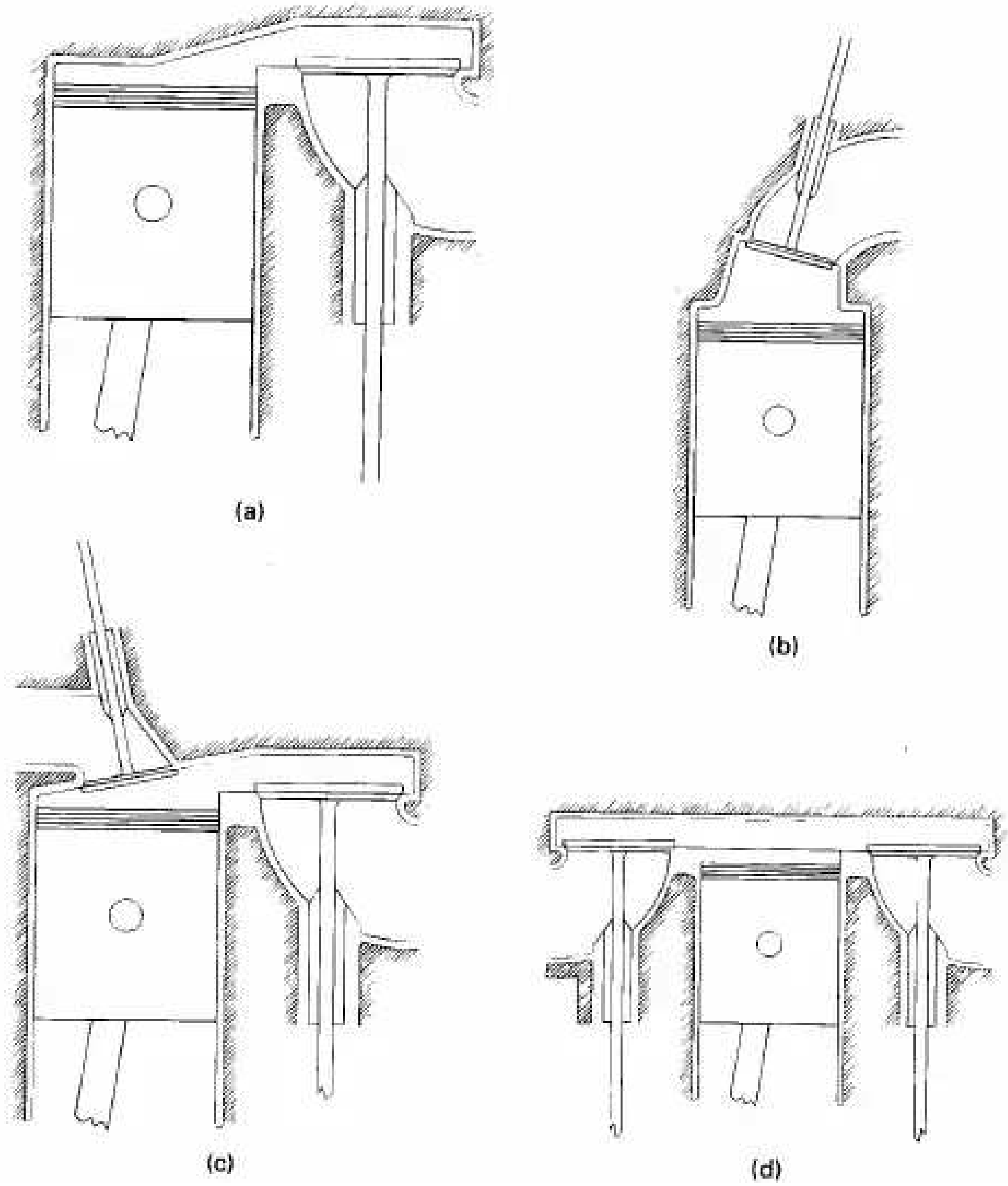


Figure 1-4 Engine Classification by Valve Location. **(a)** Valve in block, L head. Older automobiles and some small engines. **(b)** Valve in head, I head. Standard on modern automobiles. **(c)** One valve in head and one valve in block, F head. Older, less common automobiles. **(d)** Valves in block on opposite sides of cylinder, T head. Some historic automobile engines.

Three-stroke cycles and six-stroke cycles were also tried in early engine development [29].

3. Valve Location (see Fig. 1-4)

(a) Valves in head (overhead valve), also called **I Head engine.**

(b) Valves in block (flat head), also called **L Head engine.** Some historic engines with valves in block had the intake valve on one side of the cylinder and the exhaust valve on the other side. These were called **T Head engines.**

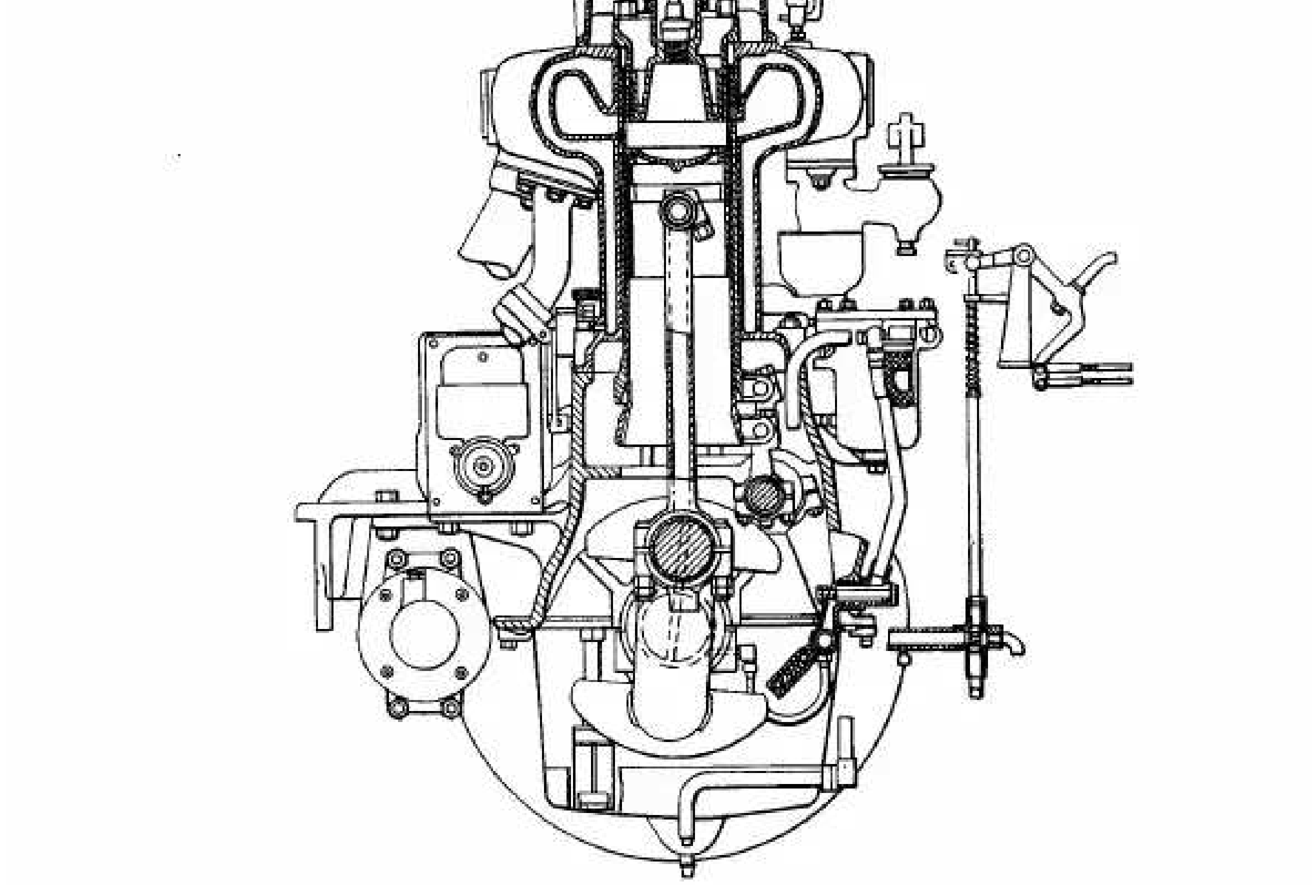


Figure 1-5 Sectional view of Willy-Knight sleeve valve engine of 1926. Reprinted with permission from © 1995 Automotive Engineering magazine. Society of Automotive Engineers, Inc.

(c) One valve in head (usually intake) and one in block, also called **F Head engine;** this is much less common.

4. Basic Design
   (a) **Reciprocating.** Engine has one or more cylinders in which pistons reciprocate back and forth. The combustion chamber is located in the closed end of each cylinder. Power is delivered to a rotating output crankshaft by mechanical linkage with the pistons.

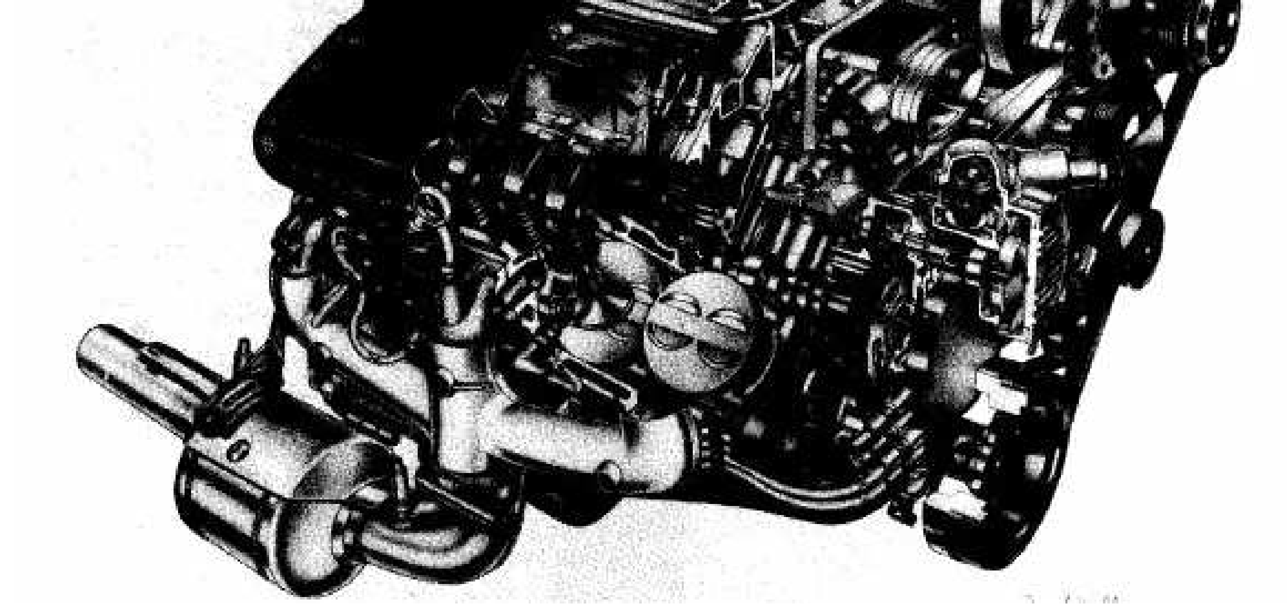


Figure 1·6 Chevrolet LT4 V8, four-stroke cycle, spark ignition engine with 5.7liter displacement. This fuel-injected, overhead valve engine was an option in the 1986 Corvette. Copyright General Motors Corp., used with permission.

(b) **Rotary.** Engine is made of a block (stator) built around a large non-concentric rotor and crankshaft. The combustion chambers are built into the nonrotating block.

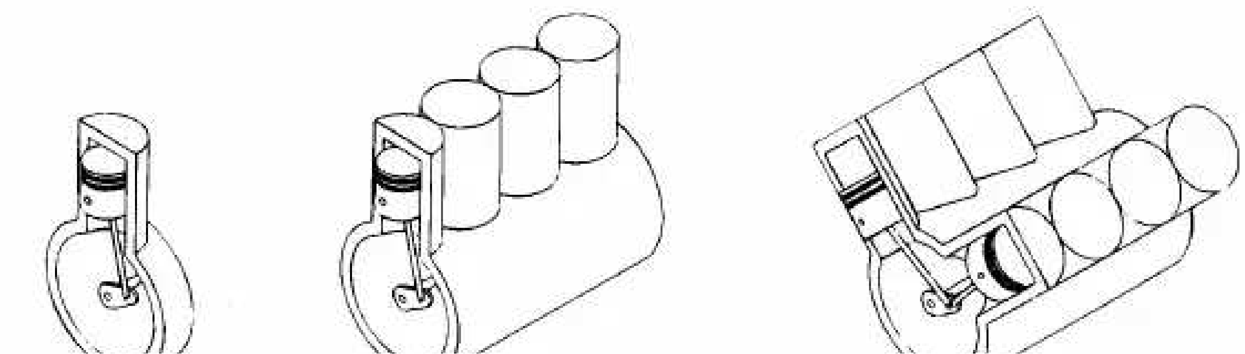
5. Position and Number of Cylinders of Reciprocating Engines (Fig. 1-7)

(a) **Single Cylinder.** Engine has one cylinder and piston connected to the crankshaft.

(b) **In-Line.** Cylinders are positioned in a straight line, one behind the other along the length of the crankshaft. They can consist of 2 to 11 cylinders or possibly more. In-line four-cylinder engines are very common for automobile and other applications. In-line six and eight cylinders are historically common automobile engines. In-line engines are sometimes called **straight** (e.g., straight six or straight eight).

(c) **V Engine.** Two banks of cylinders at an angle with each other along a single crankshaft. The angle between the banks of cylinders can be anywhere from 15° to 120°, with 60°-90° being common. V engines have even numbers of cylinders from 2 to 20 or more. V6s and V8s are common automobile engines, with V12s and V16s (historic) found in some luxury and high-performance vehicles.

(d) **Opposed Cylinder Engine.** Two banks of cylinders opposite each other on a single crankshaft (a V engine with a 180°V). These are common on small



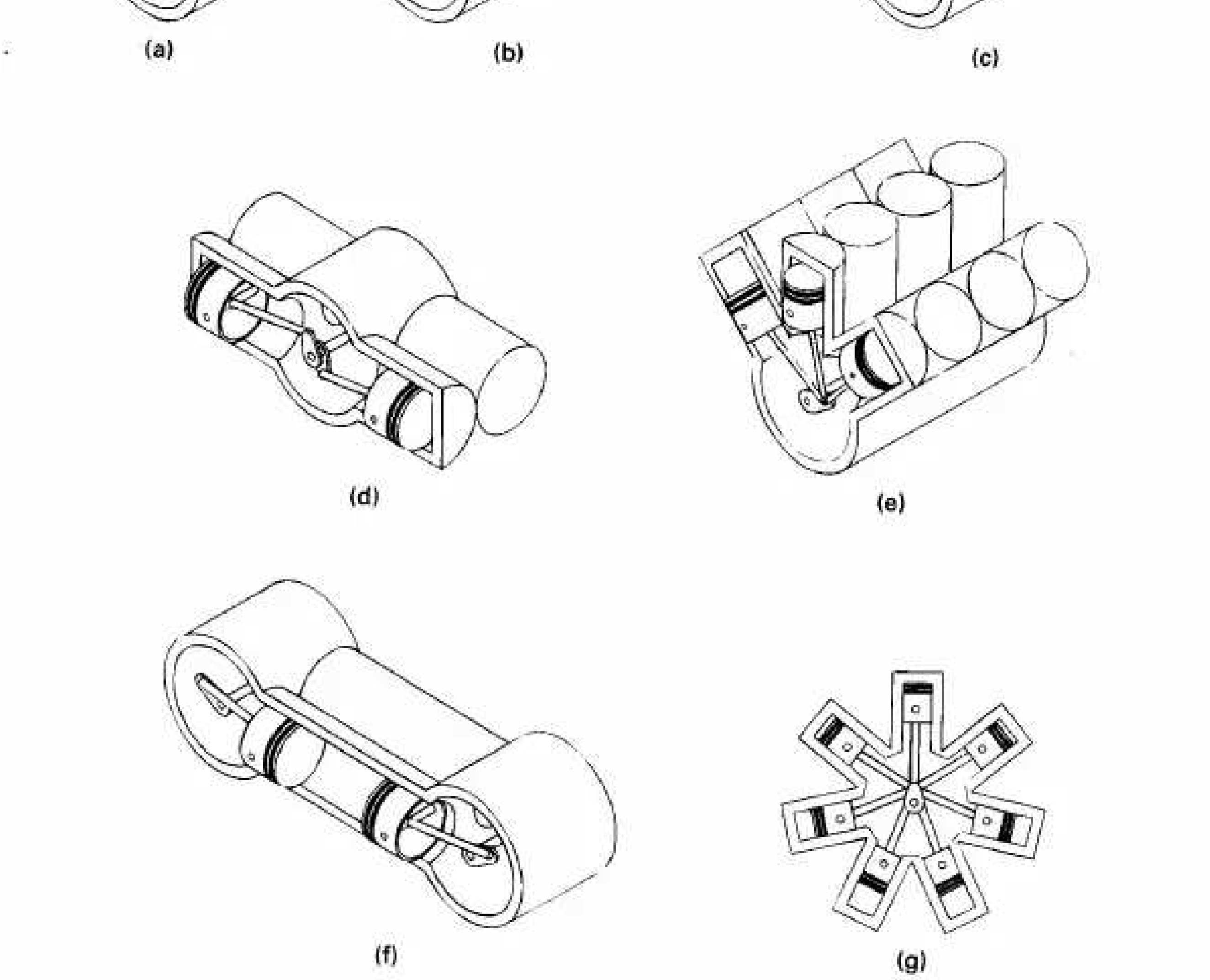


Figure 1-7 Engine Classification by Cylinder Arrangement. **(a)** Single cylinder. **(b)** In-line, or straight. **(c)** V engine. **(d)** Opposed cylinder. **(e)** W engine. **(f)** Opposed piston. **(g)** Radial.

aircraft and some automobiles with an even number of cylinders from two to eight or more. These engines are often called flat engines (e.g., flat four).

(e) W Engine. Same as a V engine except with three banks of cylinders on the same crankshaft. Not common, but some have been developed for racing automobiles, both modern and historic. Usually 12 cylinders with about a 60° angle between each bank.

(1) Opposed Piston Engine. Two pistons in each cylinder with the combustion chamber in the center between the pistons. A single-combustion process causes two power strokes at the same time, with each piston being pushed away from the center and delivering power to a separate crankshaft at each end of the cylinder. Engine output is either on two rotating crankshafts or

on one crankshaft incorporating complex mechanical linkage.

(g) Radial Engine. Engine with pistons positioned in a circular plane around the central crankshaft. The connecting rods of the pistons are connected to a master rod which, in turn, is connected to the crankshaft. A bank of cylinders on a radial engine always has an odd number of cylinders ranging from 3 to 13 or more. Operating on a four-stroke cycle, every other cylinder fires and has a power stroke as the crankshaft rotates, giving a smooth operation. Many medium- and large-size propeller-driven aircraft use radial engines. For large aircraft, two or more banks of cylinders are mounted together, one behind the other on a single crankshaft, making one powerful, smooth engine. Very large ship engines exist with up to 54 cylinders, six banks of 9 cylinders each.

HISTORIC-RADIAL ENGINES

There are at least two historic examples of radial engines being mounted with the crankshaft fastened to the vehicle while the heavy bank of radial cylinders rotated around the stationary crank. The Sopwith Camel, a very successful World War I fighter aircraft, had the engine so mounted with the propeller fastened to the rotating bank of cylinders. The gyroscopic forces generated by the large rotating engine mass allowed these planes to do some maneuvers which were not possible with other airplanes, and restricted them from some other maneuvers. Snoopy has been flying a Sopwith Camel in his battles with the Red Baron for many years.

The little-known early Adams-Farwell automobiles had three- and five-cylinder radial engines rotating in a horizontal plane with the stationary crankshaft mounted vertically. The gyroscopic effects must have given these automobiles very unique steering characteristics [45].

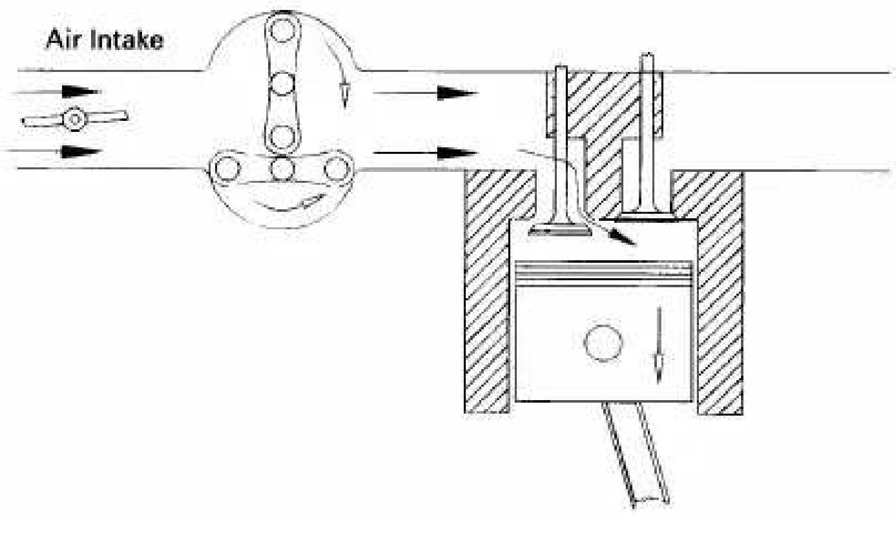


Figure 1-8 Supercharger used to increase inlet air pressure to engine. Compressor is driven off engine crankshaft, which gives fast response to speed changes but adds parasitic load to engine.

6. Air Intake Process
   (a) Naturally Aspirated. No intake air pressure boost system.
   (b) Supercharged. Intake air pressure increased with the compressor driven off of the engine crankshaft (Fig. 1-8).
   (c) Turbocharged. Intake air pressure increased with the turbine-compressor driven by the engine exhaust gases (Fig. 1-9).
   (d) Crankcase Compressed. Two-stroke cycle engine which uses the crankcase as the intake air compressor. Limited development work has also been done on design and construction of four-stroke cycle engines with crankcase compression.

7. Method of Fuel Input for SI Engines
   (a) Carbureted.
   (b) Multipoint Port Fuel Injection. One or more injectors at each cylinder intake.
   (c) Throttle Body Fuel Injection. Injectors upstream in intake manifold.

8. Fuel Used
   (a) Gasoline.
   (b) Diesel Oil or Fuel Oil.
   (c) Gas, Natural Gas, Methane.
   (d) LPG.
   (e) Alcohol-Ethyl, Methyl.
   (f) Dual Fuel. There are a number of engines that use a combination of two or more fuels. Some, usually large, CI engines use a combination of methane and diesel fuel. These are attractive in developing third-world countries because of the high cost of diesel fuel. Combined gasoline-alcohol fuels

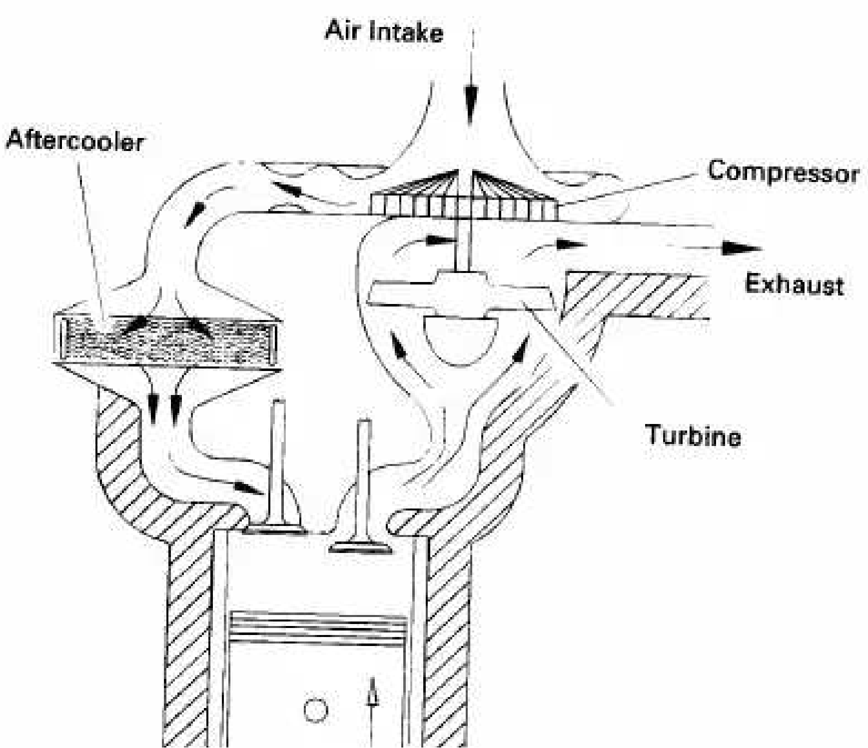


Figure 1-9 Turbocharger used to increase inlet air pressure to engine. Turbine that drives compressor is powered by exhaust flow from engine. This adds no load to the engine but results in turbo lag, a slower response to engine speed changes.

are becoming more common as an alternative to straight gasoline automobile engine fuel.

(g) **Gasohol.** Common fuel consisting of 90% gasoline and 10% alcohol.

9. Application
- **(a) Automobile, Truck, Bus.**
- **(b) Locomotive.**
- **(c) Stationary.**
- **(d) Marine.**
- **(e) Aircraft.**
- **(f) Small Portable, Chain Saw, Model Airplane.**

10. Type of Cooling
- **(a) Air Cooled.**
- **(b) Liquid Cooled, Water Cooled.**

Several or all of these classifications can be used at the same time to identify a given engine. Thus, a modern engine might be called a turbocharged, reciprocating, spark ignition, four-stroke cycle, overhead valve, water-cooled, gasoline, multipoint fuel-injected, V8 automobile engine.

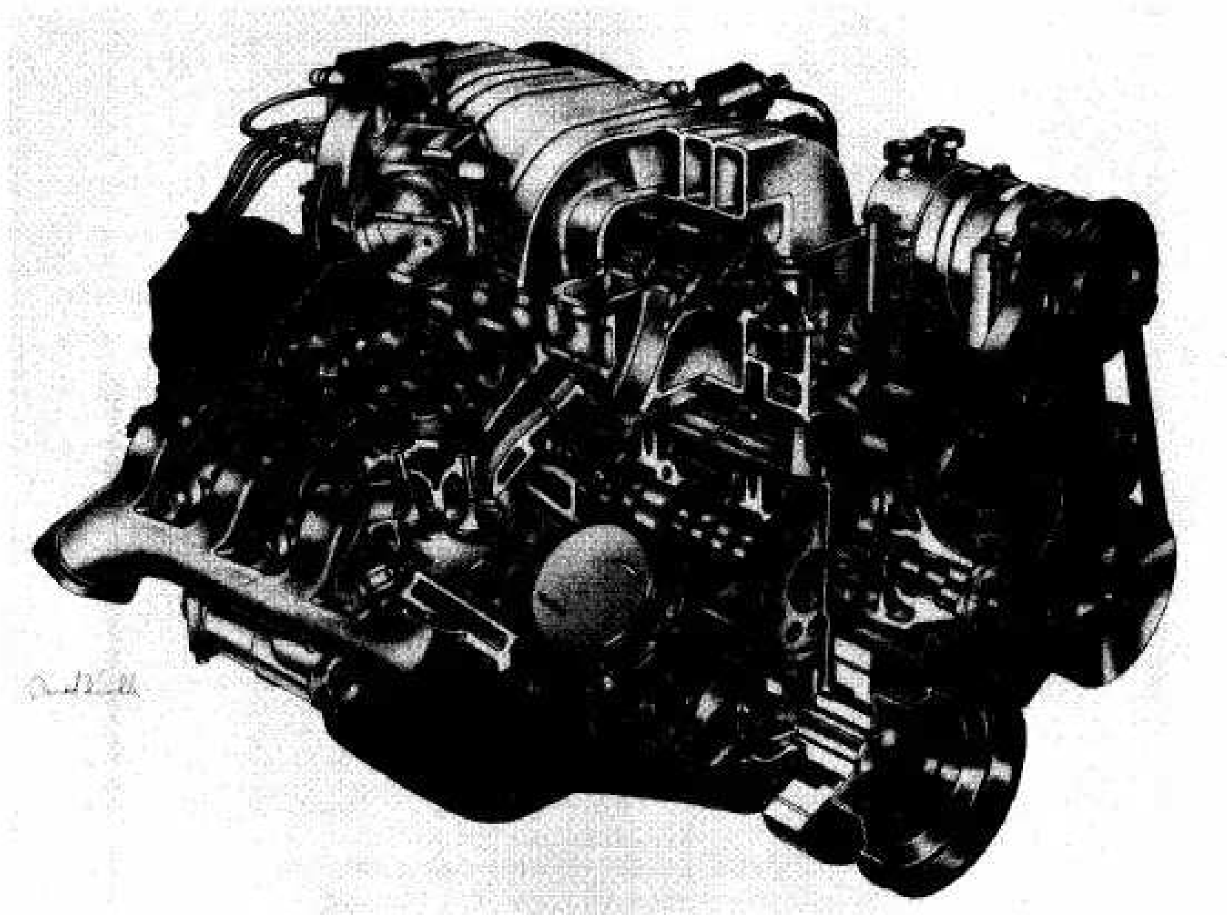


Figure 1-10 General Motors 7.4 liter L29, V8, four-stroke cycle, spark ignition,

truck engine. Displacement is 454 in.^{3} (7.44 L) with 4.25 in. bore (10.80 cm) and 4.00 in. stroke (10.16 cm). The engine has a maximum speed of 5000 RPM, with a compression ratio of 9.0: 1, overhead valves, and multipoint port fuel injection. This engine was used in several models of 1996 Chevrolet and GMC trucks. Copyright General Motors Corp., used with permission.

'-4 TERMINOLOGY AND ABBREVIATIONS

The following terms and abbreviations are commonly used in engine technology literature and will be used throughout this book. These should be learned to assure maximum understanding of the following chapters.

Internal Combustion (le)

Spark Ignition (81) An engine in which the combustion process in each cycle is started by use of a spark plug.

Compression Ignition (CI) An engine in which the combustion process starts when the air-fuel mixture self-ignites due to high temperature in the combustion chamber caused by high compression. CI engines are often called **Diesel** engines, especially in the non-technical community.

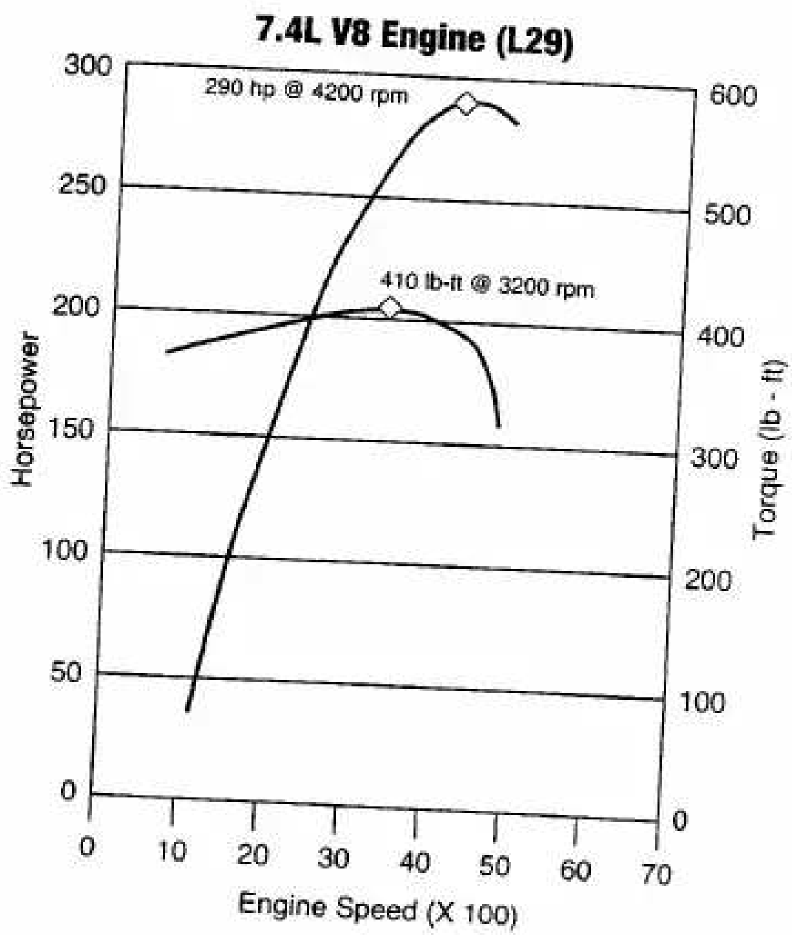


Figure 1-11 Power and torque curves of GM 7.4 liter L29 truck engine shown in Fig. 1-10. The engine has a power rating of 290 hp (216 kW) at 4200 RPM and a torque rating of 410 lbf-ft (556 N-m) at 3200 RPM. Copyright General Motors Corp., used with permission.

Top-Dead-Center (TDC) Position of the piston when it stops at the furthest point away from the crankshaft. *Top* because this position is at the top of most engines (not always), and *dead* because the piston stops at this point. Because in some engines top-dead-center is not at the top of the engine (e.g., horizontally opposed engines, radial engines, etc.), some Sources call this position **Head-End-Dead-Center** (HEDC). Some sources call this position **Top-Center** (TC). When an occurrence in a cycle happens before TDC, it is often abbreviated bTDC or bTe. When the occurrence happens after TDC, it will be abbreviated aTDC or aTe. When the piston is at TDC, the volume in the cylinder is a minimum called the *clearance volume.*

Bottom-Dead-Center (BDC) Position of the piston when it stops at the point closest to the crankshaft. Some sources call this **Crank-End-Dead-Center** (CEDC) because it is not always at the bottom of the engine. Some sources call this point **Bottom-Center** (BC). During an engine cycle things can happen before bottom-dead-center, bBDC or bBC, and after bottom-dead-center, aBDC or aBe.

Direct Injection (DI) Fuel injection into the main combustion chamber of an engine. Engines have either one main combustion chamber (open chamber)

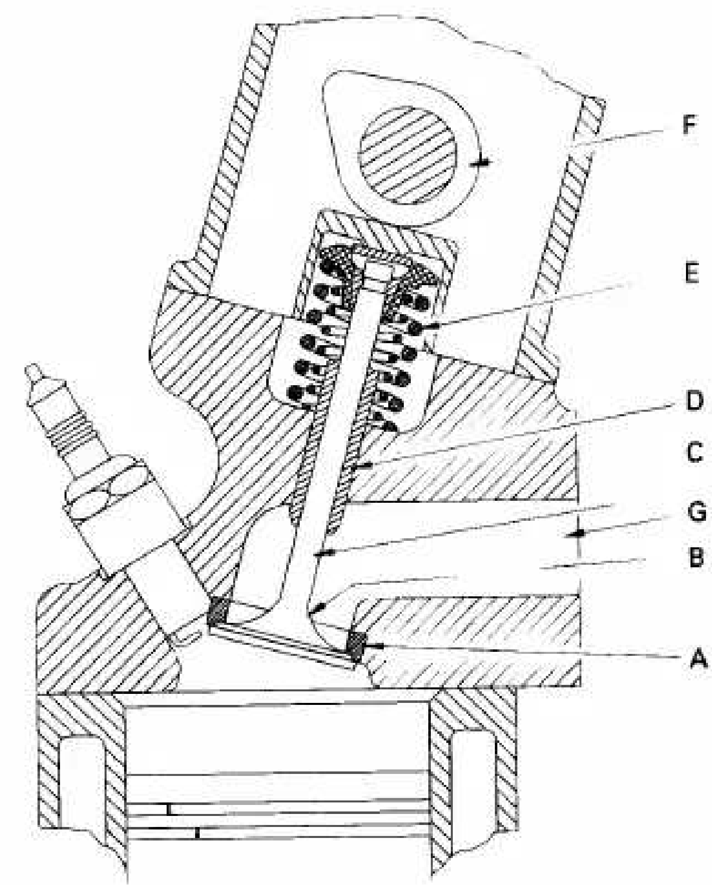


Figure 1-12 Poppet valve is spring loaded closed, and pushed open by cam action at proper time in cycle. Most automobile engines and other reciprocating engines use poppet valves. Much less common are sleeve valves and rotary valves. Components include: (A) valve seat, (B) head, (C) stem, (D) guide, (E) spring, (F) camshaft, (G) manifold.

or a divided combustion chamber made up of a main chamber and a smaller connected secondary chamber.

Indirect Injection (IDI) Fuel injection into the secondary chamber of an engine with a divided combustion chamber.

Bore Diameter of the cylinder or diameter of the piston face, which is the same minus a very small clearance.

Stroke Movement distance of the piston from one extreme position to the other

Stroke Movement distance of the piston from one extreme position to the other. TDC to BDC or BDC to TDC.

Clearance Volume Minimum volume in the combustion chamber with piston at TDC.

Displacement or Displacement Volume Volume displaced by the piston as it travels through one stroke. Displacement can be given for one cylinder or for the entire engine (one cylinder times number of cylinders). Some literature calls this *swept volume.*

Smart Engine Engine with computer controls that regulate operating characteristics such as air-fuel ratio, ignition timing, valve timing, exhaust control, intake tuning, etc. Computer inputs come from electronic, mechanical, thermal, and chemical sensors located throughout the engine. Computers in some automo-

biles are even programmed to adjust engine operation for things like valve wear and combustion chamber deposit buildup as the engine ages. In automobiles the same computers are used to make *smart cars* by controlling the steering, brakes, exhaust system, suspension, seats, anti-theft systems, sound-entertainment systems, shifting, doors, repair analysis, navigation, noise suppression, environment, comfort, etc. On some systems engine speed is adjusted at the instant when the transmission shifts gears, resulting in a smoother shifting process. At least one automobile model even adjusts this process for transmission fluid temperature to assure smooth shifting at cold startup.

Engine Management System (EMS) Computer and electronics used to control smart engines.

Wide-Open Throttle (WOT) Engine operated with throttle valve fully open when maximum power and/or speed is desired.

Ignition Delay (ID) Time interval between ignition initiation and the actual start of combustion.

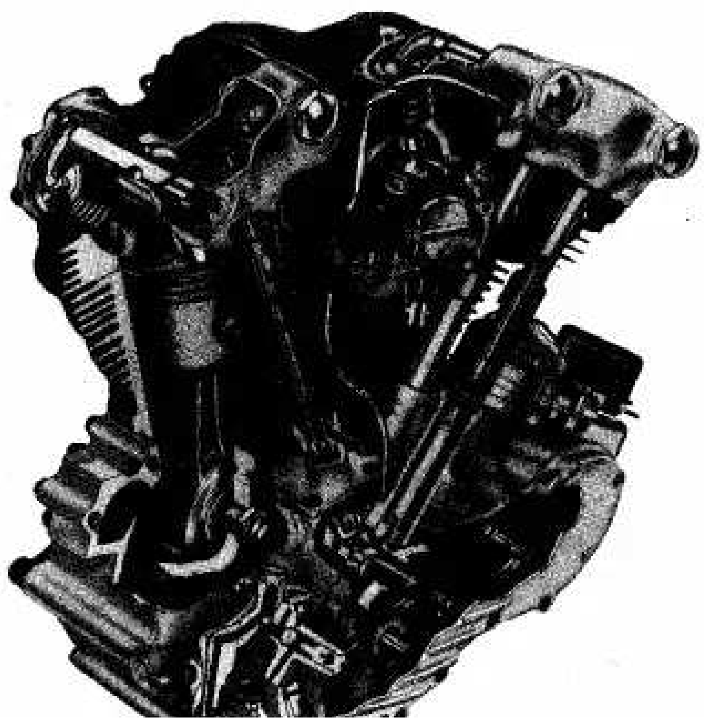


Figure 1-13 Harley-Davidson two-cylinder, air-cooled, overhead valve "Knucklehead" motorcycle engine first introduced in 1936. The 45°V engine had displacement of 60 cubic inches with 3.3125 inch bore and 3.500 inch stroke. Operating on a four-stroke cycle with a compression ratio of 7: 1 the engine was rated at 40 bhp at 4800 RPM. Ignition was by Harley-Davidson generator-battery system. Photograph courtesy of the Harley-Davidson Juneau Avenue Archives.

Figure 1-14 Harley-Davidson motorcycle of 1936 powered by "Knucklehead" engine shown in Fig. 1-13. The motorcycle had a rated top speed of 90-95 MPH with a fuel economy of 35-50 MPG. Photograph courtesy of the Harley-Davidson Juneau Avenue Archives.

Air-Fuel Ratio (AF) Ratio of mass of air to mass of fuel input into engine.

Fuel-Air Ratio (FA) Ratio of mass of fuel to mass of air input into engine.

Brake Maximum Torque (BMT) Speed at which maximum torque occurs.

Overhead Valve (ORV) Valves mounted in engine head.

Overhead Cam (aRC) Camshaft mounted in engine head, giving more direct control of valves which are also mounted in engine head.

Fuel Injected (FI)

'-5 ENGINE COMPONENTS

The following is a list of major components found in most reciprocating internal combustion engines (see Fig. 1-15).

Block Body of engine containing the cylinders, made of cast iron or aluminum. **In** many older engines, the valves and valve ports were contained in the block. The block of water-cooled engines includes a water jacket cast around the cylinders. On air-cooled engines, the exterior surface of the block has cooling cylinders. On air-cooled engines, the exterior surface of the block has cooling fins.

Camshaft Rotating shaft used to push open valves at the proper time in the engine cycle, either directly or through mechanical or hydraulic linkage (push rods,

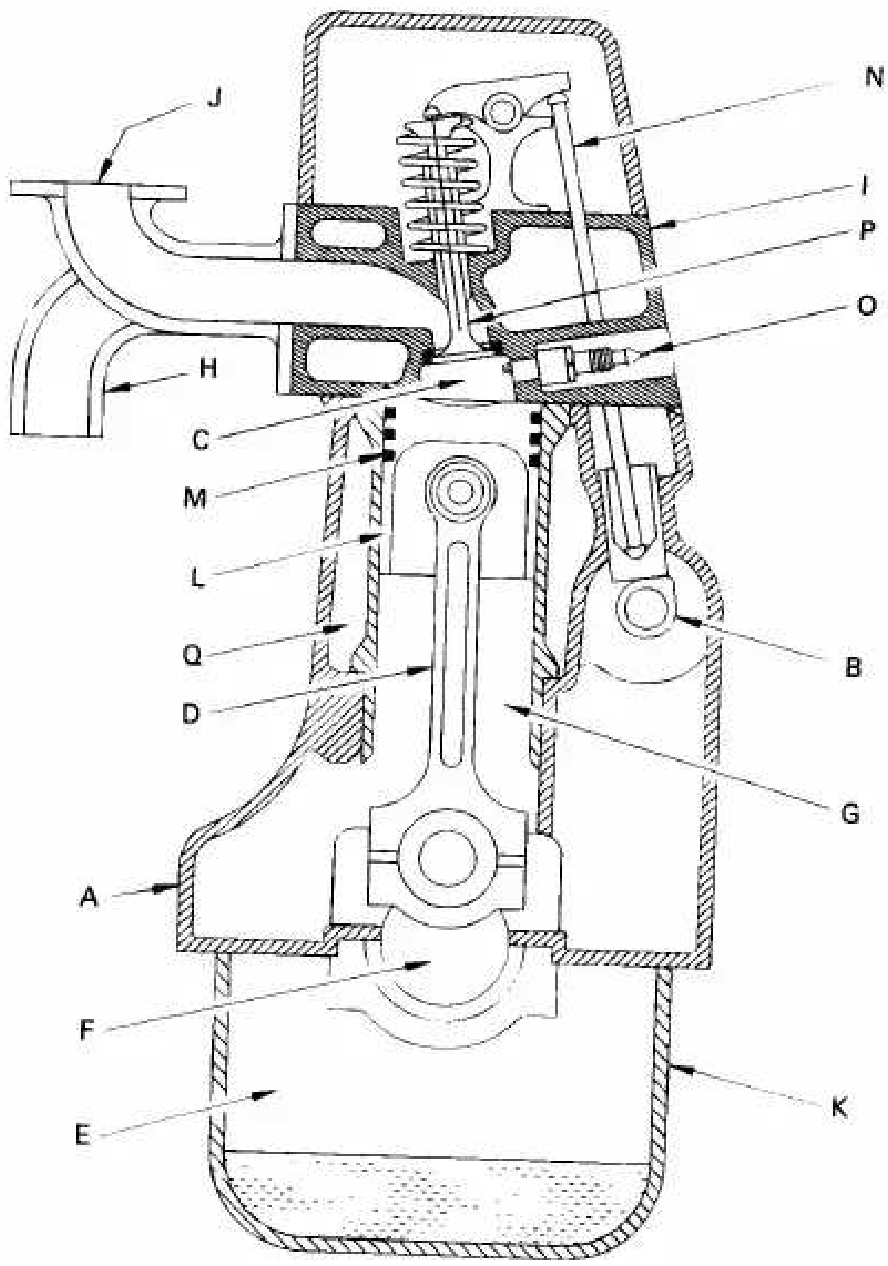


Figure 1-15 Cross-section of four-stroke cycle Sl engine showing engine components; (A) block, (B) camshaft, (C) combustion chamber, (D) connecting rod, (E) crankcase, (F) crankshaft, (G) cylinder, (H) exhaust manifold, (I) head, (J) intake manifold, (K) oil pan, (L) piston, (M) piston rings, (N) push rod, (0) spark plug, (P) valve, (Q) water jacket.

rocker arms, tappets). Most modern automobile engines have one or more camshafts mounted in the engine head (overhead cam). Most older engines had camshafts in the crankcase. Camshafts are generally made of forged steel or cast iron and are driven off the crankshaft by means of a belt or chain (timing chain). To reduce weight, some cams are made from a hollow shaft with

the cam lobes press-fit on. In four-stroke cycle engines, the camshaft rotates at half engine speed.

Carburetor Venturi flow device which meters the proper amount of fuel into the air flow by means of a pressure differential. For many decades it was the basic fuel metering system on all automobile (and other) engines. It is still used on low-cost small engines like lawn mowers, but is uncommon on new automobiles.

Catalytic converter Chamber mounted in exhaust flow containing catalytic material that promotes reduction of emissions by chemical reaction.

Combustion chamber The end of the cylinder between the head and the piston face where combustion occurs. The size of the combustion chamber continuously changes from a minimum volume when the piston is at TDC to a maximum when the piston is at BDC. The term "cylinder" is sometimes synonymous with "combustion chamber" (e.g., "the engine was firing on all cylinders"). Some engines have *open* combustion chambers which consist of one chamber for each cylinder. Other engines have *divided* chambers which consist of dual chambers on each cylinder connected by an orifice passage.

Connecting rod Rod connecting the piston with the rotating crankshaft, usually made of steel or alloy forging in most engines but may be aluminum in some small engines.

Connecting rod bearing Bearing where connecting rod fastens to crankshaft.

Cooling fins Metal fins on the outside surfaces of cylinders and head of an air-cooled engine. These extended surfaces cool the cylinders by conduction and convection.

Crankcase Part of the engine block surrounding the rotating crankshaft. In many engines, the oil pan makes up part of the crankcase housing.

Crankshaft Rotating shaft through which engine work output is supplied to external systems. The crankshaft is connected to the engine block with the *main bearings*. It is rotated by the reciprocating pistons through connecting rods connected to the crankshaft, offset from the axis of rotation. This offset is sometimes called *crank throw* or *crank radius*. Most crankshafts are made of forged steel, while some are made of cast iron.

Cylinders The circular cylinders in the engine block in which the pistons reciprocate back and forth. The walls of the cylinder have highly polished hard surfaces. Cylinders may be machined directly in the engine block, or a hard metal (drawn steel) sleeve may be pressed into the softer metal block. Sleeves may be dry sleeves, which do not contact the liquid in the water jacket, or wet sleeves, which form part of the water jacket. In a few engines, the cylinder walls are given a knurled surface to help hold a lubricant film on the walls. In some very rare cases, the cross section of the cylinder is not round.

Exhaust manifold Piping system which carries exhaust gases away from the engine cylinders, usually made of cast iron.

Exhaust system Flow system for removing exhaust gases from the cylinders, treating them, and exhausting them to the surroundings. It consists of an exhaust manifold which carries the exhaust gases away from the engine, a thermal or catalytic converter to reduce emissions, a muffler to reduce engine noise, and a tailpipe to carry the exhaust gases away from the passenger compartment.

Fan Most engines have an engine-driven fan to increase air flow through the radiator and through the engine compartment, which increases waste heat removal from the engine. Fans can be driven mechanically or electrically, and can run continuously or be used only when needed.

Flywheel Rotating mass with a large moment of inertia connected to the crankshaft of the engine. The purpose of the flywheel is to store energy and furnish a large angular momentum that keeps the engine rotating between power strokes and smooths out engine operation. On some aircraft engines the propeller serves as the flywheel, as does the rotating blade on many lawn mowers.

Fuel injector A pressurized nozzle that sprays fuel into the incoming air on SI engines or into the cylinder on CI engines. On SI engines, fuel injectors are located at the intake valve ports on multipoint port injector systems and upstream at the intake manifold inlet on throttle body injector systems. In a few SI engines, injectors spray directly into the combustion chamber.

Fuel pump Electrically or mechanically driven pump to supply fuel from the fuel tank (reservoir) to the engine. Many modern automobiles have an electric fuel pump mounted submerged in the fuel tank. Some small engines and early automobiles had no fuel pump, relying on gravity feed.

HISTORIC-FUEL PUMPS

Lacking a fuel pump, it was necessary to back Model T Fords (1909-1927) up high-slope hills because of the location of the fuel tank relative to the engine.

Glow plug Small electrical resistance heater mounted inside the combustion chamber of many CI engines, used to preheat the chamber enough so that combustion will occur when first starting a cold engine. The glow plug is turned off after the engine is started.

Head The piece which closes the end of the cylinders, usually containing part of the clearance volume of the combustion chamber. The head is usually cast iron or aluminum, and bolts to the engine block. In some less common engines, the head is one piece with the block. The head contains the spark plugs in SI engines and the fuel injectors in CI engines and some SI engines. Most modern engines have the valves in the head, and many have the camshaft(s) positioned there also (overhead valves and overhead cam).

Head gasket Gasket which serves as a sealant between the engine block and head where they bolt together. They are usually made in sandwich construction of metal and composite materials. Some engines use liquid head gaskets.

Intake manifold Piping system which delivers incoming air to the cylinders, usually made of cast metal, plastic, or composite material. In most SI engines, fuel is added to the air in the intake manifold system either by fuel injectors or with a carburetor. Some intake manifolds are heated to enhance fuel evaporation. The individual pipe to a single cylinder is called a *runner.*

Main bearing The bearings connected to the engine block in which the crankshaft rotates. The maximum number of main bearings would be equal to the number of pistons plus one, or one between each set of pistons plus the two ends. On some less powerful engines, the number of main bearings is less than this maximum.

Oil pan Oil reservoir usually bolted to the bottom of the engine block, making up part of the crankcase. Acts as the oil sump for most engines.

Oil pump Pump used to distribute oil from the oil sump to required lubrication points. The oil pump can be electrically driven, but is most commonly mechanically driven by the engine. Some small engines do not have an oil pump and are lubricated by splash distribution.

Oil sump Reservoir for the oil system of the engine, commonly part of the crankcase. Some engines (aircraft) have a separate closed reservoir called a *dry sump.*

Piston The cylindrical-shaped mass that reciprocates back and forth in the cylinder, transmitting the pressure forces in the combustion chamber to the rotating crankshaft. The top of the piston is called the *crown* and the sides are called the *skirt.* The face on the crown makes up one wall of the combustion chamber and may be a flat or highly contoured surface. Some pistons contain an indented bowl in the crown, which makes up a large percent of the clearance volume. Pistons are made of cast iron, steel, or aluminum. Iron and steel pistons can have sharper corners because of their higher strength. They also have lower thermal expansion, which allows for tighter tolerances and less crevice volume. Aluminum pistons are lighter and have less mass inertia. Sometimes synthetic or composite materials are used for the body of the piston, with only the crown made of metal. Some pistons have a ceramic coating on the face.

Piston rings Metal rings that fit into circumferential grooves around the piston and form a sliding surface against the cylinder walls. Near the top of the piston are usually two or more compression rings made of highly polished hard chrome steel. The purpose of these is to form a seal between the piston and cylinder walls and to restrict the high-pressure gases in the combustion chamber from leaking past the piston into the crankcase (blowby). Below the compression rings on the piston is at least one oil ring, which assists in lubricating the cylinder walls and scrapes away excess oil to reduce oil consumption.

Push rods Mechanical linkage between the camshaft and valves on overhead valve engines with the camshaft in the crankcase. Many push rods have oil passages through their length as part of a pressurized lubrication system.

Radiator Liquid-to-air heat exchanger of honeycomb construction used to remove heat from the engine coolant after the engine has been cooled. The radiator is usually mounted in front of the engine in the flow of air as the automobile moves forward. An engine-driven fan is often used to increase air flow through the radiator.

Spark plug Electrical device used to initiate combustion in an SI engine by creating a high-voltage discharge across an electrode gap. Spark plugs are usually made of metal surrounded with ceramic insulation. Some modern spark plugs have built-in pressure sensors which supply one of the inputs into engine control.

Speed control-cruise control Automatic electric-mechanical control system that keeps the automobile operating at a constant speed by controlling engine speed.

Starter Several methods are used to start IC engines. Most are started by use of an electric motor (starter) geared to the engine flywheel. Energy is supplied from an electric battery.

On some very large engines, such as those found in large tractors and construction equipment, electric starters have inadequate power, and small IC engines are used as starters for the large IC engines. First the small engine is started with the normal electric motor, and then the small engine engages gearing on the flywheel of the large engine, turning it until the large engine starts.

Early aircraft engines were often started by hand spinning the propeller, which also served as the engine flywheel. Many small engines on lawn mowers and similar equipment are hand started by pulling a rope wrapped around a pulley connected to the crankshaft.

Compressed air is used to start some large engines. Cylinder release valves are opened, which keeps the pressure from increasing in the compression strokes. Compressed air is then introduced into the cylinders, which rotates the engine in a *free-wheeling* mode. When rotating inertia is established, the release valves are closed and the engine is fired.

HISTORIC—STARTERS

Early automobile engines were started with hand cranks that connected with the crankshaft of the engine. This was a difficult and dangerous process, sometimes resulting in broken fingers and arms when the engine would fire and snap back the hand crank. The first electric starters appeared on the 1912 Cadillac automobiles, invented by C. Kettering, who was motivated when his friend was killed in the process of hand starting an automobile [45].

Supercharger Mechanical compressor powered off of the crankshaft, used to compress incoming air of the engine.

Throttle Butterfly valve mounted at the upstream end of the intake system, used to control the amount of air flow into an SI engine. Some small engines and stationary constant-speed engines have no throttle.

Turbocharger Turbine-compressor used to compress incoming air into the engine. The turbine is powered by the exhaust flow of the engine and thus takes very little useful work from the engine.

Valves Used to allow flow into and out of the cylinder at the proper time in the cycle. Most engines use *poppet valves,* which are spring loaded closed and pushed open by camshaft action (Fig. 1-12). Valves are mostly made of forged steel. Surfaces against which valves close are called valve seats and are made of hardened steel or ceramic. *Rotary valves* and *sleeve valves* are sometimes used, but are much less common. Many two-stroke cycle engines have *ports* (slots) in the side of the cylinder walls instead of mechanical valves.

Water jacket System of liquid flow passages surrounding the cylinders, usually constructed as part of the engine block and head. Engine coolant flows through the water jacket and keeps the cylinder walls from overheating. The coolant is usually a water-ethylene glycol mixture.

Water pump Pump used to circulate engine coolant through the engine and radiator. It is usually mechanically run off of the engine.

Wrist pin Pin fastening the connecting rod to the piston (also called the *piston pin).*

1-6 BASIC ENGINE CYCLES

Most internal combustion engines, both spark ignition and compression ignition, operate on either a four-stroke cycle or a two-stroke cycle. These basic cycles are fairly standard for all engines, with only slight variations found in individual designs

Four-Stroke SI Engine Cycle (Fig. 1-16)

1. *First Stroke: Intake Stroke or Induction* The piston travels from TDC to BDC with the intake valve open and exhaust valve closed. This creates an increasing volume in the combustion chamber, which in turn creates a vacuum. The resulting pressure differential through the intake system from atmospheric pressure on the outside to the vacuum on the inside causes air to be pushed into the cylinder. As the air passes through the intake system, fuel is added to it in the desired amount by means of fuel injectors or a carburetor.

2. *Second Stroke: Compression Stroke* When the piston reaches BDC, the intake valve closes and the piston travels back to TDC with all valves closed. This compresses the air-fuel mixture, raising both the pressure and temperature in the cylinder. The finite time required to close the intake valve means that actual compression doesn't start until sometime aBDC. Near the end of the compression stroke, the spark plug is fired and combustion is initiated.

3. *Combustion* Combustion of the air-fuel mixture occurs in a very short but finite length of time with the piston near TDC (i.e., nearly constant-volume combustion). It starts near the end of the compression stroke slightly bTDC and lasts into the power stroke slightly aTDC. Combustion changes the composition of the gas mixture to that of exhaust products and increases the temperature in the cylinder to a very high peak value. This, in turn, raises the pressure in the cylinder to a very high peak value.

4. *Third Stroke: Expansion Stroke or Power Stroke* With all valves closed, the high pressure created by the combustion process pushes the piston away from TDC. This is the stroke which produces the work output of the engine cycle. As the piston travels from TDC to BDC, cylinder volume is increased, causing pressure and temperature to drop.

5. *Exhaust Blowdown* Late in the power stroke, the exhaust valve is opened and exhaust blowdown occurs. Pressure and temperature in the cylinder are still high relative to the surroundings at this point, and a pressure differential is created through the exhaust system which is open to atmospheric pressure. This pressure differential causes much of the hot exhaust gas to be pushed out of the cylinder and through the exhaust system when the piston is near BDC. This exhaust gas carries away a high amount of enthalpy, which lowers the cycle thermal efficiency. Opening the exhaust valve before BDC reduces the work obtained during the power stroke but is required because of the finite time needed for exhaust blowdown.

6. *Fourth Stroke: Exhaust Stroke* By the time the piston reaches BDC, exhaust blowdown is complete, but the cylinder is still full of exhaust gases at approximately atmospheric pressure. With the exhaust valve remaining open, the piston now travels from BDC to TDC in the exhaust stroke. This pushes most of the remaining exhaust gases out of the cylinder into the exhaust system at about atmospheric pressure, leaving only that trapped in the clearance volume when the piston reaches TDC. Near the end of the exhaust stroke bTDC, the intake valve starts to

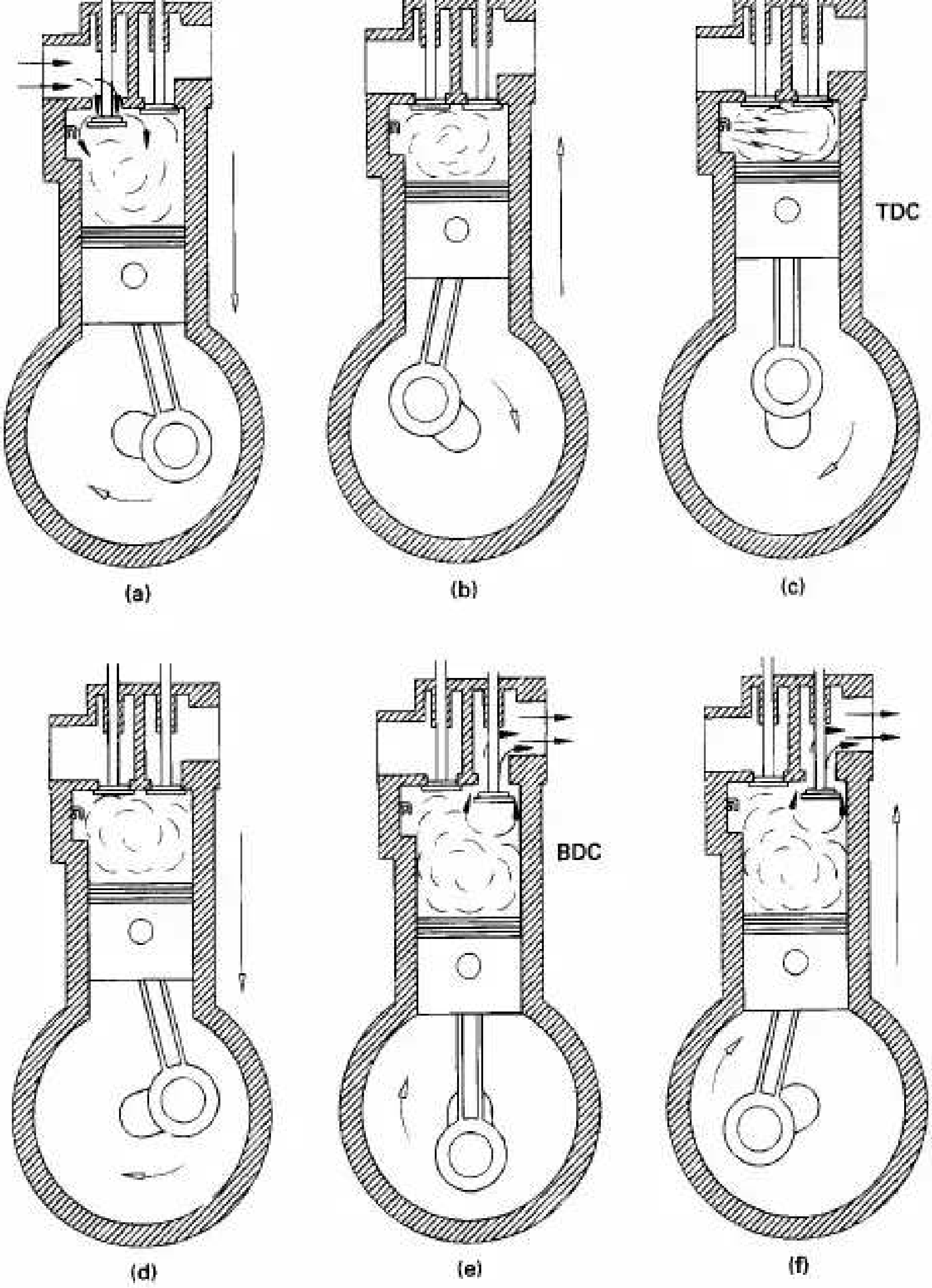


Figure 1-16 Four-stroke SI engine operating cycle. **(a)** Intake stroke. Ingress of air-fuel as piston moves from TDC to BDC. **(b)** Compression stroke. Piston moves from BDC to TDC. Spark ignition occurs near end of compression stroke. **(c)** Combustion at almost constant volume near TDC. **(d)** Power or expansion stroke. High cylinder pressure pushes piston from TDC towards BDC. **(e)** Exhaust blowdown when exhaust valve opens near end of expansion stroke. **(f)** Exhaust stroke. Remaining exhaust pushed from cylinder as piston moves from BDC to TDC.

open, so that it is fully open by TDC when the new intake stroke starts the next cycle. Near TDC the exhaust valve starts to close and finally is fully closed sometime aTDC. This period when both the intake valve and exhaust valve are open is called valve overlap.

valve overlap.

Four-Stroke CI Engine Cycle

1. First Stroke: Intake Stroke The same as the intake stroke in an SI engine with one major difference: no fuel is added to the incoming air.

2. Second Stroke: Compression Stroke The same as in an SI engine except that only air is compressed and compression is to higher pressures and temperature. Late in the compression stroke fuel is injected directly into the combustion chamber, where it mixes with the very hot air. This causes the fuel to evaporate and self-ignite, causing combustion to start.

3. Combustion Combustion is fully developed by TDC and continues at about constant pressure until fuel injection is complete and the piston has started towards BDC.

4. Third Stroke: Power Stroke The power stroke continues as combustion ends and the piston travels towards BDC.

5. Exhaust Blowdown Same as with an SI engine.

6. Fourth Stroke: Exhaust Stroke Same as with an SI engine.

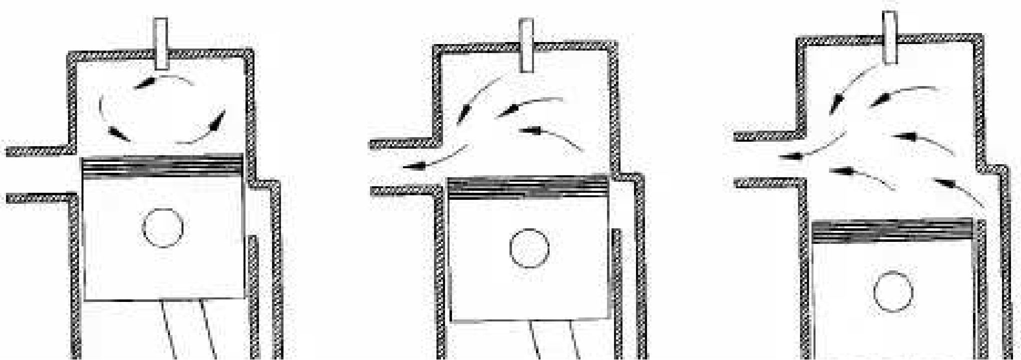
Two-Stroke SI Engine Cycle (Fig. 1-17)

1. Combustion With the piston at TDC combustion occurs very quickly, raising the temperature and pressure to peak values, almost at constant volume.

2. First Stroke: Expansion Stroke or Power Stroke Very high pressure created by the combustion process forces the piston down in the power stroke. The expanding volume of the combustion chamber causes pressure and temperature to decrease as the piston travels towards BDC.

3. Exhaust Blowdown At about 75° bBDC, the exhaust valve opens and blowdown occurs. The exhaust valve may be a poppet valve in the cylinder head, or it may be a slot in the side of the cylinder which is uncovered as the piston approaches BDC. After blowdown the cylinder remains filled with exhaust gas at lower pressure.

4. Intake and Scavenging When blowdown is nearly complete, at about 50° bBDC, the intake slot on the side of the cylinder is uncovered and intake air-fuel enters under pressure. Fuel is added to the air with either a carburetor or fuel injection. This incoming mixture pushes much of the remaining exhaust gases out the open exhaust valve and fills the cylinder with a combustible air-fuel mixture, a process called scavenging. The piston passes BDC and very quickly covers the intake port and then the exhaust port (or the exhaust valve closes). The higher pres-



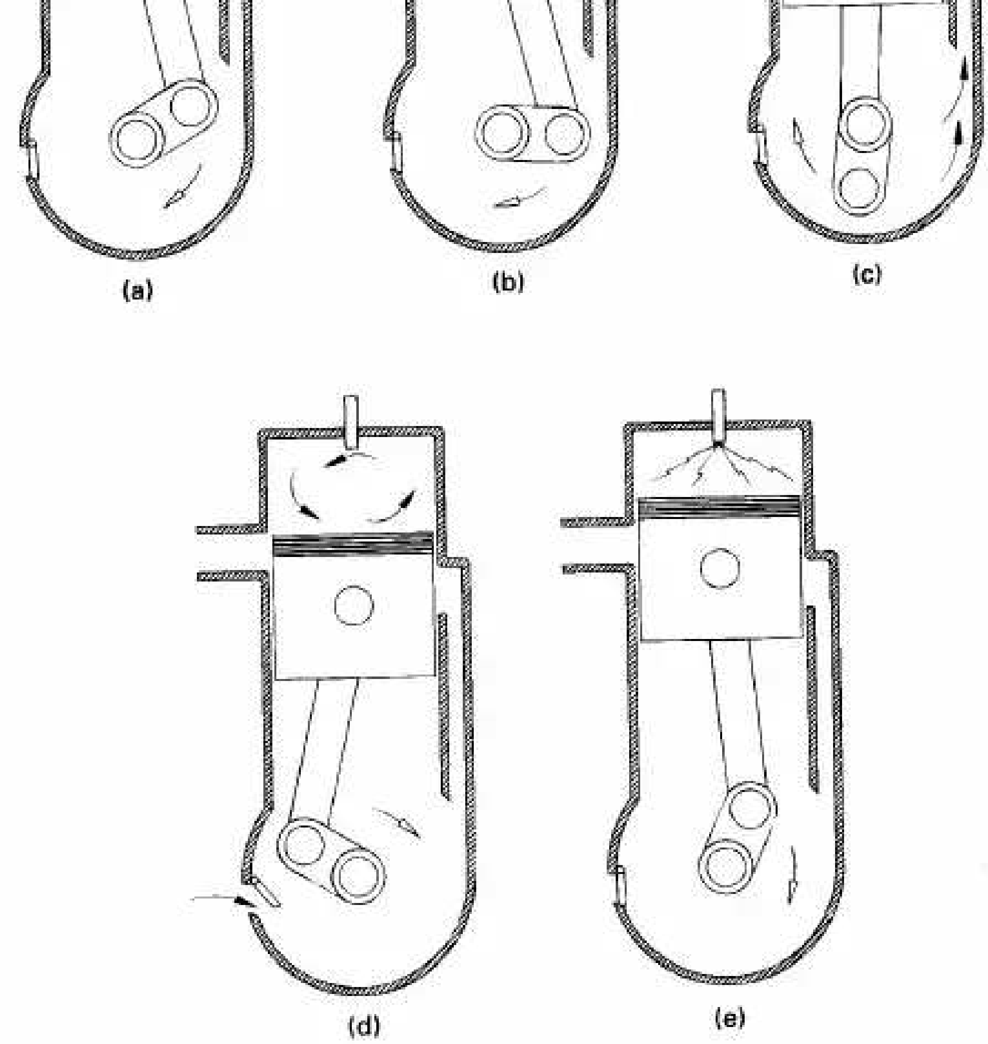


Figure 1-17 Two-stroke SI engine operating cycle with crankcase compression. **(a)** Power or expansion stroke. High cylinder pressure pushes piston from TDC towards BDC with all ports closed. Air in crankcase is compressed by downward motion of piston. **(b)** Exhaust blowdown when exhaust port opens near end of power stroke. **(c)** Cylinder scavenging when intake port opens and airfuel is forced into cylinder under pressure. Intake mixture pushes some of the remaining exhaust out the open exhaust port. Scavenging lasts until piston passes BDC and closes intake and exhaust ports. **(d)** Compression stroke. Piston moves from BDC to TDC with all ports closed. Intake air fills crankcase. Spark ignition occurs near end of compression stroke. **(e)** Combustion at almost constant volume near TDC.

sure at which the air enters the cylinder is established in one of two ways. Large two-stroke cycle engines generally have a supercharger, while small engines will intake the air through the crankcase. On these engines the crankcase is designed to serve as a compressor in addition to serving its normal function.

5. *Second Stroke: Compression Stroke* With all valves (or ports) closed, the piston travels towards TDC and compresses the air-fuel mixture to a higher pressure and temperature. Near the end of the compression stroke, the spark plug is fired; by the time the piston gets to IDC, combustion occurs and the next engine cycle begins.

Two-Stroke CI Engine Cycle

Two-Stroke CI Engine Cycle

The two-stroke cycle for a CI engine is similar to that of the SI engine, except for two changes. No fuel is added to the incoming air, so that compression is done on air only. Instead of a spark plug, a fuel injector is located in the cylinder. Near the end

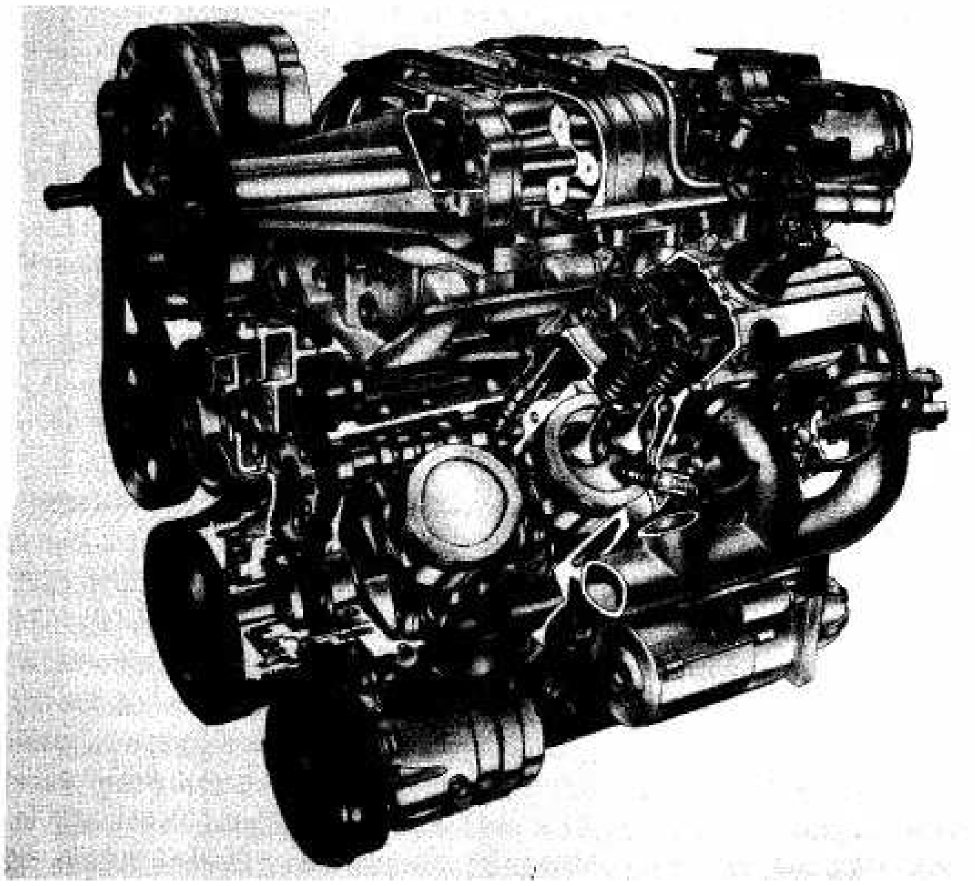


Figure 1-18 1996 General Motors L67 3800 Series II spark ignition, four-stroke cycle, overhead valve, 3.8 liter, V6 engine. This supercharged engine has two valves per cylinder and has power and torque ratings of 240 hp (179 kW) at 5200 RPM and 280 lbf-ft (380 N-m) at 3600 RPM. Copyright General Motors Corp., used with permission.

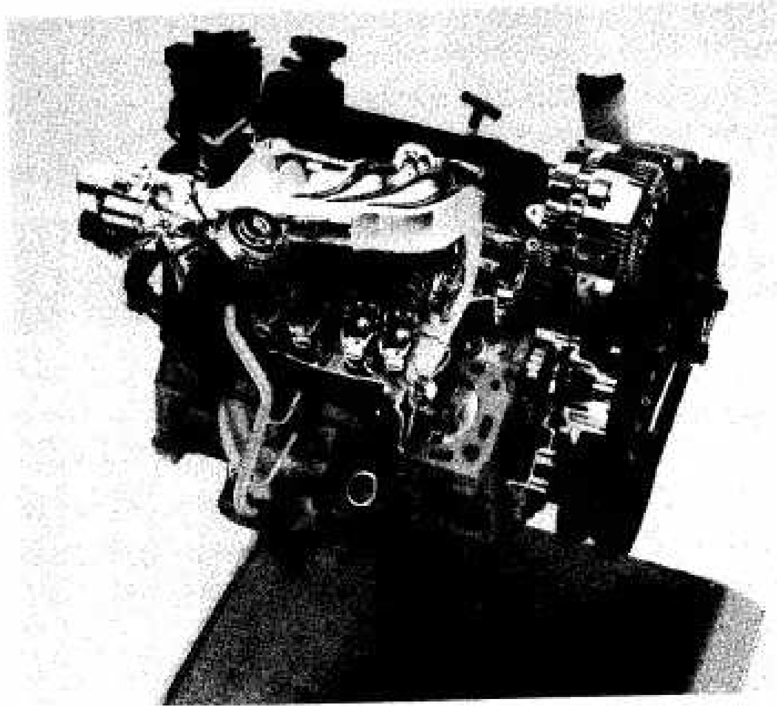


Figure 1-19 Ford 3.0 liter Vulcan V6, spark ignition, four-stroke cycle engine. This was the standard engine of the 1996 Ford Taurus and Mercury Sable automobiles. It is rated at 108 kW at 5250 RPM and develops 230 N-m of torque at 3250 RPM. Courtesy Ford Motor Company.

of the compression stroke, fuel is injected into the hot compressed air and combustion is initiated by self-ignition.

'-7 ENGINE EMISSIONS AND AIR POLLUTION

The exhaust of automobiles is one of the major contributors to the world's air pollution problem. Recent research and development has made major reductions in engine emissions, but a growing population and a greater number of automobiles means that the problem will exist for many years to come.

During the first half of the 1900s, automobile emissions were not recognized as a problem, mainly due to the lower number of vehicles. As the number of automobiles grew along with more power plants, home furnaces, and population in general, air pollution became an ever-increasing problem. During the 1940s, the problem was first seen in the Los Angeles area due to the high density of people and automobiles, as well as unique weather conditions. By the 1970s, air pollution was recognized as a major problem in most cities of the United States as well as in many large urban areas around the world.

Laws were passed in the United States and in other industrialized countries which limit the amount of various exhaust emissions that are allowed. This put a major restriction on automobile engine development during the 1980s and 1990s.

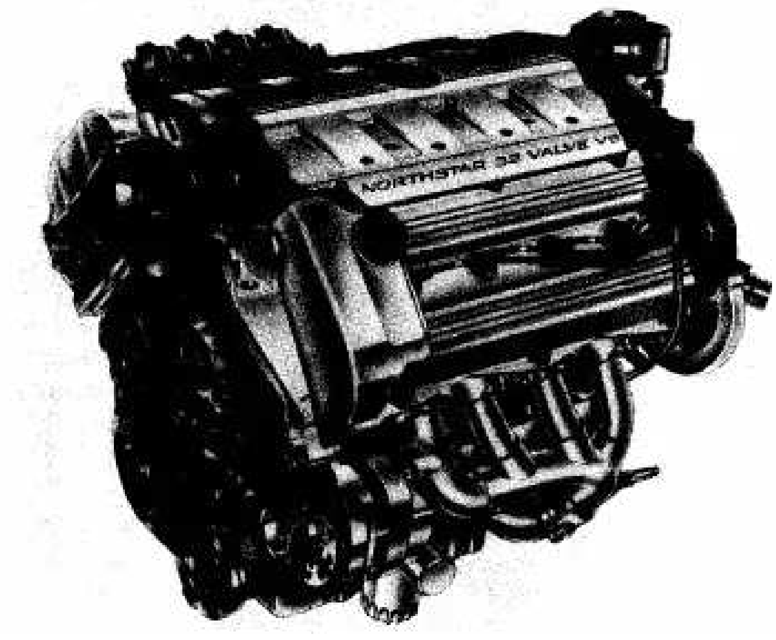


Figure 1·20 General Motors Northstar VB engine used in 1995 Cadillac automobiles. This four-stroke cycle, spark ignition, 32 valve, double overhead cam engine has a 4.6 L displacement and multipoint port fuel injection. If the cooling system of this engine has a leak, the automobile can be driven at moderate speed for up to fifty miles without coolant fluid, without damage to the engine. Copyright General

Motors Corp., used with permission.

Although harmful emissions produced by engines have been reduced by over 90% since the 1940s, they are still a major environmental problem.

Four major emissions produced by internal combustion engines are hydrocarbons (He), carbon monoxide (CO), oxides of nitrogen (NOx), and solid particulates. Hydrocarbons are fuel molecules which did not get burned and smaller nonequilibrium particles of partially burned fuel. Carbon monoxide occurs when not enough oxygen is present to fully react all carbon to CO_2 or when incomplete air-fuel mixing occurs due to the very short engine cycle time. Oxides of nitrogen are created in an engine when high combustion temperatures cause some normally stable N_2 to dissociate into monatomic nitrogen N, which then combines with reacting oxygen. Solid particulates are formed in compression ignition engines and are seen as black smoke in the exhaust of these engines. Other emissions found in the exhaust of engines include aldehydes, sulfur, lead, and phosphorus.

Two methods are being used to reduce harmful engine emissions. One is to improve the technology of engines and fuels so that better combustion Occurs and fewer emissions are generated. The second method is aftertreatment of the exhaust gases. This is done by using thermal converters or catalytic converters that promote chemical reactions in the exhaust flow. These chemical reactions convert the harmful emissions to acceptable CO_2, H_2O, and N_2.

In Chapter 2, methods of classifying emissions will be introduced. Chapter 9 studies emissions and aftertreatment methods in detail.

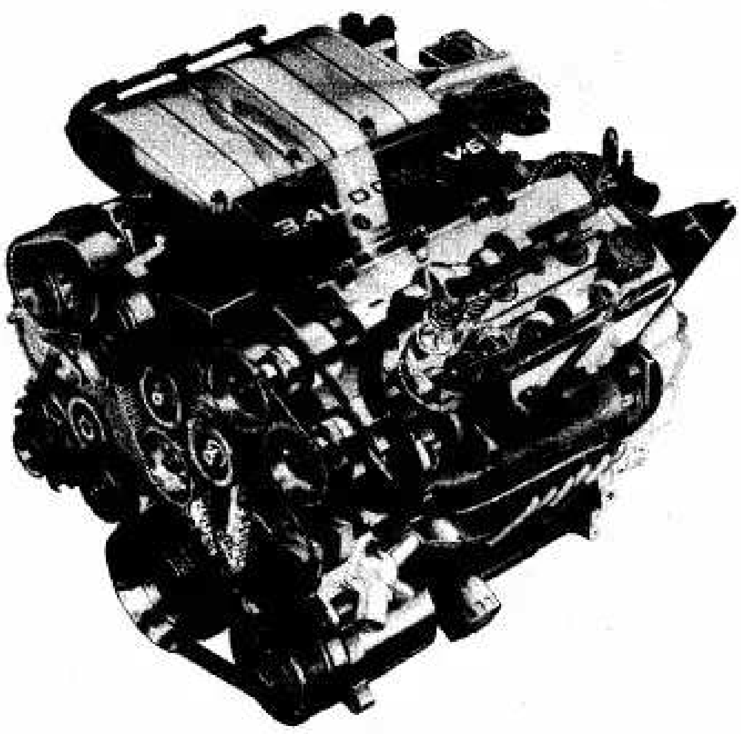
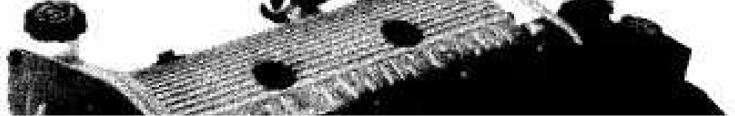


Figure 1-21 General Motors LQ1 3.4 liter V6 engine used in 1996 Chevrolet Lumina, Pontiac Grand Prix, and Oldsmobile Cutless automobiles. This engine has a compression ratio of 9.7:1, bore of 9.203 cm (3.62 in.), stroke of 8.400 cm (3.31 in.), and a maximum speed of 6750 RPM. It has a power rating of 215 hp (160 kW) at 5200 RPM and a torque rating of 220 lbf-ft (298 N-m) at 4000 RPM. Four valves per cylinder are actuated by two overhead camshafts per cylinder bank. The engine heads and intake manifold are made of cast aluminum while the engine block, exhaust manifolds, camshafts, and crankshaft are made of cast iron. Copyright General Motors Corp., used with permission.



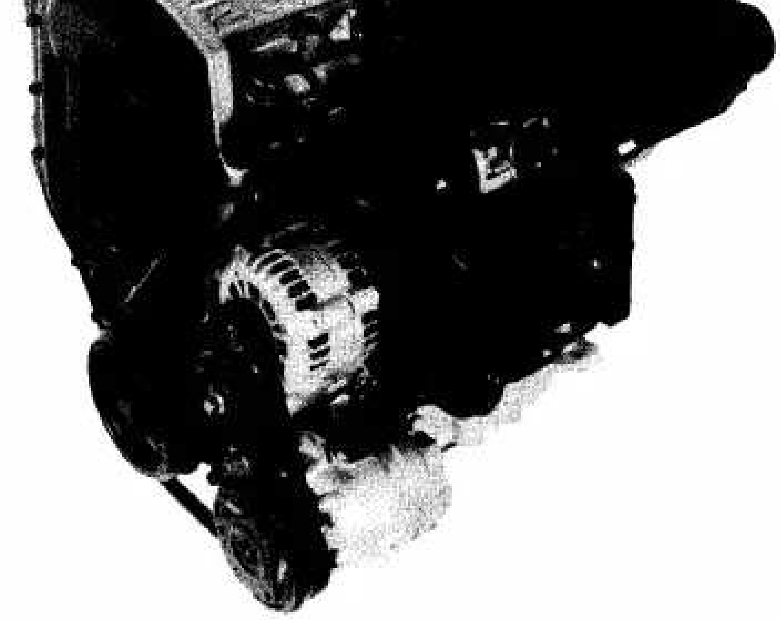


Figure 1-22 General Motors LD9 2.4 liter, in-line, four-cylinder, spark ignition, four-stroke cycle engine. The engine has double overhead camshafts and four valves per cylinder. Copyright General Motors Corp., used with permission.

PROBLEMS

1-1. List five differences between SI engines and CI engines.

1-2. A four-stroke cycle engine mayor may not have a pressure boost (supercharger, turbocharger) in the intake system. Why must a two-stroke cycle engine always have an intake pressure boost?

1-3. List two advantages of a two-stroke cycle engine over a four-stroke cycle engine. List two advantages of a four-stroke cycle engine over a two-stroke cycle engine.

1-4. (a) Why do most very small engines operate on a two-stroke cycle? (b) Why do most very large engines operate on a two-stroke cycle? (c) Why do most automobile engines operate on a four-stroke cycle? (d) Why would it be desirable to operate automobile engines on a two-stroke cycle?

1-5. A single-cylinder vertical atmospheric engine with a 1.2 m bore and a piston of 2700 kg mass is used to lift a weight. Pressure in the cylinder after combustion and cooling is 22 kPa, while ambient pressure is 98 kPa. Assume piston motion is frictionless.

Calculate: (a) Mass which can be lifted if the vacuum is at the top of the cylinder and the piston moves up. [kg]

(b) Mass which can be lifted if the vacuum is at the bottom of the cylinder and the piston moves down. [kg]

1-6. An early atmospheric engine has a single horizontal cylinder with a 3.2-ft bore, a 9.0-ft stroke, and no clearance volume. After a charge of gunpowder is set off in the open cylinder, the conditions in the cylinder are ambient pressure and a temperature of 540°F. The piston is now locked in position and the end of the cylinder is closed. After cooling to ambient temperature, the piston is unlocked and allowed to move. The power stroke is at constant temperature and lasts until pressure equilibrium is obtained. Assume the gas in the cylinder is air and piston motion is frictionless. Ambient conditions are 70°F and 14.7 psia.

Calculate: (a) Possible lifting force at start of power stroke. [lb£]
(b) Length of effective power stroke. [ft]
(c) Cylinder volume at end of power stroke. [ft3]

1-7. Two automobile engines have the same total displacement volume and the same total power produced within the cylinders.

List the possible advantages of: (a) A V6 over a straight six.
(b) A V8 over a V6.
(c) A V6 over a V8.
(d) An opposed cylinder four over a straight four.
(e) An in-line six over an in-line four.

1-8. A nine cylinder, four-stroke cycle, radial SI engine operates at 900 RPM.

Calculate: (a) How often ignition occurs, in degrees of engine rotation.
(b) How many power strokes per revolution.
(c) How many power strokes per second.

DESIGN PROBLEMS

1-10. Design a single-cylinder atmospheric engine capable of lifting a mass of 1000 kg to a height of three meters. Assume reasonable values of cylinder temperature and pressure after combustion. Decide which direction the cylinder will move, and give the bore, piston travel distance, mass of piston, piston material, and clearance volume. Give a sketch of the mechanical linkage to lift the mass.

1-20. Design an alternate fuel engine to be used in a large truck by designating all engine classifications used in Section 1-3.

1-30. Design a four-stroke cycle for an SI engine using crankcase compression. Draw schematics of the six basic processes: intake, compression, combustion, expansion, blowdown, and exhaust. Describe fully the intake of air, fuel, and oil.

2

Operating Characteristics

This chapter examines the operating characteristics of reciprocating internal combustion engines. These include the mechanical output parameters of work, torque, and power; the input requirements of air, fuel, and combustion; efficiencies; and emission measurements of engine exhaust.

2-1 ENGINE PARAMETERS

For an engine with bore B (see Fig. 2-1), crank offset a, stroke length S, turning at an engine speed of N:

$$S = 2a \tag{2-1}$$

Average piston speed is:

$$U_p = 2SN \tag{2-2}$$

N is generally given in RPM (revolutions per minute), U_p in m/sec (ft/sec), and B, a, and S in m or cm (ft or in.).

Average piston speed for all engines will normally be in the range of 5 to 15 m/sec (15 to 50 ft/sec), with large diesel engines on the low end and high-performance automobile engines on the high end. There are two reasons why engines

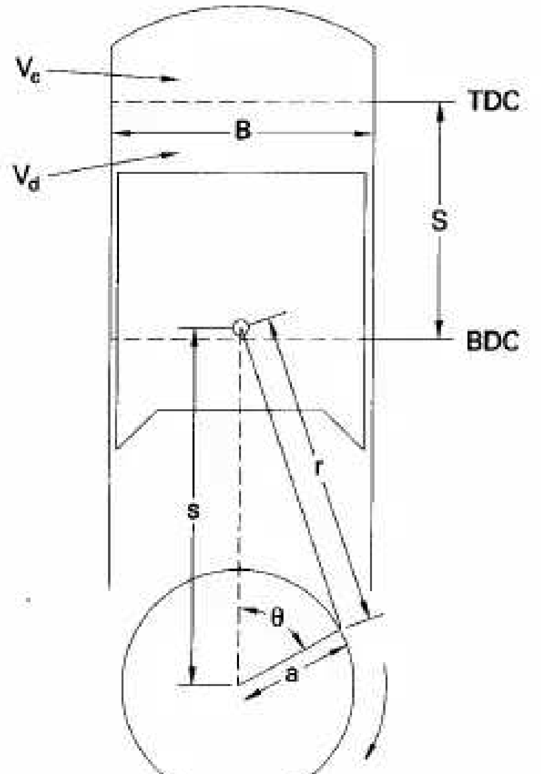


Figure 2-1 Piston and cylinder geometry of reciprocating engine. B = bore; S = stroke; r = connecting rod length; a = crank offset; s = piston position; θ = crank angle; V_c = clearance volume; V_d = displacement volume.

operate in this range. First, this is about the safe limit which can be tolerated by material strength of the engine components. For each revolution of the engine, each piston is twice accelerated from stop to a maximum speed and back to stop. At a typical engine speed of 3000 RPM, each revolution lasts 0.02 sec (0.005 sec at 12,000 RPM). If engines operated at higher speeds, there would be a danger of material failure in the pistons and connecting rods as the piston is accelerated and decelerated during each stroke. From Eq. (2-2) it can be seen that this range of acceptable piston speeds places a range on acceptable engine speeds also, depending on engine size. There is a strong inverse correlation between engine size and operating speed. Very large engines with bore sizes on the order of 0.5 m (1.6 ft) typically operate in the 200- to 400-RPM range, while the very smallest engines (model airplane) with bores on the order of 1 cm (0.4 in.) operate at speeds of 12,000 RPM and higher. Table 2-1 gives representative values of engine speeds and other operating variables for various-sized engines. Automobile engines usually operate in a speed range of 500 to 5000 RPM, with cruising at about 2000 RPM. Under certain conditions using special materials and design, high-performance experimental engines have been operated with average piston speeds up to 25 m/sec.

The second reason why maximum average piston speed is limited is because of

The second reason why maximum average piston speed is limited is because of the gas flow into and out of the cylinders. Piston speed determines the instantaneous flow rate of air-fuel into the cylinder during intake and exhaust flow out of the cylinder during the exhaust stroke. Higher piston speeds would require larger valves to

TABLE 2-1 TYPICAL ENGINE OPERATING PARAMETERS

	Model Airplane Two-Stroke Cycle	Automobile Four-Stroke Cycle	Large Stationary Two-Stroke Cycle
BORE (cm)	2.00	9.42	50.0
STROKE (cm)	2.04	9.89	161
DISPLACEMENT/cyl (L)	0.0066	0.69	316
SPEED (RPM)	13,000	5200	125
POWER/cyl (kW)	0.72	35	311
AVERAGE PISTON SPEED (m/sec)	8.84	17.1	6.71
POWER/DISPLACEMENT (kW/L)	109	50.7	0.98
bmep (kPa)	503	1170	472

allow for higher flow rates. In most engines, valves are at a maximum size with no room for enlargement.

Bore sizes of engines range from 0.5 m down to 0.5 cm (20 in. to 0.2 in.). The ratio of bore to stroke, B/S, for small engines is usually from 0.8 to 1.2. An engine with $B = S$ is often called a *square* engine. If stroke length is longer than bore diameter the engine is *under square*, and if stroke length is less than bore diameter the engine is *over square*. Very large engines are always under square, with stroke lengths up to four times bore diameter.

The distance s between crank axis and wrist pin axis is given by

$$s = a\cos\theta + \sqrt{r^2 - a^2\sin^2\theta} \tag{2-3}$$

where: a = crankshaft offset

r = connecting rod length

θ = crank angle, which is measured from the cylinder centerline and is zero when the piston is at TDC

When s is differentiated with respect to time, the instantaneous piston speed U_p is obtained:

$$U_p = ds/dt \tag{2-4}$$

The ratio of instantaneous piston speed divided by the average piston speed can then be written as

$$U_p/\bar{U}_p = (\pi/2)\sin\theta[1 + (\cos\theta/\sqrt{R^2 - \sin^2\theta})] \tag{2-5}$$

where:

$$R = r/a \tag{2-6}$$

R is the ratio of connecting rod length to crank offset and usually has values of 3 to 4 for small engines, increasing to 5 to 10 for the largest engines. Figure 2-2 shows the effect of R on piston speed.

Displacement, or displacement volume V_d, is the volume displaced by the piston as it travels from BDC to TDC:

$$V_d = V_{BDC} - V_{TDC} \qquad (2\text{-}7)$$

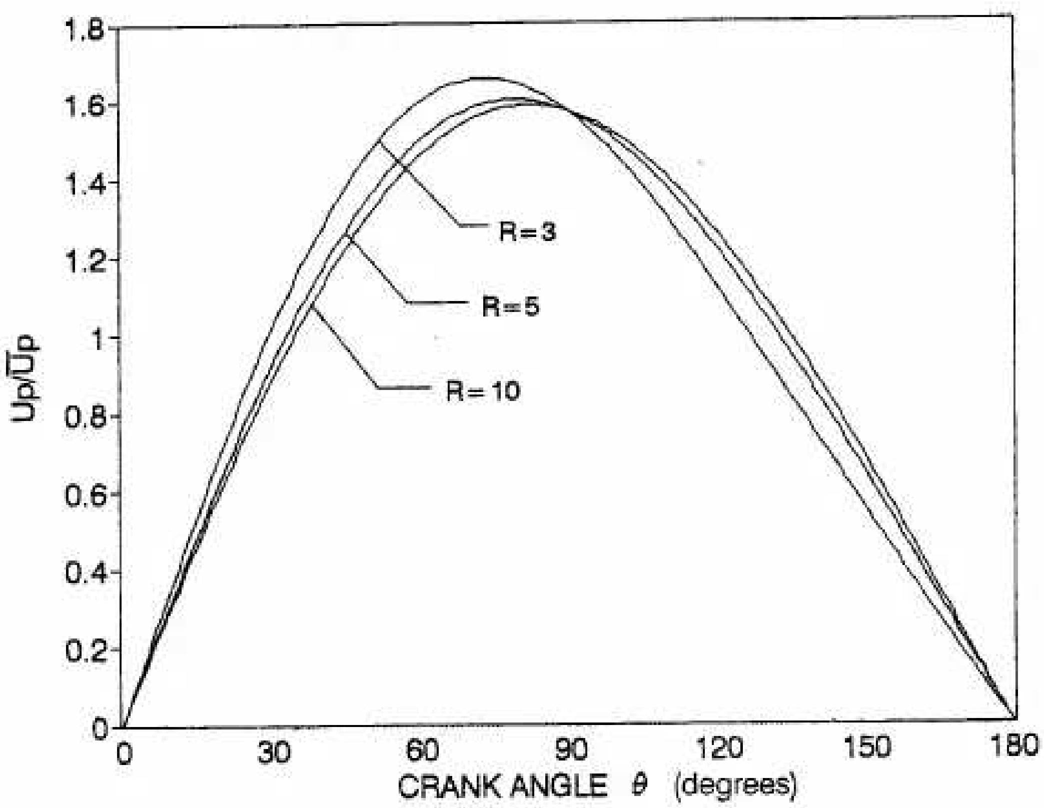


Figure 2-2 Instantaneous piston speed relative to average piston speed as a function of crank angle for various R values, where $R = r/a$, r = connecting rod length, a = crankshaft offset.

Some books call this swept volume. Displacement can be given for one cylinder or for the entire engine. For one cylinder:

$$V_d = (\pi/4)B^2S \qquad (2\text{-}8)$$

For an engine with N_c cylinders:

$$V_d = N_c(\pi/4)B^2S \qquad (2\text{-}9)$$

where: B = cylinder bore
S = stroke
N_c = number of engine cylinders

Engine displacements can be given in m^3, cm^3, $in.^3$, and, most commonly, in liters (L).

$$1\ L = 10^{-3}\ m^3 = 10^3\ cm^3 \approx 61\ in.^3$$

Typical values for engine displacement range from 0.1 cm^3 (0.0061 $in.^3$) for small model airplanes to about 8 L (490 $in.^3$) for large automobiles to much larger numbers for large ship engines. The displacement of a modern average automobile engine is about two to three liters.

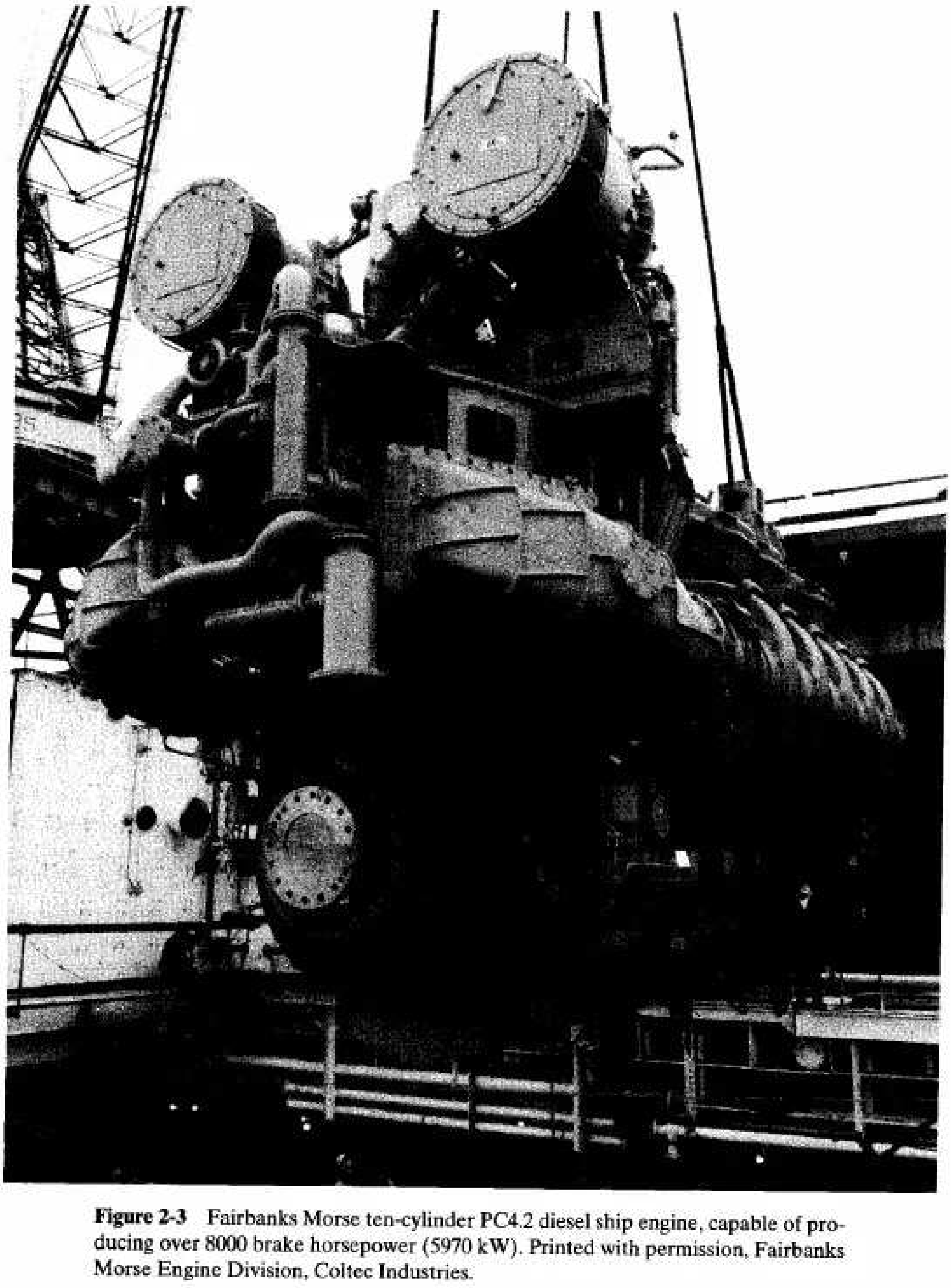


Figure 2-3 Fairbanks Morse ten-cylinder PC4.2 diesel ship engine, capable of producing over 8000 brake horsepower (5970 kW). Printed with permission, Fairbanks Morse Engine Division, Coltec Industries.

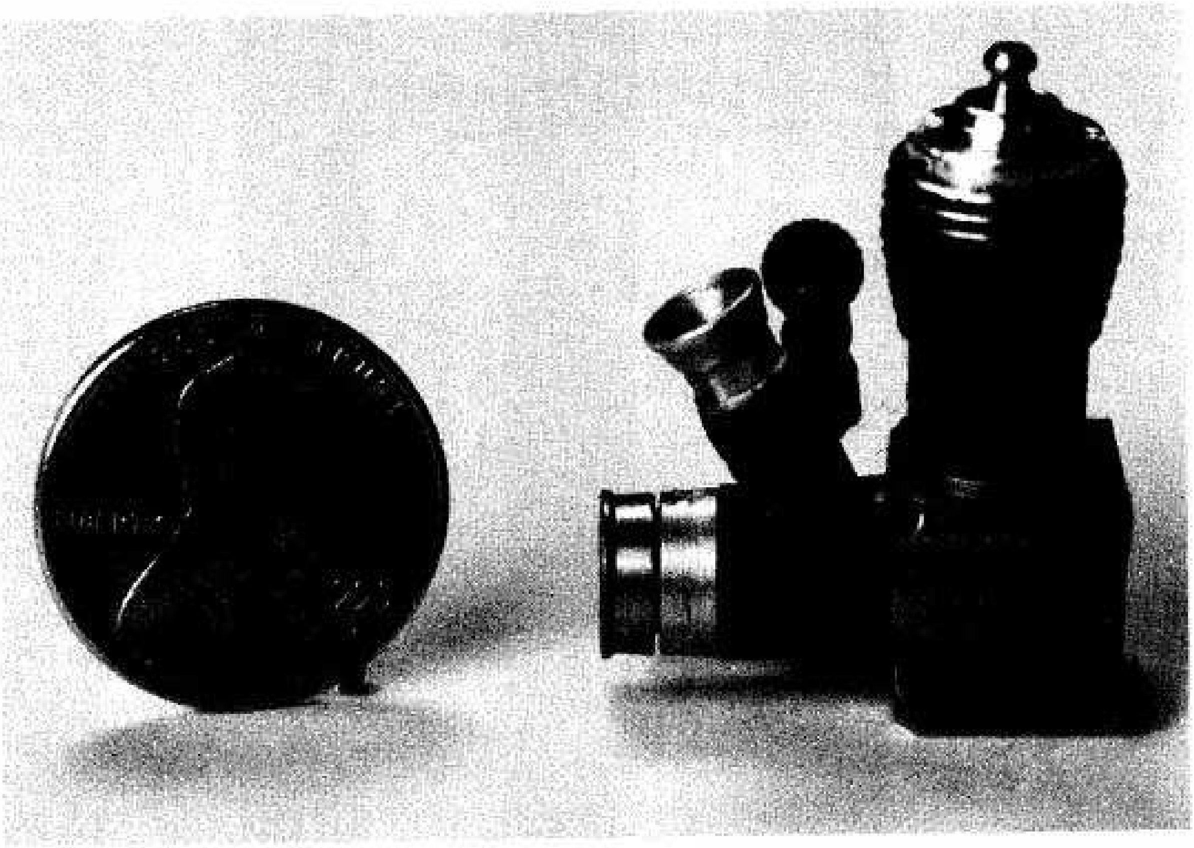


Figure 2-4 Cox air-cooled, single cylinder, two-stroke cycle model airplane engine. Engine has displacement of 0.01 cubic inches (0.164 cm^3). Photo by Tuescher Photography, Platteville, Wisconsin.

HISTORIC—CHRISTIE RACE CAR

A 1908 Christie race car had a V4 engine of 2799 $in.^3$ (46 L) displacement [45].

For a given displacement volume, a longer stroke allows for a smaller bore (under square), resulting in less surface area in the combustion chamber and correspondingly less heat loss. This increases thermal efficiency within the combustion chamber. However, the longer stroke results in higher piston speed and higher friction losses that reduce the output power which can be obtained off the crankshaft. If the stroke is shortened, the bore must be increased, and the engine will be over square. This decreases friction losses but increases heat transfer losses. Most modern automobile engines are near square, with some slightly over square and some slightly under square. This is dictated by design compromises and the technical philosophy of the manufacturer. Very large engines have long strokes with stroke-to-bore ratios as high as 4:1.

Minimum cylinder volume occurs when the piston is at TDC and is called the clearance volume V_c.

$$V_c = V_{TDC} \tag{2-10}$$

$$V_{BDC} = V_c + V_d \tag{2-11}$$

HISTORIC—SMALL HIGH-SPEED ENGINES

Engines for model airplanes and boats have been built with total displacement volume as low as 0.075 cm^3 (0.0046 in.^{3}). Some of these engines, which are commercially available, can obtain speeds up to 38,000 RPM and have power output on the order of 0.15 to 1.5 kW (0.2 to 2.0 hp). It is interesting that the average piston speeds of these engines at high speed still fall within the general range of 5 to 15 m/sec. At least one small experimental automobile engine has been developed to run at speeds up to 28,000 RPM.

The compression ratio of an engine is defined as:

$$r_e = V_{BDC}/V_{TDC} = (V_e + V_d)/V_e \tag{2-12}$$

Modern spark ignition (SI) engines have compression ratios of 8 to 11, while compression ignition (CI) engines have compression ratios in the range 12 to 24. Engines with superchargers or turbochargers usually have lower compression ratios than naturally aspirated engines. Because of limitations in engine materials, technology, and fuel quality, very early engines had low compression ratios, on the order of 2 to 3. Figure 2-5 shows how values of r_e increased over time to the 8-11 range used on modem spark ignition automobile engines. This limit of 8 to 11 is imposed mainly by gasoline fuel properties (see Section 4-4) and force limitations allowable in smaller high-speed engines.

Various attempts have been made to develop engines with a variable compression ratio. One such system uses a split piston that expands due to changing hydraulic pressure caused by engine speed and load. Some two-stroke cycle engines have been built which have a sleeve-type valve that changes the slot opening on the exhaust port. The position where the exhaust port is fully closed can be adjusted by several degrees of engine rotation. This changes the *effective compression ratio* of the engine.

HISTORIC—CRANKSHAFTS

One interesting four-cylinder engine made by Austin of England had only two main bearings, one on each end of the crankshaft. To counteract the effect of crankshaft bending during operation, the two center cylinders were given a slightly higher static compression ratio than the two end cylinders [11].

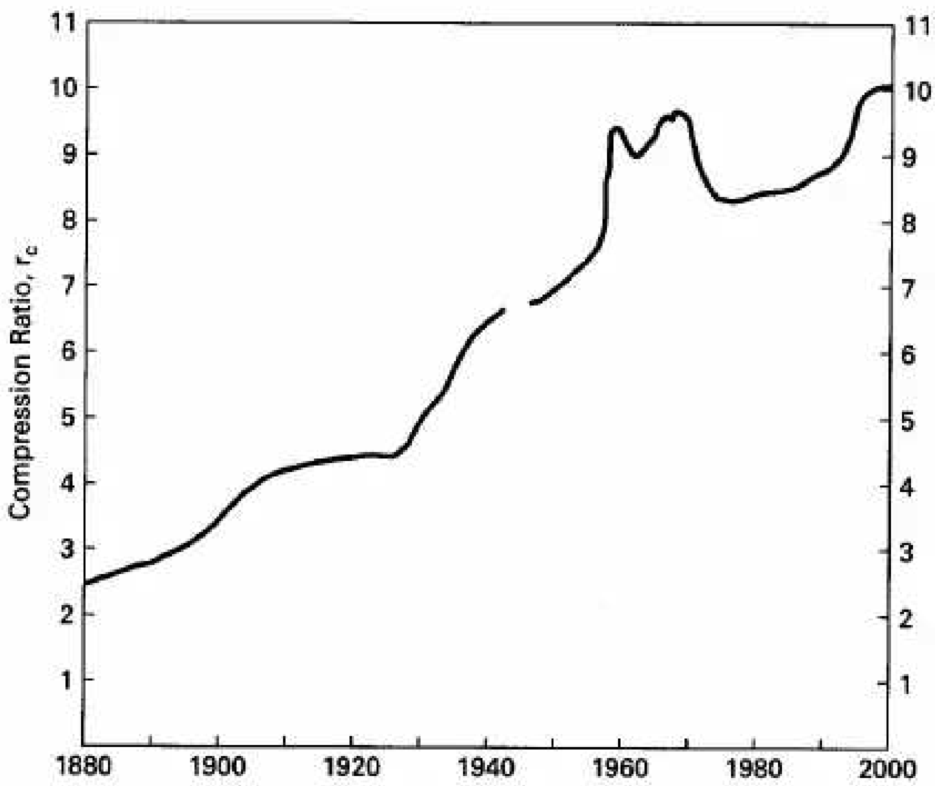


Figure 2-5 Average compression ratio of American spark ignition automobile engines as a function of year. During the first forty years compression ratios slowly increased from 2.5 to 4.5, limited mainly by low octane numbers of the available fuels. In 1923TEL was introduced as a fuel additive and this was followed by a rapid increase in compression ratios. No automobiles were manufactured during 1942-1945 when production was converted to war vehicles during World War II. A rapid rise in compression ratios occurred during the 1950s when muscle cars became popular. During the 1970s TEL was phased out as a fuel additive, pollution laws were enacted, and gasoline became expensive due to an oil embargo imposed by some oil producing countries. These resulted in lower compression ratios during this time. In the 1980s and 19908better fuels and combustion chamber technology is allowing for higher compression ratios. Adapted from [5].

The cylinder volume V at any crank angle is:

$$V = V_c + (\pi B^2/4)(r + a - s) \tag{2-13}$$

where: V_c = clearance volume
B = bore
r = connecting rod length
a = crank offset
s = piston position shown in Fig. 2-1

This can also be written in a non-dimensional form by dividing by V_c, substituting for r, a, and s, and employing the definition of R:

$$V/V_c = 1 + \tfrac{1}{2}(r_c - 1)\left[R + 1 - \cos\theta - \sqrt{R^2 - \sin^2\theta}\right] \tag{2-14}$$

where: r_c = compression ratio
$R = r/a$

The cross-sectional area of a cylinder and the surface area of a flat-topped piston are each given by:

$$A_p = (\pi/4)B^2 \tag{2-15}$$

The combustion chamber surface area is:

$$A = A_{ch} + A_p + \pi B(r + a - s) \tag{2-16}$$

where A_{ch} is the cylinder head surface area, which will be somewhat larger than A_p.

Then if the definitions for r, a, s, and R are used, Eq. (2-16) can be rewritten as:

$$A = A_{ch} + A_p + (\pi BS/2)[R + 1 - \cos\theta - \sqrt{R^2 - \sin^2\theta}] \tag{2-17}$$

EXAMPLE PROBLEM 2-1

John's automobile has a three-liter SI V6 engine that operates on a four-stroke cycle at 3600 RPM. The compression ratio is 9.5, the length of connecting rods is 16.6 cm, and the engine is square ($B = S$). At this speed, combustion ends at 20° aTDC.
Calculate:

1. cylinder bore and stroke length
2. average piston speed
3. clearance volume of one cylinder
4. piston speed at the end of combustion
5. distance the piston has traveled from TDC at the end of combustion
6. volume in the combustion chamber at the end of combustion

1) For one cylinder, using Eq. (2-8) with $S = B$:

$$V_d = V_{total}/6 = 3\text{L}/6 = 0.5\text{ L} = 0.0005\text{ m}^3 = (\pi/4)B^2S = (\pi/4)B^3$$

$$\underline{B = 0.0860\text{ m} = 8.60\text{ cm} = S}$$

2) Using Eq. (2-2) to find average piston speed:

$$\bar{U}_p = 2SN = (2\text{ strokes/rev})(0.0860\text{ m/stroke})(3600/60\text{ rev/sec})$$

$$\underline{= 10.32\text{ m/sec}}$$

3) Using Eq. (2-12) to find the clearance volume of one cylinder:

$$r_c = 9.5 = (V_d + V_c)/V_c = (0.0005 + V_c)/V_c$$

$$\underline{V_c = 0.000059\text{ m}^3 = 59\text{ cm}^3}$$

4) Crank offset, $a = S/2 = 0.0430\text{ m} = 4.30\text{ cm}$

$$R = r/a = 16.6\text{ cm}/4.30\text{ cm} = 3.86$$

Using Eq. (2-5) to find instantaneous piston speed:

$$U_p/\bar{U}_p = (\pi/2)\sin\theta[1 + (\cos\theta/\sqrt{R^2 - \sin^2\theta})]$$

$= (\pi/2)\sin(20°)\{1 + [\cos(20°)/\sqrt{(3.86)^2 - \sin^2(20°)}]\}$

$= 0.668$

$U_p = 0.668\,\overline{U}_p = (0.668)(10.32 \text{ m/sec}) = \underline{6.89 \text{ m/sec}}$

5) Using Eq. (2-3) to find piston position:

$$s = a\cos\theta + \sqrt{r^2 - a^2\sin^2\theta}$$

$$= (0.0430 \text{ m})\cos(20°) + \sqrt{(0.166 \text{ m})^2 - (0.0430 \text{ m})^2\sin^2(20°)}$$

$$= 0.206 \text{ m}$$

Distance from TDC:

$$x = r + a - s = (0.166 \text{ m}) + (0.043 \text{ m}) - (0.206 \text{ m})$$

$$= \underline{0.003 \text{ m} = 0.3 \text{ cm}}$$

6) Using Eq. (2-14) to find instantaneous volume:

$$V/V_c = 1 + \tfrac{1}{2}(r_c - 1)[R + 1 - \cos\theta - \sqrt{R^2 - \sin^2\theta}]$$

$$= 1 + \tfrac{1}{2}(9.5 - 1)[3.86 + 1 - \cos(20°) - \sqrt{(3.86)^2 - \sin^2(20°)}]$$

$$= 1.32$$

$$V = 1.32\,V_c = (1.32)(59 \text{ cm}^3) = \underline{77.9 \text{ cm}^3 = 0.0000779 \text{ m}^3}$$

This indicates that, during combustion, the volume in the combustion chamber has only increased by a very small amount and shows that combustion in an SI engine occurs at almost constant volume at TDC.

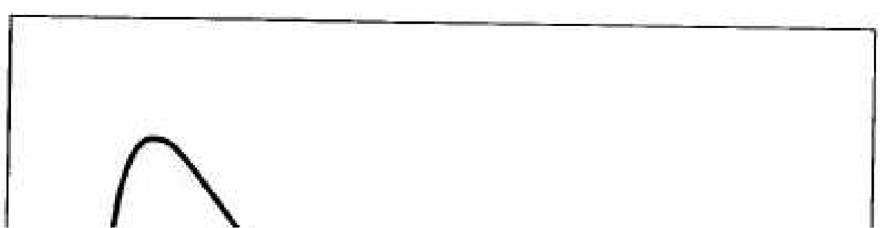
2-2 WORK

Work is the output of any heat engine, and in a reciprocating IC engine this work is generated by the gases in the combustion chamber of the cylinder. Work is the result of a force acting through a distance. Force due to gas pressure on the moving piston generates the work in an IC engine cycle.

$$W = \int F\,dx = \int PA_p\,dx \qquad (2\text{-}18)$$

where: P = pressure in combustion chamber
A_P = area against which the pressure acts (i.e., the piston face)
x = distance the piston moves

and:



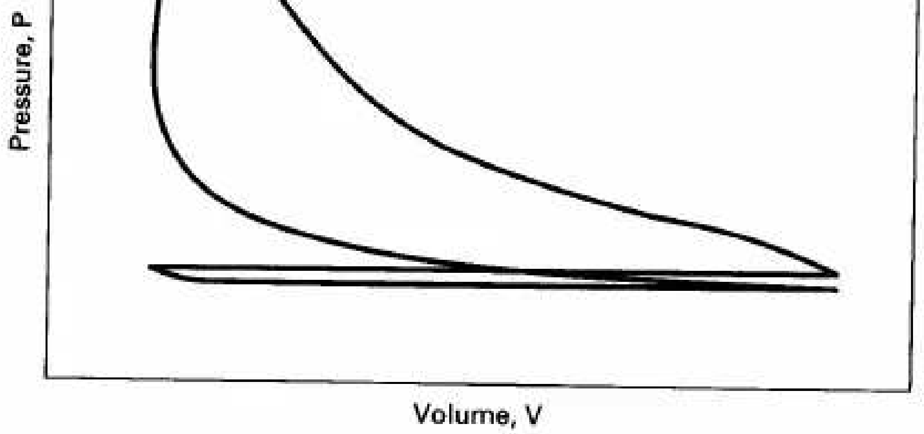


Figure 2-6 Indicator diagram for a typical four stroke cycle SI engine. An indicator diagram plots cylinder pressure as a function of combustion chamber volume over a 720° cycle. The diagram is generated on an oscilloscope using a pressure transducer mounted in the combustion chamber and a position sensor mounted on the piston or crankshaft.

$$A_p\, dx = dV \tag{2-19}$$

dV is the differential volume displaced by the piston, so work done can be written:

$$W = \int P\, dV \tag{2-20}$$

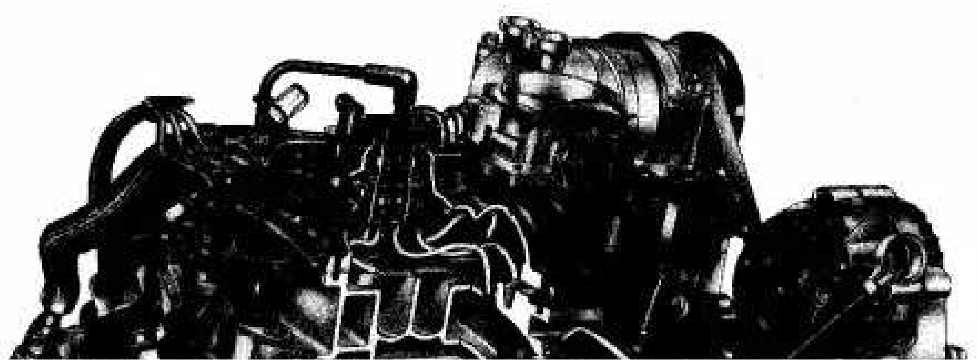
Figure 2-6, which plots the engine cycle on P–V coordinates, is often called an indicator diagram. Early indicator diagrams were generated by mechanical plotters linked directly to the engine. Modern P–V indicator diagrams are generated on an oscilloscope using a pressure transducer mounted in the combustion chamber and an electronic position sensor mounted on the piston or crankshaft.

Because engines are often multicylinder, it is convenient to analyze engine cycles per unit mass of gas m within the cylinder. To do so, volume V is replaced with specific volume v and work is replaced with specific work:

$$w = W/m \qquad v = V/m \tag{2-21}$$

$$w = \int P\, dv \tag{2-22}$$

Specific work w is equal to the area under the process lines on the P–v coordinates of Fig. 2-9.



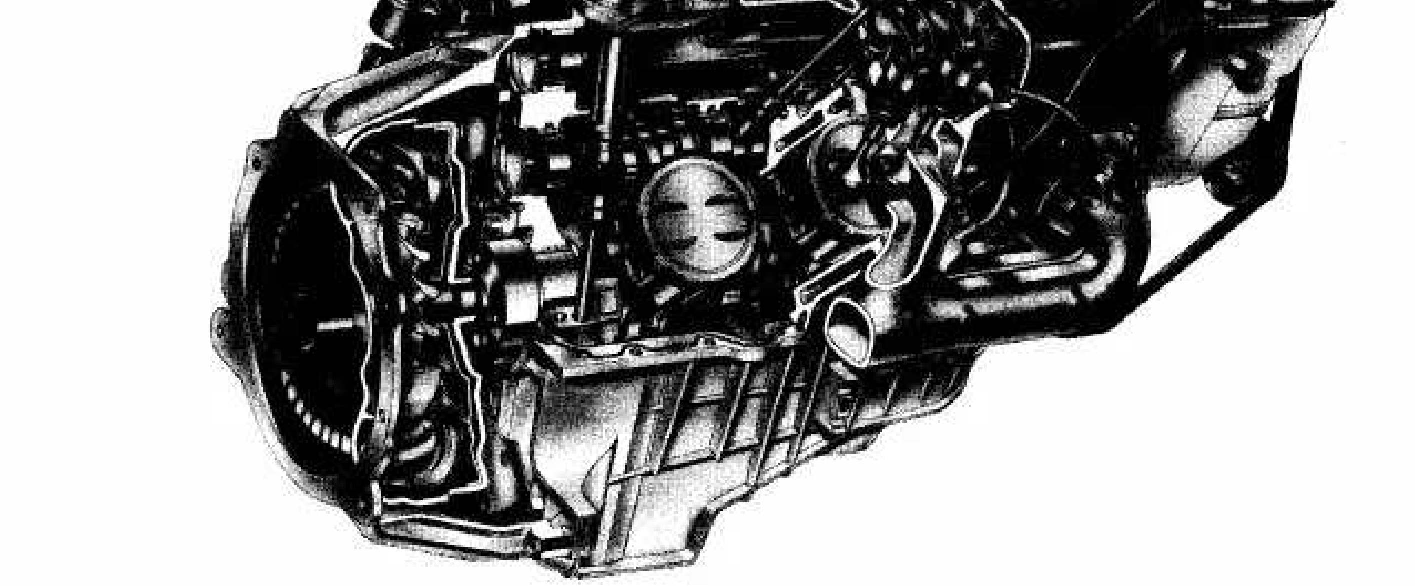


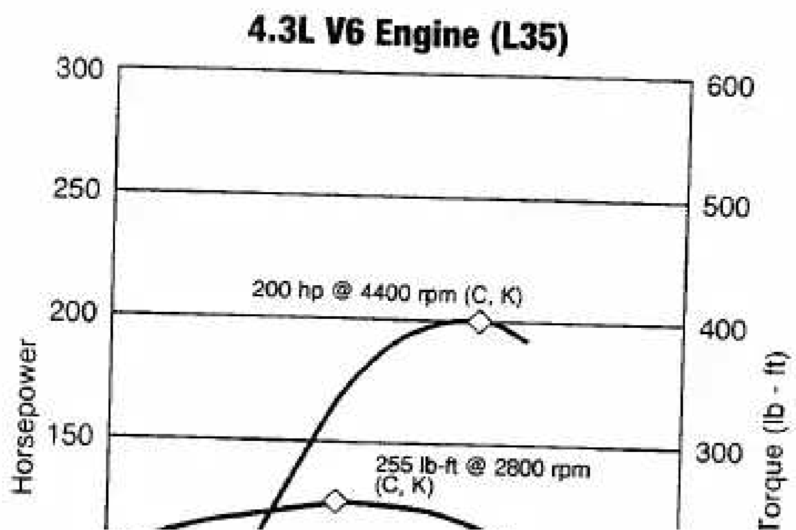
Figure 2-7 1996 General Motors L35 4300 Vortec V6 spark ignition engine. The engine has multipoint port fuel injection and overhead valves with a displacement of 4.3 L (262 in.^{3}), bore of 10.160 cm (4.00 in.) and stroke of 8.839 cm (3.48 in.). Copyright General Motors Corp., used with permission.

If P represents the pressure inside the cylinder combustion chamber, then Eq. (2-22) and the areas shown in Fig. 2-9 give the work inside the combustion chamber. This is called **indicated work.** Work delivered by the crankshaft is less than indicated work due to mechanical friction and parasitic loads of the engine. Parasitic loads include the oil pump, supercharger, air conditioner compressor, alternator, etc. Actual work available at the crankshaft is called **brake work** w_b. Units of specific work will be kJ/kg or BTU/lbm.

$$w_b = w_i - w_t \tag{2-23}$$

where: w_i = indicated specific work generated inside combustion chamber
w_t = specific work lost due to friction and parasitic loads

The upper loop of the engine cycle in Fig. 2-9 consists of the compression and power strokes where output work is generated and is called the gross **indicated work** (areas A and C in Fig. 2-9). The lower loop, which includes the intake and exhaust



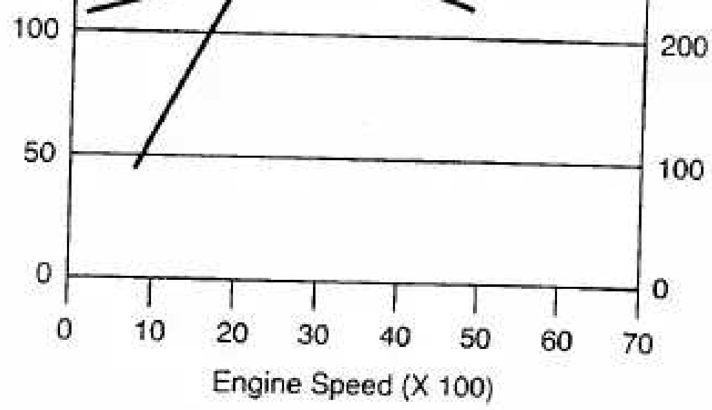


Figure 2-8 Power and torque curves of General Motors L35 Vortec V6 engine shown in Fig. 2-7. The engine has a compression ratio of 9.2:1 and a listed maximum speed of 5600 RPM. Maximum rated power is 200 hp (149 kW) at 4400 RPM and maximum rated torque is 255 lbf-ft (346 N-m) at 2800 RPM. Copyright General Motors Corp., used with permission.

strokes, is called **pump work** and absorbs work from the engine (areas B and C). **Net indicated work** is:

$$w_{net} = w_{gross} + w_{pump} \tag{2-24}$$

Pump work w_{pump} is negative for engines without superchargers:

$$w_{net} = (\text{Area A}) - (\text{Area B}) \tag{2-25}$$

Engines with superchargers or turbochargers can have intake pressure greater than exhaust pressure, giving a positive pump work (Fig. 2-10). When this occurs:

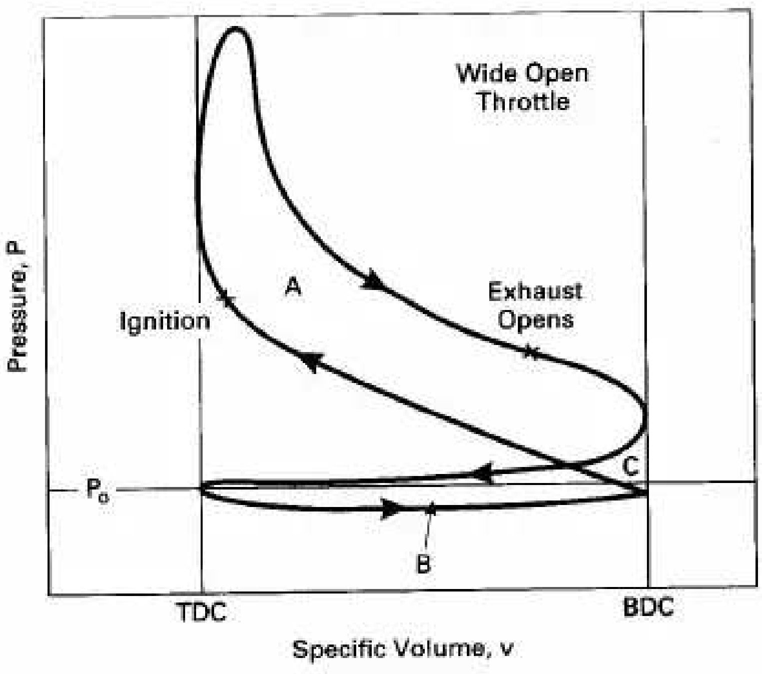
$$w_{net} = (\text{Area A}) + (\text{Area B}) \tag{2-26}$$

Superchargers increase net indicated work but add to the friction work of the engine since they are driven by the crankshaft.

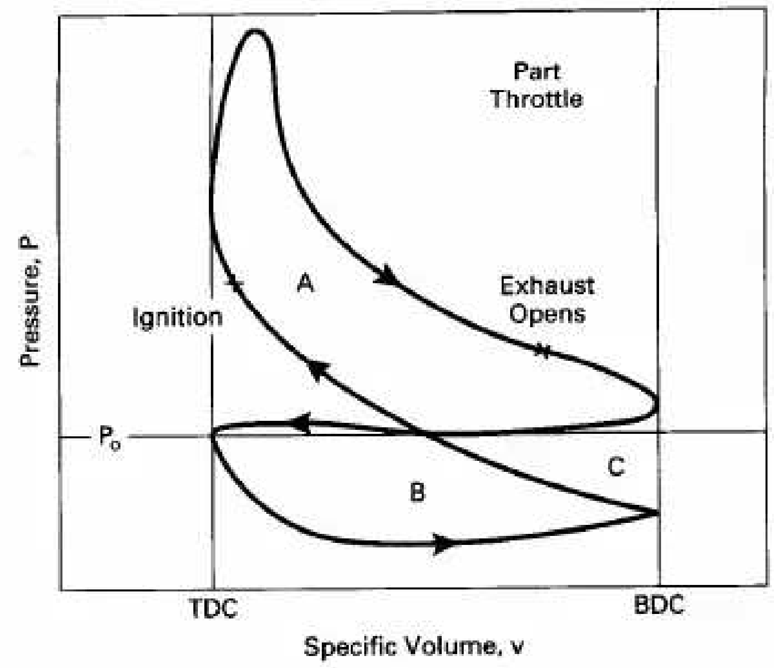
The ratio of brake work at the crankshaft to indicated work in the combustion chamber defines the **mechanical efficiency** of an engine:

$$\eta_m = w_b / w_i = W_b / W_i \tag{2-27}$$

Mechanical efficiency will be on the order of 75% to 95%, at high speed for modern automobile engines operating at wide-open throttle. It then decreases with



(a)



(b)

Figure 2-9 Four-stroke cycle of typical SI engine plotted on P-v coordinates at (a) wide open throttle, and (b) part throttle. The upper loop consists of the compression stroke and power stroke and the area represents gross indicated work. The lower loop represents negative work of the intake stroke and exhaust stroke. This is called indicated pump work.

decreasing engine speed to zero at idle conditions, when no work is taken off the crankshaft.

Care should be taken when using the terms "gross work" and "net work". In some older literature and textbooks, net work (or net power) meant the output of an engine with all components, while gross work (or gross power) meant the output of the engine with fan and exhaust system removed.

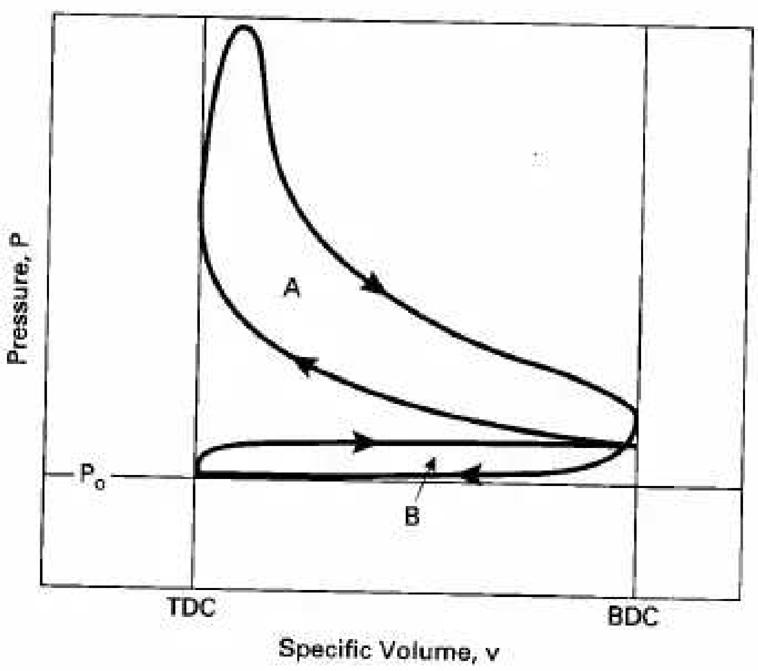


Figure 2-10 Four-stroke cycle of a SI engine equipped with a supercharger or turbocharger, plotted on P-v coordinates. For this cycle intake pressure is greater than exhaust pressure and the pump work loop represents positive work.

2-3 MEAN EFFECTIVE PRESSURE

From Fig. 2-9 it can be seen that pressure in the cylinder of an engine is continuously changing during the cycle. An average or **mean effective pressure** (mep) is defined by:

$$w = (\text{mep})\,\Delta v \tag{2-28}$$

or:

$$\text{mep} = w/\Delta v = W/V_d \tag{2-29}$$

$$\Delta v = v_{\text{BDC}} - v_{\text{TDC}} \tag{2-30}$$

where: W = work of one cycle
w = specific work of one cycle
V_d = displacement volume

Mean effective pressure is a good parameter to compare engines for design or output because it is independent of engine size and/or speed. If torque is used for engine comparison, a larger engine will always look better. If power is used as the comparison, speed becomes very important.

Various mean effective pressures can be defined by using different work terms in Eq. (2-29). If brake work is used, **brake mean effective pressure** is obtained:

$$\text{bmep} = w_b/\Delta v \tag{2-31}$$

Indicated work gives **indicated mean effective pressure**:

$$\text{imep} = w_i/\Delta v \tag{2-32}$$

The imep can further be divided into gross indicated mean effective pressure and net indicated mean effective pressure:

$$(\text{imep})_{\text{gross}} = (w_i)_{\text{gross}}/\Delta v \tag{2-33}$$

$$(\text{imep})_{\text{net}} = (w_i)_{\text{net}}/\Delta v \tag{2-34}$$

Pump mean effective pressure (which can have negative values):

$$\text{pmep} = w_{\text{pump}}/\Delta v \tag{2-35}$$

Friction mean effective pressure:

$$\text{fmep} = w_f/\Delta v \tag{2-36}$$

The following equations relate some of the previous definitions:

$$\begin{aligned} \text{nmep} &= \text{gmep} + \text{pmep} && \text{(a)} \\ \text{bmep} &= \text{nmep} - \text{fmep} && \text{(b)} \\ \text{bmep} &= \eta_m \text{imep} && \text{(c)} \\ \text{bmep} &= \text{imep} - \text{fmep} && \text{(d)} \end{aligned} \tag{2-37}$$

where: nmep = net mean effective pressure
η_m = mechanical efficiency of engine

Typical maximum values of bmep for naturally aspirated SI engines are in the range of 850 to 1050 kPa (120 to 150 psi). For CI engines, typical maximum values are 700 to 900 kPa (100 to 130 psi) for naturally aspirated engines and 1000 to 1200 kPa (145 to 175 psi) for turbocharged engines [58].

2-4 TORQUE AND POWER

Torque is a good indicator of an engine's ability to do work. It is defined as force acting at a moment distance and has units of N-m or lbf-ft. Torque τ is related to work by:

$$2\pi\tau = W_b = (\text{bmep})V_d/n \qquad (2\text{-}38)$$

where: W_b = brake work of one revolution
V_d = displacement volume
n = number of revolutions per cycle

For a two-stroke cycle engine with one cycle for each revolution:

$$2\pi\tau = W_b = (\text{bmep})V_d \qquad (2\text{-}39)$$

$$\tau = (\text{bmep})V_d/2\pi \quad \text{two-stroke cycle} \qquad (2\text{-}40)$$

For a four-stroke cycle engine which takes two revolutions per cycle:

$$\tau = (\text{bmep})V_d/4\pi \quad \text{four-stroke cycle} \qquad (2\text{-}41)$$

In these equations, bmep and brake work W_b are used because torque is measured off the output crankshaft.

Many modern automobile engines have maximum torque in the 200 to 300 N-m range at engine speeds usually around 4000 to 6000 RPM. The point of maximum torque is called maximum brake torque speed (MBT). A major goal in the design of a modern automobile engine is to *flatten* the torque-versus-speed curve as shown in Fig. 2-11, and to have high torque at both high and low speed. CI engines generally have greater torque than SI engines. Large engines often have very high torque values with MBT at relatively low speed.

Power is defined as the rate of work of the engine. If n = number of revolutions per cycle, and N = engine speed, then:

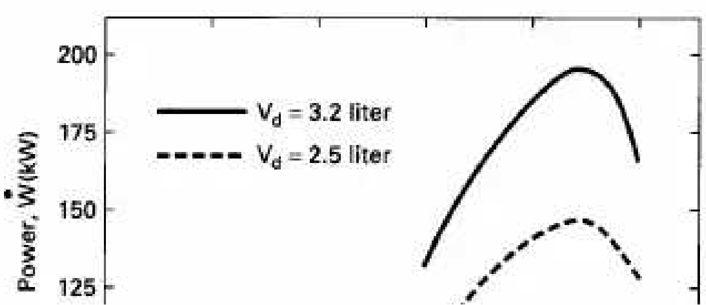
$$\dot{W} = WN/n \qquad (2\text{-}42)$$

$$\dot{W} = 2\pi N\tau \qquad (2\text{-}43)$$

$$\dot{W} = (1/2n)(\text{mep})A_p\overline{U}_p \qquad (2\text{-}44)$$

$$\dot{W} = (\text{mep})A_p\overline{U}_p/4 \quad \text{four-stroke cycle} \qquad (2\text{-}45)$$

$$\dot{W} = (\text{mep})A_p\overline{U}_p/2 \quad \text{two-stroke cycle} \qquad (2\text{-}46)$$



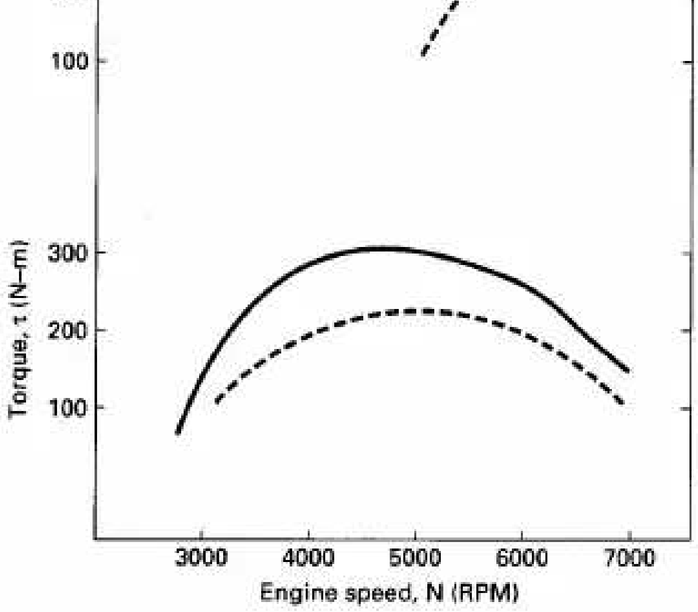


Figure 2-11 Brake power and torque of a typical automobile reciprocating engine as a function of engine speed. Speed at which peak torque occurs is called maximum brake torque (MBT) (or maximum best torque). Indicated power increases with speed while brake power increases to a maximum and then decreases. This is because friction power increases with engine speed to a higher power and becomes dominant at higher speeds. Fig. 2-8 shows power and torque curves for a specific engine.

where: W = work per cycle
A_p = piston face area of all pistons
$\overline{U}_p$ = average piston speed

Depending upon which definition of work or mep is used in Eqs. (2-42)–(2-46), power can be defined as brake power, net indicated power, gross indicated power, pumping power, and even friction power. Also:

$$\dot{W}_b = \eta_m \dot{W}_i \tag{2-47}$$

$$(\dot{W}_i)_{\text{net}} = (\dot{W}_i)_{\text{gross}} - (\dot{W}_i)_{\text{pump}} \tag{2-48}$$

$$\dot{W}_b = \dot{W}_i - \dot{W}_f \tag{2-49}$$

where η_m is the mechanical efficiency of the engine.

Power is normally measured in kW, but horsepower (hp) is still common.

$$\begin{aligned} 1\ \text{hp} &= 0.7457\ \text{kW} = 2545\ \text{BTU/hr} = 550\ \text{ft-lbf/sec} \\ 1\ \text{kW} &= 1.341\ \text{hp} \end{aligned} \tag{2-50}$$

Engine power can range from a few watts in small model airplane engines to thousands of kW per cylinder in large multiple-cylinder stationary and ship engines. There is a large commercial market for engines in the 1.5- to 5-kW (2-7 hp) range for lawn mowers, chain saws, snowblowers, etc. Power for outboard motors (engines) for small boats typically ranges from 2 to 40 kW (3-50 hp), with much larger ones available. Modern automobile engines range mostly from 40 to 220 kW (50-300 hp). It is interesting to note that a modern midsize aerodynamic automobile only requires about 5 to 6 kW (7-8 hp) to cruise at 55 mph on level roadway.

Both torque and power are functions of engine speed. At low speed, torque increases as engine speed increases. As engine speed increases further, torque reaches a maximum and then decreases as shown in Figs. 2-8 and 2-11. Torque decreases because the engine is unable to ingest a full charge of air at higher speeds. Indicated power increases with speed, while brake power increases to a maximum and then decreases at higher speeds. This is because friction losses increase with speed and become the dominant factor at very high speeds. For many automobile engines, maximum brake power occurs at about 6000 to 7000 RPM, about one and a half times the speed of maximum torque.

Greater power can be generated by increasing displacement, mep, and/or speed. Increased displacement increases engine mass and takes up space, both of which are contrary to automobile design trends. For this reason, most modern engines are smaller but run at higher speeds, and are often turbocharged or supercharged to increase mep.

Other ways which are sometimes used to classify engines are shown in Eqs. (2-51)-(2-54).

specific power $\quad SP = \dot{W}_b / A_p \quad$ (2-51)

output per displacement $\quad OPD = \dot{W}_b / V_d \quad$ (2-52)

specific volume $\quad SV = V_d / \dot{W}_b \quad$ (2-53)

specific weight $\quad SW = (\text{engine weight}) / \dot{W}_b \quad$ (2-54)

where: $\dot{W}_b$ = brake power
A_p = piston face area of all pistons
V_d = displacement volume

These parameters are important for engines used in transportation vehicles such as boats, automobiles, and especially airplanes, where keeping weight to a minimum is necessary. For large stationary engines, weight is not as important.

Modern automobile engines usually have brake power output per displacement in the range of 40 to 80 kW/L. The Honda eight-valve-per-cylinder V4 motorcycle engine generates about 130 kW/L, an extreme example of a high-performance racing engine [22]. One main reason for continued development to return to two-stroke cycle automobile engines is that they have up to 40% greater power output per unit weight.

HISTORIC-EIGHT-VALVES-PER-CYLINDER MOTORCYCLE ENGINE

In the early 1990s, Honda produced a racing motorcycle with a V4 engine, of which each cylinder had four intake valves and four exhaust valves. The engine was developed by modifying a V8 engine so that the motorcycle could be raced under rules restricting engines to four cylinders. A four-valve-per-cylinder V8 engine block was modified by removing the metal between each set of two cylinders. Special pistons were built to fit into the resulting non-round, oblong cylinders. This resulted in each cylinder having eight valves and a piston with two connecting rods using a common piston pin.

The final product was a very fast, very expensive motorcycle with an aluminum block, 90° V4 engine having a displacement of 748 cm^3. It produced 96 kW at 14,000 RPM and maximum torque of 71 N-m at 11,600 RPM [22, 143].

2-5 DYNAMOMETERS

Dynamometers are used to measure torque and power over the engine operating ranges of speed and load. They do this by using various methods to absorb the energy output of the engine, all of which eventually ends up as heat.

Some dynamometers absorb energy in a mechanical friction brake (prony brake). These are the simplest dynamometers but are not as flexible and accurate as others at higher energy levels.

Fluid or hydraulic dynamometers absorb engine energy in water or oil pumped through orifices or dissipated with viscous losses in a rotor-stator combination. Large amounts of energy can be absorbed in this manner, making this an attractive type of dynamometer for the largest of engines.

Eddy current dynamometers use a disk, driven by the engine being tested, rotating in a magnetic field of controlled strength. The rotating disk acts as an electrical conductor cutting the lines of magnetic flux and producing eddy currents in the disk. With no external circuit, the energy from the induced currents is absorbed in the disk.

One of the best types of dynamometers is the **electric dynamometer,** which absorbs energy with electrical output from a connected generator. In addition to having an accurate way of measuring the energy absorbed, the load is easily varied by changing the amount of resistance in the circuit connected to the generator output. Many electric dynamometers can also be operated in reverse, with the generator used as a motor to drive (or motor) an unfired engine. This allows the engine to be tested for mechanical friction losses and air pumping losses, quantities that are hard to measure on a running fired engine; see Section 11-2.

EXAMPLE PROBLEM 2-2

The engine in Example Problem 2-1 is connected to a dynamometer which gives a brake output torque reading of 205 N-m at 3600 RPM. At this speed air enters the cylinders at 85 kPa and 60°C, and the mechanical efficiency of the engine is 85%.

Calculate:

1. brake power
2. indicated power
3. brake mean effective pressure
4. indicated mean effective pressure
5. friction mean effective pressure
6. power lost to friction
7. brake work per unit mass of gas in the cylinder
8. brake specific power
9. brake output per displacement
10. engine specific volume

1) Using Eq. (2-43) to find brake power:

$$\dot{W}_b = 2\pi N\tau = (2\pi \text{ radians/rev})(3600/60 \text{ rev/sec})(205 \text{ N-m})$$
$$= 77{,}300 \text{ N-m/sec} = \underline{77.3 \text{ kW} = 104 \text{ hp}}$$

2) Using Eq. (2-47) to find indicated power:

$$\dot{W}_i = \dot{W}_b/\eta_m = (77.3 \text{ kW})/(0.85) = \underline{90.9 \text{ kW} = 122 \text{ hp}}$$

3) Using Eq. (2-41) to find the brake mean effective pressure:

3) Using Eq. (2-41) to find the brake mean effective pressure:

$$\text{bmep} = 4\pi\tau / V_d = (4\pi \text{ radians/cycle})(205 \text{ N-m}) / (0.003 \text{ m}^3\text{/cycle})$$
$$= \underline{859{,}000 \text{ N/m}^2 = 859 \text{ kPa} = 125 \text{ psia}}$$

4) Equation (2-37c) gives indicated mean effective pressure:

$$\text{imep} = \text{bmep} / \eta_m = (859 \text{ kPa}) / (0.85) \underline{= 1010 \text{ kPa} = 146 \text{ psia}}$$

5) Equation (2-37d) is used to calculate friction mean effective pressure:

$$\text{fmep} = \text{imep} - \text{bmep} = 1010 - 859 \underline{= 151 \text{ kPa} = 22 \text{ psia}}$$

6) Equations (2-15) and (2-44) are used to find friction power lost:

$$A_p = (\pi/4)B^2 = (\pi/4)(0.086 \text{ m})^2 = 0.00581 \text{ m}^2 \text{ for one cylinder}$$
$$\dot{W}_f = (1/2n)(\text{fmep})A_p\bar{U}_p$$
$$= (1/4)(151 \text{ kPa})(0.00581 \text{ m}^2\text{/cyl})(10.32 \text{ m/sec})(6 \text{ cyl})$$
$$\underline{= 13.6 \text{ kW} = 18 \text{ hp}}$$

Or, it can be obtained from Eq. (2-49):

$$\dot{W}_f = \dot{W}_i - \dot{W}_b = 90.9 - 77.3 = 13.6 \text{ kW}$$

7) First brake work is found for one cylinder for one cycle using Eq. (2-29):

$$W_b = (\text{bmep})V_d = (859 \text{ kPa})(0.0005 \text{ m}^3) = 0.43 \text{ kJ}$$

It can be assumed the gas entering the cylinders at BDC is air:

$$m_a = PV_{\text{BDC}} / RT = P(V_d + V_c)/RT$$
$$= (85 \text{ kPa})(0.0005 + 0.000059)\text{m}^3 / (0.287 \text{ kJ/kg-K})(333 \text{ K})$$
$$= 0.00050 \text{ kg}$$

Brake specific work per unit mass:

$$w_b = W_b / m_a = (0.43 \text{ kJ}) / (0.00050 \text{ kg}) \underline{= 860 \text{ kJ/kg} = 370 \text{ BTU/lbm}}$$

8) Equation (2-51) gives brake specific power:

$$\text{BSP} = \dot{W}_b / A_p = (77.3 \text{ kW}) / [(\pi/4)(0.086 \text{ m})^2(6 \text{ cylinders})]$$
$$\underline{= 2220 \text{ kW/m}^2 = 0.2220 \text{ kW/cm}^2 = 1.92 \text{ hp/in.}^2}$$

9) Equation (2-52) gives brake output per displacement:

$$\text{BOPD} = \dot{W}_b / V_d = (77.3 \text{ kW}) / (3 \text{ L})$$
$$\underline{= 25.8 \text{ kW/L} = 35 \text{ hp/L} = 0.567 \text{ hp/in.}^3}$$

10) Equation (2-53) gives engine specific volume:

$$\text{BSV} = V_d / \dot{W}_b = 1/\text{BOPD} = 1/25.8$$
$$\underline{= 0.0388 \text{ L/kW} = 0.0286 \text{ L/hp} = 1.76 \text{ in.}^3\text{/hp}}$$

2-6 AIR–FUEL RATIO AND FUEL–AIR RATIO

Energy input to an engine Q_{in} comes from the combustion of a hydrocarbon fuel. Air is used to supply the supply the oxygen needed for this chemical reaction. For combustion

reaction to occur, the proper relative amounts of air (oxygen) and fuel must be present.

Air–fuel ratio (AF) and fuel–air ratio (FA) are parameters used to describe the mixture ratio:

$$AF = m_a / m_f = \dot{m}_a / \dot{m}_f \tag{2-55}$$

$$FA = m_f / m_a = \dot{m}_f / \dot{m}_a = 1/AF \tag{2-56}$$

where: m_a = mass of air
$\dot{m}_a$ = mass flow rate of air
m_f = mass of fuel
$\dot{m}_f$ = mass flow rate of fuel

The ideal or stoichiometric AF for many gasoline-type hydrocarbon fuels is very close to 15:1, with combustion possible for values in the range 6 to 19. AF less than 6 is too *rich* to sustain combustion and AF greater than 19 is too *lean*. The fuel input system of an engine, fuel injectors or carburetor, must be able to regulate the proper amount of fuel for any given air flow. Gasoline-fueled engines usually have AF input in the range of 12 to 18 depending on the operating conditions at the time (e.g., accelerating, cruising, starting, etc.).

CI engines typically have AF input in the range of 18 to 70, which appears to be outside the limits where combustion is possible. Combustion occurs because the cylinder of a CI engine, unlike an SI engine, has a very non-homogeneous air–fuel mixture, with reaction only occurring in those regions where a combustible mixture exists, other regions being too rich or too lean.

Equivalence ratio ϕ is defined as the actual ratio of fuel–air to ideal or stoichiometric fuel–air:

$$\phi = (FA)_{act} / (FA)_{stoich} = (AF)_{stoich} / (AF)_{act} \tag{2-57}$$

2-7 SPECIFIC FUEL CONSUMPTION

Specific fuel consumption is defined by:

$$sfc = \dot{m}_f / \dot{W} \tag{2-58}$$

where: $\dot{m}_f$ = rate of fuel flow into engine
$\dot{W}$ = engine power

Brake power gives **brake specific fuel consumption:**

$$bsfc = \dot{m}_f / \dot{W}_b \tag{2-59}$$

Indicated power gives **indicated specific fuel consumption**:

$$isfc = \dot{m}_f / \dot{W}_i \tag{2-60}$$

Other examples of specific fuel consumption parameters can be defined as follows:

fsfc = friction specific fuel consumption

igsfc = indicated gross specific fuel consumption

insfc = indicated net specific fuel consumption

psfc = pumping specific fuel consumption

It also follows that:

$$\eta_m = \dot{W}_b / \dot{W}_i = (\dot{m}_f/\dot{W}_i) / (\dot{m}_f/\dot{W}_b) = (\text{isfc}) / (\text{bsfc}) \tag{2-61}$$

where: η_m = mechanical efficiency of engine

Brake specific fuel consumption decreases as engine speed increases, reaches a minimum, and then increases at high speeds (Fig. 2-12). Fuel consumption increases at high speed because of greater friction losses. At low engine speed, the longer time per cycle allows more heat loss and fuel consumption goes up. Figure 2-13 shows how bsfc also depends on compression ratio and fuel equivalence ratio. It decreases with higher compression ratio due to higher thermal efficiency. It is lowest when combustion occurs in a mixture with a fuel equivalence ratio near one, ($\phi = 1$). The further from stoichiometric combustion, either rich or lean, the higher will be the fuel consumption.

Brake specific fuel consumption generally decreases with engine size, being best (lowest) for very large engines (see Fig. 2-14).

Specific fuel consumption is generally given in units of gm/kW-hr or lbm/hp-hr. For transportation vehicles it is common to use **fuel economy** in terms of distance traveled per unit of fuel, such as miles per gallon (mpg). In SI units it is common to

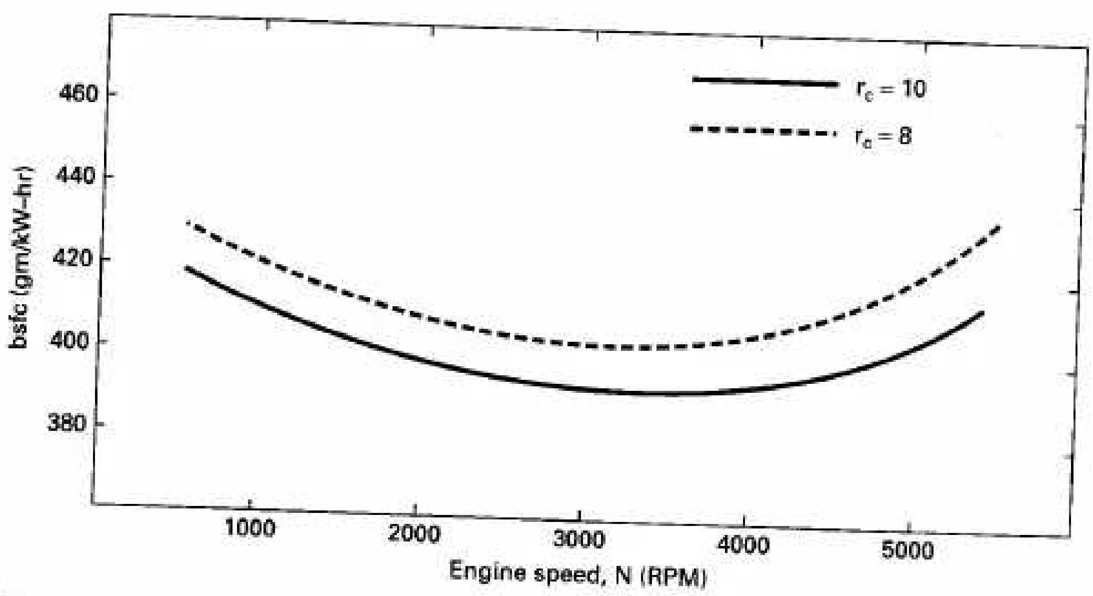


Figure 2-12 Brake specific fuel consumption as a function of engine speed. Fuel consumption decreases as engine speed increases due to the shorter time for heat loss during each cycle. At higher engine speeds fuel consumption again increases because of high friction losses. As compression ratio is increased fuel consumption decreases due to greater thermal efficiency.

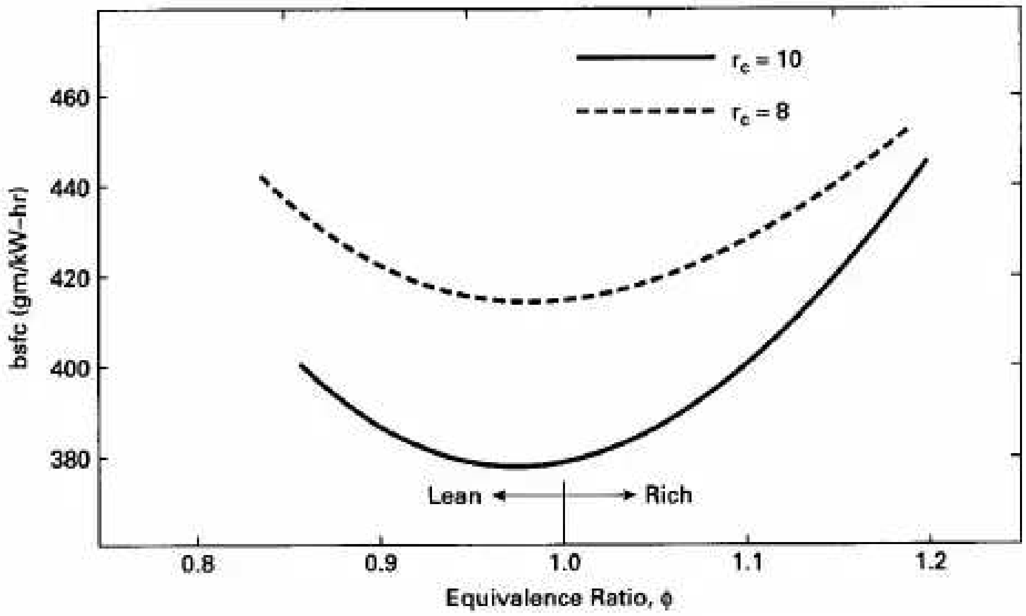


Figure 2-13 Brake specific fuel consumption as a function of fuel equivalence ratio. Consumption is minimum at a slightly lean condition, increasing with both richer and leaner mixtures.

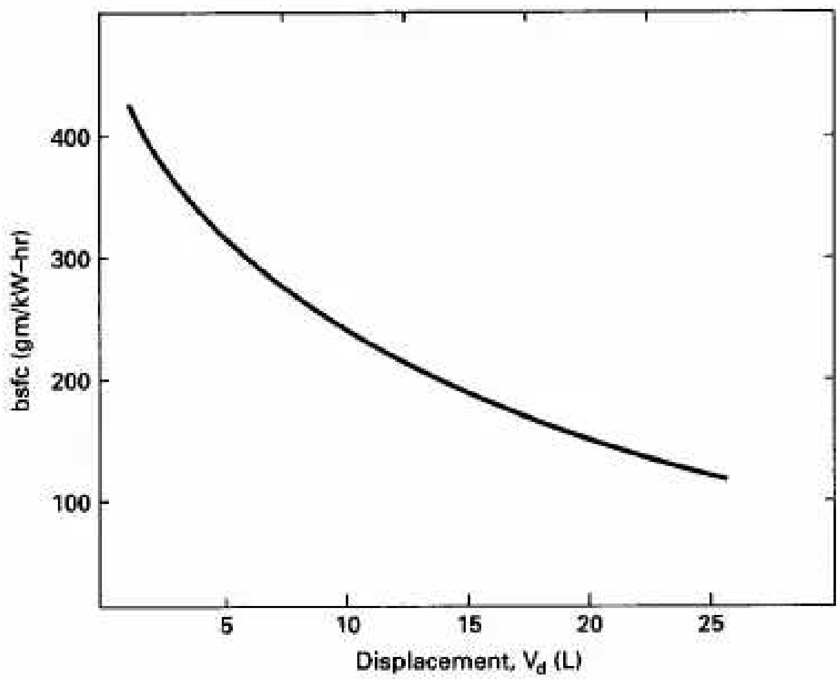


Figure 2-14 Brake specific fuel consumption as a function of engine displacement. Generally, average fuel consumption is less with larger engines. One reason for this is less heat loss due to the higher volume to surface area ratio of the combustion chamber in a large engine. Also, larger engines operate at lower speeds which reduces friction losses. Adapted from [123].

use the inverse of this, with (L/100 km) being a common unit. To decrease air pollution and depletion of fossil fuels, laws have been enacted requiring better vehicle fuel economy. Since the early 1970s, when most automobiles got less than 15 mpg (15.7 L/100 km) using gasoline, great strides have been made in improving fuel economy. Many modern automobiles now get between 30 and 40 mpg (7.8 and 5.9 L/100 km), with some small vehicles as high as 60 mpg (3.9 L/100 km).

2-8 ENGINE EFFICIENCIES

The time available for the combustion process of an engine cycle is very brief, and not all fuel molecules may find an oxygen molecule with which to combine, or the local temperature may not favor a reaction. Consequently, a small fraction of fuel does not react and exits with the exhaust flow. A **combustion efficiency** η_c is defined to account for the fraction of fuel which burns. η_c typically has values in the range 0.95 to 0.98 when an engine is operating properly. For one engine cycle in one cylinder, the heat added is:

$$Q_{in} = m_f Q_{HV} \eta_c \tag{2-62}$$

For steady state:

$$\dot{Q}_{in} = \dot{m}_f Q_{HV} \eta_c \tag{2-63}$$

and **thermal efficiency** is:

$$\eta_t = W/Q_{in} = \dot{W}/\dot{Q}_{in} = \dot{W}/\dot{m}_f Q_{HV} \eta_c = \eta_f / \eta_c \tag{2-64}$$

where: W = work of one cycle
$\dot{W}$ = power
m_f = mass of fuel for one cycle
$\dot{m}_f$ = mass flow rate of fuel
Q_{HV} = heating value of fuel
η_f = fuel conversion efficiency (see Eq. 2-66)

Thermal efficiency can be given as indicated or brake, depending on whether indicated power or brake power is used in Eq. (2-64). It follows that engine mechanical efficiency:

$$\eta_m = (\eta_t)_b / (\eta_t)_i \tag{2-65}$$

Engines can have indicated thermal efficiencies in the range of 50% to 60%, with brake thermal efficiency usually about 30%. Some large, slow CI engines can have brake thermal efficiencies greater than 50%.

Fuel conversion efficiency is defined as:

$$\eta_f = W / m_f Q_{HV} = \dot{W} / \dot{m}_f Q_{HV} \tag{2-66}$$

$$\eta_f = 1/(\text{sfc}) Q_{HV} \tag{2-67}$$

For a single cycle of one cylinder the thermal efficiency can be written:

$$\eta_t = W / m_f Q_{HV} \eta_c \quad (2\text{-}68)$$

This is the thermal efficiency introduced in basic thermodynamic textbooks, sometimes called *enthalpy efficiency*.

2-9 VOLUMETRIC EFFICIENCY

One of the most important processes that governs how much power and performance can be obtained from an engine is getting the maximum amount of air into the cylinder during each cycle. More air means more fuel can be burned and more energy can be converted to output power. Getting the relatively small volume of liquid fuel into the cylinder is much easier than getting the large volume of gaseous air needed to react with the fuel. Ideally, a mass of air equal to the density of atmospheric air times the displacement volume of the cylinder should be ingested for each cycle. However, because of the short cycle time available and the flow restrictions presented by the air cleaner, carburetor (if any), intake manifold, and intake valve(s), less than this ideal amount of air enters the cylinder. **Volumetric efficiency** is defined as:

$$\eta_v = m_a / \rho_a V_d \quad (2\text{-}69)$$

$$\eta_v = n\dot{m}_a / \rho_a V_d N \quad (2\text{-}70)$$

where: m_a = mass of air into the engine (or cylinder) for one cycle
$\dot{m}_a$ = steady-state flow of air into the engine
ρ_a = air density evaluated at atmospheric conditions outside the engine
V_d = displacement volume
N = engine speed
n = number of revolutions per cycle

Unless better values are known, standard values of surrounding air pressure and temperature can be used to find density:

$$P_o(\text{standard}) = 101 \text{ kPa} = 14.7 \text{ psia}$$

$$T_o(\text{standard}) = 298 \text{ K} = 25°\text{C} = 537°\text{R} = 77°\text{F}$$

$$\rho_a = P_o / RT_o \quad (2\text{-}71)$$

where: P_o = pressure of surrounding air
T_o = temperature of surrounding air
R = gas constant for air = 0.287 kJ/kg-K
= 53.33 ft-lbf/lbm-°R

At standard conditions, the density of air $\rho_a = 1.181 \text{ kg/m}^3 = 0.0739 \text{ lbm/ft}^3$.

When volumetric efficiency is measured experimentally, corrections can be made for temperature and humidity when other than standard conditions are experienced.

Sometimes (less common) the air density in Eqs. (2-69) and (2-70) is evaluated at conditions in the intake manifold immediately before it enters the cylinder. The conditions at this point will usually be hotter and at a lower pressure than surrounding atmospheric conditions.

Typical values of volumetric efficiency for an engine at wide-open throttle (WOT) are in the range 75% to 90%, going down to much lower values as the throttle is closed. Restricting air flow into an engine (closing the throttle) is the primary means of power control for a spark ignition engine.

EXAMPLE PROBLEM 2-3

The engine in Example Problem 2-2 is running with an air-fuel ratio AF = 15, a fuel heating value of 44,000 kJ/kg, and a combustion efficiency of 97%. Calculate:

1. rate of fuel flow into engine
2. brake thermal efficiency
3. indicated thermal efficiency
4. volumetric efficiency
5. brake specific fuel consumption

1) From Example Problem 2-2, the mass of air in one cylinder for one cycle is m_a = 0.00050 kg. Then:

$$m_f = m_a / \text{AF} = 0.00050/15 = 0.000033 \text{ kg of fuel per cylinder per cycle}$$

Therefore, the rate of fuel flow into the engine is:

$$\dot{m}_f = (0.000033 \text{ kg/cyl-cycle})(6 \text{ cyl})(3600/60 \text{ rev/sec})(1 \text{ cycle}/2 \text{ rev})$$
$$\underline{= 0.0060 \text{ kg/sec} = 0.0132 \text{ lbm/sec}}$$

2) Using Eq. (2-64) to find brake thermal efficiency:

$$(\eta_t)_b = \dot{W}_b / \dot{m}_f Q_{HV}\, \eta_c = (77.3 \text{ kW}) / (0.0060 \text{ kg/sec})(44{,}000 \text{ kJ/kg})(0.97)$$
$$\underline{= 0.302 = 30.2\%}$$

Or, using Eq. (2-68) for one cycle of one cylinder:

$$(\eta_t)_b = W_b / m_f Q_{HV}\, \eta_c = (0.43 \text{ kJ})/(0.000033 \text{ kg})(44{,}000 \text{ kJ/kg})(0.97)$$
$$= 0.302$$

3) Indicated thermal efficiency using Eq. (2-65):

$$(\eta_t)_i = (\eta_t)_b / \eta_m = 0.302/0.85 \underline{= 0.355 = 35.5\%}$$

4) Using Eq. (2-69) with standard air density for volumetric efficiency:

$$\eta_v = m_a / \rho_a V_d = (0.00050 \text{ kg}) / (1.181 \text{ kg/m}^3)(0.0005 \text{ m}^3)$$
$$\underline{= 0.847 = 84.7\%}$$

5) Using Eq. (2-59) for brake specific fuel consumption:

$$\text{bsfc} = \dot{m}_f / \dot{W}_b = (0.0060 \text{ kg/sec}) / (77.3 \text{ kW})$$

$$\text{bsfc} = \dot{m}_f/\dot{W}_b = (0.0060\ \text{kg/sec})/(77.3\ \text{kW})$$

$$= 7.76 \times 10^{-5}\ \text{kg/kW-sec} = \underline{279\ \text{gm/kW-hr} = 0.459\ \text{lbm/hp-hr}}$$

2-10 EMISSIONS

The four main engine exhaust emissions which must be controlled are oxides of nitrogen (NOx), carbon monoxide (CO), hydrocarbons (He), and solid particulates (part). Two common methods of measuring the amounts of these pollutants are **specific emissions** (SE) and the **emissions index** (EI). Specific emissions typically has units of gm/kW-hr, while the emissions index has units of emissions flow per fuel flow.

Specific Emissions:

$$\begin{aligned}
(\text{SE})_{\text{NOx}} &= \dot{m}_{\text{NOx}}/\dot{W}_b \\
(\text{SE})_{\text{CO}} &= \dot{m}_{\text{CO}}/\dot{W}_b \\
(\text{SE})_{\text{HC}} &= \dot{m}_{\text{HC}}/\dot{W}_b \\
(\text{SE})_{\text{part}} &= \dot{m}_{\text{part}}/\dot{W}_b
\end{aligned} \tag{2-72}$$

where: $\dot{m}$ = flow rate of emissions in gm/hr
$\dot{W}_b$ = brake power

Emissions Index:

$$\begin{aligned}
(\text{EI})_{\text{NOx}} &= \dot{m}_{\text{NOx}}[\text{gm/sec}]/\dot{m}_f[\text{kg/sec}] \\
(\text{EI})_{\text{CO}} &= \dot{m}_{\text{CO}}[\text{gm/sec}]/\dot{m}_f[\text{kg/sec}] \\
(\text{EI})_{\text{HC}} &= \dot{m}_{\text{HC}}[\text{gm/sec}]/\dot{m}_f[\text{kg/sec}] \\
(\text{EI})_{\text{part}} &= \dot{m}_{\text{part}}[\text{gm/sec}]/\dot{m}_f[\text{kg/sec}]
\end{aligned} \tag{2-73}$$

2-11 NOISE ABATEMENT

In recent years a lot of research and development has been directed towards reduc-
ing engine and exhaust noise. This can be done in one of three ways: *passive, semi-active,* or *active.* Noise reduction is accomplished passively by correct design and the use of proper materials. The use of ribs and stiffeners, composite materials,

and sandwich construction is now routine. This type of construction reduces noise vibrations in the various engine components.

Hydraulics are often used in semiactive noise abatement systems. Some engines are equipped with flywheels which have hydraulic passages through which fluid flows. At idle and other constant-speed operation, the system is designed to give the flywheel the proper stiffness to absorb engine vibrations for frequencies at that condition. When acceleration occurs the flywheel fluid flows to other locations, changing the overall stiffness of the flywheel and making it more absorbent to the new vibration frequency. Some automobiles have hydraulic engine mounts connecting the engine to the automobile body. Fluid in these mounts acts to absorb and dampen engine vibrations and isolate them from the passenger compartment. Engine mounts using electrorheological fluid are being developed which will allow better vibration dampening at all frequencies. The viscosity of these fluids can be changed by as much as a factor of 50:1 with the application of an external voltage. Engine noise (vibration) is sensed by accelerometers which feed this information into the engine management system (EMS). Here the frequency is analyzed, and proper voltage is applied to the engine mounts to best dampen that frequency [38]. Response time is on the order of 0.005 second.

Active noise abatement is accomplished by generating *antinoise* to cancel out engine or exhaust noise. This is done by sensing the noise with a receiver, analyzing the frequency of the noise, and then generating noise of equal frequency, but out of phase with the original noise. If two noises are at the same frequency but 180° out of phase, the wave fronts cancel each other and the noise is eliminated. This method works well with constant-speed engines and other rotating equipment but is only partially successful with variable-speed automobile engines. It requires additional electronic equipment (receiver, frequency analyzer, transmitter) than that used with normal EMS computers. Some automobiles have receivers and transmitters mounted under the seats in the passenger compartment as an active engine noise abatement system. Similar systems are used near the end of the tailpipe, a major source of engine-related noise.

Noise reduction has been so successful that some automobiles are now equipped with a safety switch on the starter. At idle speed, the engine is so quiet that the safety switch is required to keep drivers from trying to start the engine when it is already running.

2-12 CONCLUSIONS-WORKING EQUATIONS

In this chapter equations relating the working parameters of engine operation have been developed, giving tools by which these parameters can be used for engine design and characterization. By combining earlier equations from the chapter, the following additional working equations are obtained. These are given as general equations and as specific equations to be used either with SI units or with English

units. In the specific equations, units which must be used to satisfy the equality are given in brackets.

Torque:

$$\tau = \eta_f \eta_v V_d Q_{HV} \rho_a (FA)/2\pi n \tag{2-74}$$

For SI units:

$$\tau[\text{N-m}] = 159.2\, \dot{W}[\text{kW}]/N[\text{rev/sec}] \tag{2-75}$$

For English units:

$$\tau[\text{lbf-ft}] = 5252\ \dot{W}[\text{hp}]/\text{N}[\text{RPM}] \quad (2\text{-}76)$$

Power:

$$\dot{W}_b = \dot{m}_f/(\text{bsfc}) = (\text{FA})\dot{m}_a/(\text{bsfc}) \quad (2\text{-}77)$$

$$\dot{W}_b = \eta_f \eta_v N V_d Q_{\text{HV}} \rho_a (\text{FA})/n \quad (2\text{-}78)$$

For SI units:

$$\dot{W}_b[\text{kW}] = N[\text{rev/sec}]\,\tau[\text{N-m}]/159.2 \quad (2\text{-}79)$$

$$\dot{W}_b[\text{kW}] = \text{bmep}[\text{kPa}]V_d[\text{L}]N[\text{rev/sec}]/1000\,n[\text{rev/cycle}] \quad (2\text{-}80)$$

For English units:

$$\dot{W}_b[\text{hp}] = N[\text{RPM}]\,\tau[\text{lbf-ft}]/5252 \quad (2\text{-}81)$$

$$\dot{W}_b[\text{hp}] = \text{bmep}[\text{psia}]\,V_d[\text{in.}^3]N[\text{RPM}]/396{,}000\,n[\text{rev/cycle}] \quad (2\text{-}82)$$

Mechanical Efficiency:

$$\eta_m = \dot{W}_b/\dot{W}_i = \text{bmep/imep} = 1 - \dot{W}_f/\dot{W}_i \quad (2\text{-}83)$$

Mean Effective Pressure:

$$\text{bmep} = 2\pi n\tau/V_d \quad (2\text{-}84)$$

$$\text{mep} = n\dot{W}/V_d N \quad (2\text{-}85)$$

For SI units:

$$\text{bmep}[\text{kPa}] = 6.28\,n[\text{rev/cycle}]\,\tau[\text{N-m}]/V_d[\text{L}] \quad (2\text{-}86)$$

$$\text{mep}[\text{kPa}] = 1000\,\dot{W}[\text{kW}]n[\text{rev/cycle}]/V_d[\text{L}]N[\text{rev/sec}] \quad (2\text{-}87)$$

For English units:

$$\text{bmep}[\text{psia}] = 75.4\,n[\text{rev/cycle}]\,\tau[\text{lb}_f\text{-ft}]/V_d[\text{in.}^3] \quad (2\text{-}88)$$

$$\text{mep}[\text{psia}] = 396{,}000\,\dot{W}[\text{hp}]n[\text{rev/cycle}]/V_d[\text{in.}^3]N[\text{RPM}] \quad (2\text{-}89)$$

Specific Power:

$$\dot{W}/A_p = \eta_f \eta_v N S Q_{\text{HV}} \rho_a (\text{FA})/n \quad (2\text{-}90)$$

$$\dot{W}/A_p = \eta_f \eta_v \overline{U}_p Q_{\text{HV}} \rho_a (\text{FA})/2n \quad (2\text{-}91)$$

PROBLEMS

2-1. As Becky was driving "Old Betsy," the family station wagon, the engine finally quit, being worn out after 171,000 miles. It can be assumed that the average speed over its lifetime was 40 mph at an engine speed of 1700 RPM. The engine is a five-liter V8 operating on a four-stroke cycle.

ating on a four-stroke cycle.

Calculate: **(a)** How many revolutions has the engine experienced?
(b) How many spark plug firings have occurred in the entire engine?
(c) How many intake strokes have occurred in one cylinder?

2-2. A four-cylinder, two-stroke cycle diesel engine with 10.9-cm bore and 12.6-cm stroke produces 88 kW of brake power at 2000 RPM. Compression ratio $r_c = 18{:}1$.
Calculate: **(a)** Engine displacement. [cm^3, L]
(b) Brake mean effective pressure. [kPa]
(c) Torque. [N-m]
(d) Clearance volume of one cylinder. [cm^3]

2-3. A four-cylinder, 2.4-liter engine operates on a four-stroke cycle at 3200 RPM. The compression ratio is 9.4:1, the connecting rod length $r = 18$ cm, and the bore and stroke are related as $S = 1.06\ B$.
Calculate: **(a)** Clearance volume of one cylinder in cm^3, L, and $in.^3$.
(b) Bore and stroke in cm and in.
(c) Average piston speed in m/sec and ft/sec.

2-4. What are the advantages of an over square engine? What are the advantages of an under square engine?

2-5. In Problem 2-3, what is the average piston speed and what is the piston speed when the crank angle $\theta = 90°$ aTDC? [m/sec]

2-6. A five-cylinder, 3.5-liter SI engine operates on a four-stroke cycle at 2500 RPM. At this condition, the mechanical efficiency of the engine is 62% and 1000 J of indicated work are produced each cycle in each cylinder.
Calculate: **(a)** Indicated mean effective pressure. [kPa]
(b) Brake mean effective pressure. [kPa]
(c) Friction mean effective pressure. [kPa]
(d) Brake power in kW and hp.
(e) Torque. [N-m]

2-7. The engine operating at the conditions in Problem 2-6 is square, with $S = B$.
Calculate: **(a)** Specific power. [kW/cm^2]
(b) Output per displacement. [kW/cm^3]
(c) Specific volume. [cm^3/kW]
(d) Power lost to friction in kW and hp.

2-8. The engine operating at the conditions in Example Problem 2-3 has a combustion efficiency of 97%.

Calculate: (a) Rate of unburned hydrocarbon fuel which is expelled into the exhaust system. [kg/hr]
(b) Specific emissions of He. [(gm/kW-hr]
(c) Emissions index of He.

2-9. A construction vehicle has a diesel engine with eight cylinders of 5.375-inch bore and 8.0-inch stroke, operating on a four-stroke cycle. It delivers 152-shaft horsepower at 1000 RPM, with a mechanical efficiency of 0.60.
Calculate: (a) Total engine displacement. [$in.^3$]
(b) Brake mean effective pressure. [psia]
(c) Torque at 1000 RPM. [lbf-ft]
(d) Indicated horsepower.
(e) Friction horsepower.

2-10. A 1500-cm^3, four-stroke cycle, four-cylinder CI engine, operating at 3000 RPM, produces 48 kW of brake power. Volumetric efficiency is 0.92 and air-fuel ratio AF = 21:1.
Calculate: (a) Rate of air flow into engine. [kg/see]
(b) Brake specific fuel consumption. [gm/kW-hr]
(c) Mass rate of exhaust flow. [kg/hr]

(d) Brake output per displacement. [kW/L]

2-11. A pickup truck has a five-liter, V6, SI engine operating at 2400 RPM. The compression ratio rc = 10.2:1, the volumetric efficiency TIv = 0.91, and the bore and stroke are related as stroke S = 0.92 B.

Calculate: (a) Stroke length. [em]
(b) Average piston speed. [rnlsec]
(c) Clearance volume of one cylinder. [cm^3]
(d) Air flow rate into engine. [kg/see]

2-12. A small single-cylinder, two-stroke cycle SI engine operates at 8000 RPM with a volumetric efficiency of TIv = 0.85. The engine is square (bore = stroke) and has a displacement of 6.28 em3. The fuel-air ratio FA = 0.067.

Calculate: (a) Average piston speed. [m/sec]
(b) Flow rate of air into engine. [kg/see]
(c) Flow rate of fuel into engine. [kg/see]
(d) Fuel input for one cycle. [kg/cycle]

2-13. A single-cylinder, four-stroke cycle CI engine with 12.9-cm bore and 18.0-cm stroke, operating at 800 RPM, uses 0.113 kg of fuel in four minutes while developing a torque of76 N-m.

Calculate: (a) Brake specific fuel consumption. [grnlkW-hr]
(b) Brake mean effective pressure. [kpa]
(c) Brake power. [kW]
(d) Specific power. [kW/cm^2]
(e) Output per displacement. [kW/L]
(f) Specific volume. [L/kW]

2-14. A 302-in.3 displacement, V8, four-stroke cycle SI engine mounted on a hydraulic dynamometer has an output of 72 hp at 4050 RPM. Water absorbs the energy output of the engine as it flows through the dynamometer at a rate of 30 gallons per minute. The dynamometer has an efficiency of 93% and the water enters at a temperature of 46°F.

Calculate: (a) Exit temperature of the water. [OF]

(b) Torque output of the engine at this condition. [lbf-ft]
(c) What is bmep at this condition? [psia]

2-15. A 3.1-liter, four-cylinder, two-stroke cycle SI engine is mounted on an electrical generator dynamometer. When the engine is running at 1200 RPM, output from the 220-volt DC generator is 54.2 amps. The generator has an efficiency of 87%.

Calculate: **(a)** Power output of the engine in kW and hp.
(b) Engine torque. [N-m]
(c) What is engine bmep? [kPa]

2-16. An SI, six-liter, V8 race car engine operates at WOT on a four-stroke cycle at 6000 RPM using stoichiometric nitromethane. Fuel enters the engine at a rate of 0.198 kg/sec and combustion efficiency is 99%.

Calculate: **(a)** Volumetric efficiency of engine. [%]
(b) Flow rate of air into engine. [kg/sec]
(c) Heat added per cycle per cylinder. [kJ]
(d) Chemical energy from unburned fuel in the exhaust. [kW]

DESIGN PROBLEMS

2-10. Design a six-liter race car engine that operates on a four-stroke cycle. Decide what the design speed will be, and then give the number of cylinders, bore, stroke, piston rod length, average piston speed, imep, brake torque, fuel used, AF, and brake power, all at design speed. All parameter values should be within typical, reasonable values and should be consistent with the other values. State what assumptions you make (e.g., mechanical efficiency, volumetric efficiency, etc.)

2-20. Design a six-horsepower engine for a snowblower. Decide on the operating speed, number of strokes in cycle, carburetor or fuel injectors, and total displacement. Give the number of cylinders, bore, stroke, connecting rod length, average piston speed, brake torque, and brake power. What special considerations must be made knowing that this engine must start in very cold weather? All parameter values should be within typical, reasonable values and should be consistent with the other values. State all assumptions you make.

2-30. Design a small four-stroke cycle Diesel engine to produce 50 kW of brake power at design speed when installed in a small pickup truck. Average piston speed should not exceed 8 m/sec at design conditions. Give the design speed, displacement, number of cylinders, bore, stroke, bmep, and torque. All parameter values should be within typical, reasonable values and should be consistent with the other values. State all assumptions you make.

3

Engine Cycles

This chapter studies the basic cycles used in reciprocating internal combustion engines, both four-stroke and two-stroke. The most common four-stroke SI and CI engines, both four stroke and two stroke. The most common four-stroke SI and CI cycles are analyzed in detail using air-standard analysis. Lesser used cycles, including some historic, are analyzed in less detail.

3-' AIR-STANDARD CYCLES

The cycle experienced in the cylinder of an internal combustion engine is very complex. First, air (CI engine) or air mixed with fuel (SI engine) is ingested and mixed with the slight amount of exhaust residual remaining from the previous cycle. This mixture is then compressed and combusted, changing the composition to exhaust products consisting largely of CO_2, H_2O, and N_2 with many other lesser components. Then, after an expansion process, the exhaust valve is opened and this gas mixture is expelled to the surroundings. Thus, it is an open cycle with changing composition, a difficult system to analyze. To make the analysis of the engine cycle much more manageable, the real cycle is approximated with an ideal **air-standard cycle** which differs from the actual by the following:

1. The gas mixture in the cylinder is treated as air for the entire cycle, and property values of air are used in the analysis. This is a good approximation during the first half of the cycle, when most of the gas in the cylinder is air with only up to about 7% fuel vapor. Even in the second half of the cycle, when the gas composition is mostly CO_2, H_2O, and N_2, using air properties does not create large errors in the analysis. Air will be treated as an ideal gas with constant specific heats.
2. The real open cycle is changed into a closed cycle by assuming that the gases being exhausted are fed back into the intake system. This works with ideal air-standard cycles, as both intake gases and exhaust gases are air. Closing the cycle simplifies the analysis.
3. The combustion process is replaced with a heat addition term Q_{in} of equal energy value. Air alone cannot combust.
4. The open exhaust process, which carries a large amount of enthalpy out of the system, is replaced with a closed system heat rejection process Q_{out} of equal energy value.

S. Actual engine processes are approximated with ideal processes.

(a) The almost-constant-pressure intake and exhaust strokes are assumed to be constant pressure. At WOT, the intake stroke is assumed to be at a pressure P_0 of one atmosphere. At partially closed throttle or when supercharged, inlet pressure will be some constant value other than one atmosphere. The exhaust stroke pressure is assumed constant at one atmosphere.

(b) Compression strokes and expansion strokes are approximated by isentropic processes. To be truly isentropic would require these strokes to be reversible and adiabatic. There is some friction between the piston and cylinder walls but, because the surfaces are highly polished and lubricated, this friction is kept to a minimum and the processes are close to frictionless and reversible. If this were not true, automobile engines would wear out long before the 150-200 thousand miles which they now last if properly maintained. There is also fluid friction because of the gas motion within the cylinders during these strokes. This too is minimal. Heat transfer for anyone stroke will be negligibly small due to the very short time involved for that single process. Thus, an almost reversible and almost adiabatic process can quite accurately be approximated with an isentropic process.

(c) The combustion process is idealized by a constant-volume process (SI cycle), a constant-pressure process (CI cycle), or a combination of both (CI Dual cycle).

(d) Exhaust blowdown is approximated by a constant-volume process.

(e) All processes are considered reversible.

In air-standard cycles, air is considered an ideal gas such that the following ideal gas relationships can be used:

$$
\left.
\begin{aligned}
Pv &= RT && \text{(a)}\\
PV &= mRT && \text{(b)}\\
P &= \rho RT && \text{(c)}\\
dh &= c_p dT && \text{(d)}\\
du &= c_v dT && \text{(e)}\\
Pv^k &= \text{constant} \quad \text{isentropic process} && \text{(f)}\\
Tv^{k-1} &= \text{constant} \quad \text{isentropic process} && \text{(g)}\\
TP^{(1-k)/k} &= \text{constant} \quad \text{isentropic process} && \text{(h)}\\
w_{1-2} &= (P_2v_2 - P_1v_1)/(1-k) \quad \text{isentropic work in closed system} && \text{(i)}\\
&= R(T_2 - T_1)/(1-k) &&\\
c &= \sqrt{kRT} \quad \text{speed of sound} && \text{(j)}
\end{aligned}
\right\} \quad (3\text{-}1)
$$

where:
P = gas pressure in cylinder
V = volume in cylinder
v = specific volume of gas
R = gas constant of air
T = temperature
m = mass of gas in cylinder
ρ = density
h = specific enthalpy
u = specific internal energy
c_p, c_v = specific heats
$k = c_p/c_v$
w = specific work
c = speed of sound

In addition to these, the following variables are used in this chapter for cycle analysis:

$$AF = \text{air–fuel ratio}$$
$$\dot{m} = \text{mass flow rate}$$
$$q = \text{heat transfer per unit mass for one cycle}$$
$$\dot{q} = \text{heat transfer rate per unit mass}$$
$$Q = \text{heat transfer for one cycle}$$
$$\dot{Q} = \text{heat transfer rate}$$
$$Q_{HV} = \text{heating value of fuel}$$

$$r_c = \text{compression ratio}$$
$$W = \text{work for one cycle}$$
$$\dot{W} = \text{power}$$
$$\eta_c = \text{combustion efficiency}$$

Subscripts used include:

$$a = \text{air}$$
$$f = \text{fuel}$$
$$ex = \text{exhaust}$$
$$m = \text{mixture of all gases}$$

For thermodynamic analysis the specific heats of air can be treated as functions of temperature, which they are, or they can be treated as constants, which simplifies calculations at a slight loss of accuracy. In this textbook, constant specific heat analysis will be used. Because of the high temperatures and large temperature range experienced during an engine cycle, the specific heats and ratio of specific heats k do vary by a fair amount (see Table A-I in the Appendix). At the low-temperature end of a cycle during intake and start of compression, a value of $k = 1.4$ is correct. However, at the end of combustion the temperature has risen such that $k = 1.3$ would be more accurate. A constant average value between these extremes is found to give better results than a standard condition (25°C) value, as is often used in elementary thermodynamics textbooks. When analyzing what occurs within engines during the operating cycle and exhaust flow, this book uses the following air property values:

$$c_p = 1.108 \text{ kJ/kg-K} = 0.265 \text{ BTU/lbm-°R}$$
$$c_v = 0.821 \text{ kJ/kg-K} = 0.196 \text{ BTU/lbm-°R}$$
$$k = c_p/c_v = 1.108/0.821 = 1.35$$
$$R = c_p - c_v = 0.287 \text{ kJ/kg-K}$$
$$= 0.069 \text{ BTU/lbm-°R} = 53.33 \text{ ft-lbf/lbm-°R}$$

Air flow before it enters an engine is usually closer to standard temperature, and for these conditions a value of $k = 1.4$ is correct. This would include processes such as inlet flow in superchargers, turbochargers, and carburetors, and air flow through the engine radiator. For these conditions, the following air property values are used:

$$c_p = 1.005 \text{ kJ/kg-K} = 0.240 \text{ BTU/lbm-°R}$$
$$c_v = 0.718 \text{ kJ/kg-K} = 0.172 \text{ BTU/lbm-°R}$$
$$k = c_p/c_v = 1.005/0.718 = 1.40$$
$$R = c_p - c_v = 0.287 \text{ kJ/kg-K}$$

HISTORIC—SIX-STROKE CYCLES

During the second half of the 19th century, when development of the modern reciprocating internal combustion engine was in its early stages, many types of engines operating on many different cycles were tried. These included various two-, four-, and even six-stroke cycles. Six-stroke cycles were similar to four-stroke cycles with two added strokes for additional exhaust removal (i.e., three revolutions per cycle instead of two). With poor fuel quality, low compression ratios, and large clearance volumes, early engines had problems with excessive exhaust residual. After the exhaust stroke, an additional intake stroke was added which ingested only air. The air mixed with the exhaust residual and was then expelled with a second exhaust stroke. Compare this with the concept of EGR, which adds exhaust gas to the incoming air of all modern automobile engines [29].

3-2 OTTO CYCLE

The cycle of a four-stroke, SI, naturally aspirated engine at WOT is shown in Fig. 2-6. This is the cycle of most automobile engines and other four-stroke SI engines. For analysis, this cycle is approximated by the air-standard cycle shown in Fig. 3-1. This ideal air-standard cycle is called an **Otto cycle,** named after one of the early developers of this type of engine.

The intake stroke of the Otto cycle starts with the piston at TDC and is a constant-pressure process at an inlet pressure of one atmosphere (process 6-1 in Fig. 3-1). This is a good approximation to the inlet process of a real engine at WOT, which will actually be at a pressure slightly less than atmospheric due to pressure losses in the inlet air flow. The temperature of the air during the inlet stroke is increased as the air passes through the hot intake manifold. The temperature at point 1 will generally be on the order of 25° to 35°C hotter than the surrounding air temperature.

The second stroke of the cycle is the compression stroke, which in the Otto cycle is an isentropic compression from BDC to TDC (process 1-2). This is a good approximation to compression in a real engine, except for the very beginning and the very end of the stroke. In a real engine, the beginning of the stroke is affected by the intake valve not being fully closed until slightly after BDC. The end of compression is affected by the firing of the spark plug before TDC. Not only is there an increase in pressure during the compression stroke, but the temperature within the cylinder is increased substantially due to compressive heating.

The compression stroke is followed by a constant-volume heat input process 2-3 at TDC. This replaces the combustion process of the real engine cycle, which occurs at close to constant-volume conditions. In a real engine combustion is started slightly bTDC, reaches its maximum speed near TDC, and is terminated a

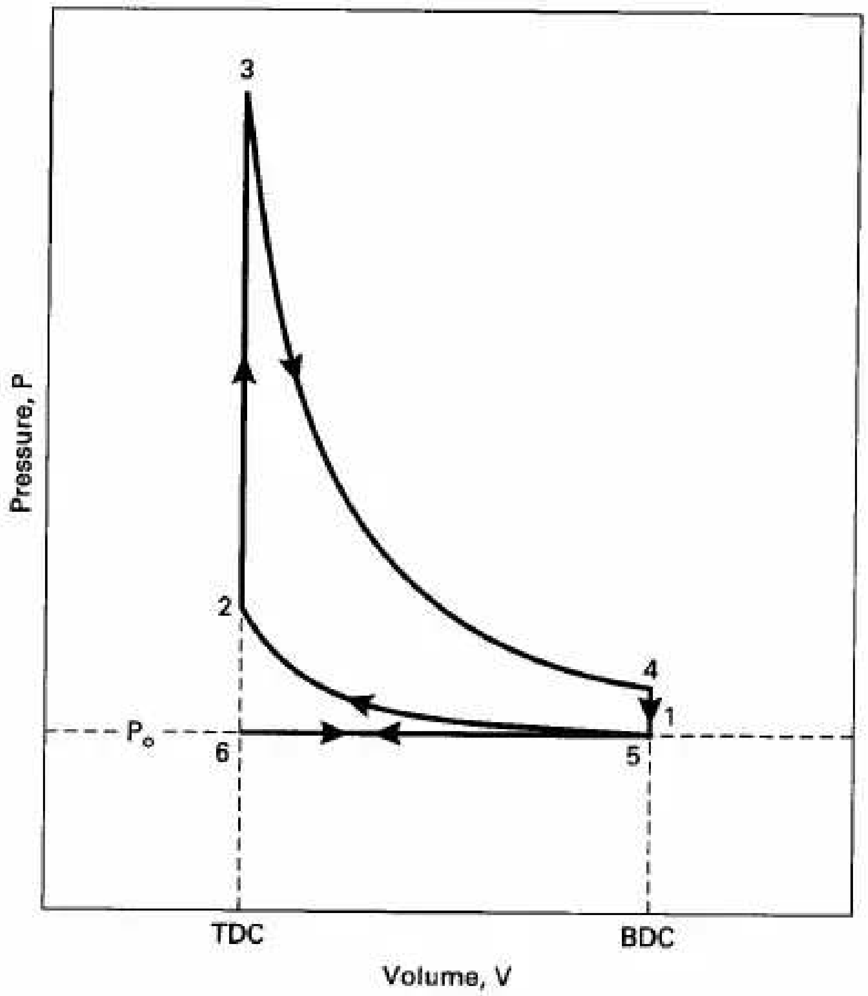


Figure 3-1 Ideal air-standard Otto cycle, 6-1-2-3-4-5-6, which approximates the four-stroke cycle of an SI engine on P-V coordinates.

little aTDC. During combustion or heat input, a large amount of energy is added to the air within the cylinder. This energy raises the temperature of the air to very high values, giving peak cycle temperature at point 3. This increase in temperature during a closed constant-volume process results in a large pressure rise also. Thus, peak cycle pressure is also reached at point 3.

The very high pressure and enthalpy values within the system at TDC generate the power stroke (or expansion stroke) which follows combustion (process 3-4). High pressure on the piston face forces the piston back towards BDC and produces the work and power output of the engine. The power stroke of the real engine cycle is approximated with an isentropic process in the Otto cycle. This is a good approximation, subject to the same arguments as the compression stroke on being frictionless and adiabatic. In a real engine, the beginning of the power stroke is affected by the last part of the combustion process. The end of the power stroke is affected by the exhaust valve being opened before BDC. During the power stroke, values of both the temperature and pressure within the cylinder decrease as volume increases from TDC to BDC.

Near the end of the power stroke of a real engine cycle, the exhaust valve is opened and the cylinder experiences exhaust blowdown. A large amount of exhaust gas is expelled from the cylinder, reducing the pressure to that of the exhaust manifold. The exhaust valve is opened bBDC to allow for the finite time of blowdown to occur. It is desirable for blowdown to be complete by BDC so that there is no high pressure in the cylinder to resist the piston in the following exhaust stroke. Blowdown in a real engine is therefore almost, but not quite, constant volume. A large quantity of enthalpy is carried away with the exhaust gases, limiting the thermal efficiency of the engine. The Otto cycle replaces the exhaust blowdown open-system process of the real cycle with a constant-volume pressure reduction, closed-system process 4-5. Enthalpy loss during this process is replaced with heat rejection in the engine analysis. Pressure within the cylinder at the end of exhaust blowdown has been reduced to about one atmosphere, and the temperature has been substantially reduced by expansion cooling.

The last stroke of the four-stroke cycle now occurs as the piston travels from BDC to TDC. Process 5-6 is the exhaust stroke that occurs at a constant pressure of one atmosphere due to the open exhaust valve. This is a good approximation to the real exhaust stroke, which occurs at a pressure slightly higher than the surrounding pressure due to the small pressure drop across the exhaust valve and in the exhaust system.

At the end of the exhaust stroke the engine has experienced two revolutions, the piston is again at TDC, the exhaust valve closes, the intake valve opens, and a new cycle begins. '

When analyzing an Otto cycle, it is more convenient to work with specific properties by dividing by the mass within the cylinder. Figure 3-2 shows the Otto cycle in P-v and T-s coordinates. It is not uncommon to find the Otto cycle shown with processes 6-1 and 5-6 left off the figure. The reasoning to justify this is that these two processes cancel each other thermodynamically and are not needed in analyzing the cycle.

Thermodynamic Analysis of Air-Standard Otto Cycle

Process 6-1—constant-pressure intake of air at P_o.
Intake valve open and exhaust valve closed:

$$P_1 = P_6 = P_o \tag{3-2}$$

$$w_{6-1} = P_o(v_1 - v_6) \tag{3-3}$$

Process 1-2—isentropic compression stroke.
All valves closed:

$$T_2 = T_1(v_1/v_2)^{k-1} = T_1(V_1/V_2)^{k-1} = T_1(r_c)^{k-1} \tag{3-4}$$

$$P_2 = P_1(v_1/v_2)^k = P_1(V_1/V_2)^k = P_1(r_c)^k \tag{3-5}$$

$$q_{1-2} = 0 \tag{3-6}$$

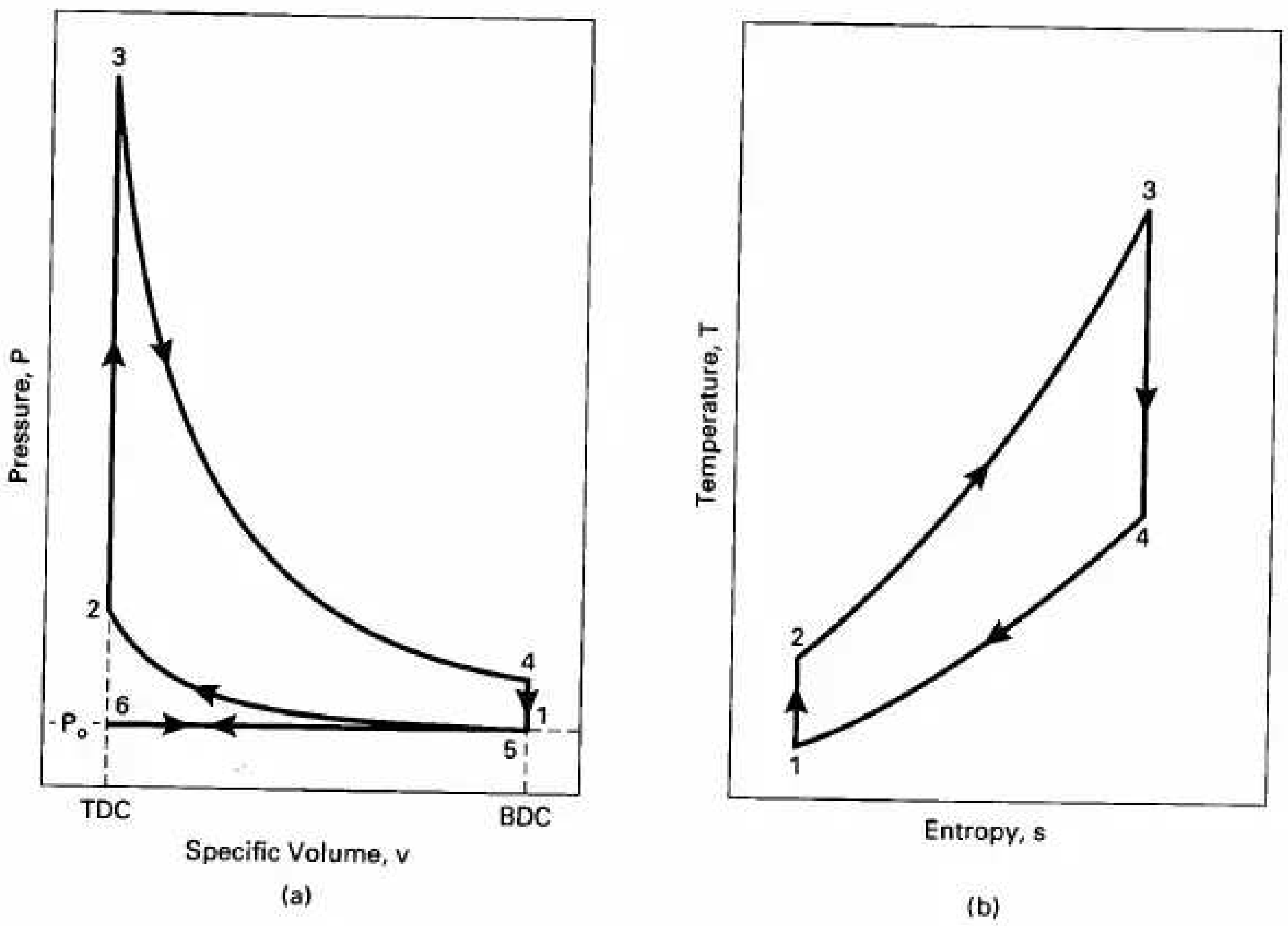


Figure 3-2 Otto cycle, 6-1-2-3-4-5-6, on (a) pressure-specific volume coordinates, and (b) temperature-entropy coordinates.

$$w_{1-2} = (P_2 v_2 - P_1 v_1)/(1 - k) = R(T_2 - T_1)/(1 - k) \tag{3-7}$$
$$= (u_1 - u_2) = c_v(T_1 - T_2)$$

Process 2-3—constant-volume heat input (combustion).
All valves closed:

$$v_3 = v_2 = v_{\text{TDC}} \tag{3-8}$$

$$w_{2-3} = 0 \tag{3-9}$$

$$Q_{2-3} = Q_{\text{in}} = m_f Q_{\text{HV}} \eta_c = m_m c_v (T_3 - T_2) \tag{3-10}$$
$$= (m_a + m_f) c_v (T_3 - T_2)$$

$$Q_{\text{HV}} \eta_c = (\text{AF} + 1) c_v (T_3 - T_2) \tag{3-11}$$

$$q_{2-3} = q_{\text{in}} = c_v (T_3 - T_2) = (u_3 - u_2) \tag{3-12}$$

$$T_3 = T_{\max} \tag{3-13}$$

$$P_3 = P_{\max} \tag{3-14}$$

Process 3-4—isentropic power or expansion stroke.
All valves closed:

$$q_{3-4} = 0 \tag{3-15}$$

$$T_4 = T_3(v_3/v_4)^{k-1} = T_3(V_3/V_4)^{k-1} = T_3(1/r_c)^{k-1} \tag{3-16}$$

$$P_4 = P_3(v_3/v_4)^k = P_3(V_3/V_4)^k = P_3(1/r_c)^k \tag{3-17}$$

$$w_{3-4} = (P_4v_4 - P_3v_3)/(1-k) = R(T_4 - T_3)/(1-k) \tag{3-18}$$

$$= (u_3 - u_4) = c_v(T_3 - T_4)$$

Process 4-5—constant-volume heat rejection (exhaust blowdown).
Exhaust valve open and intake valve closed:

$$v_5 = v_4 = v_1 = v_{BDC} \tag{3-19}$$

$$w_{4-5} = 0 \tag{3-20}$$

$$Q_{4-5} = Q_{out} = m_m c_v(T_5 - T_4) = m_m c_v(T_1 - T_4) \tag{3-21}$$

$$q_{4-5} = q_{out} = c_v(T_5 - T_4) = (u_5 - u_4) = c_v(T_1 - T_4) \tag{3-22}$$

Process 5-6—constant-pressure exhaust stroke at P_o.
Exhaust valve open and intake valve closed:

$$P_5 = P_6 = P_o \tag{3-23}$$

$$w_{5-6} = P_o(v_6 - v_5) = P_o(v_6 - v_1) \tag{3-24}$$

Thermal efficiency of Otto cycle:

$$(\eta_t)_{OTTO} = |w_{net}|/|q_{in}| = 1 - (|q_{out}|/|q_{in}|) \tag{3-25}$$

$$= 1 - [c_v(T_4 - T_1)/c_v(T_3 - T_2)]$$

$$= 1 - [(T_4 - T_1)/(T_3 - T_2)]$$

Only cycle temperatures need to be known to determine thermal efficiency. This can be simplified further by applying ideal gas relationships for the isentropic compression and expansion strokes and recognizing that $v_1 = v_4$ and $v_2 = v_3$:

$$(T_2/T_1) = (v_1/v_2)^{k-1} = (v_4/v_3)^{k-1} = (T_3/T_4) \tag{3-26}$$

Rearranging the temperature terms gives:

$$T_4/T_1 = T_3/T_2 \tag{3-27}$$

Equation (3-25) can be rearranged to:

$$(\eta_t)_{OTTO} = 1 - (T_1/T_2)\{[(T_4/T_1) - 1]/[(T_3/T_2) - 1]\} \tag{3-28}$$

Using Eq. (3-27) gives:

$$(\eta_t)_{OTTO} = 1 - (T_1/T_2) \tag{3-29}$$

Combining this with Eq. (3-4):

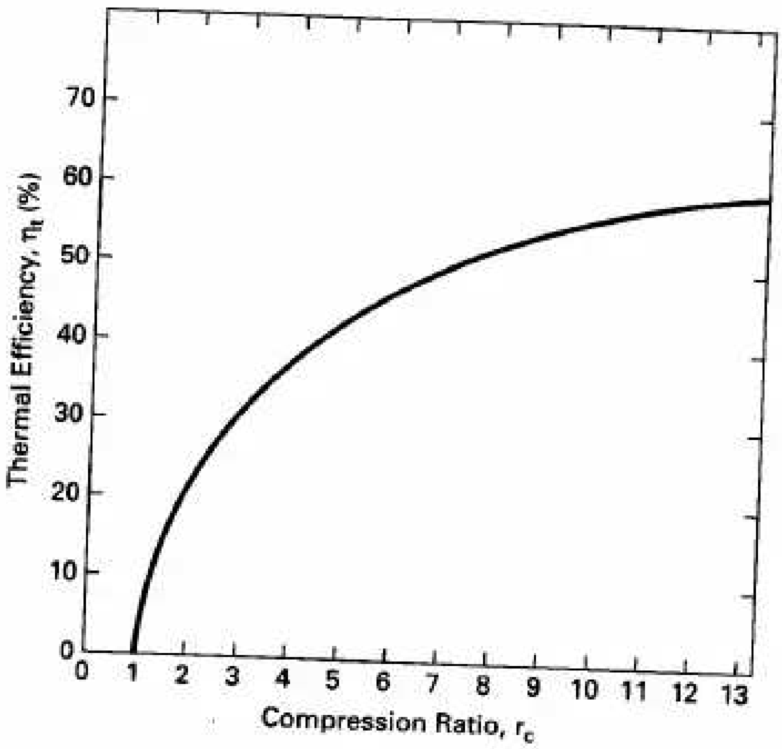


Figure 3-3 Indicated thermal efficiency as a function of compression ratio for SI engines operating at WOT on air-standard Otto cycle ($k = 1.35$).

$$(\eta_t)_{\mathrm{OTTO}} = 1 - [1/(v_1/v_2)^{k-1}] \tag{3-30}$$

With $v_1/v_2 = r_c$, the compression ratio:

$$(\eta_t)_{\mathrm{OTTO}} = 1 - (1/r_c)^{k-1} \tag{3-31}$$

Only the compression ratio is needed to determine the thermal efficiency of the Otto cycle at WOT. As the compression ratio goes up, the thermal efficiency goes up as seen in Fig. 3-3. This efficiency is the **indicated thermal efficiency**, as the heat transfer values are those to and from the air within the combustion chamber.

EXAMPLE PROBLEM 3-1

A four-cylinder, 2.5-liter, SI automobile engine operates at WOT on a four-stroke air-standard Otto cycle at 3000 RPM. The engine has a compression ratio of 8.6:1, a mechanical efficiency of 86%, and a stroke-to-bore ratio $S/B = 1.025$. Fuel is isooctane with AF = 15, a heating value of 44,300 kJ/kg, and combustion efficiency $\eta_c = 100\%$. At the start of the compression stroke, conditions in the cylinder combustion chamber are 100 kPa and 60°C. It can be assumed that there is a 4% exhaust residual left over from the previous cycle.

Do a complete thermodynamic analysis of this engine.

For one cylinder:

Displacement volume:

$$V_d = 2.5 \text{ liter}/4 = \underline{0.625 \text{ L} = 0.000625 \text{ m}^3}$$

Using Eq. (2-12) to find clearance volume:

$$r_c = V_1/V_2 = (V_c + V_d)/V_c = 8.6 = (V_c + 0.000625)/V_c$$

$$\underline{V_c = 0.0000822 \text{ m}^3 = 0.0822 \text{ L} = 82.2 \text{ cm}^3}$$

Using Eq. (2-8) to find bore and stroke:

$$V_d = (\pi/4)B^2S = (\pi/4)B^2(1.025B) = 0.000625 \text{ m}^3$$

$$\underline{B = 0.0919 \text{ m} = 9.19 \text{ cm}}$$

$$S = 1.025B = \underline{0.0942 \text{ m} = 9.42 \text{ cm}}$$

State 1:

$\underline{T_1 = 60°\text{C} = 333 \text{ K}}$ given in problem statement

$\underline{P_1 = 100 \text{ kPa}}$ given

$$V_1 = Vd + V_c = 0.000625 + 0.0000822 = \underline{0.000707 \text{ m}^3}$$

Mass of gas mixture in cylinder can be calculated at State 1. The mass within the cylinder will then remain the same for the entire cycle.

$$m_m = P_1V_1/RT_1 = (100 \text{ kPa})(0.000707 \text{ m}^3)/(0.287 \text{ kJ/kg-K})(333 \text{ K})$$
$$= 0.000740 \text{ kg}$$

State 2: The compression stroke 1-2 is isentropic. Using Eqs. (3-4) and (3-5) to find pressure and temperature:

$$P_2 = P_1(r_c)^k = (100 \text{ kPa})(8.6)^{1.35} = \underline{1826 \text{ kPa}}$$

$$T_2 = T_1(r_c)^{k-1} = (333 \text{ K})(8.6)^{0.35} = \underline{707 \text{ K} = 434°\text{C}}$$

$$V_2 = mRT_2/P_2 = (0.000740 \text{ kg})(0.287 \text{ kJ/kg-K})(707 \text{ K})/(1826 \text{ kPa})$$
$$= \underline{0.0000822 \text{ m}^3 = V_c}$$

This is the clearance volume of one cylinder, which agrees with the above. Another way of getting this value is to use Eq. (2-12):

$$V_2 = V_1/r_c = 0.000707 \text{ m}^3/8.6 = 0.0000822 \text{ m}^3$$

The mass of gas mixture m_m in the cylinder is made up of air m_a, fuel m_f, and exhaust residual m_{ex}:

mass of air	$m_a = (15/16)(0.96)(0.000740)$	$= 0.000666$ kg
mass of fuel	$m_f = (1/16)(0.96)(0.000740)$	$= 0.000044$ kg
mass of exhaust	$m_{ex} = (0.04)(0.000740)$	$= 0.000030$ kg
Total		$m_m = 0.000740$ kg

State 3: Using Eq. (3-10) for the heat added during one cycle:

$$Q_{in} = m_f Q_{HV} \eta_c = m_m c_v(T_3 - T_2)$$

$$= (0.000044\ \text{kg})(44{,}300\ \text{kJ/kg})(1.00)$$
$$= (0.000740\ \text{kg})(0.821\ \text{kJ/kg-K})(T_3 - 707\ \text{K})$$

Solving this for T_3:

$$\underline{T_3 = 3915\ \text{K} = 3642°\text{C} = T_{max}}$$
$$V_3 = V_2 = \underline{0.0000822\ \text{m}^3}$$

For constant volume:

$$P_3 = P_2(T_3/T_2) = (1826\ \text{kPa})(3915/707) \underline{= 10{,}111\ \text{kPa} = P_{max}}$$

State 4: Power stroke 3-4 is isentropic. Using Eq. (3-16) and (3-17) to find temperature and pressure:

$$T_4 = T_3(1/r_c)^{k-1} = (3915\ \text{K})(1/8.6)^{0.35} \underline{= 1844\ \text{K} = 1571°\text{C}}$$
$$P_4 = P_3(1/r_c)^k = (10{,}111\ \text{kPa})(1/8.6)^{1.35} \underline{= 554\ \text{kPa}}$$
$$V_4 = mRT_4/P_4 = (0.000740\ \text{kg})(0.287\ \text{kJ/kg-K})(1844\ \text{K})/(554\ \text{kPa})$$
$$\underline{= 0.000707\ \text{m}^3 = V_1}$$

This agrees with the value of V_1 found earlier.

Work produced in isentropic power stroke for one cylinder during one cycle:

$$W_{3-4} = mR(T_4 - T_3)/(1 - k)$$
$$= (0.000740\ \text{kg})(0.287\ \text{kJ/kg-K})(1844 - 3915)\text{K}/(1 - 1.35)$$
$$= 1.257\ \text{kJ}$$

Work absorbed during isentropic compression stroke for one cylinder during one cycle:

$$W_{1-2} = mR(T_2 - T_1)/(1 - k)$$
$$= (0.000740\ \text{kg})(0.287\ \text{kJ/kg-K})(707 - 333)\text{K}/(1 - 1.35)$$
$$= -0.227\ \text{kJ}$$

Work of the intake stroke is canceled by work of the exhaust stroke.

Net indicated work for one cylinder during one cycle is:

$$W_{net} = W_{1-2} + W_{3-4} = (+1.257) + (-0.227) \underline{= +1.030\ \text{kJ}}$$

Using Eq. (3-10) to find heat added for one cylinder during one cycle:

$$Q_{in} = m_f Q_{HV}\,\eta_c = (0.000044\ \text{kg})(44{,}300\ \text{kJ/kg})(1.00) \underline{= 1.949\ \text{kJ}}$$

Indicated thermal efficiency:

$$\eta_t = W_{net}/Q_{in} = 1.030/1.949 \underline{= 0.529 = 52.9\%}$$

or using Eqs. (3-29) and (3-31):

$$\eta_t = 1 - (T_1/T_2) = 1 - (1/r_c)^{k-1}$$
$$= 1 - (333/707) = 1 - (1/8.6)^{0.35} = 0.529$$

Equation (2-29) is used to find indicated mean effective pressure:

$$\text{imep} = W_{net}/(V_1 - V_2) = (1.030\ \text{kJ})/(0.000707 - 0.0000822)\text{m}^3 = \underline{1649\ \text{kPa}}$$

Indicated power at 3000 RPM is obtained using Eq. (2-42):

$$\dot{W}_i = WN/n$$

$$= [(1.030\ \text{kJ/cyl-cycle})(3000/60\ \text{rev/sec})/(2\ \text{rev/cycle})](4\ \text{cyl})$$

$$= \underline{103\ \text{kW} = 138\ \text{hp}}$$

Equation (2-2) is used to find mean piston speed:

$$\bar{U}_p = 2SN = (2\ \text{strokes/rev})(0.0942\ \text{m/stroke})(3000/60\ \text{rev/sec})$$

$$= \underline{9.42\ \text{m/sec}}$$

Equation (2-27) gives net brake work for one cylinder during one cycle:

$$W_b = \eta_m W_i = (0.86)(1.030\ \text{kJ}) = \underline{0.886\ \text{kJ}}$$

Brake power at 3000 RPM:

$$\dot{W}_b = (3000/60\ \text{rev/sec})(0.5\ \text{cycle/rev})(0.886\ \text{kJ/cyl-cycle})(4\ \text{cyl})$$

$$= \underline{88.6\ \text{kW} = 119\ \text{hp}}$$

or:

$$\dot{W}_b = \eta_m \dot{W}_i = (0.86)(103\ \text{kW}) = 88.6\ \text{kW}$$

Torque is calculated using Eq. (2-43):

$$\tau = \dot{W}_b/2\pi N = (88.6\ \text{kJ/sec})/(2\pi\ \text{radians/rev})(3000/60\ \text{rev/sec})$$

$$= \underline{0.282\ \text{kN-m} = 282\ \text{N-m}}$$

Friction power lost using Eq. (2-49):

$$\dot{W}_f = \dot{W}_i - \dot{W}_b = 103 - 88.6 = \underline{14.4\ \text{kW} = 19.3\ \text{hp}}$$

Equation (2-37c) is used to find brake mean effective pressure:

$$\text{bmep} = \eta_m(\text{imep}) = (0.86)(1649\ \text{kPa}) = \underline{1418\ \text{kPa}}$$

This allows another way of finding torque using Eq. (2-41), which gives consistent results:

$$\tau = (\text{bmep})V_d/4\pi = (1418\ \text{kPa})(0.0025\ \text{m}^3)/4\pi = 0.282\ \text{kN-m}$$

Brake specific power using Eq. (2-51):

$$\text{BSP} = \dot{W}_b/A_p = (88.6\ \text{kW})/\{[(\pi/4)(9.19\ \text{cm})^2](4\ \text{cyl})\} = \underline{0.334\ \text{kW/cm}^2}$$

Output per displacement using Eq. (2-52):

$$\text{OPD} = \dot{W}_b/V_d = (88.6\ \text{kW})/(2.5\ \text{L}) = \underline{35.4\ \text{kW/L}}$$

Equation (2-58) is used to find brake specific fuel consumption:

$$\text{bsfc} = \dot{m}_f/\dot{W}_b$$

$$= (0.000044\ \text{kg/cyl-cycle})(50\ \text{rev/sec})(0.5\ \text{cycle/rev})(4\ \text{cyl})/(88.6\ \text{kW})$$

$$= 0.000050$$

$= 0.000050$ kg/sec/kW = 180 gm/kW-hr

Equation (2-69) is used to find volumetric efficiency using one cylinder and standard air density:

$$\eta_v = m_a/\rho_a V_d = (0.000666 \text{ kg})/(1.181 \text{ kg/m}^3)(0.000625 \text{ m}^3)$$
$$= \underline{0.902 = 90.2\%}$$

3-3 REAL AIR-FUEL ENGINE CYCLES

The actual cycle experienced by an internal combustion engine is not, in the true sense, a thermodynamic cycle. An ideal air-standard thermodynamic cycle Occurs on a closed system of constant composition. This is not what actually happens in an IC engine, and for this reason air-standard analysis gives, at best, only approximations to actl;lal conditions and outputs. Major differences include:

1. Real engines operate on an open cycle with changing composition. Not only does the inlet gas composition differ from what exits, but often the mass flow rate is not the same. Those engines which add fuel into the cylinders after air induction is complete (CI engines and some SI engines) change the amount of mass in the gas composition part way through the cycle. There is a greater gaseous mass exiting the engine in the exhaust than what entered in the induction process. This can be on the order of several percent. Other engines carry liquid fuel droplets with the inlet air which are idealized as part of the gaseous mass in air-standard analysis. During combustion, total mass remains about the same but molar quantity changes. Finally, there is a loss of mass during the cycle due to crevice flow and blowby past the pistons. Most of crevice flow is a temporary loss of mass from the cylinder, but because it is greatest at the start of the power stroke, some output work is lost during expansion. Blowby can decrease the amount of mass in the cylinders by as much as 1% during compression and combustion. This is discussed in greater detail in Chapter 6.

2. Air-standard analysis treats the fluid flow through the entire engine as air and approximates air as an ideal gas. In a real engine inlet flow may be all air, or it may be air mixed with up to 7% fuel, either gaseous or as liquid droplets, or both. During combustion the composition is then changed to a gas mixture of mostly CO_2, H_2O, and N_2, with lesser amounts of CO and hydrocarbon vapor. In CI engines there will also be solid carbon particles in the combustion products gas mixture. Approximating exhaust products as air simplifies analysis but introduces some error.

Even if all fluid in an engine cycle were air, some error would be introduced by assuming it to be an ideal gas with constant specific heats in air-standard analysis. At

the low pressures of inlet and exhaust, air can accurately be treated as an ideal gas, but at the higher pressures during combustion, air will deviate from ideal gas behavior. A more serious error is introduced by assuming constant specific heats for the analysis. Specific heats of a gas have a fairly strong dependency on temperature and can vary as much as 30% in the temperature range of an engine (for air, $c_p = 1.004$ kJ/kg-K at 300 K and $c_p = 1.292$kJ/kg-K at 3000 K [73]); see Review Problem 3-5.

3. There are heat losses during the cycle of a real engine which are neglected in air-standard analysis. Heat loss during combustion lowers actual peak temperature and pressure from what is predicted. The actual power stroke, therefore, starts at a lower pressure, and work output during expansion is decreased. Heat transfer continues during expansion, and this lowers the temperature and pressure below the

ideal isentropic process towards the end of the power stroke. The result of heat transfer is a lower indicated thermal efficiency than predicted by air-standard analysis. Heat transfer is also present during compression, which deviates the process from isentropic. However, this is less than during the expansion stroke due to the lower temperatures at this time.

4. Combustion requires a short but finite time to occur, and heat addition is not instantaneous at TDC, as approximated in an Otto cycle. A fast but finite flame speed is desirable in an engine. This results in a finite rate of pressure rise in the cylinders, a steady force increase on the piston face, and a smooth engine cycle. A supersonic detonation would give almost instantaneous heat addition to a cycle, but would result in a.rough cycle and quick engine destruction. Because of the finite time required, combustion is started before TDC and ends after IDC, not at constant volume as in air-standard analysis. By starting combustion bTDC, cylinder pressure increases late in the compression stroke, requiring greater negative work in that stroke. Because combustion is not completed until aTDC, some power is lost at the start of the expansion stroke (see Fig. 2-6). Another loss in the combustion process of an actual engine occurs because combustion efficiency is less than 100%. This happens because of less than perfect mixing, local variations in temperature and air-fuel due to turbulence, flame quenching, etc. SI engines will generally have a combustion efficiency of about 95%, while CI engines are generally about 98% efficient.

5. The blowdown process requires a finite real time and a finite cycle time, and does not occur at constant volume as in air-standard analysis. For this reason, the exhaust valve must open *40°* to *60°* bBDC, and output work at the latter end of expansion is lost.

6. In an actual engine, the intake valve is not closed until after bottom-de adcenter at the end of the intake stroke. Because of the flow restriction of the valve, air is still entering the cylinder at BDC, and volumetric efficiency would be lower if the valve closed here. Because of this, however, actual compression does not start at BDC but only after the inlet valve closes. With ignition then occurring before top-dead-center, temperature and pressure rise before combustion is less than predicted by air-standard analysis.

7. Engine valves require a finite time to actuate. Ideally, valves would open and close instantaneously, but this is not possible when using a camshaft. Cam profiles must allow for smooth interaction with the cam follower, and this results in fast but finite valve actuation. To assure that the intake valve is fully open at the start of the induction stroke, it must start to open before TDC. Likewise, the exhaust valve must remain fully open until the end of the exhaust stroke, with final closure Occurring after TDC. The resulting valve overlap period causes a deviation from the ideal cycle.

Because of these differences which real air-fuel cycles have from the ideal cycles, results from air-standard analysis will have errors and will deviate from actual conditions. Interestingly, however, the errors are not great, and property values of temperature and pressure are very representative of actual engine values, depending on the geometry and operating conditions of the real engine. By changing operating variables such as inlet temperature and/or pressure, compression ratio, peak temperature, etc., in Otto cycle analysis, good approximations can be obtained for output changes that will Occur in a real engine as these variables are changed. Good approximation of power output, thermal efficiency, and mep can be expected.

Indicated thermal efficiency of a real four-stroke SI engine is always somewhat less than what air-standard Otto cycle analysis predicts. This is caused by the heat losses,. friction, ignition timing, valve timing, finite time of combustion and blowdown, and deviation from ideal gas behavior of the real engine. Reference [120] shows that over a large range of operating variables the indicated thermal efficiency of an actual SI four-stroke cycle engine can be approximated by;

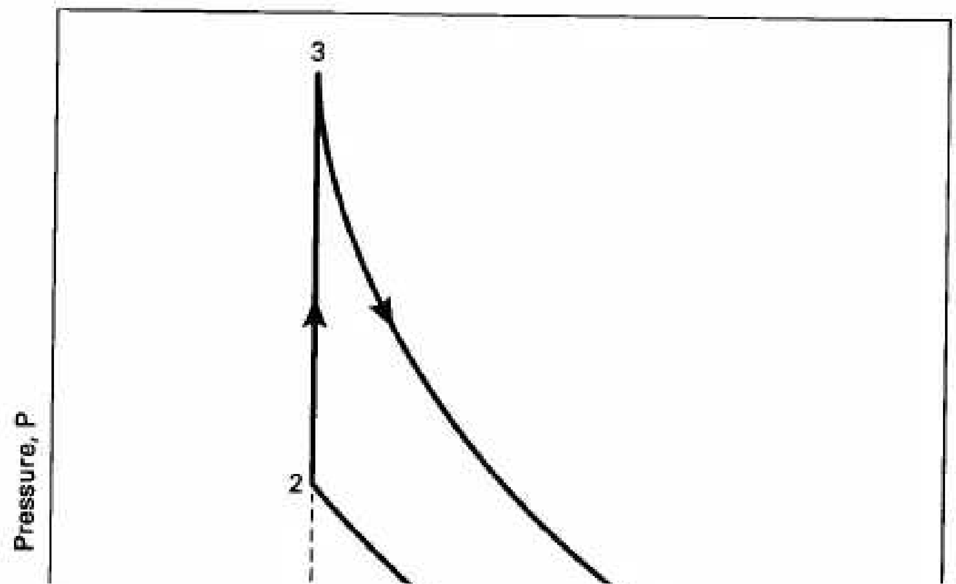
$$(\eta_t)_{actual} = 0.85\,(\eta_t)_{OTTO} \qquad (3\text{-}32)$$

This will be correct to within a few percent for large ranges of air-fuel equivalence ratio, ignition timing, engine speed, compression ratio, inlet pressure, exhaust pressure, and valve timing.

3-4 81 ENGINE CYCLE AT PART THROTTLE

When a four-stroke cycle SI engine is run at less than WOT conditions, air-fuel input is reduced by partially closing the throttle (butterfly valve) in the intake system. This creates a flow restriction and consequent pressure drop in the incoming air. Fuel input is then also reduced to match the reduction of air. Lower pressure in the intake manifold during the intake stroke and the resulting lower pressure in the cylinder at the start of the compression stroke are shown in Fig. 3-4. Although the air experiences an expansion cooling because of the pressure drop across the throttle valve, the temperature of the air entering the cylinders is about the same as at WOT because it first flows through the hot intake manifold.

Figure 3-4 shows that the net indicated work for the Otto cycle engine will be less at part throttle than at WOT. The upper loop of the cycle made up of the com-



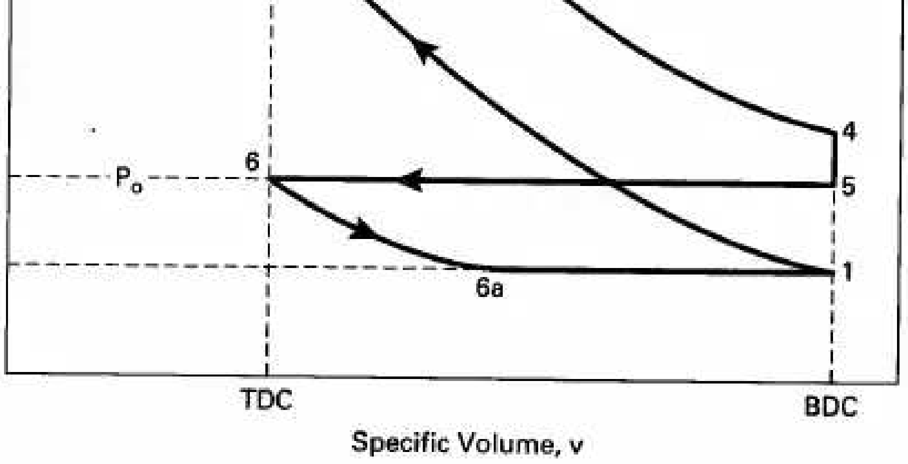
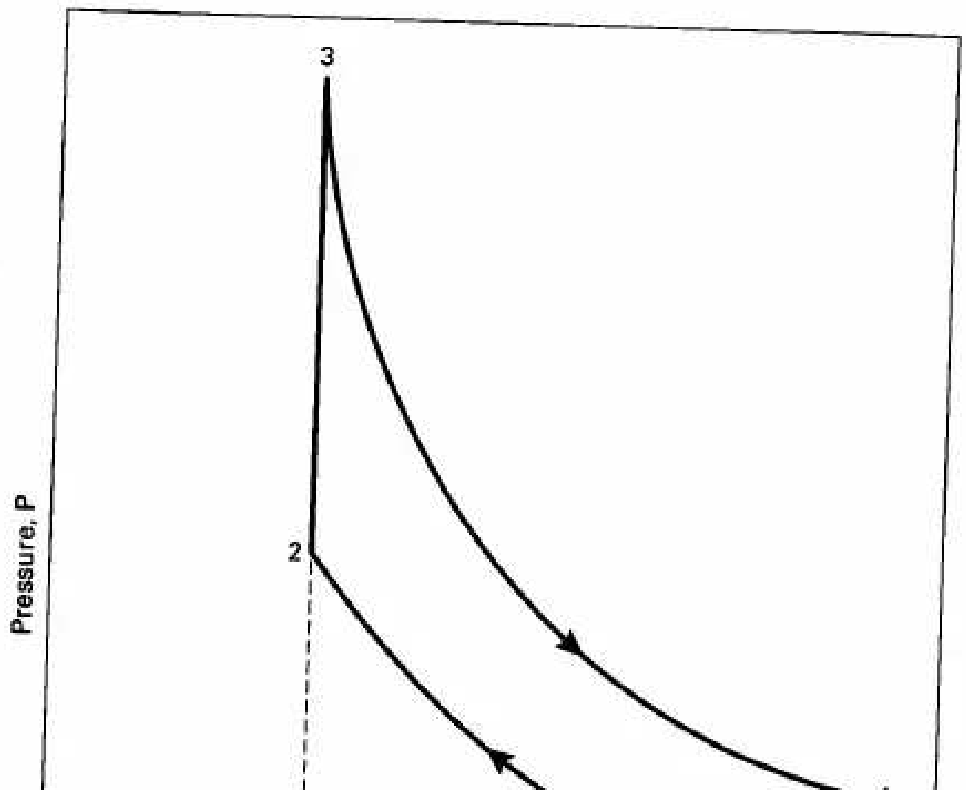


Figure 3-4 Four-stroke air-standard Otto cycle, 6-6a-1-2-3-4-5-6, for SI engine operating at part throttle.

pression and power strokes represents positive work output, while the lower loop consisting of the exhaust and intake strokes is negative work absorbed off the rotating crankshaft. The more closed the throttle position, the lower will be the pressure during the intake stroke and the greater the negative pump work. Two main factors contribute to the reduced net work at part-throttle operation. The lower pressure at the start of compression results in lower pressures throughout the rest of the cycle except for the exhaust stroke. This lowers mep and net work. In addition, when less air is ingested into the cylinders during intake because of this lower pressure, fuel input by injectors or carburetor is also proportionally reduced. This results in less thermal energy from combustion in the cylinders and less resulting work out. It should be noted that although Q_{in} is reduced, the temperature rise in process 2-3 in Fig. 3-4 is about the same. This is because the mass of fuel and the mass of air being heated are both reduced by an equal proportion.



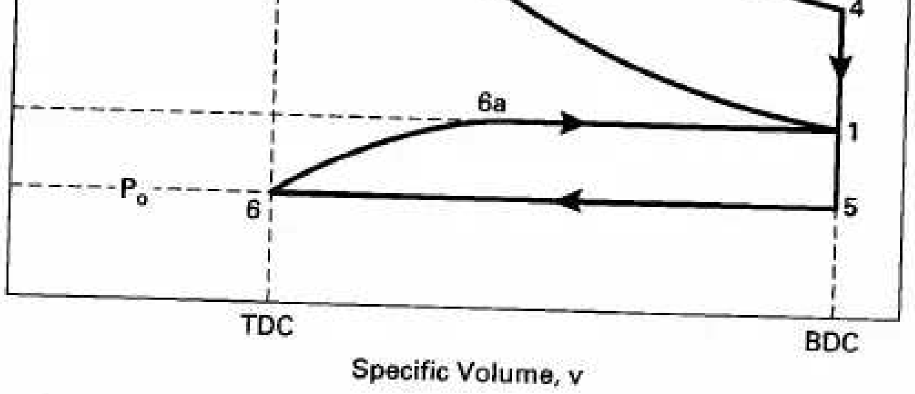


Figure 3-5 Four-stroke air-standard Otto cycle, 6-6a-1-2-3-4-5-6, for SI engine equipped with a turbocharger or supercharger.

If an engine is equipped with a supercharger or turbocharger the air-standard cycle is shown in Fig. 3-5, with intake pressure higher than atmospheric pressure. This results in more air and fuel in the combustion chamber during the cycle, and the resulting net indicated work is increased. Higher intake pressure increases all pressures though the cycle, and increased air and fuel give greater Q_{in} in process 2-3. When air is compressed to a higher pressure by a supercharger or turbocharger, the temperature is also increased due to compressive heating. This would increase air temperature at the start of the compression stroke, which in turn raises all temperatures in the remaining cycle. This can cause self-ignition and knocking problems in the latter part of compression or during combustion. For this reason, engine compressors can be equipped with an aftercooler to again lower the compressed

incoming air temperature. Aftercoolers are heat exchangers which often use outside air as the cooling fluid. In principle, aftercoolers are desirable, but cost and space limitations often make them impractical on automobile engines. Instead, engines equipped with a supercharger or turbocharger will usually have a lower compression ratio to reduce knocking problems.

When an engine is operated at WOT, it can be assumed that the air pressure in the intake manifold is P_o = one atmosphere. At part throttle the partially closed butterfly valve creates a flow restriction, resulting in a lower inlet pressure P_i in the intake manifold (point 6a in Fig. 3-4). Work done during the intake stroke is, therefore,

$$W_{6-1} = P_i(V_1 - V_6) = P_i V_d \tag{3-33}$$

where: V_d = displacement volume

Work done during the exhaust stroke where the pressure is about constant at one atmosphere is:

$$W_{5-6} = P_{ex}(V_6 - V_5) = -P_{ex} V_d \tag{3-34}$$

Net indicated pumping work for the cycle at part throttle is:

$$(W_{pump})_{net} = (P_i - P_{ex}) V_d \tag{3-35}$$

The negative value of this pump work means that it lowers the net indicated work of the cycle.

If the engine is equipped with a supercharger or turbocharger, the inlet pressure can be greater than one atmosphere, as shown in Fig. 3-5. Net indicated pump work for this cycle is still given by Eq. (3-35), but now $P_i > P_{ex}$, pump work is positive, and net indicated work is increased.

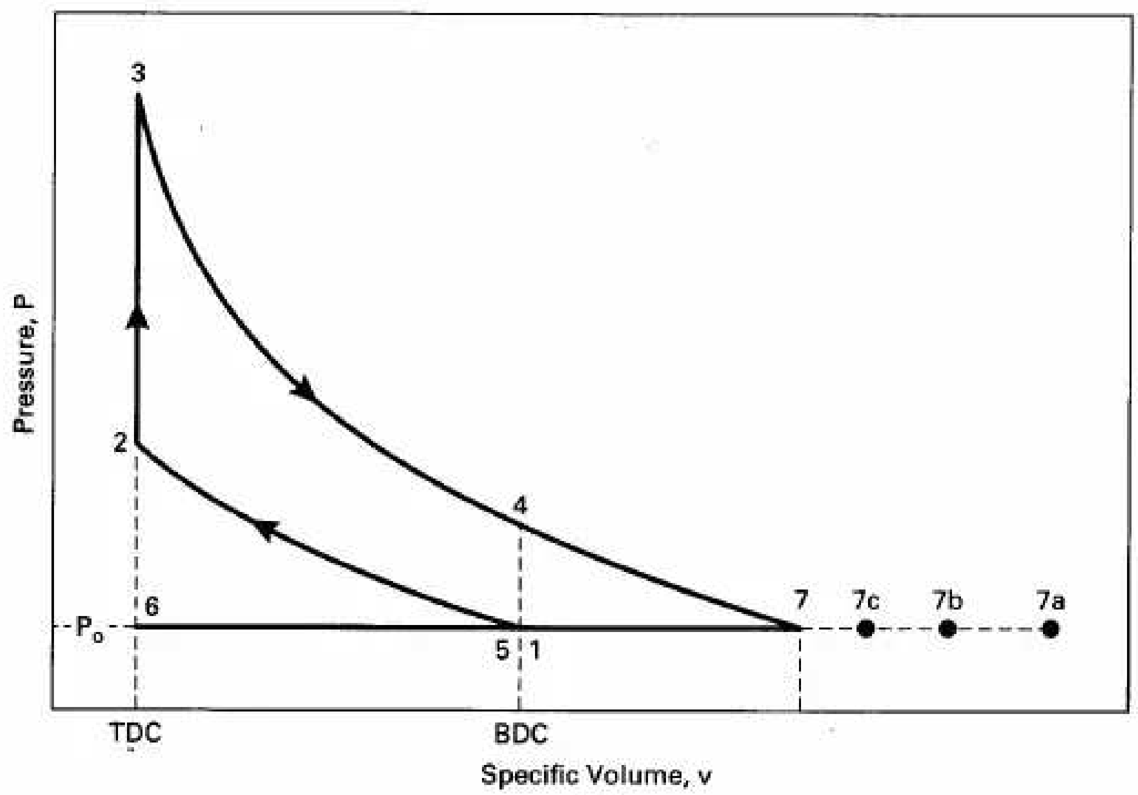
Using Eqs. (2-29) and (3-35) for pump mean effective pressure:

$$\text{pmep} = (W_{\text{pump}})_{\text{net}}/V_d = (P_i - P_{\text{ex}}) \tag{3-36}$$

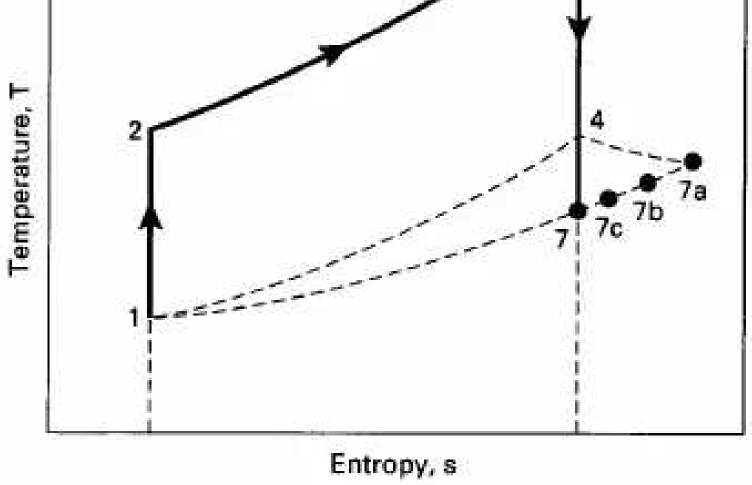
This can have positive or negative values.

3-5 EXHAUST PROCESS

The exhaust process consists of two steps: blowdown and exhaust stroke. When the exhaust valve opens near the end of the expansion stroke (point 4 in Fig. 3-6), the high-temperature gases are suddenly subjected to a pressure decrease as the resulting blowdown occurs. A large percentage of the gases leaves the combustion chamber during this blowdown process, driven by the pressure differential across the open exhaust valve. When the pressure across the exhaust valve is finally equalized, the cylinder is still filled with exhaust gases at the exhaust manifold pressure of about one atmosphere. These gases are then pushed out of the cylinder through the still open exhaust valve by the piston as it travels from BDC to TDC during the exhaust stroke.



(a)



(b)

Figure 3-6 Air-standard Otto cycle for engine at WOT, showing process 4-7 experienced by exhaust during blowdown.

Temperature of the exhaust gases is cooled by expansion cooling when the pressure is suddenly reduced during blowdown. Although this expansion is not reversible, the ideal gas isentropic relationship between pressure and temperature serves as a good model to approximate exhaust temperature T_7 in the hypothetical process 4-7 of Fig. 3-6.

$$T_7 = T_4(P_7/P_4)^{(k-1)/k} = T_3(P_7/P_3)^{(k-1)/k} \tag{3-37}$$
$$= T_4(P_{ex}/P_4)^{(k-1)/k} = T_4(P_0/P_4)^{(k-1)/k}$$

where: $P_7 = P_{ex} = P_0$

P_{ex} = exhaust pressure, which generally can be considered equal to surrounding pressure

P_7 is the pressure in the exhaust system and is almost always very close to one atmosphere in value.

Gas leaving the combustion chamber during the blowdown process will also have kinetic energy due to high velocity flow through the exhaust valve. This kinetic energy will very quickly be dissipated in the exhaust manifold, and there will be a subsequent rise in enthalpy and temperature. The first elements of gas leaving the combustion chamber will have the highest velocity and will therefore reach the highest temperature when this velocity is dissipated (point 7a in Fig. 3-6). Each subsequent element of gas will have less velocity and will thus experience less temperature rise (points 7b, 7c, etc.). The last elements of gas leaving the combustion chamber during blowdown and the gas pushed out during the exhaust stroke will have relatively low kinetic energy and will have a temperature very close to T_7. Choked flow (sonic velocity) will be experienced across the exhaust valve at the start of blowdown, and this determines the resulting gas velocity and kinetic energy. If possible, it is desirable to mount the turbine of a turbocharger very close to the exhaust manifold. This is done so that exhaust kinetic energy can be utilized in the turbine.

The state of the exhaust gas during the exhaust stroke is best approximated by a pressure of one atmosphere, a temperature of T_7 given in Eq. (3-37), and a specific volume shown at point 7 in Fig. 3-6. It will be noted that this is inconsistent with Fig. 3-6 for the exhaust stroke process 5-6. The figure would suggest that the specific volume v changes during process 5-6. This inconsistency occurs because Fig. 3-6 uses a closed system model to represent an open system process, the exhaust stroke. Also, it should be noted that point 7 is a hypothetical state and corresponds to no actual physical piston position.

At the end of the exhaust stroke, there is still a residual of exhaust gas trapped in the clearance volume of the cylinder. This exhaust residual gets mixed with the new incoming charge of air and fuel and is carried into the new cycle. Exhaust residual is defined as:

$$x_r = m_{ex}/m_m \tag{3-38}$$

where m_{ex} is the mass of exhaust gas carried into the next cycle and m_m is the mass of gas mixture within the cylinder for the entire cycle. Values of exhaust residual

range from 3% to 7% at full load, increasing to as much as 20% at part-throttle light loads. CI engines generally have less exhaust residual because their higher compression ratios give them smaller relative clearance volumes. In addition to clearance volume, the amount of exhaust residual is affected by the location of the valves and the amount of valve overlap.

In Fig. 3-6, if the blowdown process 4-7 is modeled as an isentropic expansion, then:

$$P_4/P_7 = (V_7/V_4)^k = P_4/P_{ex} = P_4/P_0 \tag{3-39}$$

$$P_3/P_7 = (V_7/V_3)^k = P_3/P_{ex} = P_3/P_0 \tag{3-40}$$

Mass of exhaust in the cylinder after blowdown but before the exhaust stroke will be:

$$m_7 = V_5/v_{ex} = V_5/v_7 = V_1/v_7 \tag{3-41}$$

Mass of exhaust in the cylinder at the end of the exhaust stroke will be:

$$m_{ex} = V_6/V_7 = V_2/v_7 \tag{3-42}$$

where v_7 is calculated using either Eq. (3-39) or (3-40) and represents the constant specific volume of exhaust gas within the cylinder for the entire exhaust stroke 5-6. The mass of gas mixture in Eq. (3-38) can be obtained from:

$$m_m = V_1/v_1 = V_2/v_2 = V_3/V_3 = V_4/V_4 = V_7/V_7 \tag{3-43}$$

Combining this with Eqs. (3-38) and (3-42):

$$x_r = (V_2/V_7)/(V_7/V_7) = V_2/V_7 \tag{3-44}$$

V_7 is the hypothetical volume of m_m expanded to P_0 after combustion. Using Eqs. (3-42) and (3-43), the exhaust residual can also be written as:

$$x_r = (V_6/V_7)/(V_4/V_4) = (V_6/V_4)(V_4/V_7) = (1/r_c)(v_4/v_7) \tag{3-45}$$

$$= (1/r_c)[(RT_4/P_4)/(RT_7/P_7)]$$

$$x_r = (1/r_c)(T_4/T_{ex})(P_{ex}/P_4) \tag{3-46}$$

where: r_c = compression ratio
$P_{ex} = P_7 = P_0$ = one atmosphere under most conditions
$T_{ex} = T_7$ from Eq. (3-37)
and T_4 and P_4 are conditions in the cylinder when the exhaust valve opens

When the intake valve opens, a new charge of inlet air m_a enters the cylinder and mixes with the remaining exhaust residual from the previous cycle. The mixing occurs such that total enthalpy remains constant and:

+=~ ~

where *hex, ha,* and *hm* are the specific enthalpy values of exhaust, air, and mixture, all of which are treated as air in air-standard analysis. If specific enthalpy values are referenced to zero value at an absolute temperature value of zero, then $h = c_p T$ and:

$$m_{ex} c_p T_{ex} + m_a c_p T_a = m_m c_p T_m \tag{3-48}$$

Canceling c_p and dividing by m_m:

$$(m_{ex}/m_m) T_{ex} + (m_a/m_m) T_a = T_m \tag{3-49}$$

Combining this with Eq. (3-38) gives the temperature of the gas mixture in the cylinder at the start of compression in terms of the exhaust residual x_r:

$$(T_m)_1 = x_r T_{ex} + (1 - x_r) T_a \tag{3-50}$$

where $T_{ex} = T_7$ and T_a is the temperature of the incoming air in the intake manifold.

As air enters the cylinder it mixes with the small charge of hot exhaust residual, heating the air and reducing its density. This, in turn, reduces the volumetric efficiency of the engine. Part of this is gained back by the substantial cooling of the small amount of exhaust residual, which increases its density. The partial vacuum this creates in the clearance volume can then be filled with additional intake air.

EXAMPLE PROBLEM 3-2

The engine operating at the conditions of Example Problem 3-1 has an exhaust pressure of 100 kPa.
Calculate:

1. exhaust temperature
2. exhaust residual
3. temperature of air entering cylinder

1) Using Fig. 3-6 and Eq. (3-37) for exhaust temperature:

$$T_{ex} = T_7 = T_3(P_7/P_3)^{(k-1)/k}$$
$$= (3915\ \text{K})(100/10{,}111)^{(1.35-1)/1.35} = \underline{1183\ \text{K} = 910°\text{C}}$$

2) Using Eq. (3-46) to find exhaust residual:

$$x_r = (1/r_c)(T_4/T_{ex})(P_{ex}/P_4) = (1/8.6)(1844/1183)(100/554) = \underline{0.033}$$

It was assumed that $x_r = 0.04$ when the engine was analyzed in Example Problem 3-1. That analysis should now be redone using this better value of $x_r = 0.033$. When this is done, the following corrected values are obtained:

$$P_3 = 10{,}300\ \text{kPa}$$
$$T_3 = 3988\ \text{K}$$
$$P_4 = 564\ \text{kPa}$$
$$T_4 = 1878\ \text{K}$$
$$T_{ex} = 1199\ \text{K}$$

$$x_r = 0.033$$

The consistent value for the exhaust residual means that an additional iteration is not needed. With a reasonable exhaust residual approximation to start with, two itera-

tions in the analysis will normally be sufficient. Other parameters (e.g., power, mep, etc.) should now be recalculated, with slight changes in their values to be expected.

3) Equation (3-50) is now used to find the temperature of the air entering the cylinder from the intake manifold:

$$T_1 = x_r T_{ex} + (1 - x_r) T_a$$

$$333 = (0.033)(1199) + (1 - 0.033) T_a$$

$$\underline{T_a = 303 \text{ K} = 30°\text{C}}$$

EXAMPLE PROBLEM 3-3

The engine in Example Problems 3-1 and 3-2 is now run at part throttle such that the intake pressure is 50 kPa. Calculate the temperature in the cylinder at the start of the compression stroke.

The temperature of the intake air can be assumed to be the same even though it has experienced a pressure reduction expansion when passing the throttle valve. This is because it still flows through the same hot intake manifold after the throttle. However, the temperature of the exhaust residual will be reduced due to the expansion cooling it undergoes when the intake valve opens and the pressure in the cylinder drops to 50 kPa. The temperature of the exhaust residual after expansion can be approximated using Fig. 3-4 and the isentropic expansion model such that:

$$T_{6a} = T_{ex}(P_{6a}/P_6)^{(k-1)/k} = (1199 \text{ K})(50/100)^{(1.35-1)/1.35} = \underline{1002 \text{ K} = 729°\text{C}}$$

Equation (3-50) is now used to find the temperature at the start of compression (point 1):

$$T_1 = x_r T_{6a} + (1 - x_r) T_a$$

$$T_1 = (0.033)(1002 \text{ K}) + (1 - 0.033)(303 \text{ K}) = \underline{326 \text{ K} = 53°\text{C}}$$

This temperature and pressure of 50 kPa should now be used as a starting point and a complete thermodynamic analysis be done on the cycle with iterations until consistent results are obtained. This is left as an exercise for the student.

3-6 DIESEL CYCLE

Early CI engines injected fuel into the combustion chamber very late in the compression stroke, resulting in the indicator diagram shown in Fig. 3-7. Due to ignition delay and the finite time required to inject the fuel, combustion lasted into the expansion stroke. This kept the pressure at peak levels well past TDC. This combustion process is best approximated as a constant-pressure heat input in an air-standard cycle, resulting in the Diesel cycle shown in Fig. 3-8. The rest of the cycle is similar to the air-standard Otto cycle. The diesel cycle is sometimes called a Constant· Pressure cycle.

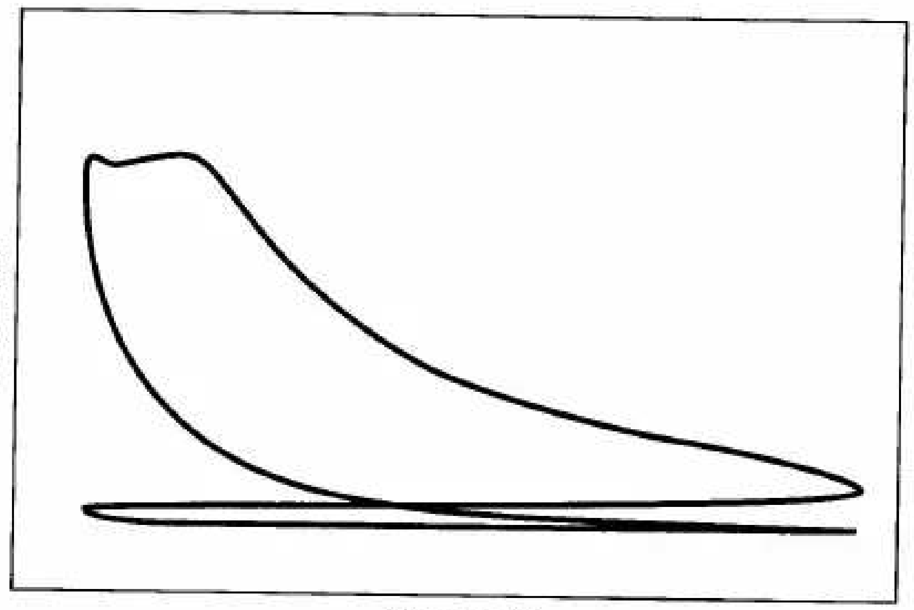


Figure 3-7 Indicator diagram of a historic CI engine operating on an early four-stroke cycle.

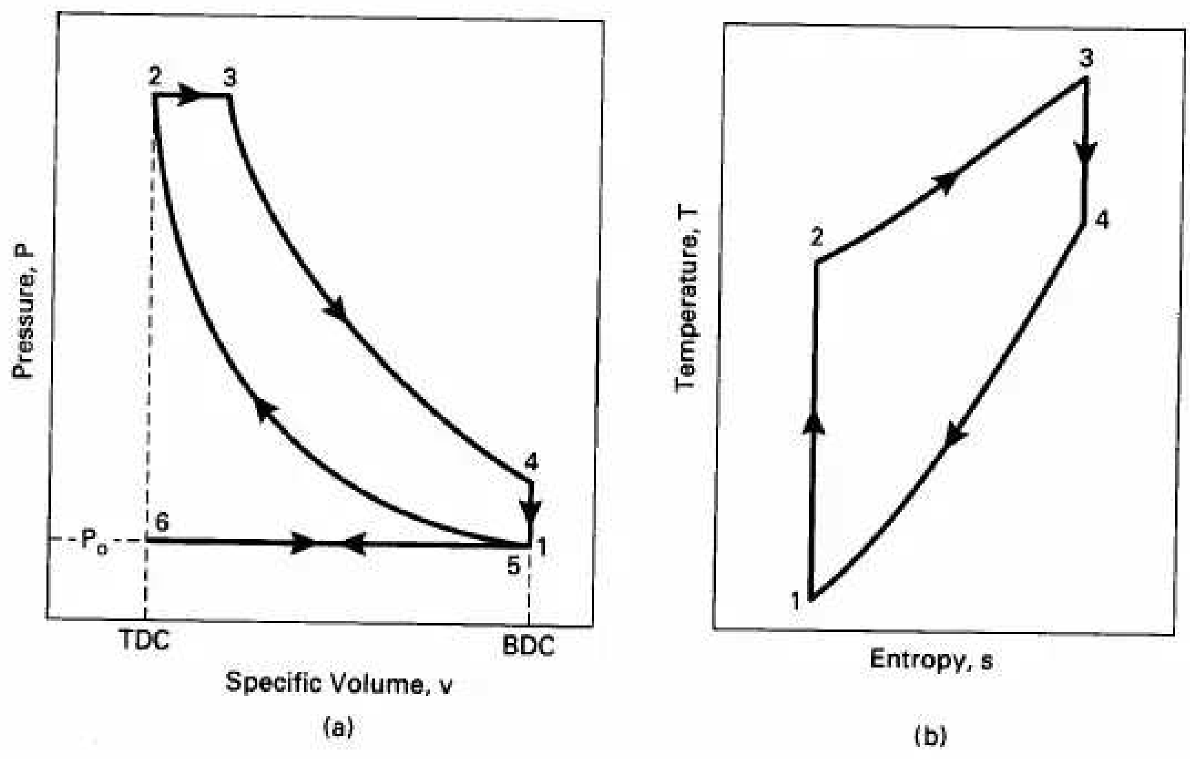


Figure 3-8 Air-standard diesel cycle, 6-1-2-3-4-5-6, which approximates the four-stroke cycle of an early CI engine on **(a)** pressure-specific volume coordinates, and **(b)** temperature-entropy coordinates.

Thermodynamic Analysis of Air-Standard Diesel Cycle

Process 6-1—constant-pressure intake of air at P_o.
Intake valve open and exhaust valve closed:

$$w_{6-1} = P_o(v_1 - v_6) \tag{3-51}$$

Process 1-2—isentropic compression stroke.
All valves closed:

$$T_2 = T_1(v_1/v_2)^{k-1} = T_1(V_1/V_2)^{k-1} = T_1(r_c)^{k-1} \tag{3-52}$$

$$P_2 = P_1(v_1/v_2)^k = P_1(V_1/V_2)^k = P_1(r_c)^k \tag{3-53}$$

$$V_2 = V_{\text{TDC}} \tag{3-54}$$

$$q_{1-2} = 0 \tag{3-55}$$

$$w_{1-2} = (P_2 v_2 - P_1 v_1)/(1 - k) = R(T_2 - T_1)/(1 - k) \tag{3-56}$$

$$= (u_1 - u_2) = c_v(T_1 - T_2)$$

Process 2-3—constant-pressure heat input (combustion).
All valves closed:

$$Q_{2-3} = Q_{\text{in}} = m_f Q_{\text{HV}}\,\eta_c = m_m c_p(T_3 - T_2) = (m_a + m_f)c_p(T_3 - T_2) \tag{3-57}$$

$$Q_{\text{HV}}\,\eta_c = (\text{AF} + 1)c_p(T_3 - T_2) \tag{3-58}$$

$$q_{2-3} = q_{\text{in}} = c_p(T_3 - T_2) = (h_3 - h_2) \tag{3-59}$$

$$w_{2-3} = q_{2-3} - (u_3 - u_2) = P_2(v_3 - v_2) \tag{3-60}$$

$$T_3 = T_{\max} \tag{3-61}$$

Cutoff ratio is defined as the change in volume that occurs during combustion, given as a ratio:

$$\beta = V_3/V_2 = v_3/v_2 = T_3/T_2 \tag{3-62}$$

Process 3-4—isentropic power or expansion stroke.
All valves closed:

$$q_{3-4} = 0 \tag{3-63}$$

$$T_4 = T_3(v_3/v_4)^{k-1} = T_3(V_3/V_4)^{k-1} \tag{3-64}$$

$$P_4 = P_3(v_3/v_4)^k = P_3(V_3/V_4)^k \tag{3-65}$$

$$w_{3-4} = (P_4 v_4 - P_3 v_3)/(1 - k) = R(T_4 - T_3)/(1 - k) \tag{3-66}$$

$$= (u_3 - u_4) = c_v(T_3 - T_4)$$

Process 4-5—constant-volume heat rejection (exhaust blowdown).
Exhaust valve open and intake valve closed:

$$v_5 = v_4 = v_1 = v_{\text{BDC}} \tag{3-67}$$

$$w_{4-5} = 0 \tag{3-68}$$

$$Q_{4-5} = Q_{out} = m_m c_v(T_5 - T_4) = m_m c_v(T_1 - T_4) \tag{3-69}$$

$$q_{4-5} = q_{out} = c_v(T_5 - T_4) = (u_5 - u_4) = c_v(T_1 - T_4) \tag{3-70}$$

Process 5-6—constant-pressure exhaust stroke at P_o.
Exhaust valve open and intake valve closed:

$$w_{5-6} = P_o(v_6 - v_5) = P_o(v_6 - v_1) \tag{3-71}$$

Thermal efficiency of diesel cycle:

$$\begin{aligned}(\eta_t)_{DIESEL} &= |w_{net}|/|q_{in}| = 1 - (|q_{out}|/|q_{in}|) \\ &= 1 - [c_v(T_4 - T_1)/c_p(T_3 - T_2)] \\ &= 1 - (T_4 - T_1)/[k(T_3 - T_2)]\end{aligned} \tag{3-72}$$

With rearrangement, this can be shown to equal:

$$(\eta_t)_{DIESEL} = 1 - (1/r_c)^{k-1}[(\beta^k - 1)/\{k(\beta - 1)\}] \tag{3-73}$$

where: r_c = compression ratio
$k = c_p/c_v$
β = cutoff ratio

If reptesentative numbers are introduced into Eq. (3-73), it is found that the value of the term in brackets is greater than one. When this equation is compared with Eq. (3-31), it can be seen that for a given compression ratio the thermal efficiency of the Otto cycle would be greater than the thermal efficiency of the Diesel cycle. Constant-volume combustion at TDC is more efficient than constant-pressure combustion. However, it must be remembered that CI engines operate with much higher compression ratios than SI engines (12 to 24 versus 8 to 11) and thus have higher thermal efficiencies.

3-7 DUAL CYCLE

If Eqs. (3-31) and (3-73) are compared, it can be seen that to have the best of both worlds, an engine ideally would be compression ignition but would operate on the Otto cycle. Compression ignition would operate on the more efficient higher compression ratios, while constant-volume combustion of the Otto cycle would give higher efficiency for a given compression ratio.

The modern high-speed CI engine accomplishes this in part by a simple operating change from early diesel engines. Instead of injecting the fuel late in the compression stroke near TDC, as was done in early engines, modern CI engines start to inject the fuel much earlier in the cycle, somewhere around 20° bTDC. The first fuel then ignites late in the compression stroke, and some of the combustion occurs almost at constant volume at TDC, much like the Otto cycle. A typical indicator diagram for a modern CI engine is shown in Fig. 3-9. Peak pressure still remains high into

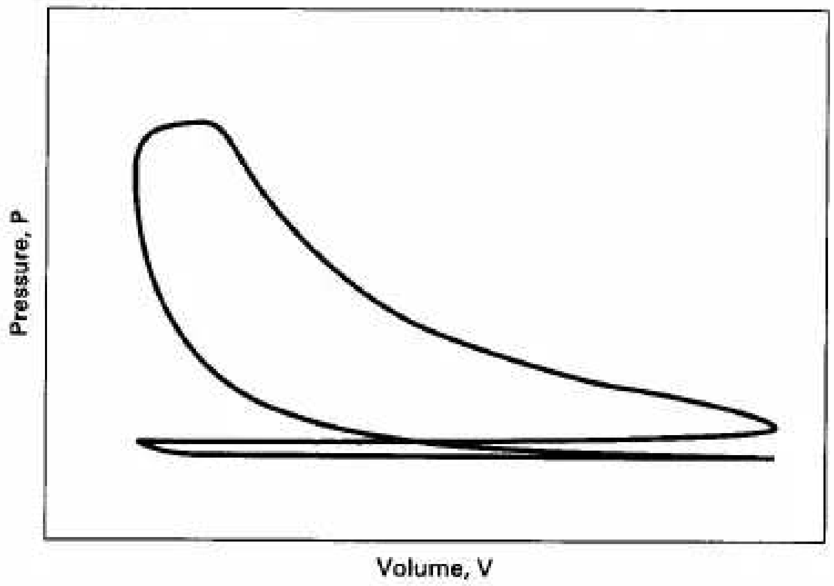


Figure 3-9 Indicator diagram of a modern CI engine operating on a four-stroke cycle.

the expansion stroke due to the finite time required to inject the fuel. The last of the fuel is still being injected at TDC, and combustion of this fuel keeps the pressure high into the expansion stroke. The resulting cycle shown in Fig. 3-9 is a cross between an SI engine cycle and the early CI cycles. The air-standard cycle used to analyze this modern CI engine cycle is called a **Dual cycle,** or sometimes a **Limited Pressure cycle** (Fig. 3-10). It is a dual cycle because the heat input process of combustion can best be approximated by a dual process of constant volume followed by constant pressure. It can also be considered a modified Otto cycle with a limited upper pressure.

Thermodynamic Analysis of Air-Standard Dual Cycle

The analysis of an air-standard Dual cycle is the same as that of the Diesel cycle except for the heat input process (combustion) *2-x-3*.

Process 2-*x*—constant-volume heat input (first part of combustion). All valves closed:

$$V_x = V_2 = V_{TDC} \tag{3-74}$$

$$w_{2-x} = 0 \tag{3-75}$$

$$Q_{2-x} = m_m c_v(T_x - T_2) = (m_a + m_f)c_v(T_x - T_2) \tag{3-76}$$

$$q_{2-x} = c_v(T_x - T_2) = (u_x - u_2) \tag{3-77}$$

$$P_x = P_{max} = P_2(T_x/T_2) \tag{3-78}$$

Pressure ratio is defined as the rise in pressure during combustion, given as a ratio:

$$\alpha = P_x/P_2 = P_3/P_2 = T_x/T_2 = (1/r_c)^k(P_3/P_1) \tag{3-79}$$

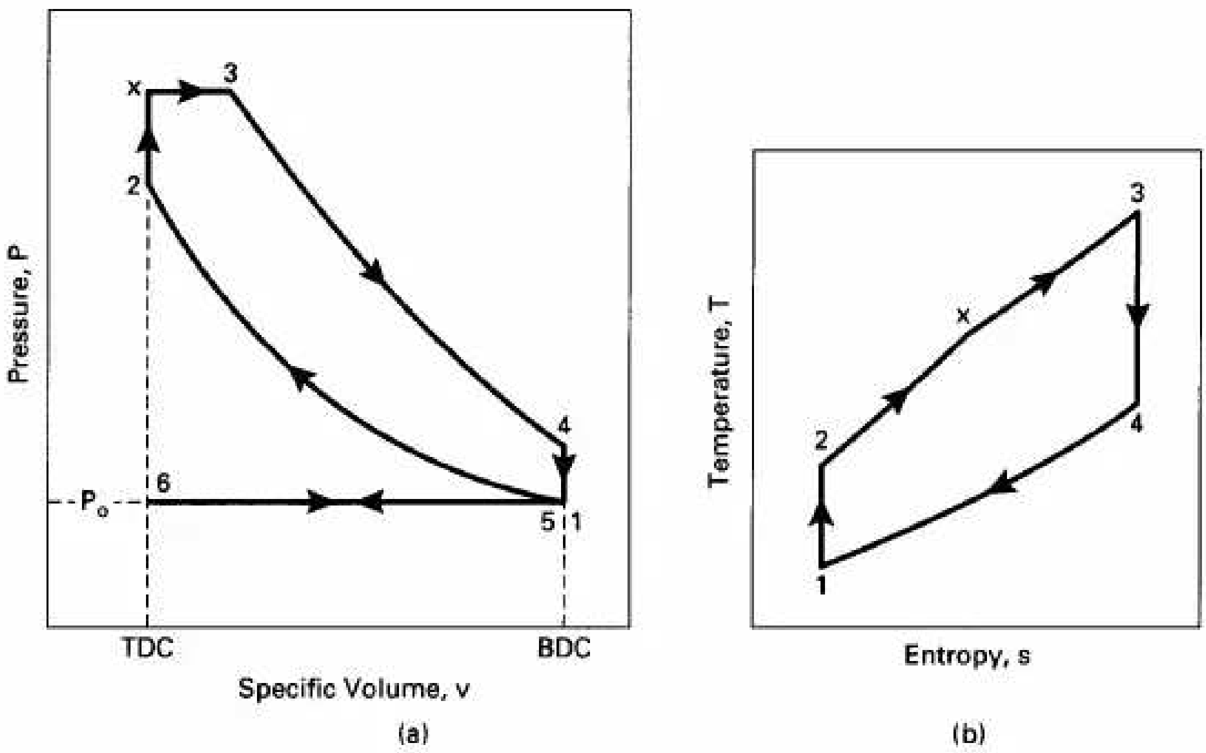


Figure 3-10 Air-standard Dual cycle, 6-1-2-x-3-4-5-6, which approximates the four-stroke cycle of a modern CI engine on **(a)** pressure-specific volume coordinates, and **(b)** temperature-entropy coordinates.

Process x-3—constant-pressure heat input (second part of combustion). All valves closed:

$$P_3 = P_x = P_{max} \tag{3-80}$$

$$Q_{x-3} = m_m c_p(T_3 - T_x) = (m_a + m_f)c_p(T_3 - T_x) \tag{3-81}$$

$$q_{x-3} = c_p(T_3 - T_x) = (h_3 - h_x) \tag{3-82}$$

$$w_{x-3} = q_{x-3} - (u_3 - u_x) = P_x(v_3 - v_x) = P_3(v_3 - v_x) \tag{3-83}$$

$$T_3 = T_{max} \tag{3-84}$$

Cutoff ratio:

$$\beta = v_3/v_x = v_3/v_2 = V_3/V_2 = T_3/T_x \tag{3-85}$$

Heat in:

$$Q_{in} = Q_{2-x} + Q_{x-3} = m_f Q_{HV}\eta_c \tag{3-86}$$

$$q_{in} = q_{2-x} + q_{x-3} = (u_x - u_2) + (h_3 - h_x) \tag{3-87}$$

Thermal efficiency of Dual cycle:

$$\begin{aligned}(\eta_t)_{DUAL} &= |w_{net}|/|q_{in}| = 1 - (|q_{out}|/|q_{in}|) \\ &= 1 - c_v(T_4 - T_1)/[c_v(T_x - T_2) + c_p(T_3 - T_x)] \\ &= 1 - (T_4 - T_1)/[(T_x - T_2) + k(T_3 - T_x)]\end{aligned} \tag{3-88}$$

This can be rearranged to give:

$$(\eta_t)_{DUAL} = 1 - (1/r_c)^{k-1}[\{\alpha\beta^k - 1\}/\{k\alpha(\beta - 1) + \alpha - 1\}] \tag{3-89}$$

where: r_c = compression ratio
$k = c_p/c_v$
α = pressure ratio
β = cutoff ratio

3-8 COMPARISON OF OTTO, DIESEL, AND DUAL CYCLES

Figure 3-11 compares Otto, Diesel, and Dual cycles with the same inlet conditions and the same compression ratios. The thermal efficiency of each cycle can be written as:

$$\eta_t = 1 - |q_{out}|/|q_{in}| \tag{3-90}$$

The area under the process lines on T-s coordinates is equal to the heat transfer, so in Fig. 3-11(b) the thermal efficiencies can be compared. For each cycle, q_{out} is the same (process 4-1). q_{in} of each cycle is different, and using Fig. 3-11(b) and Eq. (3-90) it is found for these conditions:

$$(\eta_t)_{OTTO} > (\eta_t)_{DUAL} > (\eta_t)_{DIESEL} \tag{3-91}$$

However, this is not the best way to compare these three cycles, because they

do not operate on the same compression ratio. Compression ignition engines that operate on the Dual cycle or Diesel cycle have much higher compression ratios than do spark ignition engines operating on the Otto cycle. A more realistic way to compare these three cycles would be to have the same peak pressure-an actual design limitation in engines. This is done in Fig. 3-12. When this figure is compared with Eq. (3-90), it is found:

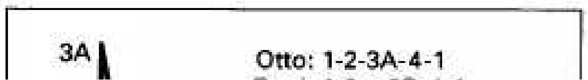
$$(\eta_t)_{DIESEL} > (\eta_t)_{DUAL} > (\eta_t)_{OTTO} \tag{3-92}$$

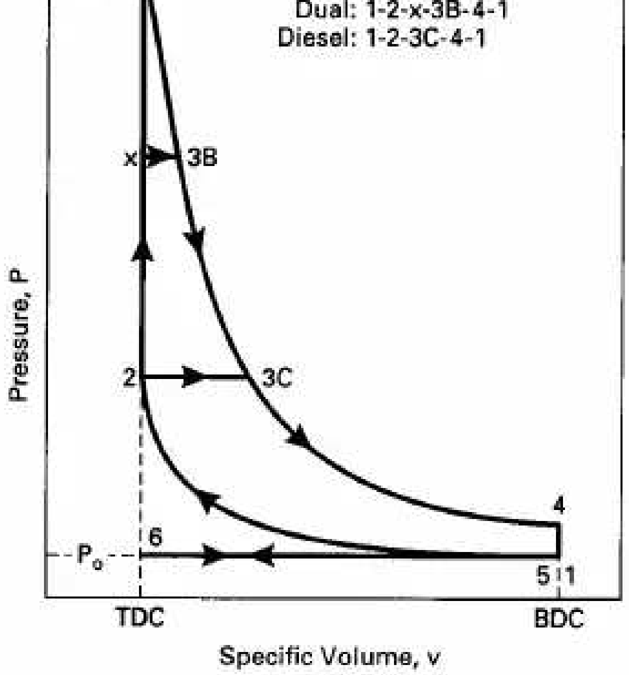
Comparing the ideas of Eqs. (3-91) and (3-92) would suggest that the most efficient engine would have combustion as close as possible to constant volume but would be compression ignition and operate at the higher compression ratios which that requires. This is an area where more research and development is needed.

EXAMPLE PROBLEM 3-4

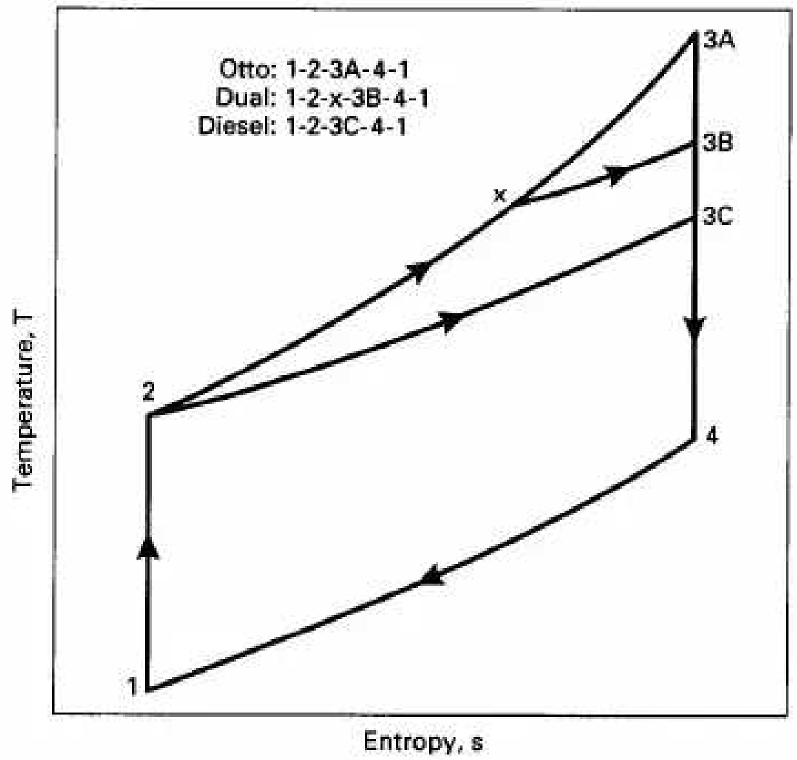
A small truck has a four-cylinder, four-liter CI engine that operates on the air-standard Dual cycle (Fig. 3-10) using light diesel fuel at an air-fuel ratio of 18. The compression ratio of the engine is 16:1 and the cylinder bore diameter is 10.0 cm. At the start of the compression stroke, conditions in the cylinders are 60°C and 100 KPa with a 2% exhaust residual. It can be assumed that half of the heat input from combustion is added at constant volume and half at constant pressure.
Calculate:

1. temperature and pressure at each state of the cycle



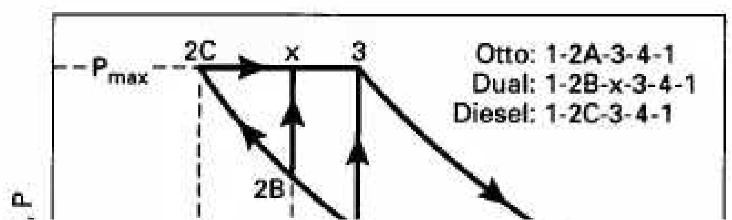


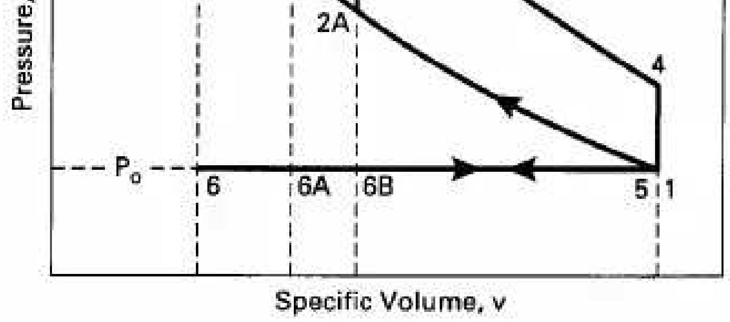
(a)



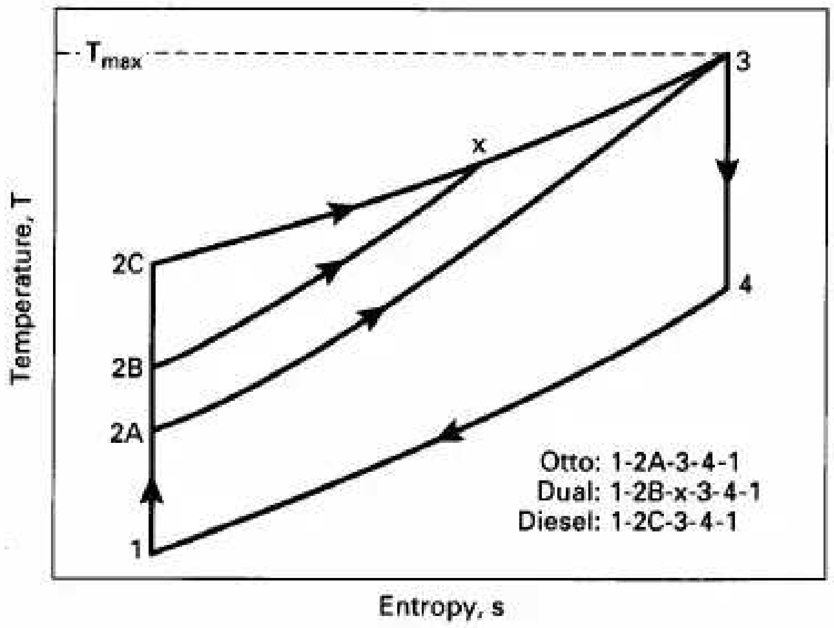
(b)

Figure 3-11 Comparison of air-standard Otto cycle, Dual cycle, and Diesel cycle. All engines have the same cylinder input conditions and same compression ratio.





(a)



(b)

Figure 3-12 Comparison of air-standard Otto cycle, Dual cycle, and Diesel cycle. All engines have the same cylinder input conditions and the same maximum temperature and pressure.

2. indicated thermal efficiency
3. exhaust temperature
4. air temperature in intake manifold
5. engine volumetric efficiency

For one cylinder:

$$V_d = (4\ \mathrm{L})/4 = 1\ \mathrm{L} = 0.001\ \mathrm{m}^3 = 1000\ \mathrm{cm}^3$$

Using Eq. (2-12):

$$r_c = V_{\mathrm{BDC}}/V_{\mathrm{TDC}} = (V_d + V_c)/V_c = 16 = (1000 + V_c)/V_c$$
$$V_c = 66.7\ \mathrm{cm}^3 = 0.0667\ \mathrm{L} = 0.0000667\ \mathrm{m}^3$$

Using Eq. (2-8):

$$V_d = (\pi/4)B^2S = 0.001\ \mathrm{m}^3 = (\pi/4)(0.10\ \mathrm{m})^2S$$
$$S = 0.127\ \mathrm{m} = 12.7\ \mathrm{cm}$$

State 1:

$T_1 = 60°\mathrm{C} = 333\ \mathrm{K}$ given in problem statement

$\underline{T_1 = 60°C = 333\ K}$ given in problem statement

$\underline{P_1 = 100\ kPa}$ given

$$V_1 = V_{BDC} = V_d + V_c = 0.001 + 0.0000667 = \underline{0.0010667\ m^3}$$

Mass of gas in one cylinder at start of compression:

$$m_m = P_1V_1/RT_1 = (100\ kPa)(0.0010667\ m^3)/(0.287\ kJ/kg\text{-}K)(333\ K)$$
$$= 0.00112\ kg$$

Mass of fuel injected per cylinder per cycle:

$$m_f = (0.00112)(0.98)(1/19) = 0.0000578\ kg$$

State 2: Equations (3-52) and (3-53) give temperature and pressure after compression:

$$T_2 = T_1(r_c)^{k-1} = (333\ K)(16)^{0.35} = \underline{879\ K = 606°C}$$

$$P_2 = P_1(r_c)^k = (100\ kPa)(16)^{1.35} = \underline{4222\ kPa}$$

$$V_2 = mRT_2/P_2 = (0.00112\ kg)(0.287\ kJ/kg\text{-}K)(879\ K)/(4222\ kPa)$$
$$\underline{= 0.000067\ m^3 = V_c}$$

or using Eq. (2-12):

$$V_2 = V_1/r_c = (0.0010667)/(16) = 0.0000667\ m^3$$

State x: Heating value of light diesel fuel is obtained from Table A-2 in the Appendix:

$$Q_{in} = m_fQ_{HV} = (0.0000578\ kg)(42{,}500\ kJ/kg) = 2.46\ kJ$$

If half of Q_{in} occurs at constant volume, then using Eq. (3-76):

$$Q_{2-x} = 1.23\ kJ = m_m c_v(T_x - T_2)$$
$$= (0.00112\ kg)(0.821\ kJ/kg\text{-}K)(T_x - 879\ K)$$
$$\underline{T_x = 2217\ K = 1944°C}$$

$$\underline{V_x = V_2 = 0.0000667\ m^3}$$

$$P_x = mRT_x/V_x$$
$$= (0.00112\ kg)(0.287\ kJ/kg\text{-}K)(2217\ K)/(0.0000667\ m^3)$$
$$\underline{= 10{,}650\ kPa = P_{max}}$$

or:

$$P_x = P_2(T_x/T_2) = (4222\ kPa)(2217/879) = 10{,}650\ kPa$$

State 3:

$$\underline{P_3 = P_x = 10{,}650\ kPa = P_{max}}$$

Equation (3-81) gives:

$$Q_{x-3} = 1.23 \text{ kJ} = m_m c_p(T_3 - T_x)$$
$$= (0.00112 \text{ kg})(1.108 \text{ kJ/kg-K})(T_3 - 2217 \text{ K})$$
$$\underline{T_3 = 3208 \text{ K} = 2935°\text{C} = T_{max}}$$
$$V_3 = mRT_3/P_3 = (0.00112 \text{ kg})(0.287 \text{ kJ/kg-K})(3208 \text{ K})/(10{,}650 \text{ kPa})$$
$$\underline{= 0.000097 \text{ m}^3}$$

State 4:

$$V_4 = V_1 = 0.0010667 \text{ m}^3$$

Equations (3-64) and (3-65) give temperature and pressure after expansion:

$$T_4 = T_3(V_3/V_4)^{k-1} = (3208 \text{ K})(0.000097/0.0010667)^{0.35}$$
$$\underline{= 1386 \text{ K} = 1113°\text{C}}$$
$$P_4 = P_3(V_3/V_4)^k = (10{,}650 \text{ kPa})(0.000097/0.0010667)^{1.35} \underline{= 418 \text{ kPa}}$$

Work out for process x-3 for one cylinder for one cycle using Eq. (3-83):

$$W_{x-3} = P(V_3 - V_x) = (10{,}650 \text{ kPa})(0.000097 - 0.0000667)\text{m}^3 \underline{= 0.323 \text{ kJ}}$$

Work out for process 3-4 using Eq. (3-66):

$$W_{3-4} = mR(T_4 - T_3)/(1 - k)$$
$$= (0.00112 \text{ kg})(0.287 \text{ kJ/kg-K})(1386 - 3208)\text{K}/(1 - 1.35)$$
$$\underline{= 1.673 \text{ kJ}}$$

Work in for process 1-2 using Eq. (3-56):

$$W_{1-2} = mR(T_2 - T_1)/(1 - k)$$
$$= (0.00112 \text{ kg})(0.287 \text{ kJ/kg-K})(879 - 333)\text{K}/(1 - 1.35)$$
$$= -0.501 \text{ kJ}$$
$$W_{net} = (+0.323) + (+1.673) + (-0.501) = +1.495 \text{ kJ}$$

2) Equation (3-88) gives indicated thermal efficiency:

$$(\eta_t)_{DUAL} = |W_{net}|/|Q_{in}| = (1.495 \text{ kJ})/(2.46 \text{ kJ}) \underline{= 0.607 = 60.7\%}$$

Pressure ratio:

$$\alpha = P_x/P_2 = 10{,}650/4222 = 2.52$$

Cutoff ratio:

$$\beta = V_3/V_x = 0.000097/0.0000667 = 1.45$$

Using Eq. (3-89) to find thermal efficiency:

$$(\eta_t)_{DUAL} = 1 - (1/r_c)^{k-1}[\{\alpha\beta^k - 1\}/\{k\alpha(\beta - 1) + \alpha - 1\}]$$
$$= 1 - (1/16)^{0.35}[\{(2.52)(1.45)^{1.35} - 1\}/\{(1.35)(2.52)(1.45 - 1) + 2.52 - 1\}]$$
$$= 0.607$$

3) Assuming exhaust pressure is the same as intake pressure and using Eq. (3-37) for exhaust temperature:

$$T_{ex} = T_4(P_{ex}/P_4)^{(k-1)/k} = (1386 \text{ K})(100/418)^{(1.35-1)/1.35} \underline{= 957 \text{ K} = 684°\text{C}}$$

Equation (3-46) gives exhaust residual:

$$x_r = (1/r_c)(T_4/T_{ex})(P_{ex}/P_4)$$
$$= (1/16)(1386/957)(100/418) = \underline{0.022 = 2.2\%}$$

4) Using Eq. (3-50) to find air temperature entering the cylinder:

$$(T_m)_1 = x_r T_{ex} + (1 - x_r) T_a$$
$$(333\text{ K}) = (0.022)(957\text{ K}) + (1 - 0.022)\,T_a$$
$$\underline{T_a = 319\text{ K} = 46°\text{C}}$$

5) Mass of air entering one cylinder during intake:

$$m_a = (0.00112\text{ kg})(0.98) = 0.00110\text{ kg}$$

Volumetric efficiency is found using Eq. (2-69):

$$\eta_v = m_a/\rho_a V_d = (0.00110\text{ kg})/(1.181\text{ kg/m}^3)(0.001\text{ m}^3)$$
$$= \underline{0.931 = 93.1\%}$$

HISTORIC—ATKINSON CYCLE

In Otto and Diesel cycles, when the exhaust valve is opened near the end of the expansion stroke, pressure in the cylinder is still on the order of three to five atmospheres. A potential for doing additional work during the power stroke is therefore lost when the exhaust valve is opened and pressure is reduced to atmospheric. If the exhaust valve is not opened until the gas in the cylinder is allowed to expand down to atmospheric pressure, a

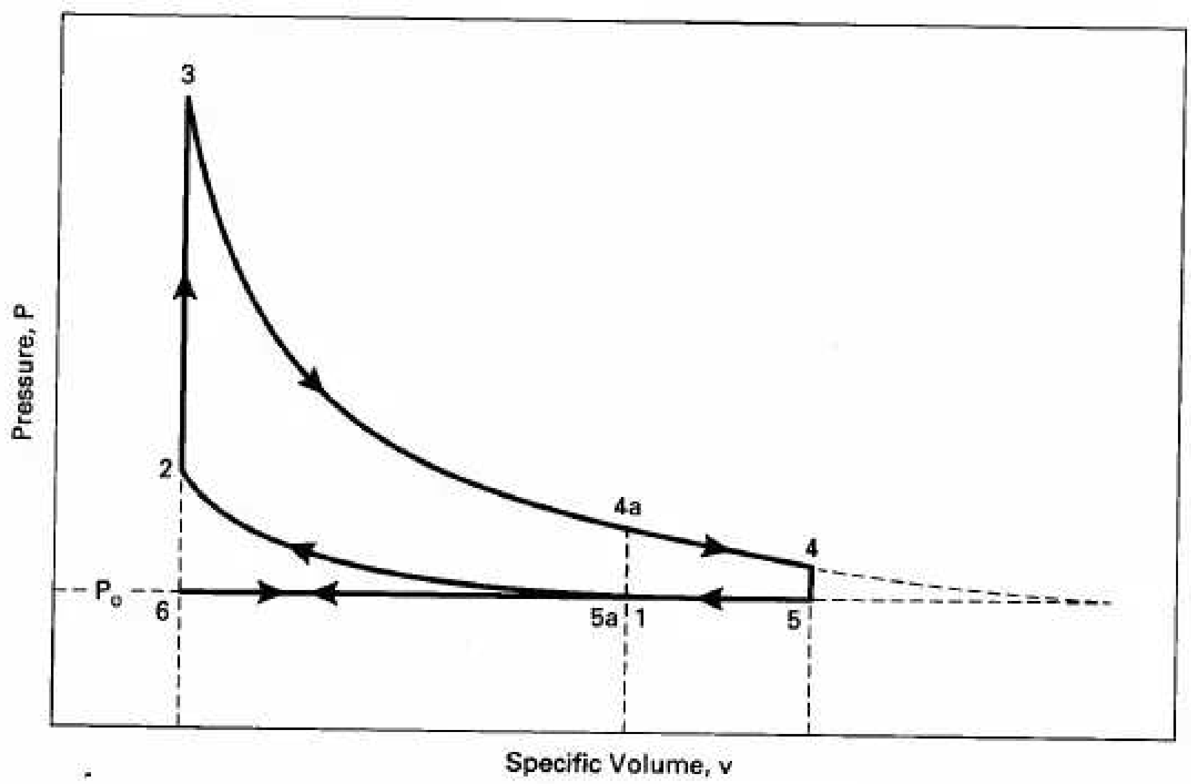


Figure 3-13 Air-standard Atkinson cycle, 6-1-2-3-4-5-6, with larger expansion ratio v_4/v_3 than compression ratio v_1/v_3. The same engine operating on an Otto cycle would follow cycle 6-1-2-3-4a-5a-6.

greater amount of work would be obtained in the expansion stroke, with an increase in engine thermal efficiency. Such an air-standard cycle is called an **Atkinson cycle** or **Overexpanded cycle** (or Complete Expansion cycle) and is shown in Fig. 3-13.

Starting in 1885 a number of crank and valve mechanisms were tried to achieve this cycle, which has a longer expansion stroke than compression stroke. No large number of these engines has ever been marketed, indicating the failure of this development [58].

3-9 MILLER CYCLE

The **Miller cycle,** named after R. H. Miller (1890-1967), is a modem modification of the Atkinson cycle and has an expansion ratio greater than the compression ratio. This is accomplished, however, in a much different way. Whereas an engine designed to operate on the Atkinson cycle needed a complicated mechanical linkage system of some kind, a Miller cycle engine uses unique valve timing to obtain the same desired results.

Air intake in a Miller cycle is unthrottled. The amount of air ingested into each cylinder is then controlled by closing the intake valve at the proper time, long before

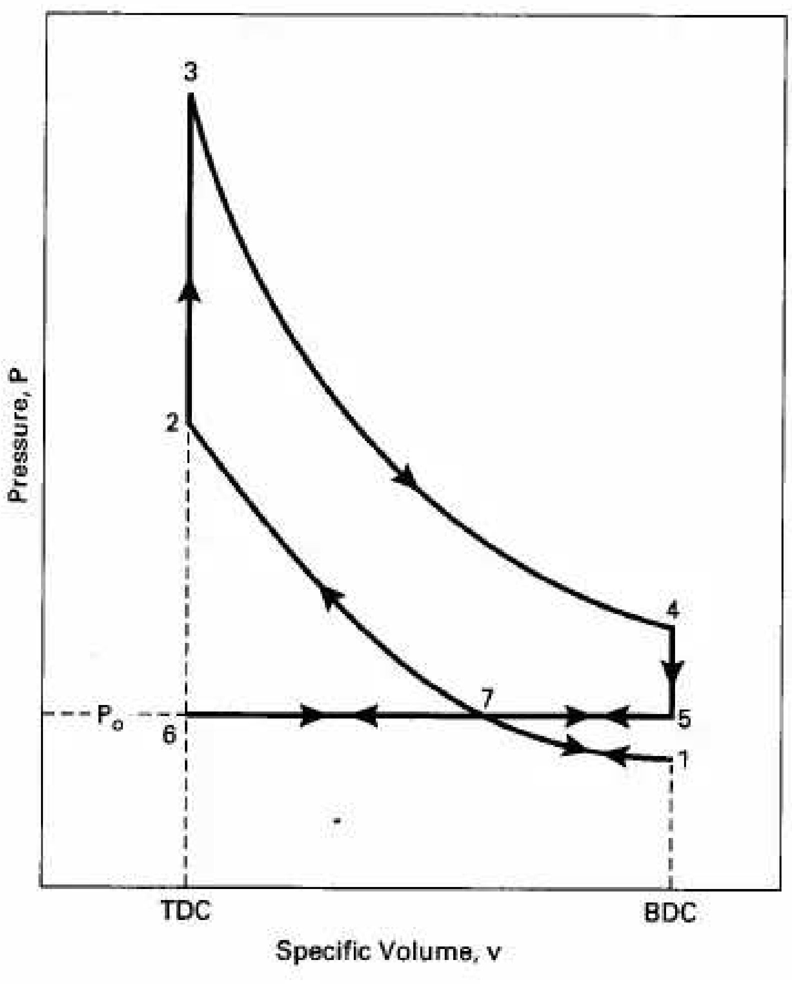


Figure 3-14 Air-standard Miller cycle for unthrottled naturally aspirated four-stroke cycle SI engine. If the engine has early intake valve closing, the cycle will be 6-7-1-7-2-3-4-5-7-6. If the engine has late intake valve closing, the cycle will be 6-7-5-7-2-3-4-5-7-6.

BDC (point 7 in Fig. 3-14). As the piston then continues towards BDC during the latter part of the intake stroke, cylinder pressure is reduced along process 7-1. When the piston reaches BDC and starts back towards TDC cylinder pressure is again increased during process 1-7. The resulting cycle is 6-7-1-7-2-3-4-5-6. The work produced in the first part of the intake process 6-7 is canceled by part of the exhaust stroke 7-6, process 7-1 is canceled by process 1-7, and the net indicated work is the area within loop 7-2-3-4-5-7. There is essentially no pump work. The compression ratio is:

$$r_c = V_7/V_2 \tag{3-93}$$

and the larger expansion ratio is:

$$r_e = V_4/V_2 = V_4/V_3 \tag{3-94}$$

The shorter compression stroke which absorbs work, combined with the longer expansion stroke which produces work, results in a greater net indicated work per cycle. In addition, by allowing air to flow through the intake system unthrottled, a major loss experienced by most SI engines is eliminated. This is especially true at part throttle, when an Otto cycle engine would experience low pressure in the intake manifold and a corresponding high negative pump work. The Miller

cycle engine has essentially no pump work (ideally none), much like a CI engine. This results in higher thermal efficiency.

The mechanical efficiency of a Miller cycle engine would be about the same as that of an Otto cycle engine, which has a similar mechanical linkage system. An Atkinson cycle engine, on the other hand, requires a much more complicated mechanical linkage system, resulting in lower mechanical efficiency.

Another variation of this cycle can be obtained if the intake air is unthrottled and the intake valve is closed after BDC. When this is done, air is ingested during the entire intake stroke, but some of it is then forced back into the intake manifold before the intake valve closes. This results in cycle 6-7-5-7-2-3-4-5-6 in Fig. 3-14. The net indicated work is again the area within loop 7-2-3-4-5-7, with the compression and expansion ratios given by Eqs. (3-93) and (3-94).

For either variation of the cycle to work efficiently, it is extremely important to be able to close the intake valve at the precise correct moment in the cycle (point 7). However, this point where the intake valve must close changes as the engine speed and/or load is changed. This control was not possible until variable valve timing was perfected and introduced. Automobiles with Miller cycle engines were first marketed in the latter half of the 1990s. A typical value of the compression ratio is about 8:1, with an expansion ratio of about 10:1.

The first production automobile engines operating on Miller cycles used both early intake valve closing methods and late intake valve closing methods. Several types of variable valve timing systems have been tried and are being developed. At present, none of these offer full flexibility, and major improvements are still needed.

If the intake valve is closed bBDC, less than full displacement volume of the cylinder is available for air ingestion. If the intake valve is closed aBDC, the full displacement volume is filled with air, but some of it is expelled out again before the valve is closed (process 5-7 in Fig. 3-14). In either case, less air and fuel end up in the cylinder at the start of compression, resulting in low output per displacement and low indicated mean effective pressure. To counteract this, Miller cycle engines are usually supercharged or turbocharged with peak intake manifold pressures of 150-200 kPa. Fig. 3-15 shows a supercharged Miller engine cycle.

EXAMPLE PROBLEM 3-5

The four-cylinder, 2.5-liter SI automobile engine of Example Problem 3-1 is converted to operate on an air-standard Miller cycle with early valve closing (cycle 6-7-1-7-2-3-4-5-6 in Fig. 3-15). It has a compression ratio of 8:1 and an expansion ratio of 10:1. A supercharger is added that gives a cylinder pressure of 160 KPa when the intake valve closes, as shown in Fig. 3-15. The temperature is again 60°C at this point. The same fuel and AF are used with combustion efficiency η_c = 100%.
Calculate:

1. temperature and pressure at all points in the cycle
2. indicated thermal efficiency
3. indicated mean effective pressure
4. exhaust temperature

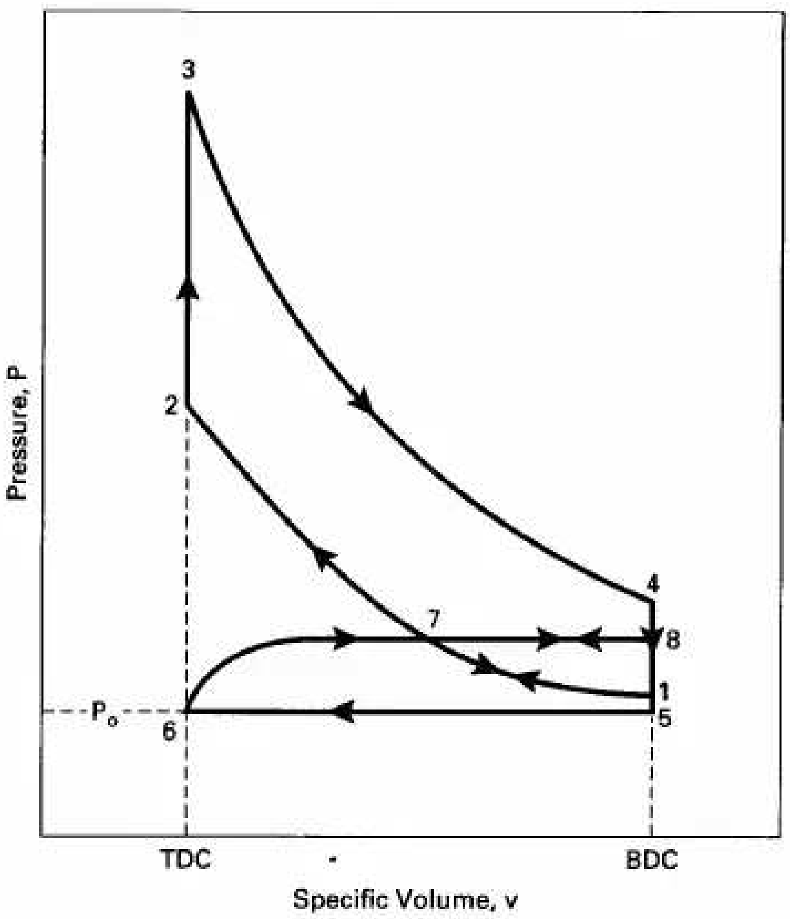


Figure 3-15 Air-standard Miller cycle for a four-stroke cycle SI engine equipped with a turbocharger or supercharger. If the engine has early intake valve closing, the cycle will be 6-7-1-7-2-3-4-5-6. If the engine has late intake valve-closing, the cycle will be 6-7-8-7-2-3-4-5-6.

From Example Problem 3-1 for one cylinder:

$$V_d = 0.000625 \text{ m}^3$$

Expansion ratio using Eq. (3-94):

$$r_e = V_4/V_3 = (V_d + V_c)/V_c = 10 = (0.000625 + V_c)/V_c$$

$$V_c = 0.000069 \text{ m}^3 = V_2 = V_3 = V_6$$

$$V_1 = V_4 = V_5 = V_d + V_c = 0.000625 + 0.000069 = 0.000694 \text{ m}^3$$

Compression ratio using Eq. (3-93):

$$V_7 = V_2 r_c = (0.000069)(8) = 0.000552 \text{ m}^3$$

1) Temperatures and Pressures:

$\underline{T_7 = 60°\text{C} = 333 \text{ K}}$ given in problem statement

$\underline{P_7 = P_8 = 160 \text{ kPa}}$ given

$$P_1 = P_7(V_7/V_1)^k = (160 \text{ kPa})(0.000552/0.000694)^{1.35} = \underline{117 \text{ kPa}}$$

$$T_1 = T_7(V_7/V_1)^{k-1} = (333 \text{ K})(0.000552/0.000694)^{0.35} = \underline{307 \text{ K} = 34°\text{C}}$$

$$T_2 = T_7(r_c)^{k-1} = (333 \text{ K})(8)^{0.35} = \underline{689 \text{ K} = 416°\text{C}}$$

$$P_2 = P_7(r_c)^k = (160 \text{ kPa})(8)^{1.35} = \underline{2650 \text{ kPa}}$$

Mass of gas in cylinder:

$$m_1 = P_1V_1/RT_1 = (117 \text{ kPa})(0.000694 \text{ m}^3)/(0.287 \text{ kJ/kg-K})(307 \text{ K})$$
$$= 0.000922 \text{ kg}$$

If AF = 15 and exhaust residual x_r = 4%, the mass of fuel will be:

$$m_f = (1/16)(0.96)(0.000922) = 0.000055 \text{ kg}$$

$$Q_{in} = m_f Q_{HV}\eta_c = (0.000055 \text{ kg})(44{,}300 \text{ kJ/kg})(1.00) = 2.437 \text{ kJ}$$

$$Q_{in} = m_m c_v(T_3 - T_2) = 2.437 \text{ kJ}$$
$$= (0.000922 \text{ kg})(0.821 \text{ kJ/kg-K})(T_3 - 689 \text{ K})$$

$$\underline{T_3 = 3908 \text{ K} = 3635°\text{C}}$$

$$P_3 = P_2(T_3/T_2) = (2650 \text{ kPa})(3908/689) = \underline{15{,}031 \text{ kPa}}$$

$$T_4 = T_3(V_3/V_4)^{k-1} = T_3(1/r_e)^{k-1} = (3908 \text{ K})(1/10)^{0.35}$$
$$\underline{= 1746 \text{ K} = 1473°\text{C}}$$

$$P_4 = P_3(1/r_e)^k = (15{,}031 \text{ kPa})(1/10)^{1.35} = \underline{671 \text{ kPa}}$$

$$V_4 = mRT_4/P_4 = (0.000922 \text{ kg})(0.287 \text{ kJ/kg-K})(1746 \text{ K})/(671 \text{ kPa})$$
$$= 0.000694 \text{ m}^3$$ This agrees with the above value for V_4.

$\underline{P_5 = P_{ex} = 100 \text{ kPa}}$

$$T_5 = T_4(P_5/P_4) = (1746 \text{ K})(100/671) = \underline{260 \text{ K}}$$

2) Indicated thermal efficiency:

$$W_{3-4} = mR(T_4 - T_3)/(1 - k)$$
$$= (0.000922 \text{ kg})(0.287 \text{ kJ/kg-K})(1746 - 3908)\text{K}/(1 - 1.35)$$
$$= 1.635 \text{ kJ}$$

$$W_{7-2} = mR(T_2 - T_7)/(1 - k)$$

$$= (0.000922 \text{ kg})(0.287 \text{ kJ/kg-K})(689 - 333)\text{K}/(1 - 1.35)$$
$$= -0.269 \text{ kJ}$$
$$W_{6-7} = P_7(V_7 - V_6) = (160 \text{ kPa})(0.000552 - 0.000069)\text{m}^3 = 0.077 \text{ kJ}$$
$$W_{5-6} = P_5(V_6 - V_5) = (100 \text{ kPa})(0.000069 - 0.000694)\text{m}^3 = -0.063 \text{ kJ}$$
$$W_{net} = (+1.635) + (-0.269) + (+0.077) + (-0.063) = +1.380 \text{ kJ}$$
$$(\eta_t)_{MILLER} = |W_{net}|/|Q_{in}| = (1.380 \text{ kJ})/(2.437 \text{ kJ}) = \underline{0.566 = 56.6\%}$$

3) Indicated mean effective pressure using Eq. (2-29):

$$\text{imep} = W_{net}/V_d = (1.380 \text{ kJ})/(0.000625 \text{ m}^3) = \underline{2208 \text{ kPa}}$$

4) Exhaust temperature:

$$T_{ex} = T_4(P_{ex}/P_4)^{(k-1)/k} = (1746 \text{ K})(100/671)^{(1.35-1)/1.35} = \underline{1066 \text{ K} = 793°\text{C}}$$

3-10 COMPARISON OF MILLER CYCLE AND OTTO CYCLE

When the Otto cycle engine of Example Problems 3-1 and 3-2 is compared with a similar engine operating on a Miller cycle as in Example Problem 3-5, the superiority of the Miller cycle can be seen. Table 3-1 gives such a comparison.

Temperatures in the two cycles are about the same, except for the exhaust temperature. It is important that the temperature at the beginning of combustion for either cycle be low enough so that self-ignition and knock do not become problems. The lower exhaust temperature of the Miller cycle is the result of greater expansion cooling that occurs from the essentially same maximum cycle temperature. Lower exhaust temperature means less energy is lost in the exhaust, with more of it used as work output in the longer expansion stroke. Pressures throughout the Miller cycle are higher than those of the Otto cycle, mainly because of the supercharged input. The output parameters of imep, thermal efficiency, and work are all higher for the Miller cycle, showing the technical superiority of this cycle. Some of the indicated work and indicated thermal efficiency of the Miller cycle will be lost due to the need to drive the supercharger. Even with this considered, however, brake work and brake thermal efficiency will be substantially greater than in an Otto cycle engine. If a turbocharger were used instead of a supercharger, brake output parameter values would be even higher. One cost of this higher output is the greater complexity of the valve system of the Miller cycle engine, and the corresponding higher manufacturing costs.

TABLE 3-1 COMPARISON OF OTTO AND MILLER CYCLES

	Miller Cycle	Otto Cycle
Temperature at start of combustion T_2:	689 K	707 K
Pressure at start of combustion P_2:	2650 kPa	1826 kPa
Maximum temperature T_3:	3908 K	3915 K
Maximum pressure P_3:	15,031 kPa	10,111 kPa
Exhaust temperature:	1066 K	1183 K
Indicated net work per cylinder per cycle for same Q_{in}:	1.380 kJ	1.030 kJ

per cycle for same Q_{in}:	1.380 kJ	1.030 kJ
Indicated thermal efficiency:	56.6 %	52.9 %
Indicated mean effective pressure:	2208 kPa	1649 kPa

3-11 TWO-STROKE CYCLES

Two-Stroke SI Engine Cycle

An air-standard approximation to a typical two-stroke SI engine cycle is shown in Fig. 3-16.

Process 1-2—isentropic power or expansion stroke.
All ports (or valves) closed:

$$T_2 = T_1(V_1/V_2)^{k-1} \tag{3-95}$$

$$P_2 = P_1(V_1/V_2)^{k} \tag{3-96}$$

$$q_{1-2} = 0 \tag{3-97}$$

$$w_{1-2} = (P_2 v_2 - P_1 v_1)/(1-k) = R(T_2 - T_1)/(1-k) \tag{3-98}$$

Process 2-3—exhaust blowdown.
Exhaust port open and intake port closed:

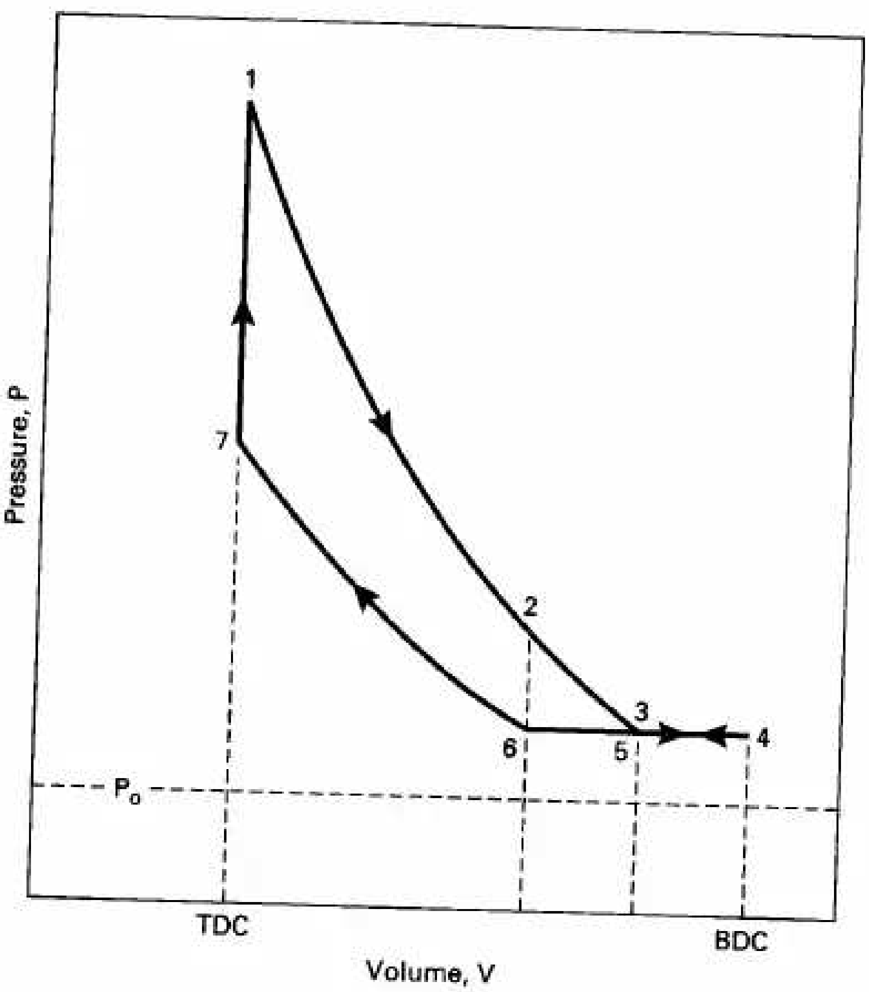


Figure 3-16 Air-standard

Figure 3-16 Air-standard approximation for a two-stroke cycle SI engine, 1-2-3-4-5-6-7-1.

Process 3-4-5-intake, and exhaust scavenging.
Exhaust port open and intake port open:

Intake air entering at an absolute pressure on the order of 140-180 kPa fills and scavenges the cylinder. Scavenging is a process in which the air pushes out most of the remaining exhaust residual from the previous cycle through the open exhaust port into the exhaust system, which is at about one atmosphere pressure. The piston uncovers the intake port at point 3, reaches BDC at point 4, reverses direction, and again closes the intake port at point 5. In some engines fuel is mixed with the incoming air. In other engines the fuel is injected later, after the exhaust port is closed.

Process 5-6—exhaust scavenging.
Exhaust port open and intake port closed:

Exhaust scavenging continues until the exhaust port is closed at point 6.

Process 6-7—isentropic compression.
All ports closed:

$$T_7 = T_6(V_6/V_7)^{k-1} \tag{3-99}$$

$$P_7 = P_6(V_6/V_7)^k \tag{3-100}$$

$$q_{6-7} = 0 \tag{3-101}$$

$$w_{6-7} = (P_7 v_7 - P_6 v_6)/(1-k) = R(T_7 - T_6)/(1-k) \tag{3-102}$$

In some engines, fuel is added very early in the compression process. The spark plug is fired near the end of process 6-7.

Process 7-1—constant-volume heat input (combustion).
All ports closed:

$$V_7 = V_1 = V_{\text{TDC}} \tag{3-103}$$

$$W_{7-1} = 0 \tag{3-104}$$

$$Q_{7-1} = Q_{\text{in}} = m_f Q_{\text{HV}} \eta_c = m_m c_v (T_1 - T_7) \tag{3-105}$$

$$T_1 = T_{\max} \tag{3-106}$$

$$P_1 = P_{\max} = P_7(T_1/T_7) \tag{3-107}$$

Two-Stroke CI Engine Cycle

Many compression ignition engines-especially large ones-operate on two-stroke cycles. These cycles can be approximated by the air-standard cycle shown in Fig. 3-17. This cycle is the same as the two-stroke SI cycle except for the fuel input and combustion process. Instead of adding fuel to the intake air or early in the compression process, fuel is added with injectors late in the compression process, the same as with four-stroke cycle CI engines. Heat input or combustion can be approximated by a two-step (dual) process .

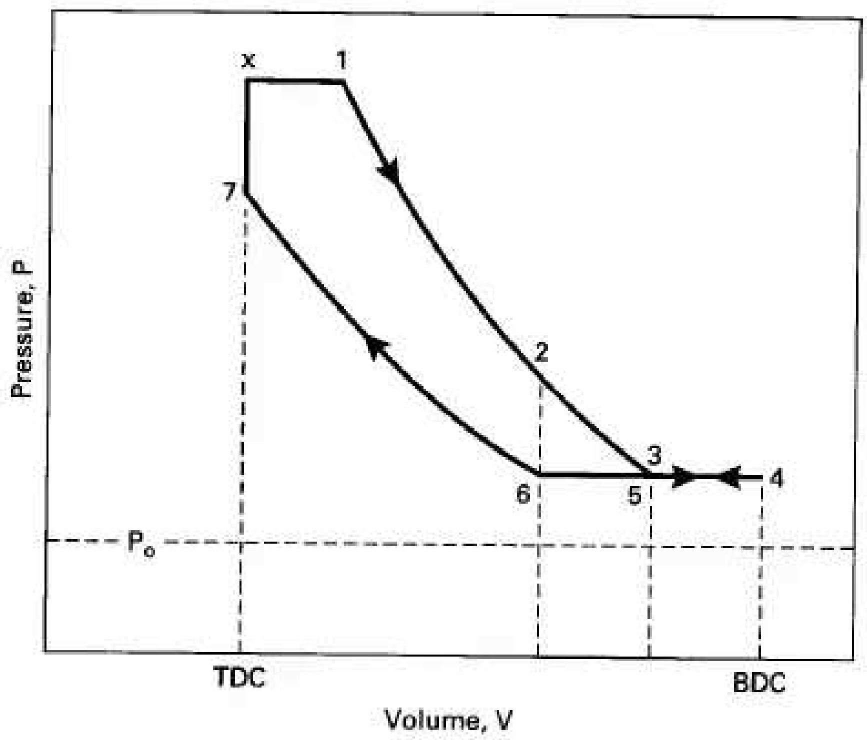


Figure 3-17 Air-standard approximation for a two-stroke cycle CI engine, 1-2-3-4-5-6-7-x-1.

Process 7-x—constant-volume heat input (first part of combustion).
All ports closed:

$$V_7 = V_x = V_{\mathrm{TDC}} \tag{3-108}$$

$$W_{7-x} = 0 \tag{3-109}$$

$$Q_{7-x} = m_m c_v(T_x - T_7) \tag{3-110}$$

$$P_x = P_{\max} = P_7(T_x/T_7) \tag{3-111}$$

Process x-1—constant-pressure heat input (second part of combustion).
All ports closed:

$$P_1 = P_x = P_{\max} \tag{3-112}$$

$$W_{x-1} = P_1(V_1 - V_x) \tag{3-113}$$

$$Q_{x-1} = m_m c_p(T_1 - T_x) \tag{3-114}$$

$$T_1 = T_{\max} \tag{3-115}$$

3-12 STIRLING CYCLE

In recent years, a number of experimental engines that operate on the **Stirling cycle** shown in Fig. 3-18 have been tested. The concept of the Stirling engine has been

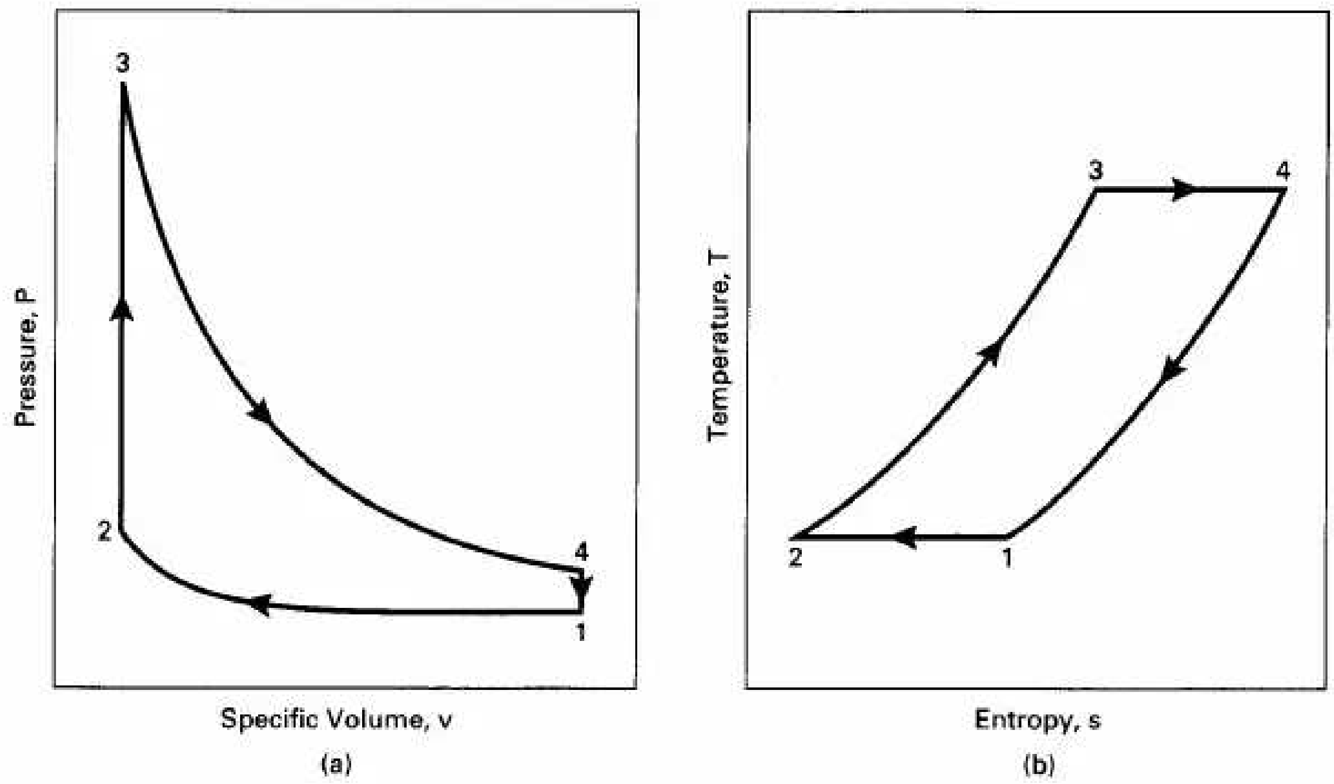


Figure 3-18 Ideal air-standard Stirling cycle, 1-2-3-4-1, on **(a)** pressure-specific volume coordinates, and **(b)** temperature-entropy coordinates.

around since 1816, and while it is not a true internal combustion engine, it is included here briefly because it is a heat engine used to propel vehicles as one of its applications. The basic engine uses a free-floating, double-acting piston with a gas chamber on both ends of the cylinder. Combustion does not occur within the cylinder, but the working gas is heated with an external combustion process. Heat input can also come from solar or nuclear sources. Engine output is usually a rotating shaft [8].

A Stirling engine has an internal regeneration process using a heat exchanger. Ideally, the heat exchanger uses the rejected heat in process 4-1 to preheat the internal working fluid in the heat addition process 2-3. The only heat transfers with the surroundings then occur with a heat addition process 3-4 at one maximum temperature T_{high}, and a heat rejection process 1-2 at one minimum temperature T_{low}. If the processes in the air-standard cycle in Fig. 3-18 can be considered reversible, the thermal efficiency of the cycle will be:

$$(\eta_t)_{STIRLING} = 1 - (T_{low}/T_{high}) \tag{3-116}$$

This is the same thermal efficiency as a Carnot cycle and is the theoretical maximum possible. Although a real engine cannot operate reversibly, a well-designed Stirling engine can have a very high thermal efficiency. This is one of the attractions which is generating interest in this type of engine. Other attractions include low emissions with no catalytic converter and the flexibility of many possible fuels that can be used. This is because heat input is from a continuous steady-state combustion in an external chamber at a relatively low temperature around 1000 K. Fuels used have included gasoline, diesel fuel, jet fuel, alcohol, and natural gas. In some engines, the fuel can be changed with no adjustments needed.

Problems with Stirling engines include sealing, warm-up time needed, and high cost. Other possible applications include refrigeration, stationary power, and heating of buildings.

HISTORIC-LENOIR ENGINE

One ofthe first successful engines developed during the second half of the 1800s was the Lenoir engine (Fig. 3-19). Several hundred of these were built in the 1860s. They operated on a two-stroke cycle and had mechanical efficiencies up to 5% and power output up to 4.5 kW (6 hp). The engines were double acting, with combustion occurring on both ends of the piston. This gave two power strokes per revolution from a single cylinder [29].

3-13 LENOIR CYCLE

The **Lenoir** cycle is approximated by the air-standard cycle shown in Fig. 3-20. The first half of the first stroke was intake, with air-fuel entering the cylinder at atmos-

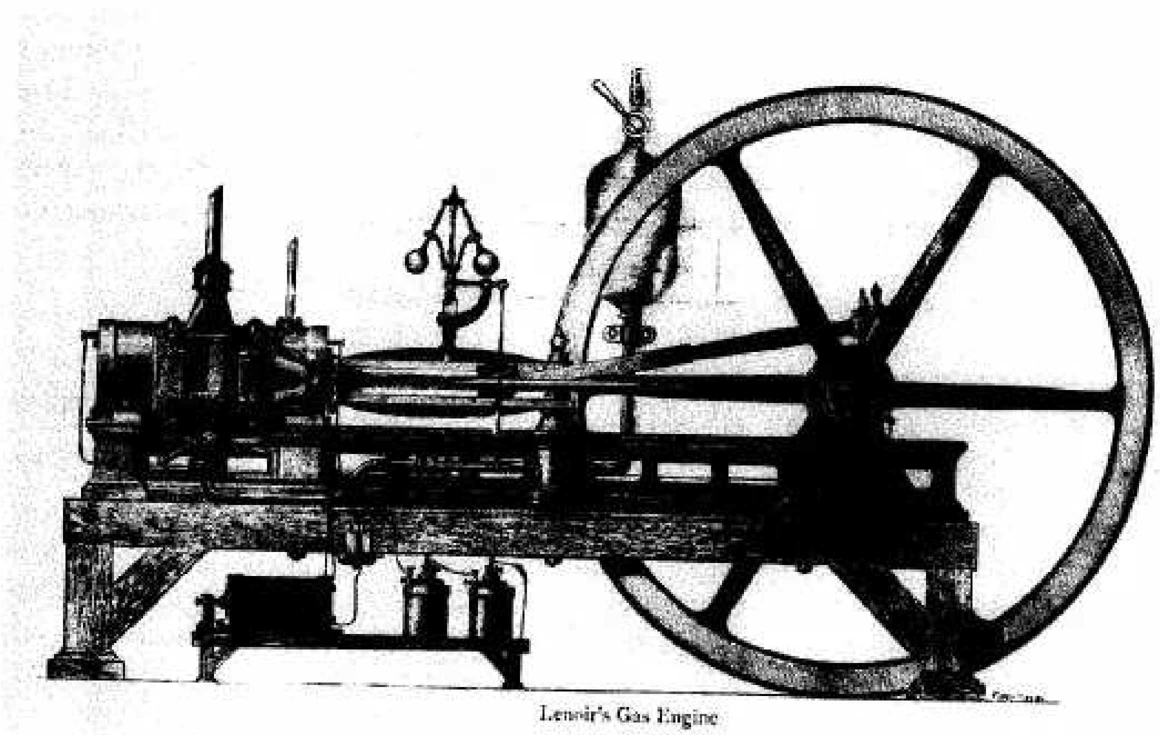


Figure 3-19 Lenoir non-compression engine of 1861. Reprinted with permission from Ref. [29], "Internal Fire" by Lyle Cummins.

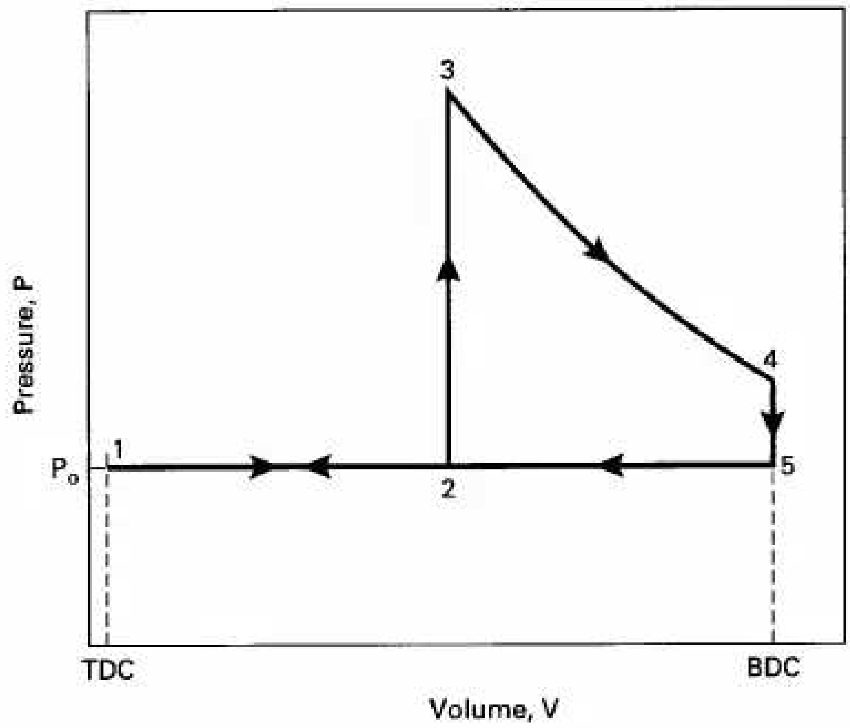


Figure 3-20 Air-standard approximation for a historic Lenoir engine cycle, 1-2-3-4-5-1.

pheric pressure (process 1-2 in Fig. 3-20). At about halfway through the first stroke, the intake valve was closed and the air-fuel mixture was ignited without any compression. Combustion raised the temperature and pressure in the cylinder almost at constant volume in the slow-moving engine (process 2-3). The second half of the first stroke then became the power or expansion process 3-4. Near BDC, the exhaust valve opened and blowdown occurred (4-5). This was followed by the exhaust stroke 5-1, completing the two-stroke cycle. There was essentially no clearance volume.

Thermodynamic Analysis of Air-Standard lenoir Cycle

The intake process 1-2 and the latter half of the exhaust stroke process 2-1 cancel each other thermodynamically on P-V coordinates and can be left out of the analysis of the Lenoir cycle. The cycle then becomes 2-3-4-5-2.

Process 2-3—constant volume heat input (combustion). All valves closed:

$$P_2 = P_1 = P_o \tag{3-117}$$

$$v_3 = v_2 \tag{3-118}$$

$$w_{2-3} = 0 \tag{3-119}$$

$$q_{2-3} = q_{in} = c_v(T_3 - T_2) = (u_3 - u_2) \tag{3-120}$$

Process 3-4—isentropic power or expansion stroke.
All valves closed:

$$q_{3-4} = 0 \tag{3-121}$$

$$T_4 = T_3(v_3/v_4)^{k-1} \tag{3-122}$$

$$P_4 = P_3(v_3/v_4)^k \tag{3-123}$$

$$w_{3-4} = (P_4 v_4 - P_3 v_3)/(1-k) = R(T_4 - T_3)/(1-k) \tag{3-124}$$

$$= (u_3 - u_4) = c_v(T_3 - T_4)$$

Process 4-5—constant-volume heat rejection (exhaust blowdown).
Exhaust valve open and intake valve closed:

$$v_5 = v_4 = v_{BDC} \tag{3-125}$$

$$w_{4-5} = 0 \tag{3-126}$$

$$q_{4-5} = q_{out} = c_v(T_5 - T_4) = (u_5 - u_4) \tag{3-127}$$

Process 5-2—constant-pressure exhaust stroke at P_o.
Exhaust valve open and intake valve closed:

$$P_5 = P_2 = P_1 = P_o \tag{3-128}$$

$$w_{5-2} = P_o(v_2 - v_5) \tag{3-129}$$

$$q_{5-2} = q_{out} = (h_2 - h_5) = c_p(T_2 - T_5) \tag{3-130}$$

Thermal efficiency of Lenoir cycle:

$$(\eta_t)_{LENOIR} = |w_{net}|/|q_{in}| = 1 - (|q_{out}|/|q_{in}|) \tag{3-131}$$

$$= 1 - [c_v(T_4 - T_5) + c_p(T_5 - T_2)]/[c_v(T_3 - T_2)]$$

$$= 1 - [(T_4 - T_5) + k(T_5 - T_2)]/(T_3 - T_2)$$

3-14 SUMMARY

This chapter reviewed the basic cycles used in internal combustion engines. Although many engine cycles have been developed, for over a century most automobile engines have operated on the basic SI four-stroke cycle developed in the 1870s by Otto and others. This can be approximated and analyzed using the ideal air-standard Otto cycle. Many small SI engines operate on a two-stroke cycle, sometimes (erroneously) called a two-stroke Otto cycle.

Early four-stroke CI engines operated on a cycle that can be approximated by the air-standard Diesel cycle. This cycle was improved in modern CI engines of the type used in automobiles and trucks. Changing the injection timing resulted in a

more efficient engine operating on a cycle best approximated by an air-standard Dual cycle. Most small CI engines and very large CI engines operate on a two-stroke cycle.

At present, most automobile engines operate on the four-stroke Otto cycle, but major research and development is resulting in two additional cycles for modem vehicles. Several companies have done major development work to try to create an automobile engine that would operate on an SI two-stroke cycle. Throughout history, two-stroke cycle automobile engines have periodically appeared with varying success. These offer greater power per unit weight, but none would pass modem emission standards. Recent development has concentrated on producing an engine that would satisfy pollution laws. The major technological change is the input of fuel by injection directly into the combustion chamber after exhaust and air intake are completed. If this development work is successful, there will be automobiles on the market with two-stroke cycle engines.

Advances in valve timing technology, including variable timing, have led to the introduction of Miller cycle engines. The Miller cycle improves on the four-stroke SI Otto cycle by closing the intake valve at a more opportune time, either early or late. This results in an expansion ratio that is greater than the compression ratio and represents the most modem of engine cycles.

PROBLEMS

3-1. Cylinder conditions at the start of compression in an SI engine operating at WOT on an air-standard Otto cycle are 60°C and 98 kPa. The engine has a compression ratio of 9.5:1 and uses gasoline with AF = 15.5. Combustion efficiency is 96%, and it can be assumed that there is no exhaust residual.

Calculate: (a) Temperature at all states in the cycle. [0C]
(b) Pressure at all states in the cycle. [kpa]
(c) Specific work done during power stroke. [kJ/kg]
(d) Heat added during combustion. [kJ/kg]
(e) Net specific work done. [kJ/kg]
(£) Indicated thermal efficiency. [%]

3.2. The engine in Problem 3-1 is a three-liter V6 engine operating at 2400 RPM. At this speed the mechanical efficiency is 84%.

Calculate: (a) Brake power. [kW]
(b) Torque. [N-m]
(c) Brake mean effective pressure. [kPa]
(d) Friction power lost. [kW]
(e) Brake specific fuel consumption. [gm/kW-hr]
(£) Volumetric efficiency. [%]
(g) Output per displacement. [kW/L]

3-3. The exhaust pressure of the engine in Problem 3-2 is 100 kPa.

Calculate: (a) Exhaust temperature. [0C]
(b) Actual exhaust residual. [%]
(c) Temperature of air entering cylinders from intake manifold. [0C]

3-4. The engine of Problems 3-2 and 3-3 is operated at part throttle with intake pressure of 75 kPa. Intake manifold temperature, mechanical efficiency, exhaust residual, and air-fuel ratio all remain the same.

Calculate: (a) Temperature in cylinder at start of compression stroke. [°C]

(b) Temperature in cylinder at start of combustion. [°C]

3-5. An SI engine operating at WOT on a four-stroke air-standard cycle has cylinder conditions at the start of compression of 100°F and 14.7 psia. Compression ratio is $r_c = 10$, and the heat added during combustion is $q_{in} = 800$ BTU/lbm. During compression the temperature range is such that a value for the ratio of specific heats $k = 1.4$ would be correct. During the power stroke the temperature range is such that a value of $k = 1.3$ would be correct. Use these values for compression and expansion, respectively, when analyzing the cycle. Use a value for specific heat of $c_v = 0.216$ BTU/lbm-°R, which best corresponds to the temperature range during combustion.

Calculate: (a) Temperature at all states in cycle. [°F]

(b) Pressure at all states in cycle. [psia]

(c) Average value of k which would give the same indicated thermal efficiency value as the analysis in parts (a) and (b).

3-6. A CI engine operating on the air-standard Diesel cycle has cylinder conditions at the start of compression of 65°C and 130 kPa. Light diesel fuel is used at an equivalence ratio of $\phi = 0.8$ with a combustion efficiency $\eta_c = 0.98$. Compression ratio is $r_c = 19$.

Calculate: (a) Temperature at each state of the cycle. [°C]

(b) Pressure at each state of the cycle. [kPa]

(c) Cutoff ratio.

(d) Indicated thermal efficiency. [%]

(e) Heat lost in exhaust. [kJ/kg]

3-7. A compression ignition engine for a small truck is to operate on an air-standard Dual cycle with a compression ratio of $r_c = 18$. Due to structural limitations, maximum allowable pressure in the cycle will be 9000 kPa. Light diesel fuel is used at a fuel-air ratio of FA = 0.054. Combustion efficiency can be considered 100%. Cylinder conditions at the start of compression are 50°C and 98 kPa.

Calculate: (a) Maximum indicated thermal efficiency possible with these conditions. [%]

(b) Peak cycle temperature under conditions of part (a). [°C]

(c) Minimum indicated thermal efficiency possible with these conditions. [%]

(d) Peak cycle temperature under conditions of part (c). [°C]

3-8. An in-line six, 3.3-liter CI engine using light diesel fuel at an air-fuel ratio of AF = 20 operates on an air-standard Dual cycle. Half the fuel can be considered burned at constant volume, and half at constant pressure with combustion efficiency $\eta_c = 100\%$. Cylinder conditions at the start of compression are 60°C and 101 kPa. Compression ratio $r_c = 14{:}1$.

Calculate: (a) Temperature at each state of the cycle. [K]

(b) Pressure at each state of the cycle. [kPa]

(c) Cutoff ratio.

(d) Pressure ratio.

(e) Indicated thermal efficiency. [%]

(f) Heat added during combustion. [kJ/kg]

(g) Net indicated work. [kJ/kg]

3-9. The engine in Problem 3-8 produces 57 kW of brake power at 2000 RPM.

Calculate: (a) Torque. [N-m]

(b) Mechanical efficiency. [%]

(c) Brake mean effective pressure. [kPa]

(d) Indicated specific fuel consumption. [gmlkW-hr]

3-10. An Otto cycle SI engine with a compression ratio of $r_c = 9$ has peak cycle temperature and pressure of 2800 K and 9000 kPa. Cylinder pressure when the exhaust valve opens is 460 kPa, and exhaust manifold pressure is 100 kPa.
Calculate: (a) Exhaust temperature during exhaust stroke. [0C]
(b) Exhaust residual after each cycle. [%]
(c) Velocity out of the exhaust valve when the valve first opens. [mlsec]
(d) Theoretical momentary maximum temperature in the exhaust. [0C]

3-11. An SI engine operates on an air-standard four-stroke Otto cycle with turbocharging. Air-fuel enters the cylinders at 70°C and 140 kPa, and heat in by combustion equals q_{in} = 1800 kJ/kg. Compression ratio r_c = 8 and exhaust pressure F_{ex} = 100 kPa.
Calculate: (a) Temperature at each state of the cycle. [0C]
(b) Pressure at each state of the cycle. [kPa]
(c) Work produced during expansion stroke. [kJ/kg]
(d) Work of compression stroke. [kJ/kg]
(e) Net pumping work. [kJ/kg]
(f) Indicated thermal efficiency. [%]
(g) Compare with Problems 3-12 and 3-13.

3-12. An SI engine operates on an air-standard four-stroke Miller cycle with turbocharging. The intake valves close late, resulting in cycle 6-7-8-7-2-3-4-5-6 in Fig. 3-15. Air-fuel enters the cylinders at 70°C and 140 kPa, and heat in by combustion equals q_{in} = 1800 kJ/kg. Compression ratio r_c = 8, expansion ratio r_e = 10, and exhaust pressure F_{ex} = 100 kPa.
Calculate: (a) Temperature at each state of the cycle. [0C]
(b) Pressure at each state of the cycle. [kPa]
(c) Work produced during expansion stroke. [kJ/kg]
(d) Work of compression stroke. [kJ/kg]
(e) Net pumping work. [kJ/kg]
(f) Indicated thermal efficiency. [%]
(g) Compare with Problems 3-11 and 3-13.

3-13. An SI engine operates on an air-standard four-stroke Miller cycle with turbocharging. The intake valves close early, resulting in cycle 6-7-1-7-2-3-4-5-6 in Fig. 3-15. Air-fuel enters the cylinders at 70°C and 140 kPa, and heat in by combustion equals q_{in} = 1800

kJ/kg. Compression ratio r_c = 8, expansion ratio r_e = 10, and exhaust pressure P_{ex} = 100 kPa.
Calculate: **(a)** Temperature at each state of the cycle. [0C]
(b) Pressure at each state of the cycle. [kPa]
(c) Work produced during expansion stroke. [kJ/kg]
(d) Work of compression stroke. [kJ/kg]
(e) Net pumping work. [kJ/kg]
(l) Indicated thermal efficiency. [%]
(g) Compare with Problems 3-11 and 3-12.

3-14. A six cylinder, two-stroke cycle CI ship engine with bore B = 35 cm and stroke S = 105 cm produces 3600 kW of brake power at 210 RPM.
Calculate: **(a)** Torque at this speed. [kN-m]
(b) Total displacement. [L]
(c) Brake mean effective pressure. [kPa]

(d) Average piston speed. [m/sec]

3-15. A single-cylinder, two-stroke cycle model airplane engine with a 7.54-cm^3 displacement produces 1.42 kW of brake power at 23,000 RPM using glow plug ignition. The square engine (bore = stroke) uses 31.7 gm/min of castor oil-methanol-nitromethane fuel at an air-fuel ratio AF = 4.5. During intake scavenging, 65% of the incoming air-fuel mixture gets trapped in the cylinder, while 35% of it is lost with the exhaust before the exhaust port closes. Combustion efficiency $\eta_c = 0.94$.

Calculate: **(a)** Brake specific fuel consumption. [gm/kW-hr]
(b) Average piston speed. [m/sec]
(c) Unburned fuel exhausted to atmosphere. [gm/min]
(d) Torque. [N-m]

3-16. A historic single-cylinder engine with a mechanical efficiency $\eta_m = 5\%$ operates at 140 RPM on the Lenoir cycle shown in Fig. 3-20. The cylinder has a double acting piston with a 12-in. bore and a 36-in. stroke. The fuel has a heating value $Q_{LHV} = 12{,}000$ BTU/lbm and is used at an air-fuel ratio AF = 18. Combustion occurs at constant volume half way through the intake-power stroke when cylinder conditions equal 70°F and 14.7 psia.

Calculate: **(a)** Temperature at each state of cycle. [°F]
(b) Pressure at each state of cycle. [psia]
(c) Indicated thermal efficiency. [%]
(d) Brake power. [hp]
(e) Average piston speed. [ft/sec]

3-17. Cylinder conditions at the start of compression of a four-stroke cycle SI engine are 27°C and 100 kPa. The engine has a compression ratio of $r_c = 8:1$, and heat addition from combustion is $q_{in} = 2000$ kJ/kg.

Calculate: **(a)** Temperature and pressure at each state of the cycle, using air-standard Otto cycle analysis with constant specific heats. [°C, kPa]
(b) Indicated thermal efficiency in part (a). [%]
(c) Temperature and pressure at each state of the cycle, using any standard air tables which are based on variable specific heats as functions of temperature (e.g., reference [73]). [°C, kPa]
(d) Indicated thermal efficiency in part (c). [%]

DESIGN PROBLEMS

3-1D. Design an SI engine to operate on a six-stroke cycle. The first four strokes of the cycle are the same as a four-stroke Otto cycle. This is followed with an additional air-only intake stroke and an air-only exhaust stroke. Draw simple schematics, and explain the speed and operation of the camshafts when the valves open and close. Also, explain the control of the ignition process.

3-2D. Design a mechanical linkage system for a four-stroke cycle, reciprocating SI engine to operate on the Atkinson cycle (i.e., normal compression stroke and a power stroke which expands until cylinder pressure equals ambient pressure). Explain using simple schematic drawings.

3-3D. An SI engine operating on an four-stroke air-standard cycle using stoichiometric gasoline is to have a maximum cylinder pressure of 11,000 kPa at WOT. Inlet pressure can be 100 kPa without supercharging, or it can be as high as 150 kPa with a supercharger. Pick a compression ratio and inlet pressure combination to give maximum indicated thermal efficiency. Pick a compression ratio and inlet pressure to give maximum imep.

4

Thermochemistry and Fuels

This chapter reviews basic thermochemistry principles as applied to IC engines. It studies ignition characteristics and combustion in engines, the *octane number* of SI fuels, and the *cetane number* of CI fuels. Gasoline and other possible *alternate fuels* are examined.

4-1 THERMOCHEMISTRY

Combustion Reactions

Most IC engines obtain their energy from the combustion of a hydrocarbon fuel with air, which converts chemical energy of the fuel to internal energy in the gases within the engine. There are many thousands of different hydrocarbon fuel components, which consist mainly of hydrogen and carbon but may also contain oxygen (alcohols), nitrogen, and/or sulfur, etc. The maximum amount of chemical energy that can be released (heat) from the fuel is when it reacts (combusts) with a *stoichiometric* amount of oxygen. Stoichiometric oxygen (sometimes called theoretical oxygen) is just enough to convert all carbon in the fuel to CO_2 and all hydrogen to

H_2O, with no oxygen left over. The balanced chemical equation of the simplest hydrocarbon fuel, methane CH_4, burning with stoichiometric oxygen is:

$$CH_4 + 2\,O_2 \rightarrow CO_2 + 2\,H_2O$$

It takes two moles of oxygen to react with one mole of fuel, and this gives one mole of carbon dioxide and two moles of water vapor. If isooctane is the fuel component, the balanced stoichiometric combustion with oxygen would be:

$$C_8H_{18} + 12.5\,O_2 \rightarrow 8\,CO_2 + 9\,H_2O$$

Molecules react with molecules, so in balancing a chemical equation, molar quantities (fixed number of molecules) are used and not mass quantities. One kgmole of a substance has a mass in kilograms equal in number to the molecular weight (molar mass) of that substance. In English units the lbmmole is used.

$$m = NM \qquad (4\text{-}1)$$

where: m = mass
N = number of moles
M = molecular weight

In SI units:

$$1 \text{ kgmole of } CH_4 = 16.04 \text{ kg}$$

$$1 \text{ kgmole of } O_2 = 32.00 \text{ kg}$$

$$1 \text{ kgmole} = 6.02 \times 10^{26} \text{ molecules}$$

In English units:

$$1 \text{ lbmmole of } CH_4 = 16.04 \text{ lbm}$$

$$1 \text{ lbmmole of } O_2 = 32.00 \text{ lbm}$$

$$1 \text{ lbmmole} = 2.73 \times 10^{26} \text{ molecules}$$

The components on the left side of a chemical reaction equation which are present before the reaction are called *reactants,* while the components on the right side of the equation which are present after the reaction are called *products* or exhaust.

Very small powerful engines could be built if fuel were burned with pure oxygen. However, the cost of using pure oxygen would be prohibitive, and thus is not done. Air is used as the source of oxygen to react with fuel. Atmospheric air is made up of about:

78% nitrogen by mole
21 % oxygen
1% argon
traces of CO_2, Ne, CH_4, He, H_2O, etc.

Nitrogen and argon are essentially chemically neutral and do not react in the combustion process. Their presence, however, does affect the temperature and pressure in the combustion chamber. To simplify calculations without causing any large error, the neutral argon in air is assumed to be combined with the neutral nitrogen, and atmospheric air then can be modeled as being made up of 21% oxygen and 79% nitrogen. For every 0.21 moles of oxygen there is also 0.79 moles of nitrogen, or for one mole of oxygen there are 0.79/0.21 moles of nitrogen. For every mole of oxygen needed for combustion, 4.76 moles of air must be supplied: one mole of oxygen plus 3.76 moles of nitrogen.

Stoichiometric combustion of methane with *air* is then:

$$CH_4 + 2\,O_2 + 2(3.76)\,N_2 \rightarrow CO_2 + 2\,H_2O + 2(3.76)\,N_2$$

and of isooctane with air is:

$$C_8H_{18} + 12.5\,O_2 + 12.5(3.76)\,N_2 \rightarrow 8\,CO_2 + 9\,H_2O + 12.5(3.76)\,N_2$$

It is convenient to balance combustion reaction equations for one kgmole of fuel. The energy released by the reaction will thus have units of energy per kgmole of fuel, which is easily transformed to total energy when the flow rate of fuel is known. This convention will be followed in this textbook. Molecular weights can be found in Table 4-1 and Table A-2 in the Appendix. The molecular weight of 29 will be used for air. Combustion can occur, within limits, when more than stoichiometric air is present (lean) or when less than stoichiometric air is present (rich) for a given amount of fuel. If methane is burned with 150% stoichiometric air, the excess oxygen ends up in the products:

$$CH_4 + 3\,O_2 + 3(3.76)\,N_2 \rightarrow CO_2 + 2\,H_2O + 3(3.76)\,N_2 + O_2$$

If isooctane is burned with 80% stoichiometric air, there is not enough oxygen to convert all the carbon to CO_2, and carbon monoxide CO is found in the products:

$$C_8H_{18} + 10\,O_2 + 10(3.76)\,N_2 \rightarrow 3\,CO_2 + 9\,H_2O + 5\,CO + 10(3.76)\,N_2$$

TABLE 4-1 MOLECULAR WEIGHTS

SUBSTANCE		MOLECULAR WEIGHT (kg/kgmole) or (lbm/lbmmole)
Air		28.97
Argon	Ar	39.95
Carbon	C	12.01
Carbon Monoxide	CO	28.01
Carbon Dioxide	CO_2	44.01
Hydrogen	H_2	2.02
Water Vapor	H_2O	18.02
Helium	He	4.00
Nitrogen	N_2	28.01

Carbon monoxide is a colorless, odorless, poisonous gas which can be further burned to form CO_2. It is formed in any combustion process when there is a deficiency of oxygen. It is also very likely that some of the fuel will not get burned when there is a deficiency of oxygen. This unburned fuel ends up as pollution in the exhaust of the engine.

Various terminology is used for the amount of air or oxygen used in combustion.

80% stoichiometric air = 80% theoretical air = 80% air
= 20% deficiency of air

133% stoichiometric oxygen = 133% theoretical oxygen
= 133% oxygen = 33% excess oxygen

For actual combustion in an engine, the equivalence ratio is a measure of the fuel-air mixture relative to stoichiometric conditions. It is defined as:

$$\phi = (FA)_{act}/(FA)_{stoich} = (AF)_{stoich}/(AF)_{act} \tag{4-2}$$

where: $FA = m_f/m_a$ = fuel–air ratio
$AF = m_a/m_f$ = air–fuel ratio
m_a = mass of air
m_f = mass of fuel

when:

$\phi < 1$ running lean, oxygen in exhaust

$\phi > 1$ running rich, CO and fuel in exhaust

$\phi = 1$ stoichiometric, maximum energy released from fuel

SI engines normally operate with an equivalence ratio in the range of 0.9 to 1.2, depending on the type of operation.

EXAMPLE PROBLEM 4-1

Isooctane is burned with 120% theoretical air in a small three-cylinder turbocharged automobile engine.

Calculate:

Calculate:

1. air–fuel ratio
2. fuel–air ratio
3. equivalence ratio

Stoichiometric reaction:

$$C_8H_{18} + 12.5\ O_2 + 12.5(3.76)\ N_2 \rightarrow 8\ CO_2 + 9\ H_2O + 12.5(3.76)\ N_2$$

With 20% excess air:

$$C_8H_{18} + 15\ O_2 + 15(3.76)\ N_2 \rightarrow 8\ CO_2 + 9\ H_2O + 15(3.76)\ N_2 + 2.5\ O_2$$

With 20% excess air, all the fuel gets burned, and the same amount of CO_2 and H_2O is found in the products. In addition, there is some oxygen and additional nitrogen in the products (the excess air).

1) Equations (2-55) and (4-1) are used to find the air–fuel ratio:

$$\text{AF} = m_a/m_f = N_a M_a/N_f M_f = [(15)(4.76)(29)]/[(1)(114)]$$
$$= \underline{18.16}$$

2) Equation (2-56) is used to find fuel–air ratio:

$$\text{FA} = m_f/m_a = 1/\text{AF} = 1/18.16 = \underline{0.055}$$

3) Fuel–air ratio of stoichiometric combustion:

$$(\text{FA})_{\text{stoich}} = [(1)(114)]/[(12.5)(4.76)(29)] = 0.066$$

Equivalence ratio is obtained using Eq. (4-2):

$$\phi = (\text{FA})_{\text{act}}/(\text{FA})_{\text{stoich}} = (0.055)/(0.066) = \underline{0.833}$$

Even when the flow of air and fuel into an engine is controlled exactly at stoichiometric conditions, combustion will not be "perfect," and components other than CO_2, H_2O, and N_2 are found in the exhaust products. One major reason for this is the extremely short time available for each engine cycle, which often means that less than complete mixing of the air and fuel is obtained. Some fuel molecules do not find an oxygen molecule to react with, and small quantities of both fuel and oxygen end up in the exhaust. Chapter 7 goes into more detail on this and other reasons that ideal combustion is not obtained. SI engines have a combustion efficiency in the range of 95% to 98% for lean mixtures and lower values for rich mixtures, where there is not enough air to react all the fuel (see Fig. 4-1). CI engines operate lean overall and typically have combustion efficiencies of about 98%.

Chemical Equilibrium

If a general chemical reaction is represented by:

$$\nu_A A + \nu_B B \rightarrow \nu_C C + \nu_D D \tag{4-3}$$

where A and B represent all reactant species, whether one or two or more, and C and D represent all products, regardless of number, then ν_A, ν_B, ν_C, and ν_D represent the stoichiometric coefficients of A, B, C, and D.

Equilibrium composition for this reaction can be found if one knows the chem-

ical equilibrium constant:

$$K_e = [(N_C^{\nu_C} N_D^{\nu_D})/(N_A^{\nu_A} N_B^{\nu_B})]\,(P/N_t)^{\Delta\nu} \qquad (4\text{-}4)$$

where: $\Delta\nu = \nu_C + \nu_D - \nu_A - \nu_B$
N_i = number of moles of component i at equilibrium
N_t = total number of moles at equilibrium
P = total absolute pressure in units of atmospheres

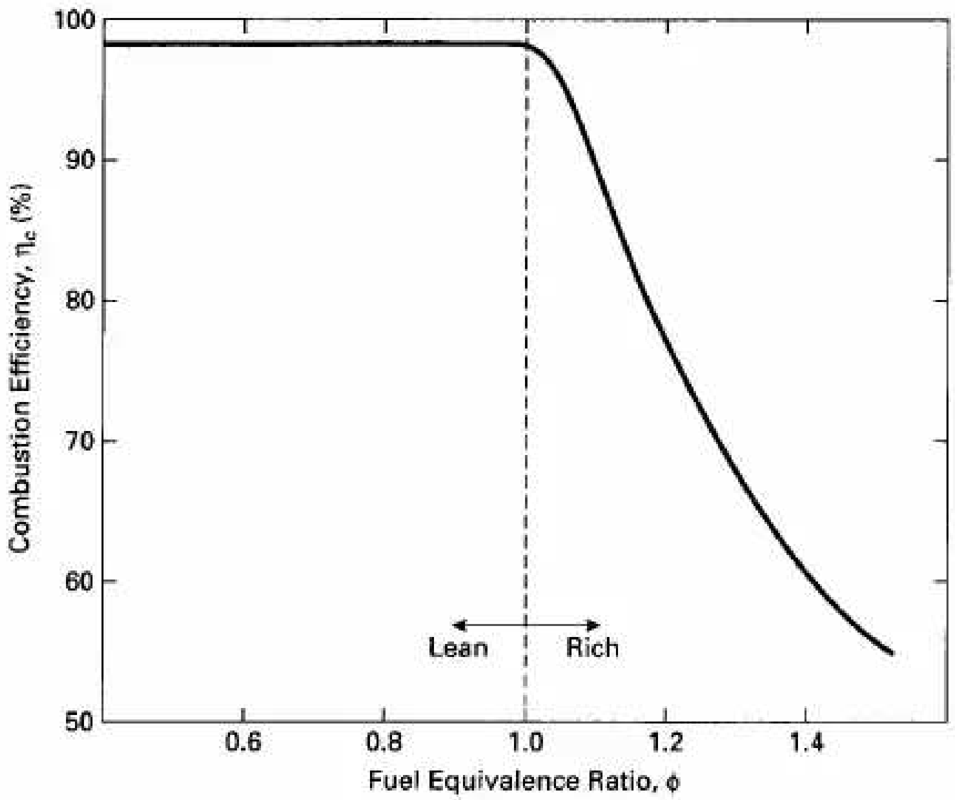


Figure 4-1 Combustion efficiency as a function of fuel equivalence ratio. Efficiency for engines operating lean is generally on the order of 98%. When an engine operates fuel rich, there is not enough oxygen to react with all the fuel, and combustion efficiency decreases. CI engines operate lean and typically have high combustion efficiency. Adapted from [58].

Equilibrium constants for many reactions can be found in thermodynamic textbooks or chemical handbooks, tabulated in logarithmic form (In or 10glO)'An abbreviated table can be found in the Appendix of this book (Table A-3).

Ke is very dependent on temperature, changing many orders of magnitude over the temperature range experienced in an IC engine. As *Ke* gets larger, equilibrium is more towards the right (products). This is the maximizing of entropy. For hydrocarbon fuels reacting with oxygen (air) at high engine temperatures, the equilibrium constant is very large, meaning that there are very few reactants (fuel and air) left at final equilibrium. However, at these high temperatures another chemical phenomenon takes place that affects the overall combustion process in the engine and what ends up in the engine exhaust.

Examination of the equilibrium constants in Table A-3 shows that dissociation of normally stable components will occur at these high engine temperatures. COz dissociates to CO and O, Oz dissociates to monatomic O, Nz dissociates to monatomic N, etc. This not only affects chemical combustion, but is a cause of one of the major emission problems of IC engines. Nitrogen, as diatomic Nz does not react with other substances, but when it dissociates to monatomic nitrogen at high temperature it readily reacts with oxygen to form nitrogen oxides, NO and NO_2, a major pollutant

from automobiles. To avoid generating large amounts of nitrogen oxides, combustion temperatures in automobile engines are lowered, which reduces the dissociation of N_2. Unfortunately, this also lowers the thermal efficiency of the engine.

Exhaust Dew Point Temperature

When exhaust gases of an IC engine are cooled below their dew point temperature, water vapor in the exhaust starts to condense to liquid. It is common to see water droplets come out of an automobile exhaust pipe when the engine is first started and the pipe is cold. Very quickly the pipe is heated above the dew point temperature, and condensing water is then seen only as vapor when the hot exhaust is cooled by the surrounding air, much more noticeable in the cold wintertime.

EXAMPLE PROBLEM 4-2

At what temperature will water start to condense out of exhaust gases of the engine in Example Problem 4-1? Exhaust pressure is one atmosphere.
Calculate this for:

1. dry inlet air
2. inlet air with relative humidity of 55%

1) Reaction equation from Example Problem 4-1:

$$C_8H_{18} + 15\,O_2 + 15(3.76)\,N_2 \rightarrow 8\,CO_2 + 9\,H_2O + 2.5\,O_2 + 15(3.76)\,N_2$$

Mole fraction of water vapor in exhaust products:

$$x_v = N_v/N_{\text{total}} = (9)/[8 + 9 + 2.5 + 15(3.76)] = 0.1186$$

Partial pressure of water vapor:

$$P_v = x_v P_{\text{total}} = (0.1186)(101\text{ kPa}) = 11.98\text{ kPa}$$

The dew point is the temperature at which this water vapor pressure becomes saturated. From steam tables [90]:

$$\underline{T_{DP} = 49°C}$$

2) If the relative humidity (rh) of the inlet air is 55% at $T = 25°C$, then the vapor pressure of water at the inlet will be

$$P_v = (rh)P_{\text{Sat at 25C}} = (0.55)(3.169\text{ kPa}) = 1.743\text{ kPa}$$

Using psychrometric equations and steam tables from any thermodynamics textbook (e.g. [90]), the specific humidity is:

$$\omega_v = m_v/m_a = 0.622[P_v/(P - P_v)] = (0.622)[(1.743)/(101 - 1.743)] = 0.0109\text{ kg}_v/\text{kg}_a$$

Changing this mass ratio to a molar ratio using the molecular weights of air (29) and water vapor (18) gives the number of moles of water carried in with the air for one mole of fuel:

$$N_v = N_{\text{air}}\,\omega_v(M_{\text{air}}/M_v) = [(15)(4.76)](0.0109)[(29)/(18)] = 1.25$$

The reaction equation then becomes:

$$C_8H_{18} + 15\,O_2 + 15(3.76)\,N_2 + 1.25\,H_2O \rightarrow$$
$$8\,CO_2 + 10.25\,H_2O + 2.5\,O_2 + 15(3.76)\,N_2$$

The mole fraction of water vapor in the exhaust is

$$x_v = (10.25)/[8 + 10.25 + 2.5 + 15(3.76)] = 0.1329$$

The partial pressure of water vapor is

$$P_v = x_v P_{total} = (0.1329)(101\ \text{kPa}) = 13.42\ \text{kPa}$$

Dew point temperature:

$$\underline{T_{DP} = 52°C}$$

Combustion Temperature

Heat liberated by the combustion reaction of a hydrocarbon fuel with air is the difference between the total enthalpy of the products and the total enthalpy of the reactants. This is called *heat of reaction, heat of combustion*, or *enthalpy of reaction* and is given by:

$$Q = \sum_{PROD} N_i h_i - \sum_{REACT} N_i h_i \qquad (4\text{-}5)$$

where: N_i = number of moles of component i

$h_i = (h_f^\circ)_i + \Delta h_i$

h_f° = enthalpy of formation, the enthalpy needed to form one mole of that component at standard conditions of 25°C and 1 atm

Δh_i = change of enthalpy from standard temperature for component i

Q will be negative, meaning that heat is given up by the reacting gases. Values of h_f° and Δh are molar-specific quantities and can be found in most thermodynamic textbooks.

Table A-2 gives *heating values* for a number of fuels. Heating value Q_{HV} is the negative of the heat of reaction for one unit of fuel, and thus is a positive number. It is calculated assuming both the reactants and the products are at 25°C. Care must be used when using heating values, which almost always are given in mass units (kJ/kg), whereas heats of reaction are obtained using molar quantities as in Eq. (4-5). Two values of heating value are given in the table; *higher heating value* is used when water in the exhaust products is in the liquid state, and *lower heating value* is used when water in the products is vapor. The difference is the heat of vaporization of the water:

$$Q_{HHV} = Q_{LHV} + \Delta h_{vap} \qquad (4\text{-}6)$$

Higher heating value is usually listed on fuel containers, the higher number making that fuel seem more attractive. For engine analysis, lower heating value is the logical value to use. All energy exchange in the combustion chamber occurs at high temperature, and only somewhere down the exhaust process, where it can no longer affect engine operation, does the product gas get cooled to the dew point temperature. Heat into the engine that gets converted to output work can be given as:

$$Q_{in} = \eta_c m_f Q_{LHV} \tag{4-7}$$

where: η_c = combustion efficiency
m_f = mass of fuel

An estimation of the maximum temperature reached in an IC engine can be obtained by calculating the *adiabatic flame temperature* of the input air–fuel mixture. This is done by using Eq. (4-5) and setting $Q = 0$, yielding

$$\sum_{PROD} N_i h_i = \sum_{REACT} N_i h_i \tag{4-8}$$

Assuming that inlet conditions of the reactants are known, it is necessary to find the temperature of the products such that this equation will be satisfied. This is the adiabatic flame temperature.

Adiabatic flame temperature is the ideal theoretical maximum temperature that can be obtained for a given fuel and air mixture. The actual peak temperature in an engine cycle will be several hundred degrees less than this. There is some heat loss even in the very short time of one cycle, combustion efficiency is less than 100% so a small amount of fuel does not get burned, and some components dissociate at the high engine temperatures. All these factors contribute to making the actual peak engine temperature somewhat less than adiabatic flame temperature.

EXAMPLE PROBLEM 4-3

Find the adiabatic flame temperature of isooctane burned with an equivalence ratio of 0.833 in dry air as in Example Problem 4-1. It can be assumed that the reactants are at a temperature of 427°C (700 K) after the compression stroke.

From Example Prob. 4-1:

$$C_8H_{18} + 15\,O_2 + 15(3.76)\,N_2 \rightarrow 8\,CO_2 + 9\,H_2O + 2.5\,O_2 + 15(3.76)\,N_2$$

Equations (4-5) and (4-8) are used for adiabatic combustion. Enthalpy values can be obtained from most thermodynamic textbooks. The values used here are from [90]:

$$\sum_{PROD} N_i(h_f^o + \Delta h)_i = \sum_{REACT} N_i(h_f^o + \Delta h)_i$$

$$8[(-393{,}522) + \Delta h_{CO_2}] + 9[(-241{,}826) + \Delta h_{H_2O}] + 2.5[0 + \Delta h_{O_2}] + 15(3.76)[0 + \Delta h_{N_2}]$$
$$= [(-259{,}280) + (73473)] + 15[0 + (12{,}499)] + 15(3.76)[0 + (11{,}937)]$$

Simplify:

$$8\Delta h_{CO_2} + 9\Delta h_{H_2O} + 2.5\Delta h_{O_2} + 56.4\Delta h_{N_2} = 5{,}999{,}535$$

By trial and error, find the temperature that satisfies this equation. Try $T = 2400$ K:

$$8(115{,}779) + 9(93{,}741) + 2.5(74{,}453) + 56.4(70{,}640) = 5{,}940{,}130$$

This is too low, so try $T = 2600$ K:

$$8(128{,}074) + 9(104{,}520) + 2.5(82{,}225) + 56.4(77{,}963) = 6{,}567{,}948$$

This is too high, so the adiabatic flame temperature is found by interpolation:

$$\underline{T_{max} = 2419 \text{ K} = 2146°\text{C}}$$

Engine Exhaust Analysis

It is common practice to analyze the exhaust of an IC engine. The control system of a modern *smart* automobile engine includes sensors that continuously monitor the exhaust leaving the engine. These sensors determine the chemical composition of the hot exhaust by various chemical, electronic, and thermal methods. This information, along with information from other sensors, is used by the engine management system (EMS) to regulate the operation of the engine by controlling the air-fuel ratio, ignition timing, inlet tuning, valve timing, etc.

Repair shops and highway check stations also routinely analyze automobile exhaust to determine operating conditions and/or emissions. This is done by taking a sample of the exhaust gases and running it through an external analyzer. When this is done, there is a high probability that the exhaust gas will cool below its dew point temperature before it is fully analyzed, and the condensing water will change the composition of the exhaust. To compensate for this, a *dry analysis* can be performed by first removing all water vapor from the exhaust, usually by some thermo-chemical means.

EXAMPLE PROBLEM 4-4

The four-cylinder engine of a light truck owned by a utility company has been converted to run on propane fuel. A dry analysis of the engine exhaust gives the following volumetric percentages:

CO_2	4.90%
CO	9.79%
O_2	2.45%

Calculate the equivalence ratio at which the engine is operating. The three components identified sum up to 4.90 + 9.79 + 2.45 = 17.14% of the total, which means that the remaining gas (nitrogen) accounts for 82.86% of the total. Volume percent equals molar percent, so if an unknown amount of fuel is burned with an unknown amount of air, the resulting reaction is:

$$x\,C_3H_8 + y\,O_2 + y(3.76)\,N_2 \rightarrow 4.90\,CO_2 + 9.79\,CO + 2.45\,O_2 + 82.86\,N_2 + z\,H_2O$$

where: z = number of moles of water vapor removed before dry analysis

Conservation of nitrogen during reaction gives:

$$y(3.76) = 82.86 \qquad \text{or} \quad y = 22.037$$

Conservation of carbon:

$$3x = 4.90 + 9.79 \qquad \text{or} \quad x = 4.897$$

Conservation of hydrogen:

$$8x = 8(4.897) = 2z \qquad \text{or} \quad z = 19.588$$

The reaction is:

$$4.90\ C_3H_8 + 22.037\ O_2 + 22.037(3.76)\ N_2 \rightarrow$$
$$4.90\ CO_2 + 9.79\ CO + 2.45\ O_2 + 82.86\ N_2 + 19.588\ H_2O$$

Dividing by 4.90:

$$C_3H_8 + 4.50\ O_2 + 4.50(3.76)N_2 \rightarrow CO_2 + 2\ CO + 0.50\ O_2 + 16.92\ N_2 + 4\ H_2O$$

Actual air–fuel ratio:

$$AF_{act} = m_a/m_f = [(4.50)(4.76)(29)]/[(1)(44)] = 14.12$$

Stoichiometric combustion:

$$C_3H_8 + 5\ O_2 + 5(3.76)\ N_2 \rightarrow 3\ CO_2 + 4\ H_2O + 5(3.76)\ N_2$$

Stoichiometric air–fuel ratio:

$$AF_{stoich} = m_a/m_f = [(5)(4.76)(29)]/[(1)(44)] = 15.69$$

Equivalence ratio using Eq. (4-2):

$$\phi = (AF)_{stoich}/(AF)_{act} = 15.69/14.12 = \underline{1.11}$$

4-2 HYDROCARBON FUELS-GASOLINE

The main fuel for SI engines is gasoline, which is a mixture of many hydrocarbon components and is manufactured from crude petroleum. Crude oil was first discovered in Pennsylvania in 1859, and the fuel product line generated from it developed along with the development of the IC engine. Crude oil is made up almost entirely of carbon and hydrogen with some traces of other species. It varies from 83% to 87% carbon and 11% to 14% hydrogen by weight. The carbon and hydrogen can combine in many ways and form many different molecular compounds. One test of a crude oil sample identified over 25,000 different hydrocarbon components [93].

The crude oil mixture which is taken from the ground is separated into component products by *cracking* and/or distillation using thermal or catalytic methods at an oil refinery. Cracking is the process of breaking large molecular components into more useful components of smaller molecular weight. Preferential distillation is used to separate the mixtures into single components or smaller ranges of components. Generally, the larger the molecular weight of a component, the higher is its boiling temperature. Low boiling temperature components (smaller molecular weights) are used for solvents and fuels (gasoline), while high boiling temperature components with their large molecular weights are used for tar and asphalt or returned to the refining process for further cracking. The component mixture of the refining process is used for many products, including:

automobile gasoline
diesel fuel
aircraft gasoline
jet fuel
home heating fuel
industrial heating fuel
natural gas
lubrication oil
asphalt
alcohol
rubber
paint
plastics
explosives

The availability and cost of gasoline fuel, then, is a result of a market competition with many other products. This becomes more critical with the depletion of the earth's crude oil reserves, which looms on the horizon.

Crude oil obtained from different parts of the world contain different amounts and combinations of hydrocarbon species. In the United States, two general classifications are identified: Pennsylvania crude and western crude. Pennsylvania crude has a high concentration of paraffins with little or no asphalt, while western crude has an asphalt base with little paraffin. The crude oil from some petroleum fields in the Mideast is made up of component mixtures that could be used immediately for IC engine fuel with little or no refining.

Figure 4-2 shows a temperature-vaporization curve for a typical gasoline mixture. The various components of different molecular weights will vaporize at different temperatures, small molecular weights boiling at low temperature and larger molecular weights at higher temperature. This makes a very desirable fuel. A small percentage of components that vaporize (boil) at low temperature is needed to assure the starting of a cold engine; fuel must vaporize before it can burn. However, too much of this *front-end volatility* can cause problems when the fuel vaporizes too quickly. Volumetric efficiency of the engine will be reduced if fuel vapor replaces air too early in the intake system. Another serious problem this can cause is *vapor lock,* which occurs when fuel vaporizes in the fuel supply lines or in the carburetor in the hot engine compartment. When this happens, the supply of fuel is cut off and the

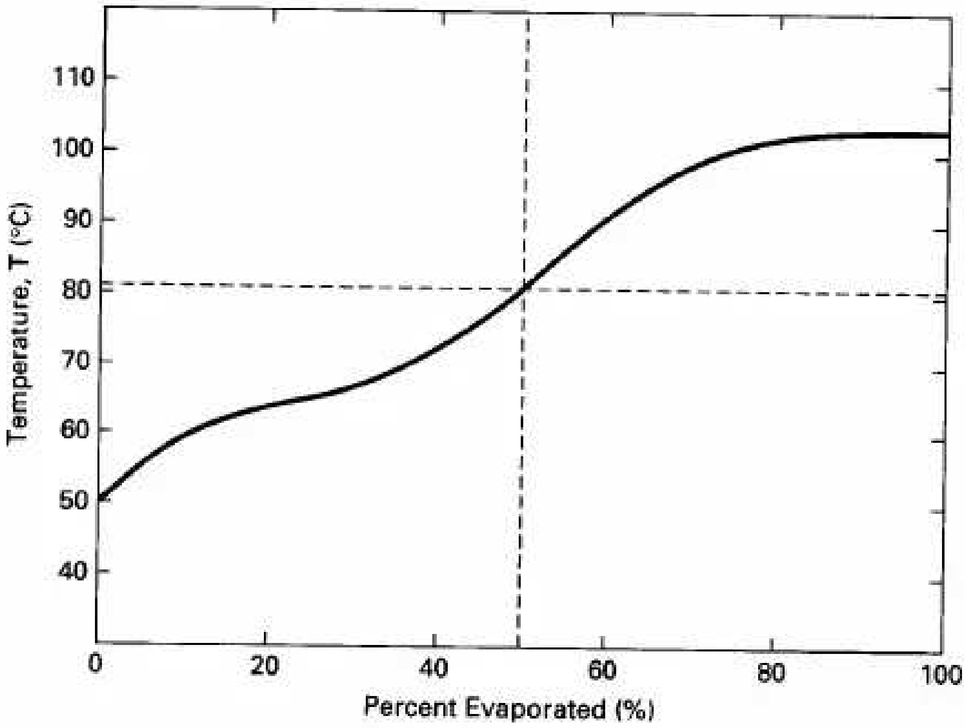


Figure 4-2 Temperature–vaporization curve for a typical gasoline mixture. The components which vaporize at low temperatures are called *front-end volatility* components and are useful for starting a cold engine. Those components which vaporize at the highest temperatures are called *high-end volatility* components and increase engine volumetric efficiency. This gasoline could be classified as 57-81-103°C. Fifty percent of the gasoline would be vaporized at 81°C.

engine stops. A large percent of fuel should be vaporized at the normal intake system temperature during the short time of the intake process. To maximize volumetric efficiency, some of the fuel should not vaporize until late into the compression stroke and even into the start of combustion. This is why some high-molecular-weight components are included in gasoline mixtures. If too much of this *high-end volatility* is included in the gasoline, however, some of the fuel never gets vaporized and ends up as exhaust pollution or condenses on the cylinder walls and dilutes the lubricating oil.

One way that is sometimes used to describe a gasoline is to use three temperatures: the temperature at which 10% is vaporized, at which 50% is vaporized, and at which 90% is vaporized. The gasoline in Fig. 4-2 could therefore be classified as 57-81-103°C.

If different commercial brands of gasoline are compared, there is found to be little difference in the volatility curves for a given season and location in the country. There is usually about a 5°C shift down in temperature on the vaporization curve for winter gasoline compared with summer.

If gasoline is approximated as a single-component hydrocarbon fuel, it would have a molecular structure of about C_8H_{15} and a corresponding molecular weight of

111. These are the values that will be used in this textbook. Sometimes, gasoline is approximated by the real hydrocarbon component isooctane C_8H_{18}, which best matches its component structure and thermodynamic properties. Table A-2 lists

properties of gasoline, isooctane, and some other common fuels.

4-3 SOME COMMON HYDROCARBON COMPONENTS

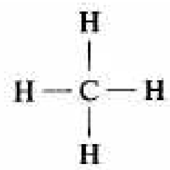
Carbon atoms form four bonds in molecular structures, while hydrogen has one bond. A *saturated* hydrocarbon molecule will have no double or triple carbon-to-carbon bonds and will have a maximum number of hydrogen atoms. An *unsaturated* molecule will have double or triple carbon-to-carbon bonds.

A number of different *families* of hydrocarbon molecules have been identified; a few of the more common ones are described.

Paraffins

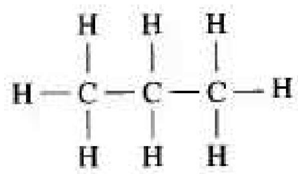
The paraffin family (sometimes called alkanes) are chain molecules with a carbon-hydrogen combination of C_nH_{2n+2}, n being any number. The simplest member of this family, and the simplest of all stable hydrocarbon molecules, is methane (CH_4), which is the main component of natural gas. It can be pictured as:

methane (CH_4)



Other species of this family include:

propane (C_3H_8)



butane (C_4H_{10})

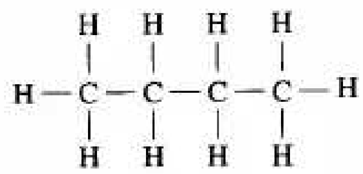


Table 4-2 gives the prefixes used to identify paraffins and other hydrocarbon families according to the number of carbon atoms in the main molecular structure. The paraffin family, then, uses the suffix ane—thus methane, propane, etc.

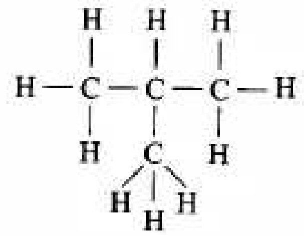
TABLE 4-2 PREFIXES FOR HYDROCARBON FUEL COMPONENTS

Number of Carbon Atoms in Main Chain or Ring	Prefix
1	meth
2	eth

3	prop
4	but
5	pent
6	hex
7	hept
8	oct
9	non
10	dec
11	undec
12	dodec

Sometimes the chains in the molecule are branched, and other molecular structures are obtained with the same number of carbon and hydrogen atoms. One such *isomer* is isobutane, which has the same chemical formula as butane (C_4H_{10}) but has a different structure:

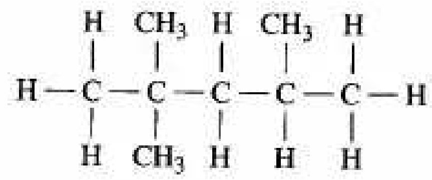
isobutane (C_4H_{10})



Isobutane can also be called methylpropane-propane because it has three carbon atoms in the main chain and one methyl *radical,* CH_3, replacing one of the hydrogen atoms. Molecules with no branches in their chain are sometimes called normal; thus butane is sometimes called normal butane or n-butane. Even though isobutane and n-butane have the same chemical formula, C_4HIO, and almost identical molecular weights, they have different thermal and physical properties. This is true for any two chemical species that have different molecular structures, even if they have the same chemical formula.

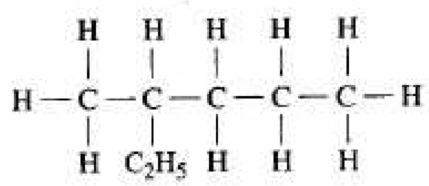
There are many ways chemical chains can be branched, giving a very large number of possible chemical species. Isooctane (C_SH_{1S}) has the following molecular structure:

isooctane (C_8H_{18})

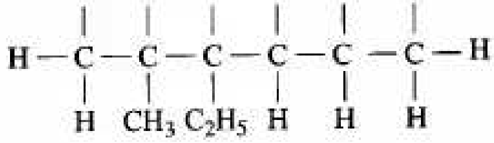


It can also be called 2,2,4-trimethylpentane, pentane because it has five carbon atoms in the main chain, trimethyl because it has three methyl radicals (CH_3), and 2,2,4 because the three radicals are on the second, second, and fourth carbon atoms in the chain. Note that 2,4,4 trimethylpentane would be the same molecule. Other examples of isomers are:

2-ethylpentane (C_7H_{16})



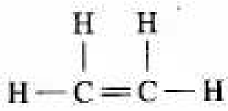
2-methyl-3-ethylhexane (C_9H_{20})



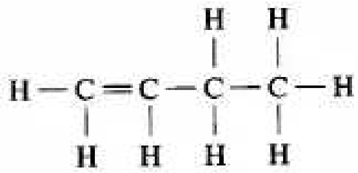
Olefins

The olefin family consists of chain molecules that contain one double carbon–carbon bond, and are therefore unsaturated. The prefixes from Table 4-2 are used with the suffix "ene." The chemical makeup is C_nH_{2n}. Examples of olefins are:

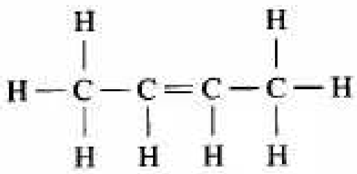
ethene (C_2H_4)



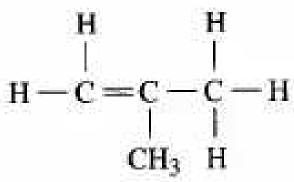
butene-1 (C_4H_8)



butene-2 (C_4H_8)



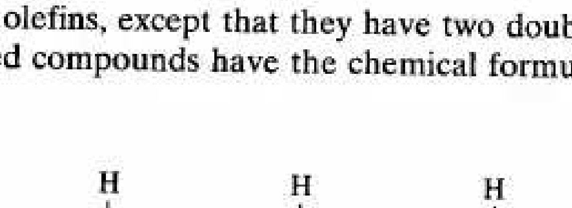
isobutene or 2-methylpropene (C_4H_8)



Diolefins

Diolefins are chain molecules similar to olefins, except that they have two double carbon–carbon bonds. These unsaturated compounds have the chemical formula C_nH_{2n-2} and use the suffix "diene."

2,5-heptadiene (C_7H_{12})



Acetylene

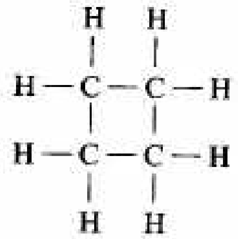
The acetylene family has unsaturated chain molecules with a triple carbon–carbon bond and the chemical formula C_nH_{2n-2}. The best known member of the family is acetylene (C_2H_2).

acetylene (C_2H_2) $$H-C\equiv C-H$$

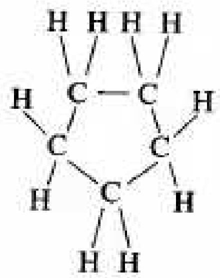
Cycloparaffins

Cycloparaffins have unsaturated molecules with a single-bond ring and a chemical formula of C_nH_{2n}.

cyclobutane (C_4H_8)



cyclopentane (C_5H_{10})

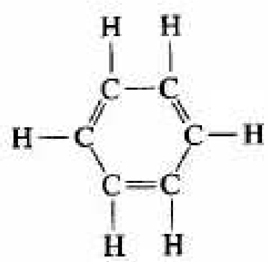


Many variations of these molecules are possible, with one or more of the attached hydrogen atoms replaced with various side radicals and/or chains. Cycloparaffins make good automobile gasoline components.

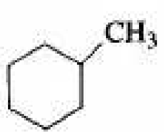
Aromatics

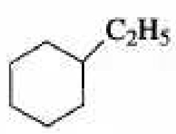
Aromatic molecules have an unsaturated ring structure with double carbon–carbon bonds and a general chemical formula of C_nH_{2n-6}. The basic molecule in this family is the benzene ring:

benzene (C_6H_6)



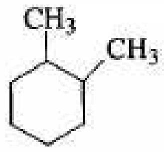
This is modified by replacing the hydrogen atoms with various groups:



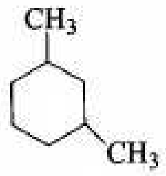


toluene (C_7H_8) ethylbenzene (C_8H_{10})

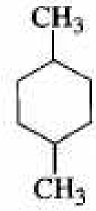
When more than one hydrogen atom is replaced, many isomers are possible:



orthoxylene (C_8H_{10})

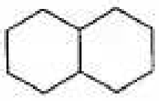


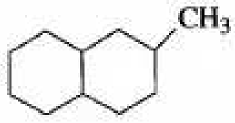
metaxylene (C_8H_{10})

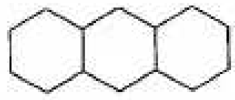


paraxylene (C_8H_{10})

When more than one ring combine in a single large molecule, many additional species are possible:

$C_{10}H_8$ 

$C_{11}H_{10}$ 

$C_{14}H_{10}$ 

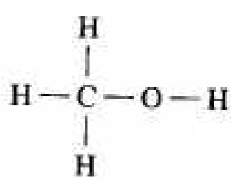
Aromatics generally make good gasoline fuel components, with some exceptions due to exhaust pollution. They have high densities in the liquid state and thus have high energy content per unit volume. Aromatics have high solvency characteristics, and care must be used in material selection for the fuel delivery system (e.g., they will dissolve or swell some gasket materials). Aromatics will dissolve a greater

amount of water than some other hydrocarbons and thus can create fuel line freezing problems when the temperature is lowered and some of the water comes out of solution. Aromatics make poor CI engine fuel.

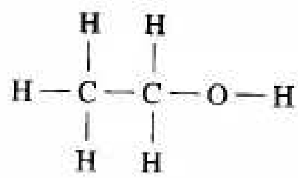
Alcohol

Alcohols are similar to paraffins with one of the hydrogen atoms replaced with the hydroxyl radical OH. The most common alcohols are:

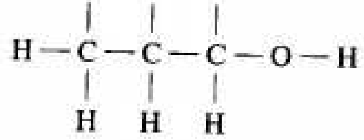
methyl alcohol (methanol), CH_3OH



ethyl alcohol (ethanol), C_2H_5OH



propyl alcohol (propanol), C_3H_7OH



4-4 SELF-IGNITION AND OCTANE NUMBER

Self-Ignition Characteristics of Fuels

If the temperature of an air-fuel mixture is raised high enough, the mixture will self-ignite without the need of a spark plug or other external igniter. The temperature above which this occurs is called the **self-ignition temperature** (SIT). This is the basic principle of ignition in a compression ignition engine. The compression ratio is high enough so that the temperature rises above SIT during the compression stroke. Self-ignition then occurs when fuel is injected into the combustion chamber. On the other hand, self-ignition (or pre-ignition, or auto-ignition) is not desirable in an SI engine, where a spark plug is used to ignite the air-fuel at the proper time in the cycle. The compression ratios of gasoline-fueled SI engines are limited to about 11:1 to avoid self-ignition. When self-ignition does occur in an SI engine higher than desirable, pressure pulses are generated. These high pressure pulses can cause dam-

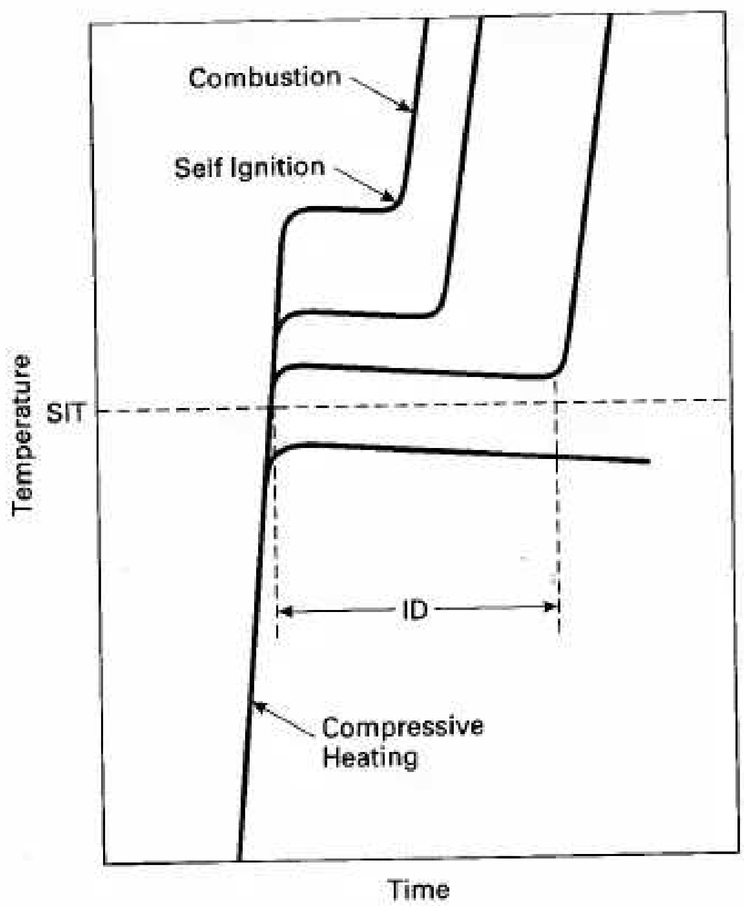


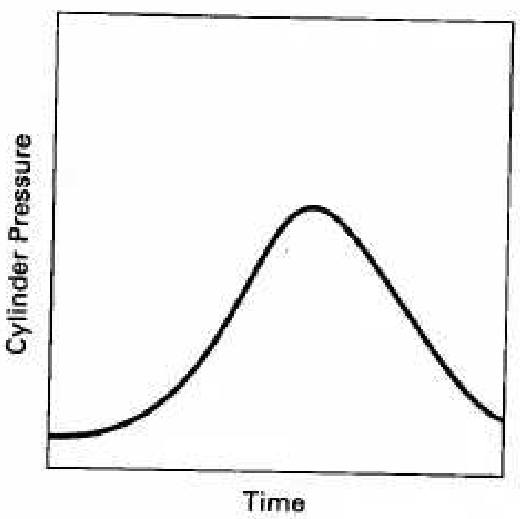
Figure 4-3 Self-ignition characteristics of fuels. If the temperature of a fuel is raised above the self-ignition temperature (SIT), the fuel will spontaneously ignite after a short ignition delay (ID) time. The higher above SIT which the fuel is heated, the shorter will be ID. Ignition delay is generally on the order of thousandths of a second. Adapted from [126].

age to the engine and quite often are in the audible frequency range. This phenomenon is often called knock or ping.

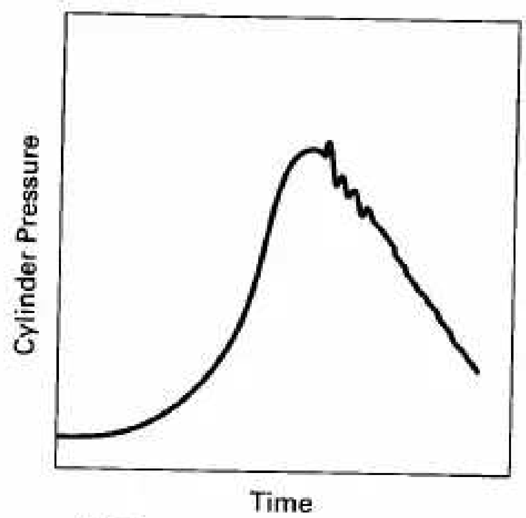
Figure 4-3 shows the basic process of what happens when self-ignition occurs. If a combustible air-fuel mixture is heated to a temperature less than SIT, no ignition will occur and the mixture will cool off. If the mixture is heated to a temperature above SIT, self-ignition will occur after a short time delay called ignition delay (ID). The higher the initial temperature rise above SIT, the shorter will be ID. The values for SIT and ID for a given air-fuel mixture are ambiguous, depending on many variables which include temperature, pressure, density, turbulence, swirl, fuel-air ratio, presence of inert gases, etc. [93].

Ignition delay is generally a very small fraction of a second. During this time, preignition reactions occur, including oxidation of some fuel components and even cracking of some large hydrocarbon components into smaller HC molecules. These preignition reactions raise the temperature at local spots, which then promotes additional reactions until, finally, the actual combustion reaction occurs.

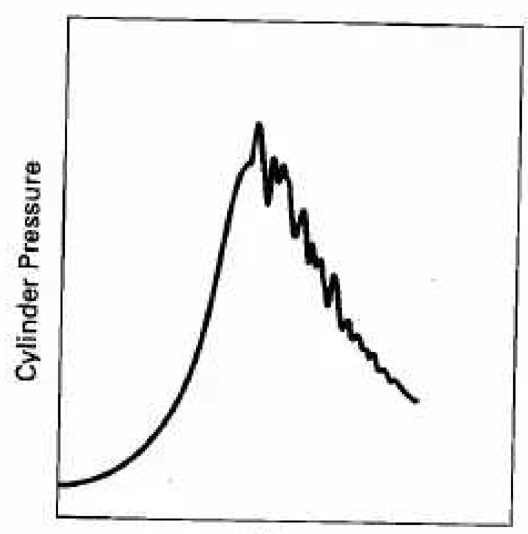
Figure 4-4 shows the pressure-time history within a cylinder of a typical SI engine. With no self-ignition the pressure force on the piston follows a smooth curve, resulting in smooth engine operation. When self-ignition does occur, pressure forces on the piston are not smooth and engine knock occurs.



(a) Normal Combustion with No Knock



(b) Combustion with Light Knock



Time

(c) Combustion with Heavy Knock

Figure 4-4 Cylinder pressure as a function of time in a typical SI engine combustion chamber showing **(a)** normal combustion, **(b)** combustion with light knock, and **(c)** combustion with heavy knock. Adapted from [33].

For illustrative reasons, a combustion chamber can be visualized schematically as a long hollow tube, shown in Fig. 4-5. Obviously, this is not the shape of a real engine combustion chamber, but it allows visualization of what happens during combustion. These ideas can then be extrapolated to real combustion engine shapes. Before combustion the chamber is divided into four equal mass units, each occupying an equal volume. Combustion starts at the spark plug on the left side, and the flame front travels from left to right. As combustion occurs, the temperature of the burned gases is increased to a high value. This, in turn, raises the pressure of the burned gases and expands the volume of that mass as shown in Fig. 4-5(b). The unburned gases in front of the flame front are compressed by this higher pressure, and compressive

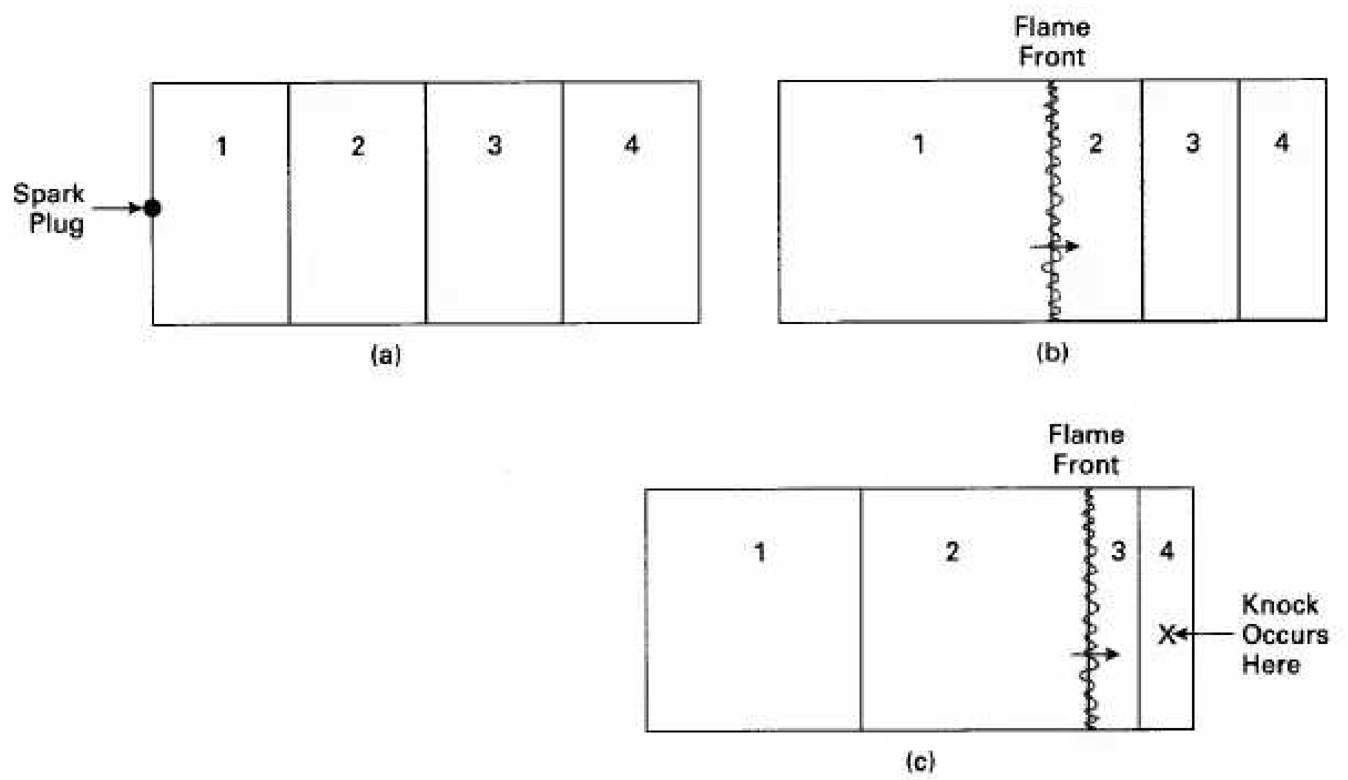


Figure 4-5 SI engine combustion chamber schematically visualized as long hollow cylinder with the spark plug located at left end. (a) Mass of air-fuel is equally distributed as spark plug is fired to start combustion. (b) As flame front moves across chamber, unburned mixture in front of flame is compressed into smaller volume. (c) Flame front continues to compress unburned mixture into smaller volume, which increases its temperature and pressure. If compression raises temperature of end gas above SIT, self-ignition and knock can occur.

heating raises the temperature of the gas. The temperature of the unburned gas is further raised by radiation heating from the flame, and this then raises the pressure even higher. Heat transfer by conduction and convection are not important during this process, due to the very short time interval involved.

The flame front moving through the second mass of air-fuel does so at an accelerated rate because of the higher temperature and pressure, which increase the reaction rate. This, in turn, further compresses and heats the unburned gases in front of the flame as shown in Fig. 4-5(c). In addition, the energy release in the combustion process raises further the temperature and pressure of the burned gases behind the flame front. This occurs both by compressive heating and radiation. Thus, the flame front continues its travel through an unburned mixture that is progressively higher in temperature and pressure. By the time the flame reaches the last portion of unburned gas, this gas is at a very high temperature and pressure. In this *end gas* near the end of the combustion process is where self-ignition and knock occur. To avoid knock, it is necessary for the flame to pass through and consume all unburned gases which have risen above self-ignition temperature before the ignition delay

time elapses. This is done by a combination of fuel property control and design of combustion chamber geometry.

At the end of the combustion process, the hottest region in the cylinder is near the spark plug where combustion was initiated. This region became hot at the start of combustion and then continued to increase in temperature due to compressive heating and radiation as the flame front passed through the rest of the combustion chamber.

By limiting the compression ratio in an SI engine, the temperature at the end of the compression stroke where combustion starts is limited. The reduced temperature at the start of combustion then reduces the temperature throughout the entire combustion process, and knock is avoided. On the other hand, a high compression ratio will result in a higher temperature at the start of combustion. This will cause all temperatures for the rest of the cycle to be higher. The higher temperature of the end gas will create a short ID time, and knock will occur.

Octane Number and Engine Knock

The fuel property that describes how well a fuel will or will not self-ignite is called the octane number or just octane. This is a numerical scale generated by comparing the self-ignition characteristics of the fuel to that of standard fuels in a specific test engine at specific operating conditions. The two standard reference fuels used are isooctane (2,2,4 trimethylpentane), which is given the octane number (ON) of 100, and n-heptane, which is given the ON of 0. The higher the octane number of a fuel, the less likely it will self-ignite. Engines with low compression ratios can use fuels with lower octane numbers, but high-compression engines must use high-octane fuel to avoid self-ignition and knock.

There are several different tests used for rating octane numbers, each of which will give a slightly different ON value. The two most common methods of rating gasoline and other automobile SI fuels are the Motor Method and the Research Method. These give the motor octane number (MaN) and research octane number (RON). Another less common method is the Aviation Method, which is used for aircraft fuel and gives an Aviation Octane Number (AON). The engine used to measure MaN and RON was developed in the 1930s. It is a single-cylinder, overhead valve engine that operates on the four-stroke Otto cycle. It has a variable compression ratio which can be adjusted from 3 to 30. Test conditions to measure MaN and RON are given in Table 4-3.

To find the ON of a fuel, the following test procedure is used. The test engine is run at specified conditions using the fuel being tested. Compression ratio is adjusted until a standard level of knock is experienced. The test fuel is then replaced with a mixture of the two standard fuels. The intake system of the engine is designed such that the blend of the two standard fuels can be varied to any percent from all isooctane to all n-heptane. The blend of fuels is varied until the same knock characteristics are observed as with the test fuel. The percent of isooctane in the fuel blend

TABLE 4-3 TEST CONDITIONS FOR OCTANE NUMBER MEASUREMENT

	RON	MON
Engine Speed (RPM):	600	900
Inlet Air Temperature (°C):	52 (125°F)	149 (300°F)
Coolant Temperature (°C):	100 (212°F)	100
Oil Temperature (°C):	57 (135°F)	57
Ignition Timing:	13° bTDC	19°–26° bTDC
Spark Plug Gap (mm):	0.508 (0.020 in.)	0.508
Inlet Air Pressure:	atmospheric pressure	
Air–Fuel Ratio:	adjusted for maximum knock	
Compression Ratio:	adjusted to get standard knock	

Adapted from [58].

is the ON given to the test fuel. For instance, a fuel that has the same knock characteristics as a blend of 87% isooctane and 13% n-heptane would have an ON of 87.

On the fuel pumps at an automobile service station is found the **anti-knock index:**

$$AKI = (MON + RON)/2 \tag{4-9}$$

This is often referred to as the octane number of the fuel.

Because the test engine has a combustion chamber designed in the 1930s and because the tests are conducted at low speed, the octane number obtained will not always totally correlate with operation in modern high-speed engines. Octane numbers should not be taken as absolute in predicting knock characteristics for a given engine. If there are two engines with the same compression ratio, but with different combustion chamber geometries, one may not knock using a given fuel while the other may experience serious knock problems with the same fuel.

Operating conditions used to measure MON are more severe than those used to measure RON. Some fuels, therefore, will have a RON greater than MON (see Table A-2). The difference between these is called **fuel sensitivity:**

$$FS = RON - MON \tag{4-10}$$

Fuel sensitivity is a good measure of how sensitive knock characteristics of a fuel will be to engine geometry. A low FS number will usually mean that knock characteristics of that fuel are insensitive to engine geometry. FS numbers generally range from 0 to 10.

For measuring octane numbers above 100, fuel additives are mixed with isooctane and other standard points are established. A common additive used for many years to raise the octane number of a fuel was tetraethyllead (TEL).

Common octane numbers (anti-knock index) for gasoline fuels used in automobiles range from 87 to 95, with higher values available for special high-performance and racing engines. Reciprocating SI aircraft engines usually use low-lead fuels with octane numbers in the 85 to 100 range.

The octane number of a fuel depends on a number of variables, some of which are not fully understood. Things that affect ON are combustion chamber geometry, turbulence, swirl, temperature, inert gases, etc. This can be seen by the difference in RON and MaN for some fuels, brought about by different operating characteristics of the test engine. Other fuels will have identical RON and MON. The higher the flame speed in an air-fuel mixture, the higher the octane number. This is because, with a higher flame speed, the air-fuel mixture that is heated above SIT will be consumed during ignition delay time, and knock will be avoided.

Generally there is a high correlation between the compression ratio and the ON of the fuel an engine requires to avoid knock (Fig. 4-6).

If several fuels of known ON are mixed, a good approximation of the mixture octane number is:

$$ON_{mix} = (\% \text{ of} A)(ON_A) + (\% \text{ of} B)(ON_B) + (\% \text{ of} C)(ON_C) \qquad (4\text{-}11)$$

where % = mass percent.

Early crude fuels for automobiles had very low octane numbers that required low compression ratios. This was not a serious handicap to early engines, which needed low compression ratios because of the technology and materials of that day. High compression ratios generate higher pressures and forces that could not be tolerated in early engines.

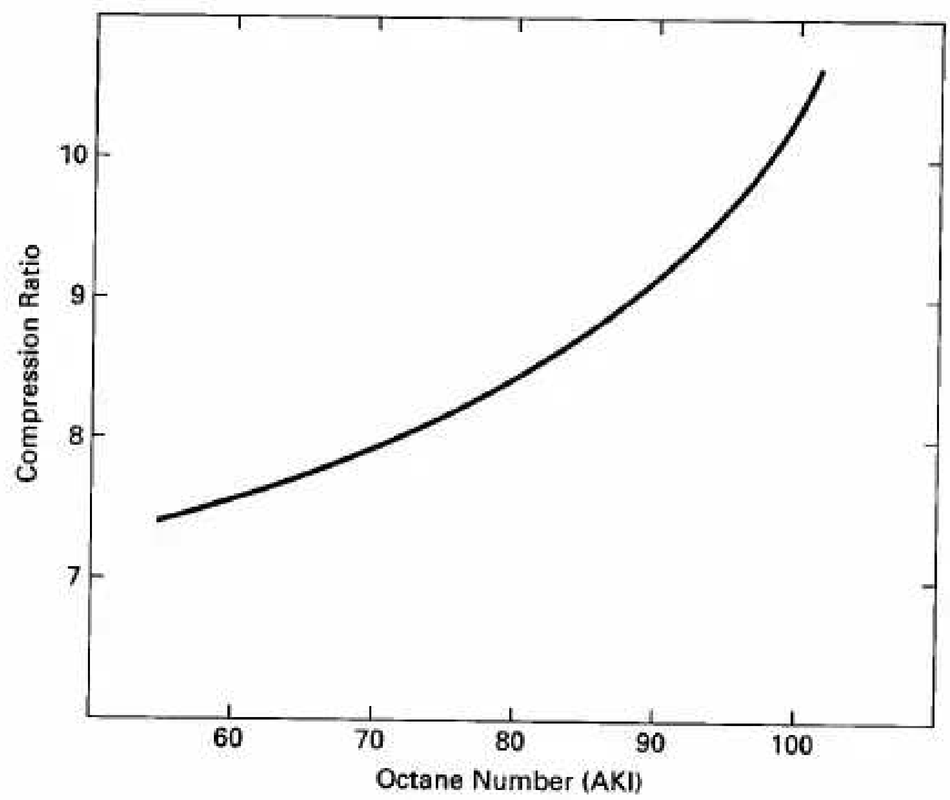


Figure 4-6 Critical compression ratio as a function of fuel octane number (AKI) used in an engine. Critical compression ratio is that above which many engines could have a knock problem. Adapted from [19].

Fuel components with long chain molecules generally have lower octane numbers: the longer the chain the lower is the ON. Components with more side chains have higher octane numbers. For a compound with a given number of carbon and hydrogen atoms, the more these atoms are combined in side chains and not in a few long chains, the higher will be the octane number. Fuel components with ring molecules have higher octane numbers. Alcohols have high octane numbers because of their high flame speeds.

There are a number of gasoline additives that are used to raise the octane number. For many years the standard additive was tetraethyllead TEL, $(C_2H_5)_4Pb$. A few milliliters of TEL in several liters of gasoline could raise the ON several points in a very predictable manner (Fig. 4-7).

When TEL was first used, it was mixed with the gasoline at the local fuel service station. The process was to pour liquid TEL into the fuel tank and then add gasoline, which would mix with the TEL due to the natural turbulence of the pouring. This was not a safe way of handling TEL, which has toxic vapors and is even harmful in contact with human skin. Soon after this, TEL was blended into the gasoline at the refineries, which made it much safer to handle. However, this created a need for additional storage tanks and gasoline pumps at the service station. High-octane and low-octane fuels were now two different gasolines and could not be blended at the service station from a common gasoline base.

Figure 2-5 shows how the compression ratios of automobile engines increased after the introduction of TEL in the 1920s.

The major problem with TEL is the lead that ends up in the engine exhaust. Lead is a very toxic engine emission. For many years, the lead emissions problem was not considered serious just due to the lower numbers of automobiles. However, by the late 1940s and in the 1950s the pollution problem of automobile exhaust was

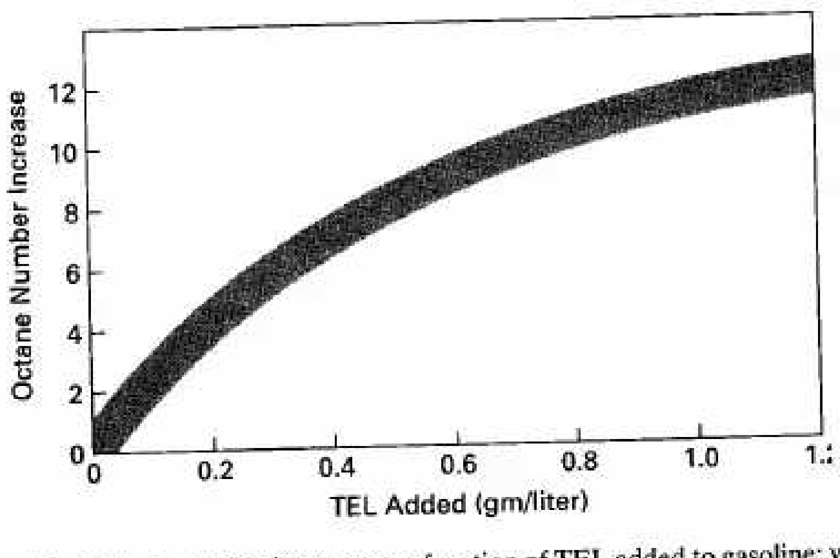


Figure 4-7 Octane number increase as a function of TEL added to gasoline; varies somewhat with gasoline blend. Octane number is MON, RON, or AKI. Adapted from [58].

recognized, first in the Los Angeles basin area of California. The reason that awareness of the problem started here was a combination of a high density of automobiles and the unique weather conditions in the basin. In the 1960s and 1970s, as the number of automobiles proliferated both in the United States and in the rest of the world, it was recognized that lead could no longer be tolerated in gasoline fuel. In the 1970s low-lead and no-lead gasolines were being marketed, and by the early 1990s lead in fuel was unlawful in the United States for most vehicles.

The elimination of lead from gasoline created a problem for older automobiles and other older engines. When TEL is consumed in the combustion process in the cylinder of an engine, one of the results is lead being deposited on the walls of the combustion chamber. This lead reacts with the hot walls and forms a very hard surface. When older engines were manufactured, softer steels were used in the cylinder walls, heads, and valve seats. It was then expected that when these engines were operated using leaded fuel, these parts would become heat treated and hardened during use. Now, when these engines are operated using unleaded fuel, they do not experience this hardening treatment with possible long-term wear problems. The wear that occurs on valve seats is the most critical, and there have been catastrophic engine failures when valve seats wore through. There are now lead substitutes available which can be added to gasoline for people who wish to operate older automobiles for extended lengths of time. Additives that are now used in gasoline to raise the octane number include alcohols and organomanganese compounds.

As an engine ages, deposits build up on the combustion chamber walls. This increases knock problems in two ways. First, it makes the clearance volume smaller and consequently increases the compression ratio. Second, the deposits act as a thermal barrier and increase the temperatures throughout the engine cycle, including peak temperature. Octane requirements can go up as an engine ages, with an average increase needed of about three or four for older engines.

Knock usually occurs at WOT when the engine is loaded (e.g., fast startup or going up a hill). Serious knock problems can be reduced by retarding the ignition spark and starting combustion slightly later in the compression stroke. Many modern smart engines have knock detection to help determine optimum operating conditions. These are usually transducers that detect knock pressure pulses. Some spark plugs are equipped with pressure transducers for this purpose. The human ear is a good knock detector.

Engine knock can also be caused by surface ignition. If any local hot spot exists on the combustion chamber wall, this can ignite the air-fuel mixture and cause the same kind of loss of cycle combustion control. This can occur on surface deposits of older engines, with hot exhaust valves, on hot spark plug electrodes, on any sharp corner in the combustion chamber. The worst kind of surface ignition is pre-ignition, which starts combustion too early in the cycle. This causes the engine to run hotter, which causes more surface hot spots, which causes more surface ignition. In extreme surface ignition problems, when the combustion chamber walls are too hot, run-on will occur. This means the engine will continue to run after the spark ignition has been turned off.

EXAMPLE PROBLEM 4-5

A gasoline-type fuel is generated by blending 15% by weight of butene-1, 70% triptane, and 15% isodecane.
Determine:

1. the anti-knock index
2. the anti-knock index if 0.4 gm of TEL is added per liter of fuel
3. if the fuel is sensitive to combustion chamber geometry

1) Using data from Table A-2 and Eq. (4-11) gives MON and RON for the mixture:

Research:

$$RON = (0.15)(99) + (0.70)(112) + (0.15)(113) = 110.2$$

Motor:

$$MON = (0.15)(80) + (0.70)(101) + (0.15)(92) = 96.5$$

Anti-knock index is obtained from Eq. (4-9):

$$AKI = (MON + RON)/2 = (96.5 + 110.2)/2 = \underline{103}$$

2) Using Fig. 4-7 gives an increase for 0.4 gm/L of about 7:

$$\underline{(AKI)_{\text{with TEL}} = 110}$$

3) Fuel sensitivity is obtained from Eq. (4-10):

$$FS = RON - MON = 110.2 - 96.5 = 13.7.$$

This is a fairly large number, which suggests that knock characteristics of this fuel are very sensitive to combustion chamber geometry.

4-5 DIESEL FUEL

Diesel fuel (diesel oil, fuel oil) is obtainable over a large range of molecular weights and physical properties. Various methods are used to classify it, some using numerical scales and some designating it for various uses. Generally speaking, the greater the refining done on a sample of fuel, the lower is its molecular weight, the lower is its viscosity, and the greater is its cost. Numerical scales usually range from one (1) to five (5) or six (6), with subcategories using alphabetical letters (e.g., AI, 2D, etc). The lowest numbers have the lowest molecular weights and lowest viscosity. These are the fuels typically used in CI engines. Higher numbered fuels are used in residential heating units and industrial furnaces. Fuels with the largest numbers are very viscous and can only be used in large, massive heating units. Each classification has acceptable limits set on various physical properties, such as viscosity, flash point, pour point, cetane number, sulfur content, etc.

Another method of classifying diesel fuel to be used in internal combustion engines is to designate it for its intended use. These designations include bus, truck, railroad, marine, and stationary fuel, going from lowest molecular weight to highest.

For convenience, diesel fuels for IC engines can be divided into two extreme categories. Light diesel fuel has a molecular weight of about 170 and can be approximated by the chemical formula $C_{12.3}H_{22.2}$ (see Table A-2). Heavy diesel fuel has a molecular weight of about 200 and can be approximated as $C_{14.6}H_{24.8}$. Most diesel fuel used in engines will fit in this range. Light diesel fuel will be less viscous and easier to pump, will generally inject into smaller droplets, and will be more costly. Heavy diesel fuel can generally be used in larger engines with higher injection pressures and heated intake systems. Often an automobile or light truck can use a less costly heavier fuel in the summer, but must change to a lighter, less viscous fuel in cold weather because of cold starting and fuel line pumping problems.

Cetane Number

In a compression ignition engine, self-ignition of the air-fuel mixture is a necessity. The correct fuel must be chosen which will self-ignite at the precise proper time in the engine cycle. It is therefore necessary to have knowledge and control of the ignition delay time of the fuel. The property that quantifies this is called the cetane number. The larger the cetane number, the shorter is the ID and the quicker the fuel will self-ignite in the combustion chamber environment. A low cetane number means the fuel will have a long ID.

Like octane number rating, cetane numbers are established by comparing the test fuel to two standard reference fuels. The fuel component n-cetane (hexadecane), $C_{16}H_{34}$, is given the cetane number value of 100, while heptamethylnonane (HMN), $C_{12}H_{34}$, is given the value of 15. The cetane number (CN) of other fuels is then obtained by comparing the ID of that fuel to the ID of a mixture blend of the two reference fuels with

$$\text{CN of fuel} = (\text{percent of n-cetane}) + (0.15)(\text{percent of HMN}) \qquad (4\text{-}12)$$

A special CI test engine is used which has the capability of having the compression ratio changed as it operates. Fuel being rated is injected into the engine cylinder late in the compression stroke at 13°bTDC. The compression ratio is then varied until combustion starts at TDC, giving an ID of 13° of engine rotation. Without changing the compression ratio, the test fuel is replaced with a blend of the two reference fuels. Using two fuel tanks and two flow controls, the blend of the fuels is varied until combustion is again obtained at TDC, an ID of 13°.

The difficulty of this method, in addition to requiring a costly test engine, is to be able to recognize the precise moment when combustion starts. The very slow rise in pressure at the start of combustion is very difficult to detect.

Normal cetane number range is about 40 to 60. For a given engine injection timing and rate, if the cetane number of the fuel is low the ID will be too long. When this occurs, more fuel than desirable will be injected into the cylinder before the first fuel particles ignite, causing a very large, fast pressure rise at the start of combustion. This results in low thermal efficiency and a rough-running engine. If the CN of the fuel is high, combustion will start too soon in the cycle. Pressure will rise before TDC, and more work will be required in the compression stroke.

Cetane numbers below 40 result in unacceptable levels of exhaust smoke and are illegal by many emission laws. The cetane number of a fuel can be raised with certain additives which include nitrates and nitrites. There is a strong inverse correlation between the cetane number of a fuel and its octane number.

Because of the cost and difficulty of measuring the CN of a fuel, several empirical approximation methods using physical properties of the fuel have been developed. One such approximation is the **cetane index** [40].

$$\begin{aligned} CI = &-420.34 + 0.016\,G^2 + 0.192\,G(\log_{10} T_{mp}) \\ &+ 65.01(\log_{10} T_{mp})^2 - 0.0001809\,T_{mp}^2 \end{aligned} \tag{4-13}$$

where: $G = (141.5/S_g) - 131.5$

S_g = specific gravity

T_{mp} = midpoint boiling temperature in °F

4-6 ALTERNATE FUELS

Sometime during the 21st century, crude oil and petroleum products will become very scarce and costly to find and produce. At the same time, there will likely be an increase in the number of automobiles and other IC engines. Although fuel economy of engines is greatly improved from the past and will probably continue to be improved, numbers alone dictate that there will be a great demand for fuel in the coming decades. Gasoline will become scarce and costly. Alternate fuel technology, availability, and use must and will become more common in the coming decades.

Although there have always been some IC engines fueled with non-gasoline or diesel oil fuels, their numbers have been relatively small. Because of the high cost of petroleum products, some third-world countries have for many years been using manufactured alcohol as their main vehicle fuel.

Many pumping stations on natural gas pipelines use the pipeline gas to fuel the engines driving the pumps. This solves an otherwise complicated problem of delivering fuel to the pumping stations, many of which are in very isolated regions. Some large displacement engines have been manufactured especially for pipeline work. These consist of a bank of engine cylinders and a bank of compressor cylinders connected to the same crankshaft and contained in a single engine block similar to a V-style engine.

Another reason motivating the development of alternate fuels for the IC engine is concern over the emission problems of gasoline engines. Combined with other air-polluting systems, the large number of automobiles is a major contributor to the air quality problem of the world. Vast improvements have been made in reducing emissions given off by an automobile engine. If a 30% improvement is made over a period of years and during the same time the number of automobiles in the world increases by 30%, there is no net gain. Actually, the net improvement in cleaning up automobile exhaust since the 1950s, when the problem became apparent, is over 95%. However, additional improvement is needed due to the ever-increasing number of automobiles.

A third reason for alternate fuel development in the United States and other industrialized countries is the fact that a large percentage of crude oil must be imported from other countries which control the larger oil fields. In recent years, up to a third of the United States foreign trade deficit has been from the purchase of crude oil, tens of billions of dollars.

Listed next are the major alternate fuels that have been and are being considered and tested for possible high-volume use in automobile and other kinds of IC engines. These fuels have been used in limited quantities in automobiles and small trucks and vans. Quite often, fleet vehicles have been used for testing (e.g., taxies, delivery vans, utility company trucks). This allows for comparison testing with similar gasoline-fueled vehicles, and simplifies fueling of these vehicles.

It must be remembered that, in just about all alternate fuel testing, the engines used are modified engines which were originally designed for gasoline fueling. They are, therefore, not the optimum design for the other fuels. Only when extensive research and development is done over a period of years will maximum performance and efficiency be realized from these engines. However, the research and development is difficult to justify until the fuels are accepted as viable for large numbers of engines (the chicken-and-egg problem).

Some diesel engines are starting to appear on the market which use dual fuel. They use methanol or natural gas and a small amount of diesel fuel that is injected at the proper time to ignite both fuels.

Most alternate fuels are very costly at present. This is often because of the quantity used. Many of these fuels will cost much less if the amount of their usage gets to the same order of magnitude as gasoline. The cost of manufacturing, distribution, and marketing all would be less.

Another problem With alternate fuels is the lack of distribution points (service stations) where the fuel is available to the public. The public will be reluctant to purchase an automobile unless there is a large-scale network of service stations available where fuel for that automobile can be purchased. On the other hand, it is difficult to justify building a network of these service stations until there are enough automobiles to make them profitable. Some cities are starting to make available a few distribution points for some of these fuels, like propane, natural gas, and methanol. The transfer from one major fuel type to another will be a slow, costly, and sometimes painful process.

In the following list, some of the drawbacks for a particular fuel may become less of a problem if large quantities of that fuel are used (i.e., cost, distribution, etc.).

Alcohol

Alcohols are an attractive alternate fuel because they can be obtained from a number of sources, both natural and manufactured. Methanol (methyl alcohol) and

ethanol (ethyl alcohol) are two kinds of alcohol that seem most promising and have had the most development as engine fuel.

The advantages of alcohol as a fuel include:

1. Can be obtained from a number of sources, both natural and manufactured.
2. Is high octane fuel with anti-knock index numbers (octane number on fuel pump) of over 100. High octane numbers result, at least in part, from the high flame speed of alcohol. Engines using high-octane fuel can run more efficiently by using higher compression ratios.
3. Generally less overall emissions when compared with gasoline.
4. When burned, it forms more moles of exhaust, which gives higher pressure and more power in the expansion stroke.
5. Has high evaporative cooling (h_{fg}) which results in a cooler intake process and compression stroke. This raises the volumetric efficiency of the engine and reduces the required work input in the compression stroke.
6. Low sulfur content in the fuel.

The disadvantages of alcohol fuels include:

1. Low energy content of the fuel as can be seen in Table A-2. This means that almost twice as much alcohol as gasoline must be burned to give the same energy input to the engine. With equal thermal efficiency and similar engine output usage, twice as much fuel would have to be purchased, and the distance which could be driven with a given fuel tank volume would be cut in half. The same amount of automobile use would require twice as much storage capacity in the distribution system, twice the number of storage facilities, twice the volume of storage at the service station, twice as many tank trucks and pipelines, etc. Even with the lower energy content of alcohol, engine power for a given displacement would be about the same. This is because of the lower air-fuel ratio needed by alcohol. Alcohol contains oxygen and thus requires less air for stoichiometric combustion. More fuel can be burned with the same amount of air.
2. More aldehydes in the exhaust. If as much alcohol fuel was consumed as gasoline, aldehyde emissions would be a serious exhaust pollution problem.
3. Alcohol is much more corrosive than gasoline on copper, brass, aluminum, rubber, and many plastics. This puts some restrictions on the design and manufacturing of engines to be used with this fuel. This should also be considered when alcohol fuels are used in engine systems deSigned to be used with gasoline. Fuel lines and tanks, gaskets, and even metal engine parts can deteriorate with long-term alcohol use (resulting in cracked fuel lines, the need for special fuel tank, etc). Methanol is very corrosive on metals.
4. Poor cold weather starting characteristics due to low vapor pressure and evaporation. Alcohol-fueled engines generally have difficulty starting at temperatures below 10°C. Often a small amount of gasoline is added to alcohol

fuel, which greatly improves cold-weather starting. The need to do this, however, greatly reduces the attractiveness of any alternate fuel.

5. Poor ignition characteristics in general.
6. Alcohols have almost invisible flames, which is considered dangerous when handling fuel. Again, a small amount of gasoline removes this danger.

handling fuel. Again, a small amount of gasoline removes this danger.

7. Danger of storage tank flammability due to low vapor pressure. Air can leak into storage tanks and create a combustible mixture.
8. Low flame temperatures generate less NOx, but the resulting lower exhaust temperatures take longer to heat the catalytic converter to an efficient operating temperature.
9. Many people find the strong odor of alcohol very offensive. Headaches and dizziness have been experienced when refueling an automobile.
10. Vapor lock in fuel delivery systems.

Methanol

Of all the fuels being considered as an alternate to gasoline, methanol is one of the more promising and has experienced major research and development. Pure methanol and mixtures of methanol and gasoline in various percentages have been extensively tested in engines and vehicles for a number of years [88, 130]. The most common mixtures are M85 (85% methanol and 15% gasoline) and M10 (10% methanol and 90% gasoline). The data of these tests which include performance and emission levels are compared to pure gasoline (MO) and pure methanol (M100). Some smart **flexible-fuel** (or **variable-fuel)** engines are capable of using any random mixture combination of methanol and gasoline ranging from pure methanol to pure gasoline. Two fuel tanks are used and various flow rates of the two fuels can be pumped to the engine, passing through a mixing chamber. Using information from sensors in the intake and exhaust, the EMS adjusts to the proper air-fuel ratio, ignition timing, injection timing, and valve timing (where possible) for the fuel mixture being used. Fast, abrupt changes in fuel mixture combinations must be avoided to allow for these adjustments to occur smoothly.

One problem with gasoline-alcohol mixtures as a fuel is the tendency for alcohol to combine with any water present. When this happens the alcohol separates locally from the gasoline, resulting in a non-homogeneous mixture. This causes the engine to run erratically due to the large AF differences between the two fuels.

At least one automobile company has been experimenting with a three-fuel vehicle which can use any combination of gasoline-methanol-ethanol [11].

Methanol can be obtained from many sources, both fossil and renewable. These include coal, petroleum, natural gas, biomass, wood, landfills, and even the ocean. However, any source that requires extensive manufacturing or processing raises the price of the fuel and requires an energy input back into the overall environmental picture, both unattractive.

In some parts of the country, MlO fuel (10% methanol and 90% gasoline) is now sold at some local service stations in place of gasoline. It is advisable to read the sometimes small print on the fuel pump to determine the type of fuel that is being used in your automobile.

Emissions from an engine using MlO fuel are about the same as those using gasoline. The advantage (and disadvantage) of using this fuel is mainly the 10% decrease in gasoline use. With M85 fuel there is a measurable decrease in HC and CO exhaust emissions. However, there is an increase in NOx and a large (=500%) increase in formaldehyde formation.

Methanol is used in some dual-fuel CI engines. Methanol by itself is not a good CI fuel because of its high octane number, but if a small amount of diesel oil is used for ignition, it can be used with good results. This is very attractive for third-world countries, where methanol can often be obtained much cheaper than diesel oil. Older CI bus engines have been converted to operate on methanol in tests conducted in California. This resulted in an overall reduction of harmful emissions compared with worn engines operating with diesel fuel [115].

Ethanol

Ethanol has been used as automobile fuel for many years in various regions of the world. Brazil is probably the leading user, where in the early 1990s, 4.5 million vehicles operated on fuels that were 93% ethanol. For a number of years **gasohol** has been available at service stations in the United States, mostly in the Midwest corn-producing states. Gasohol is a mixture of 90% gasoline and 10% ethanol. As with methanol, the development of systems using mixtures of gasoline and ethanol continues. Two mixture combinations that are important are E85 (85% ethanol) and E10 (gasohol). E85 is basically an alcohol fuel with 15% gasoline added to eliminate some of the problems of pure alcohol (i.e., cold starting, tank flammability, etc.). E10 reduces the use of gasoline with no modification needed to the automobile engine. Flexible-fuel engines are being tested which can operate on any ratio of ethanol-gasoline [122].

Ethanol can be made from ethylene or from fermentation of grains and sugar. Much of it is made from corn, sugar beets, sugar cane, and even cellulose (wood and paper). In the United States, corn is the major source. The present cost of ethanol is high due to the manufacturing and processing required. This would be reduced if larger amounts of this fuel were used. However, very high production would create a food-fuel competition, with resulting higher costs for both. Some studies show that at present in the United States, crops grown for the production of ethanol consume more energy in plowing, planting, harvesting, fermenting, and delivery than what is in the final product. This defeats one major reason for using an alternate fuel [95].

Ethanol has less HC emissions than gasoline but more than methanol.

EXAMPLE PROBLEM 4-6

A taxicab is equipped with a flexible-fuel four-cylinder SI engine running on a mixture of methanol and gasoline at an equivalence ratio of 0.95. How must the air-fuel ratio

change as the fuel flow to the engine shifts from 10% methanol (M10) to 85% methanol (M85)?

Change masses of fuel to moles at M10:

FUEL	MASS, m (kg)	MOLECULAR WEIGHT, M	MOLES $N = m/M$ (kgmoles)	MOLE FRACTION
CH_3OH	0.10	32	0.003125	0.278
C_8H_{15}	0.90	111	0.008108	0.722
	1.00		0.011233	1.000

For one kgmole of fuel reacting with stoichiometric air:

$$0.278\ CH_3OH + 0.722\ C_8H_{15} + 8.9005\ O_2 + 8.9005(3.76)\ N_2 \rightarrow$$

$$6.054\ CO_2 + 5.971\ H_2O + 8.9005(3.76)\ N_2$$

For one kgmole of fuel reacting with an equivalence ratio $\phi = 0.95$:

$$0.278\ CH_3OH + 0.722\ C_8H_{15} + (8.9005/0.95)\ O_2 + (8.9005/0.95)(3.76)\ N_2$$
$$\rightarrow 6.054\ CO_2 + 5.971\ H_2O + (8.9005/0.95)(3.76)\ N_2 + 0.468\ O_2$$

Air–fuel ratio:

$$\begin{aligned} AF &= m_a/m_f \\ &= [(8.9005/0.95)(1 + 3.76)(29)]/[(0.278)(32) + (0.722)(111)] \\ &= \underline{14.53} \end{aligned}$$

Repeating calculations for M85:

FUEL	MASS, m (kg)	MOLECULAR WEIGHT, M	MOLES $N = m/M$ (kgmoles)	MOLE FRACTION
CH_3OH	0.85	32	0.026563	0.952
C_8H_{15}	0.15	111	0.001351	0.048
	1.00		0.027914	1.000

Stoichiometric reaction:

$$0.952\ CH_3OH + 0.048\ C_8H_{15} + 1.992\ O_2 + 1.992(3.76)\ N_2 \rightarrow$$
$$1.336\ CO_2 + 2.264\ H_2O + 1.992(3.76)\ N_2$$

Reaction with an equivalence ratio $\phi = 0.95$:

$$0.952\ CH_3OH + 0.048\ C_8H_{15} + (1.992/0.95)\ O_2 + (1.992/0.95)(3.76)\ N_2$$
$$\rightarrow 1.336\ CO_2 + 2.264\ H_2O + (1.992/0.95)(3.76)\ N_2 + 0.105\ O_2$$

Air–fuel ratio:

$$\begin{aligned} AF &= m_a/m_f \\ &= [(1.992/0.95)(1 + 3.76)(29)]/[(0.952)(32) + (0.048)(111)] \\ &= \underline{8.09} \end{aligned}$$

As the fuel flow composition is changed, the engine management system (EMS) must adjust the AF from 14.53 to 8.09.

EXAMPLE PROBLEM 4-7

As an old-car collector enthusiast is cruising along in his 1950 chrome-covered Buick "dream car," he realizes he is running low on fuel, so he pulls up to the fuel pump at the local convenience store and fills the fuel tank. The car has a large-displacement straight-eight engine with a carburetor adjusted to supply stoichiometric air at normal operating conditions using gasoline fuel. The man does not notice that the fuel he is putting into his automobile contains 10% methanol (M10), but he does notice a slight loss in power as the fuel is consumed in the following days. Calculate the actual equivalence ratio the carburetor is supplying to the engine when it is adjusted to give $\phi = 1$ with gasoline but is actually operating with M10 fuel.

Stoichiometric combustion of gasoline:

$$C_8H_{15} + 11.75\ O_2 + 11.75(3.76)\ N_2 \rightarrow 8\ CO_2 + 7.5\ H_2O + 11.75(3.76)\ N_2$$

Air–fuel ratio:

$$AF = m_a/m_f = [(11.75)(1 + 3.76)(29)]/[(1)(111)] = 14.61$$

The carburetor is adjusted to give stoichiometric combustion with gasoline, so this is the actual $(AF)_{act}$ that it supplies.

When M10 fuel is used, the stoichiometric requirements would be (using result obtained in Example Problem 4-6):

$$0.278\ CH_3OH + 0.722\ C_8H_{15} + 8.9005\ O_2 + (8.9005)(3.76)\ N_2 \rightarrow$$

$$6.054\ O_2 + 5.971\ H_2O + (8.9005)(3.76)\ N_2$$

$$(AF)_{stoich} = [(8.9005)(1 + 3.76)(29)]/[(0.278)(32) + (0.722)(111)] = 13.80$$

Equivalence ratio:

$$\phi = (AF)_{stoich}/(AF)_{act} = (13.80)/(14.61) = \underline{0.945}$$

This assumes that the mass densities of gasoline and M10 fuel are the same when M10 actually has a slightly higher density. If M10 fuel were going to be used extensively, the carburetor should be readjusted for stoichiometric operation. Using alcohol fuel in older engines designed for gasoline might also create material compatibility problems between the fuel and engine components.

Hydrogen

A number of companies have built automobiles with prototype or modified engines which operate on hydrogen fuel [64].

The advantages of hydrogen as a IC engine fuel include:

1. Low emissions. Essentially no CO or HC in the exhaust as there is no carbon in the fuel. Most exhaust would be H_2O and N_2.
2. Fuel availability. There are a number of different ways of making hydrogen, including electrolysis of water.
3. Fuel leakage to environment is not a pollutant.
4. High energy content per volume when stored as a liquid. This would give a large vehicle range for a given fuel tank capacity, but see the following.

Disadvantages of using hydrogen as a fuel:

1. Heavy, bulky fuel storage, both in vehicle and at the service station. Hydrogen can be stored either as a cryogenic liquid or as a compressed gas. If stored as a liquid, it would have to be kept under pressure at a very low temperature. This would require a thermally super-insulated fuel tank. Storing in a gas phase would require a heavy pressure vessel with limited capacity.
2. Difficult to refuel.
3. Poor engine volumetric efficiency. Any time a gaseous fuel is used in an engine, the fuel will displace some of the inlet air and poorer volumetric efficiency will result.
4. Fuel cost would be high at present day technology and availability.

4. Fuel cost would be high at present-day technology and availability.
5. High NOx emissions because of high flame temperature.
6. Can detonate.

At least one automobile company (Mazda) has adapted a rotary Wankel engine to run on hydrogen fuel. It was reasoned that this is a good type of engine for this fuel. The fuel intake is on the opposite side of the engine from where combustion occurs, lowering the chance of pre-ignition from a hot engine block; hydrogen fuel ignites very easily. This same experimental car uses a metal-hydride fuel storage system [86].

Natural Gas-Methane

Natural gas is a mixture of components, consisting mainly of methane (60-98%) with small amounts of other hydrocarbon fuel components. In addition it contains various amounts of N_2, CO_2, He, and traces of other gases. Its sulfur content ranges from very little (sweet) to larger amounts (sour). It is stored as compressed natural gas (CNG) at pressures of 16 to 25 MPa, or as liquid natural gas (LNG) at pressures of 70 to 210 kPa and a temperature around -160°C. As a fuel, it works best in an engine system with a single-throttle body fuel injector. This gives a longer mixing time, which is needed by this fuel. Tests using CNG in various-sized vehicles continue to be conducted by government agencies and private industry [12, 94, 101].

Advantages of natural gas as a fuel include:

1. Octane number of 120, which makes it a very good SI engine fuel. One reason for this high octane number is a fast flame speed. Engines can operate with a high compression ratio.
2. Low engine emissions. Less aldehydes than with methanol.
3. Fuel is fairly abundant worldwide with much available in the United States. It can be made from coal but this would make it more costly.

Disadvantages of natural gas as an engine fuel:

1. Low energy density resulting in low engine performance.
2. Low engine volumetric efficiency because it is a gaseous fuel.
3. Need for large pressurized fuel storage tank. Most test vehicles have a range of only about 120 miles. There is some safety concern with a pressurized fuel tank.
4. Inconsistent fuel properties.
5. Refueling is slow process.

Some very large stationary CI engines operate on a fuel combination of methane and diesel fuel. Methane is the major fuel, amounting to more than 90% of the total. It is supplied to the engine as a gas through high-pressure pipes. A small amount of high grade, low sulfur diesel fuel is used for ignition purposes. The net result is very clean running engines. These engines would also be good power plants for large ships, except that high-pressure gas pipes are undesirable on ships.

HISTORIC-BUSES OPERATING ON NATURAL GAS

In some countries in eastern and southern Asia, buses using natural gas as fuel have a unique fuel reservoir system. The gas is stored at about one atmosphere pressure in a large inflatable rubber diaphragm on the roof of the bus. With a full load of fuel, the bus is about twice the height as when it has no fuel. No fuel gauge is needed on these buses.

Propane

Propane has been tested in fleet vehicles for a number of years. It is a good high-octane SI engine fuel and produces less emissions than gasoline: about 60% less CO, 30% less HC, and 20% less NOx.

Propane is stored as a liquid under pressure and delivered through a high-pressure line to the engine, where it is vaporized. Being a gaseous fuel, it has the disadvantage of lower engine volumetric efficiency.

Reformulated Gasoline

Reformulated gasoline is normal-type gasoline with a slightly modified formulation and additives to help reduce engine emissions. Included in the fuel are oxidation inhibitors, corrosion inhibitors, metal deactivators, detergents, and deposit control additives. Oxygenates such as methyl tertiary-butyl ether (MTBE) and alcohols are added, such that there is 1-3% oxygen by weight. This is to help reduce CO in the exhaust. Levels of benzene, aromatics, and high boiling components are reduced, as is the vapor pressure. Recognizing that engine deposits contribute to emissions, cleaning additives are included. Some additives clean carburetors, some clean fuel injectors, and some clean intake valves, each of which often does not clean other components.

On the plus side is that all gasoline-fueled engines, old and new, can use this fuel without modification. On the negative side is that only moderate emission reduction is realized, cost is increased, and the use of petroleum products is not reduced. [121].

Coal-Water Slurry

In the latter half of the 1800s, before petroleum-based fuels were perfected, many other fuels were tested and used in IC engines. When Rudolf Diesel was developing his engine, one of the fuels he used was a coal dust-water slurry. Fine particles of coal (carbon) were dispersed in water and injected and burned in early diesel engines. Although this never became a common fuel, a number of experimental engines using this fuel have been built over the last hundred years. Even today, some work continues on this fuel technology. The major improvement in this type of fuel has been the reduction of the average coal particle size. In 1894 the average particle size was on the order of 100 μ (1 μ = 1 micron = 10^{-6} m). This was reduced to about 75 μ in the 1940-1970 period and further reduced to about 10 μ by the early

1980s. The typical slurry is about 50% coal and 50% water by mass. One major problem with this fuel is the abrasiveness of the solid particles, which manifests itself in worn injectors and piston rings [27].

Coal is an attractive fuel because of the large supply which is available. However, as an engine fuel, other methods of use seem more feasible. These include liquefaction or gasification of the coal.

EXAMPLE PROBLEM 4-8

A 50% coal-water slurry (50% coal and 50% water by mass) is bumed in stoichiometric air.

Calculate:

1. air-fuel ratio
2. heating value of fuel

1) Assuming coal is carbon, the fuel mixture in molar quantities will consist of one mole of carbon to (12/18) moles of water.

Stoichiometric combustion:

$$C(s) + (12/18)\,H_2O + O_2 + (3.76)\,N_2 \rightarrow CO_2 + \tfrac{2}{3}\,H_2O + (3.76)\,N_2$$

$$AF = [(1)(4.76)(29)]/[(1)(12) + (12/18)(18)] = \underline{5.75}$$

2) Heating value of carbon from Table A-2:

$$(Q_{LHV})_c = 33{,}800 \text{ kJ/kg}$$

Heating value of water:

$$(Q_{LHV})_w = 0$$

Therefore, a fuel mixture made of 50% carbon and 50% water by mass:

$$(Q_{LHV})_{mix} = [(33{,}800 + 0)/2] = \underline{16{,}900 \text{ kJ/kg}}$$

This is a very low heating value compared to normal hydrocarbon fuels, which generally have heating values in the 40,000 to 50,000 kJ/kg range. However, the low stoichiometric AF means that a greater amount of fuel can be used with the same amount of inlet air, and the power output developed by the engine can equal that of engines using other fuels. Specific fuel consumption would be very high.

Other Fuels

Attempts to use many other types of fuel have been tried throughout the history of IC engines. Often, this was done out of necessity or to promote financial gain by some group. At present, a number of biomass fuels are being evaluated, mainly in Europe. These include CI fuel oil made from wood, barley, soy beans, rape seed, and even beef tallow. Advantages of these fuels generally include availability and low cost, low sulfur, and low emissions. Disadvantages include low energy content (heating value) and corresponding high specific fuel consumption.

HISTORIC-WHEN AUTOMOBILES RAN ON CHARCOAL

In the late 1930s and early 1940s petroleum products became very scarce, especially in Europe, due to World War II. Just about all gasoline products were claimed by the German army, leaving no fuel for civilian automobile use. Although this was an inconvenience for the civilian population, it did not stop them from using their beloved automobiles [44].

Enterprising people in several countries, mainly Sweden and Germany, developed a way to operate their automobiles using solid fuels like charcoal, wood, or coal. Using technology first researched 20 years earlier, they converted their vehicles by building a combustion chamber in the trunk of the car or on a small trailer pulled by the car. In this combustion chamber, the coal, wood, or other solid or waste fuel was burned with a restricted supply of oxygen (air), This generated a supply of carbon monoxide, which was then piped to the engine and used to fuel the

engine. With coal being basically carbon, the ideal reaction in the combustion chamber (CO generator) would be:

$$C + \tfrac{1}{2}\, O_2 + \tfrac{1}{2}(3.76)\, N_2 \rightarrow CO + \tfrac{1}{2}(3.76)\, N_2$$

This CO was then piped to the automobile engine, where, ideally, the reaction would be

$$\underbrace{CO + \tfrac{1}{2}(3.76)\, N_2}_{\text{fuel}} + \tfrac{1}{2}\, O_2 + \tfrac{1}{2}(3.76)\, N_2 \rightarrow CO_2 + 3.76\, N_2$$

Obviously the carburetor on the engine had to be modified and adjusted to supply the gaseous fuel and to give the correct AF. When this was done, the automobile could be operated, but with much less power and longevity. Impurities from the CO generator very quickly dirtied up the combustion chambers of the engine, even when extensive filtering was done. Another problem was with leakage of CO, which is a poisonous, odorless, colorless gas. When this occurred and CO got into the passenger compartment, the driver and other occupants were in danger of sickness and death. Drivers subjected to CO would react as if intoxicated in much the same way as a drunk driver, with the same results of erratic driving, accidents, and even death.

As late as the 1970s, Sweden was still working on the development of this method of using solid fuel to power automobiles [44].

EXAMPLE PROBLEM 4-9

During World War II a Swedish car lover found himself unable to obtain gasoline to operate his vehicle. Like many of his neighbors, he built himself a trailer on which he mounted a charcoal-burning carbon monoxide generator, a combustion chamber with restricted air input. The CO generated was then piped to the engine and used as fuel for the automobile pulling the trailer. Estimate the loss of power that was experienced when CO was used in this manner as fuel for the carbureted engine originally designed to operate on stoichiometric gasoline.

With gasoline:

$$C_8H_{15} + 11.75\, O_2 + 11.75(3.76)\, N_2 \rightarrow 8\, CO_2 + 7.5\, H_2O + 11.75(3.76)\, N_2$$

Thus, for 11.75(4.76) kgmoles of air, heat in is:

$$Q_{in} = m_f Q_{HV} = (1 \text{ kgmole of fuel})(111 \text{ kg/kgmole})(43.0 \text{ MJ/kg}) = 4773 \text{ MJ}$$

For one kgmole of input oxygen (4.76 kgmoles of air), heat in is:

$$Q_{in} = (4773\ \text{MJ})/(11.75) = 406.2\ \text{MJ}$$

When CO (and the N_2 which comes with it from the generator) is burned stoichiometrically with one kgmole of oxygen:

$$2\ CO + 2[\tfrac{1}{2}(3.76)]\ N_2 + O_2 + (3.76)\ N_2 \rightarrow 2\ CO_2 + 2(3.76)\ N_2$$

$$Q_{in} = (2\ \text{kgmoles of CO})(28\ \text{kg/kgmole})(10.1\ \text{MJ/kg}) = 565.6\ \text{MJ}$$

This, however, is reduced because CO is a gaseous fuel which displaces some of the air in the intake system. For each kgmole of oxygen (4.76 kgmoles of air) in the intake system, there will be $[2 + 2(\tfrac{1}{2})(3.76)] = 5.76$ kgmoles of gaseous fuel. For the same total gas volumetric flow rate, only $(4.76)/[(4.76) + (5.76)] = 0.452$ will be the fraction which is new inlet air. Therefore:

$$Q_{in} = (565.6\ \text{MJ})(0.452) = 255.7\ \text{MJ}$$

Percent loss in heat is then:

$$\%\ \text{loss of}\ Q_{in} = \{[(406.2) - (255.7)]/(406.2)\}(100) = 37.1\%\ \text{loss}$$

Assuming the same thermal efficiency and same engine mechanical efficiency with the two fuels gives brake power output:

$$(\dot{W}_b)_{CO} = 62.9\%(\dot{W}_b)_{gasoline}$$

These calculations are all based on ideal reactions. An actual engine-CO generator system operated under these conditions would experience many additional losses, including less than ideal reactions, solid impurities and filtering problems, and fuel delivery complications. All of these would significantly lower the actual engine output.

4-7 CONCLUSIONS

For most of the 20th century, the two main fuels that have been used in internal combustion engines have been gasoline (SI engines) and fuel oil (diesel oil for CI engines). During this time, these fuels have experienced an evolution of composition and additives according to the contemporary needs of the engines and environment. In the latter part of the century, alcohol fuels made from various farm products and other sources have become increasingly more important, both in the United States and in other countries. With increasing air pollution problems and a petroleum shortage looming on the horizon, major research and development programs are being conducted throughout the world to find suitable alternate fuels to supply engine needs for the coming decades.

PROBLEMS

4-1. C_4H_8 is burned in an engine with a fuel-rich air-fuel ratio. Dry analysis of the exhaust gives the following volume percents: $CO_2 = 14.95\%$, $C_4H_8 = 0.75\%$, $CO = 0\%$, $H_2 = 0\%$, $O_2 = 0\%$, with the rest being N_2. Higher heating value of this fuel is $Q_{HHV} = 46.9$ MJ/kg. Write the balanced chemical equation for one mole of this fuel at these conditions.
Calculate: (a) Air-fuel ratio.

(b) Equivalence ratio.
(c) Lower heating value of fuel. [MJ/kg]

(d) Energy released when one kg of this fuel is burned in the engine with a combustion efficiency of 98%. [MJ]

4-2. Draw the chemical structural formula of 2-methyl-2,3-ethylbutane. This is an isomer of what chemical family? Write the balanced chemical reaction equation for one mole of this fuel burning with an equivalence ratio of ϕ=0.7. Calculate the stoichiometric AF for this fuel.

4-3. Draw the chemical structural formula of (a) 3,4-dimethylhexane, (b) 2,4-diethylpentane, (c) 3-methyl-3-ethylpentane. These are isomers of what other molecules?

4-4. Hydrogen is used as a fuel in an experimental engine and is burned with stoichiometric *oxygen.* Reactants enter at a temperature of 25°C and complete combustion occurs at constant pressure. Write the balanced chemical reaction equation.
Calculate: (a) Fuel-air (fuel-oxygen) ratio.
(b) Equivalence ratio.
(c) Theoretical maximum temperature from this combustion. (use enthalpy values from a thermodynamics textbook) [K]
(d) Dew point temperature of exhaust if exhaust pressure is 101 kPa. [°C]

4-5. Isooctane is burned with air in an engine at an equivalence ratio of 0.8333. Assuming complete combustion, write the balanced chemical reaction equation.
Calculate: (a) Air-fuel ratio.
(b) How much excess air is used. [%]
(c) AKI and FS of this fuel.

4-6. A race car burns nitromethane with air at an equivalence ratio of 1.25. Except for unburned fuel, all nitrogen ends up as N_2. Write the balanced chemical equation.
Calculate: (a) Percent stoichiometric air. [%]
(b) Air-fuel ratio.

4-7. Methanol is burned in an engine with air at an equivalence ratio of ϕ=0.75. Exhaust pressure and inlet pressure are 101 kPa. Write the balanced chemical equation for this reaction.
Calculate: (a) Air-fuel ratio.
(b) Dew point temperature of the exhaust if the inlet air is dry. [°C]
(c) Dew point temperature of the exhaust if the inlet air has a relative humidity of 40% at 25°C. [°C]
(d) Antiknock index of methanol.

4-8. Compute the indicated power generated at WOT by a three-liter, four-cylinder, four-stroke cycle SI engine operating at 4800 RPM using either gasoline or methanol. For each case, the intake manifold is heated such that all fuel is evaporated before the intake ports, and the air-fuel mixture enters the cylinders at 60°C and 100 kPa. Compression ratio r_c = 8.5, fuel equivalence ratio ϕ= 1.0, combustion efficiency η_c = 98%, and volumetric efficiency η_v =100%. Calculate the indicated specific fuel consumption for each fuel. [gm/kW-hr]

4-9. A four-cylinder SI engine with a compression ratio r_c = 10 operates on an air-standard Otto cycle at 3000 RPM using ethyl alcohol as fuel. Conditions in the cylinders at the start of the compression stroke are 60°C and 101 kPa. Combustion efficiency η_c =97%. Write the balanced stoichiometric chemical equation for this fuel.
Calculate: (a) AF if the engine operates at an equivalence ratio ϕ= 1.10.
(b) Peak temperature in cycle of part (a). [°C]
(c) Peak pressure in cycle of part (a). [kPa]

4-10. Tim's 1993 Buick has a six-cylinder, four-stroke cycle SI engine with multipoint port fuel injectors operating on an Otto cycle at WOT. The fuel injectors are set to deliver an AF such that gasoline would burn at stoichiometric conditions. (approximate gasoline using isooctane properties)

Calculate:
- (a) Equivalence ratio of air-gasoline mixture.
- (b) Equivalence ratio if gasoline is replaced with ethanol without readjusting the AF delivered by the fuel injectors.
- (c) Increase or decrease in brake power using alcohol instead of gasoline under these conditions, with the same air flow rate and same thermal efficiency. Assume ethanol would burn under these conditions with the same combustion efficiency. [%]

4-11. For the sarlte air flow rate, what would be the percentage increase in engine power if stoichiometric gasoline is replaced with stoichiometric nitromethane? Assume the same thermal efficiency and same combustion efficiency. [%]

4-12. Compare the indicated power generated in an engine using stoichiometric gasoline, stoichiometric methanol, or stoichiometric nitromethane. Assume the same combustion efficiency, thermal efficiency, and air flow rate for all fuels.

4-13. Isodecane is used as a fuel.

Calculate:
- (a) Anti-knock index.
- (b) MON if 0.2 gmlL of TEL is added to the fuel.
- (c) How many gallons of butene-1 should be added to 10 gallons of isodecane to give a mixture MON of 87.

4-14. A six-liter, eight-cylinder, four-stroke cycle SI race car engine operates at 6000 RPM using stoicl1iometric nitromethane as fuel. Combustion efficiency is 99%, and the fuel input rate is 0.198 kg/sec.

Calculate:
- (a) Volumetric efficiency of the engine. [%]
- (b) Flow rate of air into the engine. [kg/sec]
- (c) Heat generated in each cylinder per cycle. [kJ]
- (d) How much chemical energy there is in unburned fuel in the exhaust. [kW]

4-15. (a) Give three reasons why methanol is a good alternate fuel for automobiles. (b) Give three reasons why it is not a good alternate fuel.

4-16. When one-half mole of oxygen and one-half mole of nitrogen are heated to 3000 K at a pressure of 5000 kPa, some of the mixture will react to form NO by the reaction equation! O_2 + ! N_2 ~ NO. Assume these are the only components which react.

Calculate:
- (a) Chemical equilibrium constant for this reaction at these conditions using Table A-3.
- (b) Number of moles of NO at equilibrium.
- (c) Number of moles of O_2 at equilibrium.
- (d) Number of moles of NO at equilibrium if total pressure is doubled.
- (e) Number of moles of NO at equilibrium if there was originally one-half mole of oxygen, one-half mole of nitrogen, and one mole of argon at 5000 kPa total pressure.

4-17. A fuel mixture consists of 20% isooctane, 20% triptane, 20% isodecane, and 40% toluene by moles. Write the chemical reaction formula for the stoichiometric combustion of one mole of this fuel.

Calculate:
- (a) Air-fuel ratio.

(b) Research octane number.

(c) Lower heating value of fuel mixture. [kJ/kg]

4-18. A flexible-fuel vehicle operates with a stoichiometric fuel mixture of one-third isooctane, one-third ethanol, and one-third methanol, by mass.
Calculate: (a) Air-fuel ratio.

(b) MaN, RON, FS, and AKI.

4-19. It is desired to find the cetane number of a fuel oil which has a density of 860 kg/m^3 and a midpoint boiling temperature of 229°C. When tested in the standard test engine, the fuel is found to have the same ignition characteristics as a mixture of 23% hexadecane and 77% heptamethylnonane.
Calculate: (a) Cetane number of fuel.

(b) Percent error if cetane index is used to approximate the cetane number. [%]

4-20. A CI engine running at 2400 RPM has an ignition delay of 15° of crankshaft rotation. What is the ID in seconds?

4-21. A fuel blend has a density of 720 kg/m^3 and a midpoint boiling temperature (temperature at which 50% will be evaporated) of 91°C. Calculate the cetane index.

4-22. The fuel which a coal-burning carbon monoxide generator supplies to an automobile engine consists of CO + [illegible](3.76)N_2.
Calculate: (a) HHV and LHV of fuel. [kJ/kg]

(b) Stoichiometric air-fuel ratio.

(c) Dew point temperature of exhaust. [0C]

DESIGN PROBLEMS

4-10. Using data from Table A-2 and boiling point data from chemistry handbooks, design a three-component gasoline blend. Give a three-temperature classification of your blend and draw a vaporization curve similar to Fig. 4-2. What is RON, MaN, and AKI of your blend?

4-20. An automobile will use hydrogen as fuel. Design a fuel *tank* (i.e., a fuel storage system for the vehicle) and a method to deliver the fuel from the tank to the engine.

4-30. An automobile will use propane as fuel. Design a fuel *tank* (i.e., a fuel storage system for the vehicle) and a method to deliver the fuel from the tank to the engine.

5

Air And Fuel Induction

This chapter describes intake systems of engines-how air and fuel are delivered into the cylinders. The object of the intake system is to deliver the proper amount of air and fuel accurately and equally to all cylinders at the proper time in the engine cycle. Flow into an engine is pulsed as the intake valves open and close, but can generally be modeled as quasi-steady state flow.

The intake system consists of an intake manifold, a throttle, intake valves, and either fuel injectors or a carburetor to add fuel. Fuel injectors can be mounted by the intake valves of each cylinder (multipoint port injection), at the inlet of the manifold (throttle body injection), or in the cylinder head (CI engines and modem two-stroke cycle and some four-stroke cycle SI automobile engines).

5-1 INTAKE MANIFOLD

The **intake manifold** is a system designed to deliver air to the engine through pipes to each cylinder, called *runners.* The inside diameter of the runners must be large enough so that a high flow resistance and the resulting low volumetric efficiency do not occur. On the other hand, the diameter must be small enough to assure high air velocity and turbulence, which enhances its capability of carrying fuel droplets and increases evaporation and air-fuel mixing.

The length of a runner and its diameter should be sized together to equalize, as much as possible, the amount of air and fuel that is delivered to each separate cylinder. Some engines have active intake manifolds with the capability of changing runner length and diameter for different engine speeds. At low speeds, the air is directed through longer, smaller diameter runners to keep the velocity high and to assure proper mixing of air and fuel. At high engine speeds, shorter, larger diameter runners are used, which minimizes flow resistance but still enhances proper mixing. The amount of air and fuel in one runner length is about the amount that gets delivered to one cylinder each cycle.

To minimize flow resistance, runners should have no sharp bends, and the interior wall surface should be smooth with no protrusions such as the edge of a gasket.

Some intake manifolds are heated to accelerate the evaporation of the fuel droplets in the air-fuel mixture flow. This is done by heating the walls with hot engine coolant flow, by designing the intake manifold to be in close thermal contact with the hot exhaust manifold, or sometimes with electrical heating.

On SI engines, air flow rate through the intake manifold is controlled by a throttle plate (butterfly valve) usually located at the upstream end. The throttle is incorporated into the carburetor for those engines so equipped.

Fuel is added to inlet air somewhere in the intake system before the manifold, in the manifold, or directly into each cylinder. The further upstream the fuel is added, the more time there is to evaporate the fuel droplets and to get proper mixing of the air and fuel vapor. However, this also reduces engine volumetric efficiency by displacing incoming air with fuel vapor. Early fuel addition also makes it more difficult to get good cylinder-to-cylinder AF consistency because of the asymmetry of the manifold and different lengths of the runners.

It is found that when fuel is added early in the intake system, the flow of fuel through the manifold occurs in three different manners. Fuel vapor mixes with the air and flows with it. Very small liquid fuel droplets are carried by the air flow, smaller droplets following the streamlines better than larger droplets. With a higher mass inertia, liquid particles will not always flow at the same velocity as the air and will not flow around corners as readily, larger droplets deviating more than smaller ones. The third way fuel flows through the manifold is in a thin liquid film along the walls. This film occurs because gravity separates some droplets from the flow, and when other droplets strike the wall where the runner executes a corner. These latter two types of liquid fuel flow make it difficult to deliver the same air-fuel ratio to each of the cylinders. The length of a runner to a given cylinder and the bends in it will influence the amount of fuel that gets carried by a given air flow rate. The liquid film on the manifold walls also makes it difficult to have precise throttle control. When the throttle position is changed quickly and the air flow rate changes, the time rate of change of fuel flow will be slower due to this liquid wall film.

Gasoline components evaporate at different temperatures and at different rates. Because of this, the composition of vapor in the air flow will not be exactly the same as that of the fuel droplets carried by the air or the liquid film on the manifold walls. The air-fuel mixture which is then delivered to each cylinder can be quite different, both in composition and in air-fuel ratio. One result of this is that the possibility of knock problems will be different in each cylinder. The minimum fuel octane number that can be used in the engine is dictated by the worst cylinder (i.e., the cylinder with the greatest knock problem). This problem is further complicated by the fact that the engine is operated over a range of throttle settings. At part throttle there is a lower total pressure in the intake manifold, and this changes the evaporation rate of the various fuel components. Most of these problems are reduced or eliminated by using multipoint port fuel injection, with each cylinder receiving its own individual fuel input.

5-2 VOLUMETRIC EFFICIENCY OF SI ENGINES

It is desirable to have maximum volumetric efficiency in the intake of any engine. This will vary with engine speed, and Fig. 5-1 represents the efficiency curve of a typical engine. There will be a certain engine speed at which the volumetric efficiency is maximum, decreasing at both higher and lower speeds. There are many physical and operating variables that shape this curve. These will be examined.

Fuel

In a naturally aspirated engine, volumetric efficiency will always be less than 100% because fuel is also being added and the volume of fuel vapor will displace some incoming air. The type of fuel and how and when it is added will determine how much the volumetric efficiency is affected. Systems with carburetors or throttle body injection add fuel early in the intake flow and generally have lower overall volumetric efficiency. This is because the fuel will immediately start to evaporate and fuel

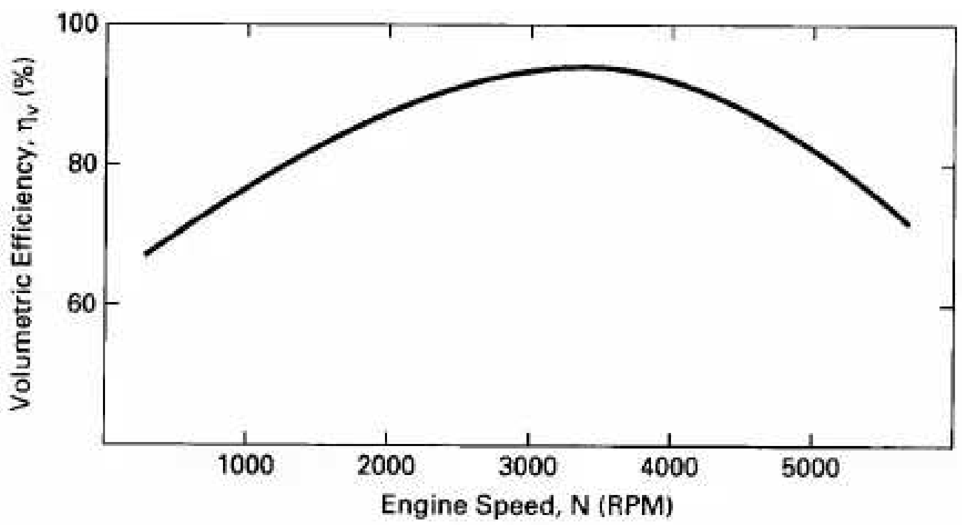


Figure 5-1 Volumetric efficiency as a function of engine speed for a typical SI engine.

vapor will displace incoming air. Multipoint injectors which add fuel at the intake valve ports will have better efficiency because no air is displaced until after the intake manifold. Fuel evaporation does not occur until the flow is entering the cylinder at the intake valve. Those engines that inject fuel directly into the cylinders after the intake valve is closed will experience no volumetric efficiency loss due to fuel evaporation. Manifolds with late fuel addition can be designed to further increase volumetric efficiency by having larger diameter runners. High velocity and turbulence to promote evaporation are not needed. They can also be operated cooler, which results in a denser inlet air flow.

Those fuels with a smaller air-fuel ratio, like alcohol, will experience a greater loss in volumetric efficiency. Fuels with high heat of vaporization will regain some of this lost efficiency due to the greater evaporation cooling that will occur with these fuels. This cooling will create a denser air-fuel flow for a given pressure, allowing for more air to enter the system. Alcohol has high heat of vaporization, so some efficiency lost due to AF is gained back again.

Gaseous fuels like hydrogen and methane displace more incoming air than liquid fuels, which are only partially evaporated in the intake system. This must be considered when trying to modify engines made for gasoline fuel to operate on these gaseous fuels. It can be assumed that fuel vapor pressure in the intake system is between 1% and 10% of total pressure when gasoline-type liquid fuel is being used. When gaseous fuels or alcohol is being used, the fuel vapor pressure is often greater than 10% of the total. Intake manifolds can be operated much cooler when gaseous fuel is used, as no vaporization is required. This will gain back some lost volumetric efficiency.

The later that fuel vaporizes in the intake system, the better is the volumetric efficiency. On the other hand, the earlier that fuel vaporizes, the better are the mixing process and cylinder-to-cylinder distribution consistency.

In older carbureted automobile engines, somewhere around 60% evaporation of the fuel in the intake manifold was considered desirable, with the rest of the evaporation taking place during the compression stroke and combustion process. If fuel is evaporated too late in the cycle, a small percent of the high-molecular-weight components may not vaporize. Some of this unvaporized fuel ends up on the cylinder walls, where it gets by the piston rings and dilutes the lubricating oil in the crankcase.

Heat Transfer-High Temperature

All intake systems are hotter than the surrounding air temperature and will consequently heat the incoming air. This lowers the density of the air, which reduces volumetric efficiency. Intake manifolds for carbureted systems or throttle body injection systems are purposely heated to enhance fuel evaporation. At lower engine speeds, the air flow rate is slower and the air remains in the intake system for a longer time. It thus gets heated to higher temperatures at low speeds, which lowers the volumetric efficiency curve in Fig. 5-1 at the low-speed end.

Some systems have been tried which inject small amounts of water into the intake manifold. This is to improve the volumetric efficiency by increasing the resulting evaporative cooling that occurs. Probably the most successful use of this principle was with large high-performance aircraft engines of World War II. Power was increased by substantial amounts when water injection was added to some of these engines.

Valve Overlap

At TDC at the end of the exhaust stroke and the beginning of the intake stroke, both intake and exhaust valves are open simultaneously for a brief moment. When this happens, some exhaust gas can get pushed through the open intake valve back into the intake system. The exhaust then gets carried back into the cylinder with the intake air-fuel charge, displacing some of the incoming air and lowering volumetric efficiency. This problem is greatest at low engine speeds, when the real time of valve overlap is greater. This effect lowers the efficiency curve in Fig. 5-1 at the low engine speed end. Other factors that affect this problem are the intake and exhaust valve location and engine compression ratio.

Fluid Friction Losses

Air moving through any flow passage or past any flow restriction undergoes a pressure drop. For this reason, the pressure of the air entering the cylinders is less than the surrounding atmospheric air pressure, and the amount of air entering the cylinder is subsequently reduced. The viscous flow friction that affects the air as it passes through the air filter, carburetor, throttle plate, intake manifold, and intake valve reduces the volumetric efficiency of the engine intake system. Viscous drag, which causes the pressure loss, increases with the square of flow velocity. This results in decreasing the efficiency on the high-speed end ofthe curve in Fig. 5-1. Much development work has been done to reduce pressure losses in air intake systems. Smooth walls in the intake manifold, the avoidance of sharp corners and bends, elimination of the carburetor, and close-fitting parts alignment with no gasket protrusions all contribute to decreasing intake pressure loss. One of the greatest flow restrictions is the flow through the intake valve. To reduce this restriction, the intake valve flow area has been increased by building multivalve engines having two or even three intake valves per cylinder.

Air-fuel flow into the cylinders is usually diverted into a rotational flow pattern within the cylinder. This is done to enhance evaporation, mixing, and flame speed and will be explained in the next chapter. This flow pattern is accomplished by shaping the intake runners and contouring the surface of the valves and valve ports. This increases inlet flow restriction and decreases volumetric efficiency.

If the diameter of the intake manifold runners is increased, flow velocity will be decreased and pressure losses will be decreased. However, a decrease in velocity

will result in poorer mixing of the air and fuel and less accurate cylinder-to-cylinder distribution. Compromises in design must be made.

In some low-performance, high fuel-efficient engines, the walls of the intake manifold are made rough to enhance turbulence to get better air-fuel mixing. High volumetric efficiency is not as important in these engines.

Choked Flow

Choked Flow

The extreme case of flow restriction is when choked flow occurs at some location in the intake system. As air flow is increased to higher velocities, it eventually reaches sonic velocity at some point in the system. This *choked flow* condition is the maximum flow rate that can be produced in the intake system regardless of how controlling conditions are changed. The result of this is a lowering of the efficiency curve on the high-speed end in Fig. 5-1. Choked flow occurs in the most restricted passage of the system, usually at the intake valve or in the carburetor throat on those engines with carburetors.

Closing Intake Valve After BDC

The timing of the closure of the intake valve affects how much air ends up in the cylinder. Near the end of the intake stroke, the intake valve is open and the piston is moving from TDC towards BDC. Air is pushed into the cylinder through the open intake valve due to the vacuum created by the additional volume being displaced by the piston. There is a pressure drop in the air as it passes through the intake valve, and the pressure inside the cylinder is less than the pressure outside the cylinder in the intake manifold. This pressure differential still exists the instant the piston reaches BDC and air is still entering the cylinder. This is why the closing of the intake valve is timed to occur aBDC. When the piston reaches BDC, it starts back towards TDC and in so doing starts to compress the air in the cylinder. Until the air is compressed to a pressure equal to the pressure in the intake manifold, air continues to enter the cylinder. The ideal time for the intake valve to close is when this pressure equalization occurs between the air inside the cylinder and the air in the manifold. If it closes before this point, air that was still entering the cylinder is stopped and a loss of volumetric efficiency is experienced. If the valve is closed after this point, air being compressed by the piston will force some air back out of the cylinder, again with a loss in volumetric efficiency.

This valve-closing point in the engine cycle, at which the pressure inside the cylinder is the same as the pressure in the intake manifold, is highly dependent on engine speed. At high engine speeds, there is a greater pressure drop across the intake valve due to the higher flow rate of air. This, plus the less real cycle time at high speed, would ideally close the intake valve at a later cycle position. On the other hand, at low engine speeds the pressure differential across the intake valve is less, and pressure equalization would occur earlier after BDC. Ideally, the valve should close at an earlier position in the cycle at low engine speeds.

The position where the intake valve closes on most engines is controlled by a camshaft and cannot change with engine speed. Thus, the closing cycle position is designed for one engine speed, depending on the use for which the engine is designed. This is no problem for a single-speed industrial engine but is a compromise for an automobile engine that operates over a large speed range. The result of this single-position valve timing is to reduce the volumetric efficiency of the engine at both high and low speeds. This is a strong argument for variable valve-timing control.

Intake Tuning

When gas flows in a pulsed manner, as in the intake manifold of an engine, pressure waves are created that travel down the length of the flow passage. The wavelength waves are created that travel down the length of the flow passage. The wavelength of these waves is dependent on pulse frequency and air flow rate or velocity. When these waves reach the end of the runner or an obstruction in the runner, they create a reflected pressure wave back along the runner. The pressure pulses of the primary waves and the reflected waves can reinforce or cancel each other, depending on whether they are in or out of phase.

If the length of the intake manifold runner and the flow rate are such that the pressure waves reinforce at the point where the air enters the cylinder through the intake valve, the pressure pushing the air in will be slightly higher, and slightly more air will enter the cylinder. When this happens, the system is **tuned** and volumetric efficiency is increaed. However, when the flow rate of air is such that the reflected pressure pulses are out of phase with the primary pulses, the pressure pushing air into the cylinder is slightly reduced and volumetric efficiency is lower. All older engines and many modern engines have passive constant-length intake runner systems that can be tuned for one engine speed (i.e., length of runner designed for one certain air flow rate and pulse timing). At other speeds the system will be out of tune, and volumetric efficiency will be less at both higher and lower engine speeds.

Some modern engines have active intake systems that can tune the manifold over a range of engine speeds. This is done by changing the length of the intake runners to match the air flow rate at various engine operating conditions. Various methods are used to accomplish this. Some systems have single-path runners that can be changed in length during operation by various mechanical methods. Other systems have dual-path runners with controlling valves and/or secondary throttle plates. As the engine speed changes, the air is directed through various-length runners which best tune the flow for that speed. All active systems are controlled by the EMS.

Exhaust Residual

During the exhaust stroke, not all of the exhaust gases get pushed out of the cylinder by the piston, a small residual being trapped in the clearance volume. The amount of this residual depends on the compression ratio, and somewhat on the location of the valves and valve overlap. In addition to displacing some incoming air, this exhaust

gas residual interacts with the air in two other ways. When the very hot gas mixes with the incoming air it heats the air, lowers the air density, and decreases volumetric efficiency. This is counteracted slightly, however, by the partial vacuum created in the clearance volume when the hot exhaust gas is in turn cooled by the incoming au.

EGR

In all modern automobile engines and in many other engines, some exhaust gas is recycled (EGR) into the intake system to dilute the incoming air. This reduces combustion temperatures in the engine, which results in less nitrogen oxides in the exhaust. Up to about 20% of exhaust gases will be diverted back into the intake manifold, depending on how the engine is being operated. Not only does this exhaust gas displace some incoming air, but it also heats the incoming air and lowers exhaust gas displace some incoming air, but it also heats the incoming air and lowers its density. Both of these interactions lower the volumetric efficiency of the engine.

In addition, engine crankcases are vented into the intake systems, displacing some of the incoming air and lowering the volumetric efficiency. Gases forced through the crankcase can amount to about 1% of the total gas flow through the engine.

5-3 INTAKE VALVES

Intake valves of most IC engines are poppet valves that are spring loaded closed and pushed open at the proper cycle time by the engine camshaft, shown schematically in Fig. 1-12. Much rarer are rotary valves or sleeve valves, found on some engines.

Most valves and valve seats against which they close are made of hard alloy steel or, in some rarer cases, ceramic. They are connected by hydromechanical or mechanical linkage to the camshaft. Ideally, they would open and close almost instantaneously at the proper times. This is impossible in a mechanical system, and slower openings and closings are necessary to avoid wear, noise, and chatter. The lobes on a camshaft are designed to give quick but smooth opening and closing without bounce at the mechanical interface. This requires some compromise in the speed of valve actuation.

Earlier engines had camshafts mounted close to the crankshaft and the valves mounted in the engine block. As combustion chamber technology progressed, valves were moved to the cylinder head (overhead valves), and a long mechanical linkage system (push rods, rocker arms, tappets) was required. This was improved by also mounting the camshaft in the engine head (i.e., overhead cam engines). Most modern automobile engines have one or two camshafts mounted in the head of each bank of cylinders. The closer the camshaft is mounted to the stems of the valves, the greater is the mechanical efficiency of the system.

The distance which a valve opens (dimension I in Fig. 5-2) is called **valve lift** and is generally on the order of a few millimeters to more than a centimeter,

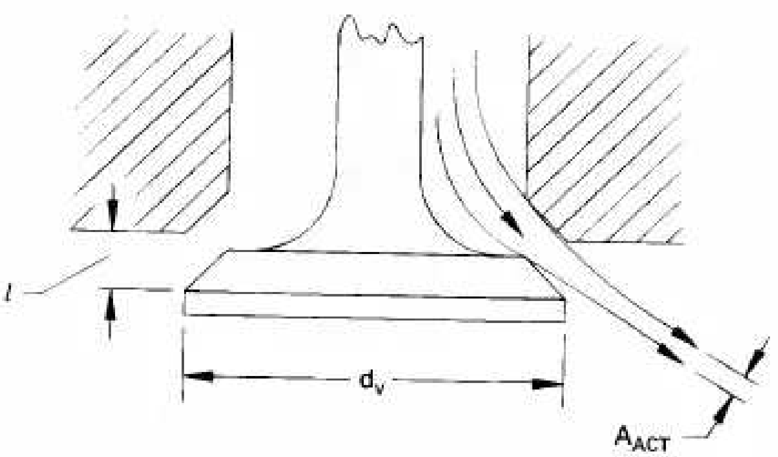


Figure 5-2 Flow through a poppet valve. When flow separates from the surface at the corners, the actual flow area is less than the geometric passage area of the valve. The ratio of these areas is called the discharge coefficient. Valve diameter is d_v and valve lift is l.

depending on the engine size, usually about 5 to 10 mm for automobile engines. Generally:

$$l_{max} < d_v/4 \qquad (5\text{-}1)$$

where: l_{max} = valve lift when valve is fully open
d_v = diameter of valve

d_v = diameter of valve

The angle of the valve surface at the interface with the valve seat is generally designed to give minimum flow restriction. As air flows around corners, the streamlines separate from the surface, and the actual cross-sectional area of flow is less than the flow passage area, as shown in Fig. 5-2. The ratio of the actual flow area to the flow passage area is called the *valve discharge coefficient*:

$$C_{Dv} = A_{act}/A_{pass} \tag{5-2}$$

The passage area of flow is:

$$A_{pass} = \pi d_v l \tag{5-3}$$

Shape and angle of valve surfaces are sometimes designed to give special mass flow patterns to improve overall engine efficiency.

Intake valves offer the greatest restriction to incoming air in most engines. This is especially true at higher engine speeds. Various empirical formulas can be found in technical engine literature for sizing intake valves. Equations giving the minimum valve intake area necessary for a modern engine can be given [40] in the form of:

$$A_i = CB^2[(\bar{U}_p)_{max}/c_i] = (\pi/4)d_v^2 \tag{5-4}$$

where: C = constant having a value of about 1.3
B = bore
$(\bar{U}_p)_{max}$ = average piston speed at maximum engine speed
c_i = speed of sound at inlet conditions
d_v = diameter of valve

Ai is the total inlet valve area for one cylinder, whether it has one, two, or three intake valves.

On many newer engines with overhead valves and small fast-burn combustion chambers, there is often not enough wall space in the combustion chambers to fit the spark plug and exhaust valve and still have room for an intake valve large enough to satisfy Eq. (5-4). For this reason, most engines are now built with more than one intake valve per cylinder. Two or three smaller intake valves give more flow area and less flow resistance than one larger valve, as was used in older engines. At the same time, these two or three intake valves, along with usually two exhaust valves, can be better fit into a given cylinder head size with enough clearance to maintain the required structural strength; see Fig. 5-3.

Multiple valves require a greater complexity of design with more camshafts and mechanical linkages. It is often necessary to have specially shaped cylinder heads and recessed piston faces just to avoid valve-to-valve or valve-to-piston contact. These designs would be difficult if not impossible without the use of computer-aided design (CAD). When two or more valves are used instead of one, the valves will be smaller and lighter. This allows the use of lighter springs and reduces forces in the linkage. Lighter valves can also be opened and closed faster. Greater voluIlli<tric efficiency of multiple valves overshadows the added cost of manufacturing and the added complexity and mechanical inefficiency.

Some engines with multiple intake valves are designed so that only one intake

Some engines with multiple intake valves are designed so that only one intake valve per cylinder operates at low speed. As speed is increased, less time per cycle is available for air intake, and the second (and sometimes third) valve actuates, giving additional inlet flow area. This allows for increased control of the flow of air within the cylinder at various speeds, which results in more efficient combustion. In some of these systems, the valves will have different timing. The low-speed valve will close at a relatively early point aBDC. When operating, the high-speed valve(s) will then

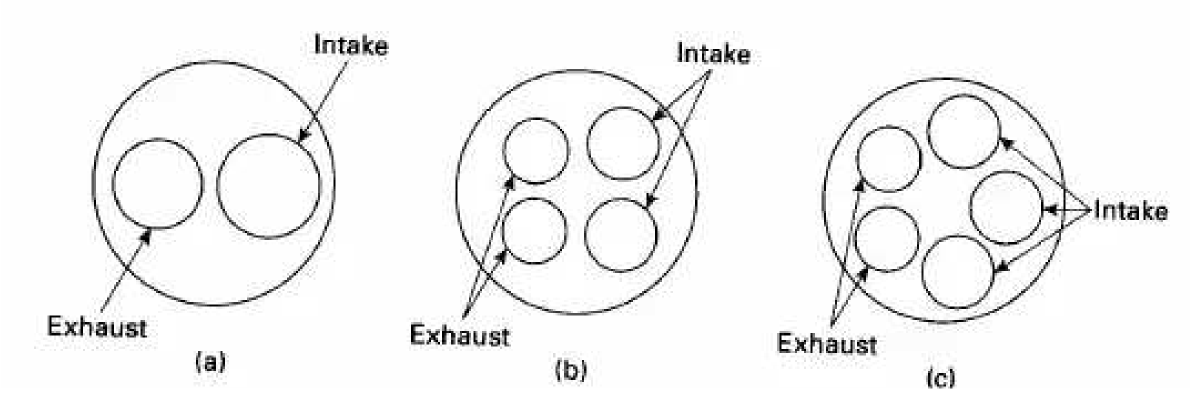


Figure 5·3 Possible valve arrangements for a modem overhead valve engine. For a given combustion chamber size, two or three smaller intake valves will give greater flow area than one larger valve. For each cylinder, the flow area of the intake valve(s) is generally about 10 percent greater than the flow area of the exhaust valve(s). (8) Most early overhead valve engines (1950s-1980s) and a few modern engines. (b) Most present-day automobile engines. (c) Some modern high-performance automobile engines.

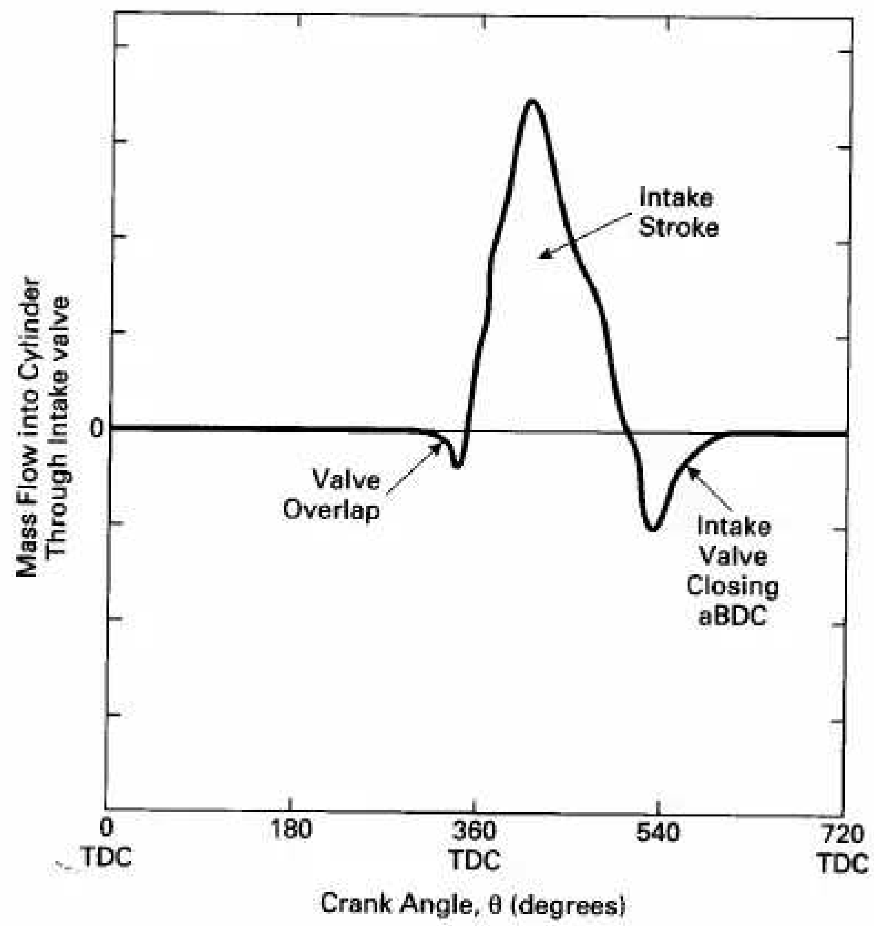


Figure 5-4 Flow of air–fuel mixture through the intake valve(s) into an engine cylinder. Possible backflow can occur during valve overlap and when the intake valve closes after BDC. Adapted from [28].